

ANOMALOUS SCALING LIMIT OF A BROWNIAN PARTICLE IN A LOG-CORRELATED POTENTIAL

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ABSTRACT. We study a Brownian particle in $\mathbb{R}^d$, $d \geq 2$, with drift given by the gradient of a log-correlated Gaussian potential. At weak disorder, we prove convergence to a scaling limit whose law is singular with respect to Brownian motion and whose invariant measure is given by Gaussian multiplicative chaos. The proof uses a renormalization group induction: at each scale, the elliptic operator is well approximated by a Laplacian with effective diffusivity decaying as a power of scale. We compute this exponent exactly in dimension two.

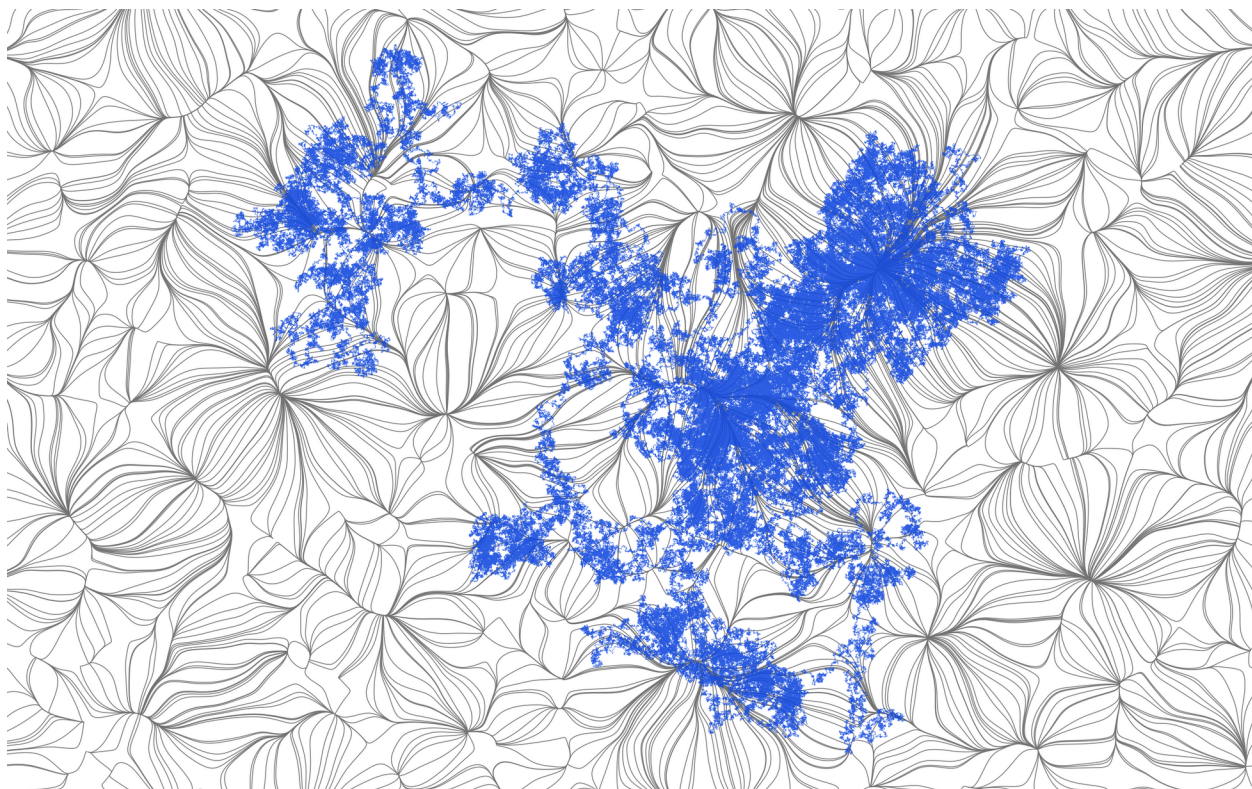


FIGURE 1. A Brownian particle advected by the gradient of a log-correlated random potential.

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2020 *Mathematics Subject Classification.* 60K37, 35R60, 82B28.

Key words and phrases. Scaling limits, stochastic homogenization, diffusions in random environment, anomalous diffusion, subdiffusion, log-correlated fields, multiplicative chaos, renormalization group.

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1. INTRODUCTION

1.1. The diffusion and its scaling limit. We study a Brownian particle in $\mathbb{R}^d$, $d \geq 2$, with drift given by the gradient of a log-correlated Gaussian potential $\log \mathbf{a}$. At weak disorder, we prove that the particle is subdiffusive and converges, after rescaling, to a diffusion whose law is singular with respect to Brownian motion.

The position X_t solves the stochastic differential equation

$$(1.1) \quad dX_t = \nabla \log \mathbf{a}(X_t) dt + \sqrt{2} dW_t, \quad X_0 = x \in \mathbb{R}^d,$$

where W is a standard Brownian motion on $\mathbb{R}^d$.

The potential $\log \mathbf{a}$ is built from independent random fields $\mathbf{g}_0, \mathbf{g}_1, \mathbf{g}_2, \dots$ at scales $1, 3, 3^2, \dots$. The field $\mathbf{g}_0$ is a smooth, stationary, centered Gaussian field with finite range of dependence: its restrictions to sufficiently distant sets are independent. Its size is measured by the **disorder strength** $\bar{\delta} \in (0, 1/2]$. The field $\mathbf{g}_k$ is a copy of $\mathbf{g}_0$ stretched by the factor 3^k : it has the law of $\mathbf{g}_0(3^{-k} \cdot)$. We set

$$(1.2) \quad \mathbf{a}(x) := \exp \left(\sum_{k=0}^{\infty} (\mathbf{g}_k(x) - \mathbf{g}_k(0)) \right).$$

The subtraction of the values at the origin makes the series converge locally in $C^{1,1}$. Each field whose scale is much smaller than $|x - y|$ takes independent values at x and y , the number of such fields grows logarithmically with $|x - y|$, and the fields whose scales are much larger than $|x - y|$ change little between x and y . Consequently, for all $x, y \in \mathbb{R}^d$ with $|x - y| \geq 2$, the variance of $\log \mathbf{a}(x) - \log \mathbf{a}(y)$ lies between two positive multiples of $\log |x - y|$ that depend only on d and the law of $\mathbf{g}_0$. An environment ω is a realization of the random fields, and we write $\mathbb{P}$ and $\mathbb{E}$ for the law and expectation of the environment. We call the law of the trajectories in a fixed environment **quenched** and its average over the environment **annealed**.

For a bounded stationary potential with sufficiently rapid decay of correlations, homogenization theory shows that, on large scales, the particle behaves like a Brownian motion with a deterministic positive diffusivity, called the effective diffusivity. Here, independent fluctuations of equal strength occur at each of the scales $1, 3, 3^2, \dots$. If the sum defining $\log \mathbf{a}$ is cut off at a large scale, the particle in the resulting potential has an effective diffusivity, and the cumulative effect of these fluctuations drives it to zero as the cutoff scale grows. We prove that it decays at least as a power of that scale, making the particle **subdiffusive**: the time needed to travel a large distance grows faster than the square of that distance.

If $\bar{\delta}$ is smaller than a threshold that depends only on d , Theorem A constructs a unique limiting diffusion Z and proves convergence to it. Let $\bar{\mathbf{a}}_N$ denote the effective diffusivity of the coefficient with infrared cutoff at scale 3^N , which keeps only the fields $\mathbf{g}_0, \dots, \mathbf{g}_N$ and is normalized to have mean one (see (1.7) and the paragraph after it), and let $\tau^2 := \log \mathbb{E}[\exp(\mathbf{g}_0(0))]$. For a particle started at the origin, the convergence and the leading weak-disorder estimate for $\log \bar{\mathbf{a}}_N$ take the informal form

$$(1.3) \quad (3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0} \longrightarrow (Z_t)_{t \geq 0} \quad \text{with} \quad \bar{\mathbf{a}}_N \approx e^{-2\tau^2(N+1)/d} \quad \text{as } N \rightarrow \infty,$$

where the displayed approximation to $\bar{\mathbf{a}}_N$ comes from an estimate of its logarithm, with error bounded in (1.9). The limit Z is a continuous, conservative, strong Markov diffusion. Its paths are rougher than Brownian paths, its law is singular with respect to Brownian motion, and its invariant measure is obtained from Gaussian multiplicative chaos: the limit, as the smoothing scale tends to

zero, of the random measures whose density is the exponential of a smoothed log-correlated field, divided by its mean. In dimension two, the estimate of $\bar{\mathbf{a}}_N$ is an identity, $\bar{\mathbf{a}}_N = e^{-(N+1)\tau^2}$, so time is rescaled by exactly $3^{2N}e^{(N+1)\tau^2}$.

The precise assumptions, which also allow non-Gaussian fields, are (g1)–(g4) below. Fix $d \geq 2$ and the disorder strength $\bar{\delta} \in (0, 1/2]$, which bounds the size of the fields in (g2). On a unit cube, the suprema of $|\mathbf{g}_0|$, $|\nabla \mathbf{g}_0|$ and $|\nabla^2 \mathbf{g}_0|$ are at most of order $\bar{\delta}$, with Gaussian tails. Let $\mathbf{g}_0, \mathbf{g}_1, \mathbf{g}_2, \dots$ be independent random fields such that, for every $k \in \mathbb{N}_0$,

$$(1.4) \quad \mathbf{g}_k \quad \text{and} \quad \mathbf{g}_0(3^{-k} \cdot) \quad \text{have the same law.}$$

For $m \in \mathbb{Z}$, let $\square_m := (-1/2 3^m, 1/2 3^m)^d$ be the cube of side 3^m ; in particular, $\square_0 = (-1/2, 1/2)^d$. We use the maximum norm on $\mathbb{R}^d$ and the dual norm on gradients; in (g2), the norm of the Hessian is the Lipschitz constant of $\nabla \mathbf{g}_0$ on $\square_0$ with respect to these norms. We assume the following.

(g1) The field $\mathbf{g}_0$ is stationary, has mean zero, and has range of dependence $\sqrt{d}$.

(g2) Almost surely, $\mathbf{g}_0 \in C_{\text{loc}}^{1,1}(\mathbb{R}^d)$, and

$$(1.5) \quad \mathbb{E} \left[\exp \left(\bar{\delta}^{-2} (\|\mathbf{g}_0\|_{L^\infty(\square_0)} + \|\nabla \mathbf{g}_0\|_{L^\infty(\square_0)} + \|\nabla^2 \mathbf{g}_0\|_{L^\infty(\square_0)})^2 \right) \right] \leq 2.$$

(g3) The law of $\mathbf{g}_0$ is invariant under signed permutations of the coordinates and under $\mathbf{g}_0 \mapsto -\mathbf{g}_0$.

(g4) The number $\tau^2 = \log \mathbb{E}[\exp(\mathbf{g}_0(0))]$ is positive.

By (g2) and the symmetry $\mathbf{g}_0 \mapsto -\mathbf{g}_0$ in (g3), there exists $C(d) < \infty$ such that

$$(1.6) \quad \tau^2 \leq C\bar{\delta}^2 \quad \text{and} \quad \left| \tau^2 - \frac{1}{2} \mathbb{E}[|\mathbf{g}_0(0)|^2] \right| \leq C\bar{\delta}^4.$$

A statement holds **at weak disorder** if it holds whenever $\bar{\delta} \leq \bar{\delta}_0$, for a threshold $\bar{\delta}_0 > 0$ depending only on d and the indicated parameters.

Throughout, X denotes the continuous Feller diffusion with generator $\mathbf{a}^{-1} \nabla \cdot (\mathbf{a} \nabla)$ that we construct in Section 9.1 from its resolvent; there we also show that, from every starting point, its law is that of the solution of (1.1). It is reversible with respect to $\mathbf{a}(x) dx$.

For $L \in \mathbb{N}_0$, the coefficient with **infrared cutoff** at scale 3^L retains only the fields with scales at most 3^L , and we call 3^L the **cutoff scale**:

$$(1.7) \quad \mathbf{a}_L(x) := \exp \left(\sum_{k=0}^L (\mathbf{g}_k(x) - \tau^2) \right).$$

The coefficient $\mathbf{a}_L$ is stationary, has range of dependence of order 3^L , and has mean one. Subtracting τ^2 , rather than the values at the origin as in (1.2), normalizes the mean and leaves the drift unchanged. Let $X^{(L)}$ be the diffusion constructed in the same way with $\mathbf{a}_L$ in place of $\mathbf{a}$; its generator is

$$(1.8) \quad \mathbf{a}_L^{-1} \nabla \cdot (\mathbf{a}_L \nabla) = \Delta + \nabla \log \mathbf{a}_L \cdot \nabla.$$

The homogenized matrix of $\mathbf{a}_L$ is the constant matrix such that, in domains much larger than 3^L , solutions of $-\nabla \cdot (\mathbf{a}_L \nabla u) = f$ are close to solutions of the same equation with this constant matrix in place of $\mathbf{a}_L$. By the symmetry (g3), it is a scalar multiple of the identity. We denote this scalar by $\bar{\mathbf{a}}_L$ and call it the **effective diffusivity**. It measures how fast the particle $X^{(L)}$ moves on the cutoff scale: at weak disorder, when started at the center of a cube of side 3^L , it takes time of order $3^{2L}/\bar{\mathbf{a}}_L$ to leave the cube, on average over the environment and the trajectories. The same holds for the particle X without cutoff (Lemma 10.2). This is why (1.3) rescales time by $3^{2N}/\bar{\mathbf{a}}_N$ when it rescales space by 3^{-N} .

Theorem 5.1 shows that $\bar{\mathbf{a}}_m$ is nonincreasing in m and that, for every $m \in \mathbb{N}_0$, it is at least $e^{-(m+1)\tau^2}$ and at most a negative power of 3^{m+1} whose exponent depends only on d and the

law of $\mathbf{g}_0$. We also estimate the logarithm of $\bar{\mathbf{a}}_m$: there exist $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then, for every $m \in \mathbb{N}_0$,

$$(1.9) \quad \left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

In dimension two, planar duality gives the exact identity $\bar{\mathbf{a}}_m = e^{-(m+1)\tau^2}$ for every $m \in \mathbb{N}_0$: rotation by a right angle exchanges gradient fields and divergence-free fields, which makes the homogenized matrices of $\mathbf{a}_m$ and $\mathbf{a}_m^{-1}$ inverse to each other, and $\mathbf{a}_m^{-1}$ has the law of $e^{2(m+1)\tau^2} \mathbf{a}_m$ by the symmetry $\mathbf{g}_0 \mapsto -\mathbf{g}_0$. When $m\bar{\delta} \geq 1$ and τ^2 is at least a fixed positive multiple of $\bar{\delta}^2$, the error in (1.9) is smaller than the leading term by a factor of order $\bar{\delta} |\log \bar{\delta}|$.

Theorem A (Scaling limit). *Assume (g1)–(g4). There exists $\bar{\delta}_0(d) > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there is a continuous strong Markov process Z on $\mathbb{R}^d$ in a random environment, with quenched laws $\mathbf{P}_x$ and expectations $\mathbf{E}_x$ when started at x , such that the following statements hold.*

- (i) Convergence: For every $x \in \mathbb{R}^d$, as the integer N tends to infinity, the annealed law of the process $(3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0}$ with $X_0 = 3^N x$ converges to the annealed law of Z started at x .
- (ii) Reversible measure: Almost surely, Z is reversible with respect to a measure μ that is singular with respect to Lebesgue measure.
- (iii) Anomalous scaling: There exist $C < \infty$ and $\eta > 0$, depending only on d and the law of $\mathbf{g}_0$, such that, if σ_r denotes the first time at which Z leaves the cube of side r centered at its starting point, then, for every $x \in \mathbb{R}^d$ and $r \in (0, 1]$,

$$\mathbb{E}[\mathbf{E}_x[\sigma_r]] \leq Cr^{2+\eta}.$$

For every $x \in \mathbb{R}^d$, almost surely, $\mathbf{P}_x$ -almost every path satisfies

$$\limsup_{t \downarrow 0} t^{-\gamma} |Z_t - Z_0| = \infty \quad \text{for every } \gamma > \frac{1}{2+\eta}.$$

In dimension two, one can take $\eta = \tau^2 / \log 3$.

- (iv) Non-Gaussian marginals: Almost surely, for every $x \in \mathbb{R}^d$ and $t > 0$, the law of Z_t under $\mathbf{P}_x$ is absolutely continuous with respect to μ and is not Gaussian.

Theorem A is a simplified form of Theorem 7.1. There, all the rescaled processes are realized in a single environment, and their quenched laws converge almost surely. Since rescaling space by 3^{-N} turns the field with scale 3^k into a field with scale 3^{k-N} , this environment consists of independent fields at all scales 3^j with $j \in \mathbb{Z}$. The measure μ is the almost-sure limit of the reversible measures of the rescaled processes, suitably normalized. Apart from a continuous positive factor that comes from the fields with scales larger than one, it is stationary; for Gaussian fields, this stationary part is a Gaussian multiplicative chaos measure.

Brownian motion needs a time of order r^2 to leave a cube of side r . The slow motion of X over large distances corresponds to fast motion of Z over small distances: by (iii), the expected time that Z needs to leave a small cube is much smaller than the square of its side. These rapid exits make the paths of Z rougher than Brownian paths: their pointwise Hölder exponent at time zero is at most $1/(2+\eta)$, below the Brownian value $1/2$. In particular, by the law of the iterated logarithm, $\mathbf{P}_x$ is singular with respect to every mixture of laws of Brownian motions. If $\mathbf{g}_0$ is Gaussian, then $\tau^2 = 1/2 \text{Var}[\mathbf{g}_0(0)]$, so in dimension two this exponent is at most $(2 + \text{Var}[\mathbf{g}_0(0)]/(2 \log 3))^{-1}$.

1.2. Quantitative homogenization. We compare solutions of the elliptic equation for the cutoff coefficient divided by its effective diffusivity with solutions of the same problem for the Laplacian. The error remains controlled as the cutoff grows, although the ratio of the supremum to the infimum

of the coefficient on the domain, which controls the constants in classical elliptic estimates, grows with the cutoff. For $0 \leq L \leq M$, define

$$A_{L,M}(x) := \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L(3^M x), \quad x \in \square_0.$$

The coefficient $A_{L,M}$ is stationary, its homogenized matrix is the identity, and its range of dependence is of order 3^{L-M} . For $f \in L^2(\square_0)$ and $h \in H^2(\square_0)$, let $u_{L,M}, u_{\text{hom}} \in H^1(\square_0)$ solve the problems with the same source and boundary values:

$$\begin{cases} -\nabla \cdot (A_{L,M} \nabla u_{L,M}) = f & \text{in } \square_0, \\ u_{L,M} = h & \text{on } \partial \square_0, \end{cases} \quad \begin{cases} -\Delta u_{\text{hom}} = f & \text{in } \square_0, \\ u_{\text{hom}} = h & \text{on } \partial \square_0, \end{cases}$$

and measure the homogenization error by

$$\mathcal{D}_{L,M} := \|u_{L,M} - u_{\text{hom}}\|_{L^2(\square_0)} + \|\nabla u_{L,M} - \nabla u_{\text{hom}}\|_{H^{-1}(\square_0)} + \|A_{L,M} \nabla u_{L,M} - \nabla u_{\text{hom}}\|_{H^{-1}(\square_0)}.$$

Both $u_{L,M}$ and $\mathcal{D}_{L,M}$ depend on the data f and h , which we suppress from the notation. For smooth data, $u_{L,M}$ has a probabilistic interpretation in terms of the diffusion with generator $\nabla \cdot (A_{L,M} \nabla)$, which is a time change of the particle $X^{(L)}$ with space rescaled by 3^{-M} . With $f = 1$ and $h = 0$, $u_{L,M}$ is the expected exit time from $\square_0$; with $f = 0$, it is the expected value of h at the exit position.

Theorem B (Quantitative homogenization of the Dirichlet problem). *Assume (g1)–(g4). The following statements hold.*

- (i) Estimate uniform in the cutoff: For every $\vartheta \in (0, 1)$ and $q \in [1, \infty)$, there exist $\bar{\delta}_0(\vartheta, q, d) > 0$ and $C(\vartheta, q, d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there exist random variables $\mathcal{Z}_{L,M} \geq 1$ satisfying

$$\sup_{0 \leq L \leq M} \|\mathcal{Z}_{L,M}\|_{L^q(\mathbb{P})} \leq C$$

such that, almost surely, for every L, M, f and h ,

$$\mathcal{D}_{L,M} \leq \mathcal{Z}_{L,M} \bar{\delta}^\vartheta (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

If $f = 0$, then the factor $\bar{\delta}^\vartheta$ may be replaced by $C \bar{\delta}$.

- (ii) Algebraic rate at a fixed cutoff: There exists $\alpha_{\text{hom}}(d) > 0$ such that, for every $L \in \mathbb{N}_0$ and $q \in [1, \infty)$, there exists a random variable $\mathcal{Y}_{L,q} \geq 1$ with $\|\mathcal{Y}_{L,q}\|_{L^q(\mathbb{P})} \leq C(L, q, d, \bar{\delta})$ such that, almost surely, for every $M \geq L$ and every f and h ,

$$\mathcal{D}_{L,M} \leq \mathcal{Y}_{L,q} 3^{-\alpha_{\text{hom}}(M-L)} (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

In (i), the error is small because the disorder is weak, not because the domain is large: the estimate covers the case $M = L$, in which the range of dependence is comparable to the size of the domain, and it is uniform in L . Part (ii) does not assume weak disorder: at a fixed cutoff, enlarging the domain gives algebraic decay of the error, with a prefactor that may depend on L and $\bar{\delta}$. Section 6 proves the estimate in (i) in Theorem 6.18 and the fixed-cutoff rate in Theorem 6.19.

In probabilistic terms, Theorem B compares these expected exit times and exit distributions with their Brownian counterparts in $L^2(\square_0)$. The generator of the particle in (1.1) also contains the factor $\mathbf{a}^{-1}$: the process is determined by its Dirichlet form, the energy $\int \mathbf{a} |\nabla u|^2 dx$, together with its reversible measure $\mathbf{a} dx$. Its scaling limit therefore requires the convergence of both the energies and the reversible measures of the rescaled processes. Theorem B does not give the convergence of the energies, since its bound in (i) does not tend to zero as the cutoff grows, and the proof of Theorem A does not use it (see Figure 1.1).

1.3. Large-scale regularity and a Liouville theorem. To show that the energies of the rescaled diffusions have subsequential limits and that all these limits coincide, we need to control the oscillation of $\mathbf{a}_L$ -harmonic functions, that is, of solutions of $\nabla \cdot (\mathbf{a}_L \nabla u) = 0$, and the energy that they carry on small cubes. We obtain this control uniformly in L , with Hölder exponents approaching one as $\bar{\delta} \rightarrow 0$. For a fixed exponent, the estimates compare a cube on which the function is $\mathbf{a}_L$ -harmonic with much smaller subcubes; the number of scales that must separate the two is random, with a tail bound uniform in the cutoff. For a measurable set U of finite positive volume and $p \in [1, \infty)$, write $(u)_U := \oint_U u$ and $\|u\|_{\underline{L}^p(U)} := (\oint_U |u|^p)^{1/p}$.

Theorem C (Large-scale Hölder regularity and a Liouville theorem). *Assume (g1)–(g4). There exist $\bar{\delta}_0(d) > 0$, $C_0(d) < \infty$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then the following statements hold with*

$$(1.10) \quad \gamma_{\text{reg}} := 1 - C_0 \bar{\delta} |\log \bar{\delta}|^{1/2}.$$

In both statements, L ranges over $\mathbb{N}_0 \cup \{\infty\}$, and $\mathbf{a}_L$ is read as $\mathbf{a}$ when $L = \infty$.

(i) Large-scale Hölder estimate: *Let $\gamma \in [1/2, \gamma_{\text{reg}}]$. For every L and $m \in \mathbb{N}$, there exists an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ such that, for every $k \in \mathbb{N}$,*

$$(1.11) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C \bar{\delta}^2 |\log \bar{\delta}|}\right).$$

Moreover, almost surely, for every L and m , if $u \in H^1(\square_m)$ is $\mathbf{a}_L$ -harmonic in $\square_m$, then, for every $n \in \mathbb{N}_0$ satisfying $n \leq m - \mathcal{L}_L(\gamma, m)$ and every $z \in 3^n \mathbb{Z}^d$ with $z + \square_n \subseteq \square_{m-1}$,

$$(1.12) \quad \|u - (u)_{z+\square_n}\|_{\underline{L}^2(z+\square_n)} \leq C 3^{-\gamma(m-n)} \|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)}$$

and

$$(1.13) \quad \left\| \mathbf{a}_L^{1/2} \nabla u \right\|_{\underline{L}^2(z+\square_n)} \leq C 3^{(1-\gamma)(m-n)} \left\| \mathbf{a}_L^{1/2} \nabla u \right\|_{\underline{L}^2(\square_m)}.$$

(ii) Liouville theorem: *Almost surely, for every L , if $u \in H_{\text{loc}}^1(\mathbb{R}^d)$ satisfies*

$$-\nabla \cdot (\mathbf{a}_L \nabla u) = 0 \quad \text{in } \mathbb{R}^d \quad \text{and} \quad \liminf_{R \rightarrow \infty} R^{-\gamma_{\text{reg}}} \inf_{c \in \mathbb{R}} \|u - c\|_{\underline{L}^2(B_R)} = 0,$$

then u is constant.

The energy estimate (1.13) prevents a fixed proportion of the energy from concentrating on a small number of the small cubes. More precisely, if the energy in $\square_m$ is positive, then every family of cubes $z + \square_n$, with n and z as in (1.13), that carries a fixed positive proportion of that energy contains at least a positive multiple of $3^{(d-2(1-\gamma))(m-n)}$ cubes, where the multiple depends on d and on that proportion. Taking $\gamma = \gamma_{\text{reg}}$ makes the exponent in this count approach d as $\bar{\delta} \rightarrow 0$.

Theorem C is proved as Theorem 6.21 in Section 6.

1.4. Comparison with previous work. Exponentials of log-correlated Gaussian fields are the basic objects of Gaussian multiplicative chaos, introduced by Kahane [1985]; see Rhodes and Vargas [2014] and Berestycki [2017] for modern accounts and Duplantier, Rhodes, Sheffield, and Vargas [2017] for an overview of log-correlated fields. Random walks and diffusions in random force fields, and the anomalous diffusion they produce, were studied in the physics literature by Fisher [1984], Kravtsov, Lerner, and Yudson [1985], Fisher, Friedan, Qiu, Shenker, and Shenker [1985] and Bouchaud, Comtet, Georges, and Le Doussal [1987]; see Bouchaud and Georges [1990] for a review. For drifts given by gradients of log-correlated Gaussian potentials, renormalization group computations predicted a subdiffusive power law with an exponent depending on the disorder strength; see Bouchaud and Georges [1990, Section 4.3.2.3].

For Gaussian fields, in the normalization of (1.1), the variance of $\log \mathbf{a}(x) - \log \mathbf{a}(y)$ is asymptotic to $4\tau^2 \log |x-y| / \log 3$ as $|x-y| \rightarrow \infty$, and the weak-disorder formula of Castillo and Le Doussal [2001,

equation (3)] predicts that the particle needs a time of order $r^{2+2\tau^2/(d\log 3)}$ to travel a distance r . The estimate (1.9) gives this leading exponent for the time scale $3^{2N}/\bar{\mathbf{a}}_N$ in (1.3), and in dimension two the identity $\bar{\mathbf{a}}_N = e^{-(N+1)\tau^2}$ gives exactly the predicted exponent $2 + \tau^2/\log 3$.

Biskup, Ding, and Goswami [2020] studied the planar lattice analogue of (1.1), a random walk among conductances given by exponentials of the discrete Gaussian free field, and proved recurrence, return-probability estimates, and exponents for expected exit times. When the field is multiplied by a small constant, Ding and Wang [2026] proved tightness of the random quenched path laws for a version of this walk whose conductances are determined by a smooth approximation to the field. In both models, the walk chooses a neighbor with probability proportional to the conductance of the edge joining them, and the reversible measure gives each vertex the sum of the conductances of its incident edges. The field therefore affects both the transition probabilities and the reversible measure, as in the continuum diffusion (1.1).

For the walk studied by Biskup et al. [2020], Biskup [2020, Problem 16.13] formulated the scaling-limit problem through a continuum diffusion whose drift is the gradient of a smooth approximation to the planar Gaussian free field, and asked for joint convergence of the field and the diffusion, after a suitable change of time, to a field and a process with continuous, nonconstant paths. Theorem A proves existence and uniqueness of a scaling limit for the continuum diffusion (1.1) with log-correlated potentials satisfying (g1)–(g4), at weak disorder and in every dimension $d \geq 2$.

Liouville Brownian motion, constructed by Garban, Rhodes, and Vargas [2016] and by Berestycki [2015], is a time change of planar Brownian motion using the Gaussian multiplicative chaos measure of the planar Gaussian free field. Its Dirichlet form is the unweighted energy $\int |\nabla u|^2 dx$ on the L^2 space of that measure. For a measure with a smooth positive density $\mathbf{a}$, this construction gives the generator $\mathbf{a}^{-1}\Delta$, so the field changes only the time parametrization of the Brownian trajectories. The diffusion (1.1) instead has energy $\int \mathbf{a}|\nabla u|^2 dx$ on $L^2(\mathbf{a} dx)$ and generator $\mathbf{a}^{-1}\nabla \cdot (\mathbf{a}\nabla)$, so the field also enters the energy. Identifying the limit of these weighted energies is the main step of this paper.

Armstrong, Bou-Rabee, and Kuusi [2026] proved power-law superdiffusion for a diffusion in a self-similar incompressible random drift with sufficiently small positive Hurst exponent, using an induction across scales for its elliptic operator. We use the same deterministic coarse-graining framework, in which the constants in elliptic estimates on a cube are given not by pointwise bounds on the coefficient but by two coarse-grained matrices, effective coefficients of the cube defined by variational problems, and we normalize these estimates by the effective diffusivity at each scale. The incompressible drift increases the effective diffusivity; the gradient drift reduces it. In addition to estimates across scales, the present paper constructs a scaling limit.

1.5. Overview of the proofs. The proof has three parts. The first is a renormalization group analysis of the divergence-form elliptic operators with the cutoff coefficients: by induction across scales, we show that, on cubes whose side is the cutoff scale, the operator is close to the effective diffusivity times the Laplacian, uniformly in the cutoff, although the pointwise ellipticity of the coefficient deteriorates as the cutoff grows. The second part, the central step of the paper, shows that the energies of the rescaled diffusions have a unique limit. This limit is a random Dirichlet form for which there is no formula, so we cannot identify it by computing it. Instead, we compare two subsequential limits through the optimal constants in a two-sided bound between them: unless the two forms are proportional, replacing a solution for one form by harmonic functions for the other on small cubes would improve these constants. The third part pairs the limit form with the limit of the reversible measures and proves convergence of the processes. In the list below, steps (1)–(3) form the first part, step (4) the second and step (5) the third, and step (6) derives the properties of the limit.

- (1) **Coarse-graining (Section 2).** We recall elliptic estimates whose constants are given by coarse-grained matrices instead of pointwise bounds on the coefficient.
- (2) **Induction across scales (Sections 3–5).** On a cube whose side is the cutoff scale, the coarse-grained matrices of the cutoff coefficient are close to the effective diffusivity times the identity, uniformly in the cutoff. Perturbative estimates for the effective diffusivities give (1.9), and a separate argument proves their power-law decay without assuming weak disorder.
- (3) **Large-scale regularity and Dirichlet homogenization (Section 6).** Approximation of solutions by harmonic functions on the scales where it holds, together with a bound on the proportion of scales where it fails, gives Theorem C and bounds the energy that a solution carries in small cubes. Theorem B, also proved there, follows from the estimates of steps (1) and (2) and does not need the regularity estimates.
- (4) **Uniqueness of the limit form (Section 8).** Two subsequential limits of the energies bound each other by deterministic constants. The optimal constants in this comparison coincide, so the limits are proportional, and homogenization on large cubes shows that the proportionality factor is one.
- (5) **Convergence of the process (Section 9).** The unique limit form, the limiting reversible measure and tightness give convergence of the path laws in probability. Comparing the energies at two finite cutoffs gives estimates with summable error probabilities, and the Borel–Cantelli lemma upgrades this to almost-sure convergence.
- (6) **Properties of the limit (Section 10).** Power-law decay of the effective diffusivities gives the exit-time bounds and the singularity of the path laws with respect to Brownian motion. The transition laws are absolutely continuous with respect to μ , which is singular with respect to Lebesgue measure and charges no hyperplane, so no transition law is Gaussian.

Figure 1.1 records how the main results depend on each other. The appendices contain moment bounds, concentration inequalities and comparisons of Besov norms.

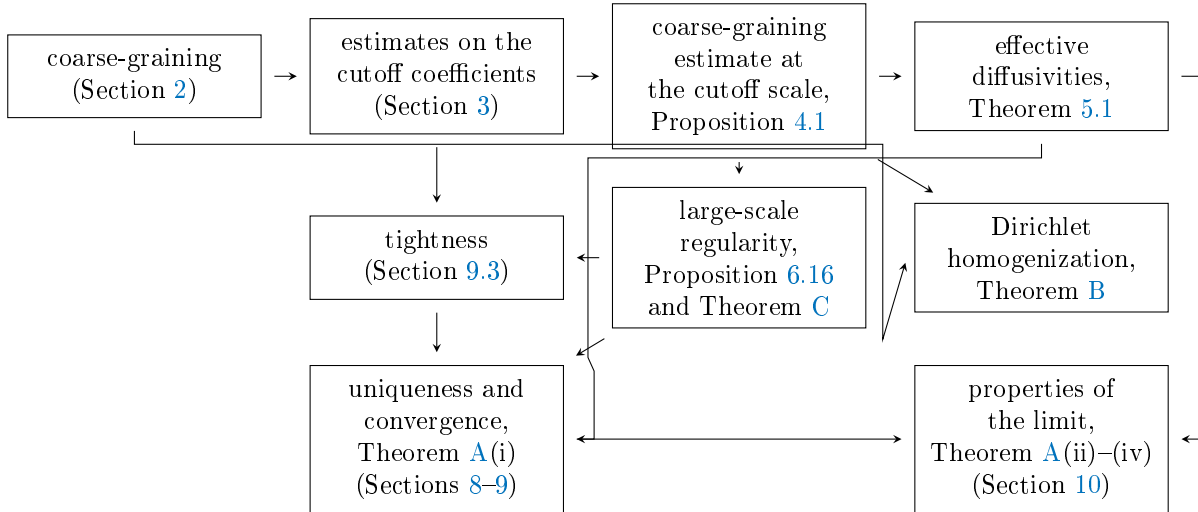


FIGURE 1.1. Logical dependence of the main results. An arrow means that the first result is used in the proof of the second.

1.5.1. *Coarse-grained matrices.* The ratio of the supremum to the infimum of the cutoff coefficient on a cube whose side is the cutoff scale grows with the cutoff, so elliptic estimates based on pointwise bounds give no control uniform in the cutoff. We use instead the two **coarse-grained matrices** of a cube, following the deterministic theory of Armstrong and Kuusi [2025] and Armstrong et al.

[2026] recalled in Section 2. The first gives the minimal energy of a function with affine boundary values of a prescribed slope. The inverse of the second gives the energy of the solution of the Neumann problem whose boundary flux is that of a prescribed constant vector field. For a constant coefficient, both matrices equal the coefficient. For a variable coefficient, their difference controls the error in approximating solutions by solutions of a constant-coefficient equation. The coarse-grained matrices of subcubes control the constants in Poincaré and Caccioppoli inequalities and in harmonic approximation. Bounding them relative to the effective diffusivity on cubes whose side is the cutoff scale gives estimates that remain useful as the effective diffusivity tends to zero.

1.5.2. Induction across scales and the effective diffusivities. Proposition 4.1 shows that, for each fixed $p < \infty$ and for $\bar{\delta}$ small enough depending on p , both coarse-grained matrices of $\mathbf{a}_m$ on $\square_m$ are within a relative distance of order $\bar{\delta}$ of $\bar{\mathbf{a}}_m \mathbf{I}_d$ in $L^p(\mathbb{P})$, uniformly in m . We prove it by induction on the scale. Suppose that the estimate holds for $\mathbf{a}_L$ on cubes of side 3^L , and let $m > L$. On $\square_m$, the coefficient $\mathbf{a}_L$ still has range of dependence of order 3^L , so its restrictions to well-separated subcubes are independent, and averaging over these subcubes contracts the error, the gap between the two coarse-grained matrices. The contraction comes from the variational structure of these matrices, as in Armstrong, Kuusi, and Mourrat [2019]. The error on a cube is bounded by two terms. The first consists of the spatial averages of the gradient of an optimizer over subcubes, measured in a negative Sobolev norm; it is small because the coefficient is independent on well-separated subcubes. The second is the decrease of the error from the subcubes to the cube. Since the expected error does not increase with the scale, these decreases add up to a bounded quantity, so the second term is small on most scales. The estimates of Armstrong and Kuusi [2025] bound both terms with constants given by coarse-grained matrices. We then add the fields $\mathbf{g}_{L+1}, \dots, \mathbf{g}_m$. They vary on scales up to 3^m and need not be nearly constant on $\square_m$, but at weak disorder the ratio $\mathbf{a}_m/\mathbf{a}_L$, normalized by $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_L$, is close to one in the moments that control the coarse-grained matrices. On a subcube, the fields whose scales are much larger than the subcube are nearly constant, and multiplying a coefficient by a constant multiplies both coarse-grained matrices by that constant. We choose $m - L$ so that the contraction absorbs these losses. The independence of the fields and the equality of laws (1.4) then let the argument repeat at the next scale with the same constants.

To estimate the effective diffusivities, we add the fields a block of scales at a time. To first order in the disorder strength, the effect of an added block on the variational problems that determine the effective diffusivity is given by the solution of a Poisson equation forced by the added fields, and the energy of this solution gives the leading decrease $2\tau^2/d$ per scale of the logarithm of the effective diffusivity. This gives (1.9), and in dimension two planar duality gives the exact formula. The perturbative error need not be smaller than the leading term for every field satisfying (g1)–(g4), so Proposition 5.6 proves power-law decay by a separate variational argument that does not assume weak disorder: since $\tau^2 > 0$, a single field lowers the expected minimal energy with affine boundary values below that of the affine function by a fixed factor smaller than one, and these factors multiply over scales separated by a fixed gap.

1.5.3. Large-scale regularity and homogenization. Theorem C is proved by comparison with harmonic functions. On a scale where the coefficient is accurately approximated by a constant matrix, as measured through the coarse-grained matrices, a solution is close to a harmonic function, which is close to an affine function on smaller cubes, so the distance of the solution to affine functions decreases from that scale to the next. On the other scales this distance can increase, but only by a bounded factor. The approximation errors on different scales are not independent, since they depend on the same fields. We call a scale **good** when the fields, and the energies of the variational problems that define the coarse-grained matrices, are small, with a tolerance that grows geometrically with the distance of their scale from the given scale. Independence of the fields then gives a concentration estimate: outside an event of exponentially small probability, only a small proportion of scales fail to be good. The improvement on good scales absorbs the losses on the others, which

gives the Hölder exponent in Theorem C, and the concentration estimate gives the tail bound (1.11). On cubes larger than the cutoff scale, no further fields enter the coefficient, so averaging over larger cubes only improves the approximation; hence the estimates are uniform in the cutoff.

The large-scale Hölder estimate and the Caccioppoli inequality bound the energy that a solution carries in small cubes. In particular, thin strips around the faces of a cube carry little energy, which we use repeatedly in Section 8. Theorem B does not use the regularity estimates: it follows from the coarse-graining estimates applied to the whole domain.

1.5.4. Uniqueness of the limit form. For each N , the rescaled process in (1.3) is reversible with respect to a measure μ_N whose density is proportional to the rescaled coefficient, and its Dirichlet form is $\int A_N |\nabla u|^2 dx$ on $L^2(\mu_N)$, where A_N is the density of μ_N divided by $\bar{\mathbf{a}}_N$. Apart from a factor that comes from the fields with scales larger than one and does not depend on N , the coefficient A_N has the law of $\bar{\mathbf{a}}_N^{-1} \mathbf{a}_N(3^N \cdot)$, as in the definition of $A_{L,M}$ with $L = M = N$; we therefore call N the cutoff. The energy determines the elliptic operator, and the reversible measure determines the time parametrization of the paths. The measures μ_N converge almost surely, by the construction of multiplicative chaos due to Kahane [1985]. Apart from the same factor, the masses that μ_N gives to a fixed cube form a nonnegative martingale in N , because the field added at each step multiplies the density by an independent factor with mean one. At weak disorder, a moment of order larger than one of these masses is bounded uniformly in N , so no mass is lost in the limit. The problem is therefore to identify the limit of the energies.

We first prove compactness. Resampling a field whose scale is much smaller than a cube changes the minimal energy on that cube very little. To see this, we partition the cube into smaller boxes and set the field to zero on thin strips around their faces. The parts of the boxes outside the strips are then farther apart than the range of dependence of the field, so the modified energy depends on the field only through its restrictions to these parts, which are independent. The Efron–Stein inequality bounds the variance of a function of independent random variables by half the sum of the expected squared changes caused by resampling them one at a time. Here it bounds the fluctuations by the energy in the boxes, and the regularity estimates make this bound and the energy in the strips decay as a power of that scale. Fields much coarser than the cube are nearly constant on it and act as multipliers. Hence a minimal energy is close, uniformly in the cutoff, to its conditional expectation given finitely many fields, and these conditional expectations form a compact family in $L^p(\mathbb{P})$ (Section 8.3). The subsequential limits of the Dirichlet Green operators, which map a source f to the solution of $-\nabla \cdot (A_N \nabla u) = f$ in a cube with zero boundary values, define Dirichlet forms, which are measurable with respect to the fields.

Let E and F be two such limits. On a small cube, we replace each field coarser than the cube by its value at the center of the cube; this multiplies the restrictions of both forms to the cube by the same random factor. Rescaling the cube, of side 3^{-k} , to unit size then turns the energy at cutoff N into an energy at cutoff $N - k$, multiplied by this random factor, by $3^{-(d-2)k}$ and by the deterministic factor $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$. The limits of these deterministic factors along the two subsequences that define E and F may differ, with a ratio that depends on k . Suppose that this ratio were small on a positive proportion of scales. On cubes of those scales, we could then replace a solution for E by F -harmonic functions, that is, by minimizers of the F -energy with the same boundary values. This would bound the F -energy of the solution by a small multiple of its E -energy, which by duality contradicts the bounds, uniform in the cutoff, on the positive and negative moments of the energy of the solution with a fixed source. Interchanging E and F excludes large ratios. On the remaining scales the ratio is bounded above and below, and the same replacement argument gives the two-sided comparison and equality of the domains (Theorem 8.41). This adapts the normalization argument of Ding and Gwynne [2023a, Section 3.4].

Consider the optimal constants in the two-sided comparison of E and F : the largest lower constant and the smallest upper constant. The energy measure of a function distributes its energy

over space; at a finite cutoff it is $A_N |\nabla u|^2 dx$. Resampling one of the fields multiplies the energy measures of both forms by the same weight, and so leaves these constants unchanged; hence they are deterministic. Suppose that they differ. On a small cube with affine boundary values, the minimal F -energy is close to a deterministic multiple of the minimal E -energy; the multiple lies between the two constants and may depend on the scale. On a positive proportion of scales, the multiple lies on the same side of their average, say below it. On a cube of such a scale, with affine boundary values, the minimal F -energy then falls below the upper constant times the minimal E -energy by at least a fixed fraction of the difference of the constants times the minimal E -energy. For any boundary values, the minimal F -energy lies between the lower and the upper constant times the minimal E -energy. Hence the amount by which the minimal F -energy falls below the upper constant times the minimal E -energy is a nonnegative quadratic form in the boundary values that is at most the difference of the constants times the minimal E -energy. Therefore, for the boundary values of a solution that is close to affine on the cube, this amount is close to its value for affine boundary values, up to an error proportional to the difference of the constants. Replacing a solution u for E by F -harmonic functions on such cubes then lowers the F -energy below the upper constant times $E(u)$ by a fixed fraction of the difference of the constants times $E(u)$, and lower semicontinuity bounds $F(u)$ by $E(u)$ times the upper constant lowered by a fixed multiple of the difference of the constants. This contradicts the optimality of the upper constant. If instead the multiple lies above the average on a positive proportion of scales, we interchange the forms and improve the inverse of the lower constant (Theorem 8.51).

Two difficulties remain. First, that amount is proportional to the difference of the optimal constants, so every error must be a small multiple of this difference, not merely small as $\bar{\delta} \rightarrow 0$. Separate concentration estimates for the two energies do not give this. We instead apply the Efron–Stein bound of Section 8.3 to the difference between the minimal F -energy and a multiple of the minimal E -energy. The E -energy of the difference of the two minimizers is at most the square of the difference of the optimal constants, divided by the square of the lower constant, times the minimal E -energy (Lemma 8.43), so the resampled quantity fluctuates by a small multiple of the difference of the constants (Proposition 8.45). Second, the cubes used for the replacements must carry a fixed fraction of the energy of u . This energy depends on the same fields as the estimates of Lemma 8.33, namely Hölder regularity, energy bounds for Hölder boundary values and harmonic approximation, so it might concentrate on the cubes where these estimates fail. On a single event that does not depend on u , we bound the number of cubes on which these estimates fail along every chain of nested cubes, and we integrate this bound against the energy measure of u .

It remains to show that the proportionality factor is one. Without the factor that comes from the fields with scales larger than one, the coefficients A_N are stationary and have homogenized matrix equal to the identity, and their coarse-grained matrices on large cubes approach the identity uniformly in N (Theorem 8.52). Both limits inherit this normalization (Theorem 8.53).

The comparison through optimal constants follows the proof by Gwynne and Miller [2021, Theorem 1.9] that weak subcritical Liouville quantum gravity metrics are unique up to a deterministic factor. These are random metrics on the plane that formally weight Euclidean length by the exponential of a multiple of the Gaussian free field; they are characterized by a list of axioms, to which the word “weak” refers. There, a shortest path plays the role of our solution, its length plays the role of the energy, and replacing a segment of the path by a shorter one corresponds to replacing our solution on a cube by the minimizer of the other form with the same boundary values. They controlled the local events encountered along a shortest path by means of the confluence of geodesics, the property that geodesics from a common point to different targets share an initial segment [Gwynne and Miller, 2021, Section 4]. For critical and supercritical metrics, Ding and Gwynne [2023b, Section 4.2] avoided confluence by modifying the field in annuli visited by a shortest path. Our estimate along chains of nested cubes takes the place of these arguments. It holds on an event that does not depend on the solution, so it can be integrated against the solution’s energy measure,

and it works in every dimension $d \geq 2$. The uniqueness results for metrics characterize the metrics satisfying a list of axioms. We instead prove the properties that the argument needs for the limits of our energies: consistency under restriction to subcubes, locality, and the rule that multiplying the coefficients A_N by a positive continuous function multiplies the energy measure of the limit form by the same function.

1.5.5. Construction of the limiting diffusion. We pair the unique limit form with the limit μ of the reversible measures. The densities of μ_N have no bound uniform in the cutoff, but the mass that μ_N gives to a ball is at most a random constant times a power of its radius, with exponent arbitrarily close to, and below, d , uniformly in N . This bound controls integrals against μ_N by the energy of the integrand, which lets us pass to the limit in the variational problems for the resolvents of the processes killed on leaving a cube. Smoothing the measure-valued sources of these problems and applying the regularity estimates gives convergence of the killed resolvents, uniformly in the starting point.

Tightness comes from a bound on the probability of leaving a small cube in a short time, which we prove from two ingredients. The first is a bound on the Laplace transform of the exit time, obtained through a resolvent problem on the cube. The second is a local maximum principle, a bound on the supremum of a subsolution by its average over a larger cube, which we obtain by Moser iteration (iterated energy estimates for powers of the subsolution) with respect to the reversible measure (Section 9.3). The same bound shows that, with high probability, the paths stay in a large cube over bounded time intervals. We can therefore recover the whole-space resolvents from the killed ones. Iterating these resolvents determines the joint laws at finitely many times, and tightness gives convergence in probability on continuous path space, locally uniformly in the starting point.

To upgrade to almost-sure convergence, we compare the energies at two finite cutoffs, adapting an argument of [Ding and Gwynne \[2023a\]](#) for random metrics in the form strengthened by [Devlin \[2024, Lemma 3.11\]](#). Suppose that, on a positive proportion of scales, the ratio of the deterministic factors that multiply the energies at the two cutoffs were smaller than one by a fixed amount. Harmonic extensions on disjoint cubes of these scales would then bound one energy by a fixed fraction, smaller than one, of its limit, which is positive; this contradicts the convergence in probability. Excluding such ratios, and interchanging the cutoffs, gives comparisons with summable error probabilities, and the Borel–Cantelli lemma gives almost-sure convergence of the energies. Almost-sure bounds uniform in the cutoff then transfer this convergence to the killed resolvents and the path laws.

The remaining properties in Theorem A have two sources: those of the paths come from power-law decay of the effective diffusivities, and those of the invariant measure and the transition laws come from singularity of μ . Power-law decay between arbitrary scales makes the limiting paths leave small cubes faster than Brownian paths. To prove that μ is singular, we weight the joint law of the environment and a spatial point by the stationary part of μ . This exponentially tilts the laws of the fields at the point, and the strong law of large numbers shows that the approximating densities diverge at almost every point with respect to μ , whereas they vanish at Lebesgue-almost every point. The transition laws are absolutely continuous with respect to the invariant measure, so its singularity and the bounds on its mass in small balls exclude Gaussian transition laws.

Use of artificial intelligence. The authors developed all the main ideas of this paper in 2024. Versions of GPT were used to check the arguments, and Claude was used to help with the writing. The authors take responsibility for the mathematical content and presentation of the paper.

Acknowledgments. S.A. was supported by NSF grants DMS-2000200 and DMS-2350340. A.B. was supported by NSF grant DMS-2202940. T.K. was supported by the Academy of Finland and by the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation programme (grant agreement No 818437).

2. COARSE-GRAINING THEORY

In this section we recall a deterministic coarse-graining theory for uniformly elliptic coefficient fields $\mathbf{a} : \mathbb{R}^d \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$. The two coarse-grained matrices of a bounded domain give two-sided bounds on the energy of solutions in that domain, and their difference measures how far the equation is from being homogenized. The main argument of this paper tracks this difference from one scale to the next.

Throughout this section, and in contrast to those that follow, no probabilistic structure is used: the coefficient field is arbitrary and fixed. We define the matrices and record their subadditivity under partitions of a cube, show that solutions satisfy Poincaré and Caccioppoli inequalities with constants given by the coarse-grained matrices, estimate how the matrices change under a perturbation of the coefficient field, and combine these into the general coarse-graining estimate used later. The definitions and estimates are from [Armstrong and Kuusi \[2025\]](#) and [Armstrong et al. \[2026\]](#).

2.1. Notation. For a matrix A , the symbol $|A|$ denotes its operator norm. We identify a positive scalar coefficient with the corresponding scalar multiple of Id . For the coarse-grained matrices $\mathbf{a}(U)$ and $\mathbf{a}_*(U)$ defined in (2.17), superscripts denote matrix powers; for example, $\mathbf{a}_*^{-1}(U) = (\mathbf{a}_*(U))^{-1}$. A coefficient field used to construct a coarse-grained matrix is written after a semicolon, as in $\mathbf{a}(U; \mathbf{a}^{-1})$.

For a permutation π of $\{1, \dots, d\}$, $n \in \mathbb{Z}$, and $z \in \mathbb{R}^d$, define

$$(2.1) \quad \Delta_n^\pi(z) := z + 3^n \left\{ (x_1, \dots, x_d) \in \mathbb{R}^d : -\frac{1}{2} < x_{\pi(1)} < x_{\pi(2)} < \dots < x_{\pi(d)} < \frac{1}{2} \right\}.$$

We call $\Delta_n^\pi(z)$ a **Kuhn simplex** of the cube $z + \square_n$; these $d!$ simplices form its **Kuhn triangulation**.

For $n \leq m$, each Kuhn simplex of a cube of side 3^m is partitioned, up to null sets, by the Kuhn simplices of the cubes of side 3^n that it contains. Moreover, the cube $z + \square_n$ is the union of the simplices $\Delta_n^\pi(z)$ over all permutations π , up to null sets, and $|\Delta_n^\pi(z)| = 3^{nd}/d!$.

Function spaces. For $s \in (0, 1)$ and $p \in [1, \infty)$, the fractional Sobolev seminorm on the cube $\square_m$ is defined by

$$(2.2) \quad [u]_{\underline{W}^{s,p}(\square_m)} := \left(s \int_{\square_m} \int_{\square_m} \frac{|u(x) - u(y)|^p}{|x - y|^{d+sp}} dx dy \right)^{1/p}.$$

At $s = 1$, the seminorm is the classical first-order one, $\|\nabla u\|_{\underline{L}^p(\square_m)}$. For a bounded Lipschitz set U , the same formulas define $[\cdot]_{\underline{W}^{s,p}(U)}$. We set

$$(2.3) \quad \|u\|_{\underline{W}^{s,p}(U)} := [u]_{\underline{W}^{s,p}(U)} + |U|^{-s/d} \|u\|_{\underline{L}^p(U)}.$$

For $s \in (0, 1]$, $q \in (1, \infty)$, $N \in \mathbb{N}$, and $F : U \rightarrow \mathbb{R}^N$, define the inhomogeneous volume-normalized negative Sobolev norm by

$$(2.4) \quad \|F\|_{\widehat{\underline{W}}^{-s,q}(U)} := \sup_{\substack{\varphi \in C^\infty(\overline{U}; \mathbb{R}^N) \\ \varphi \neq 0}} \frac{|\oint_U F \cdot \varphi|}{[\varphi]_{\underline{W}^{s,q'}(U)} + |U|^{-s/d} \|\varphi\|_{\underline{L}^{q'}(U)}}.$$

When $q = 2$ and $N = d$, we write

$$(2.5) \quad \|F\|_{\underline{H}^{-s}(U)} := \sup_{\substack{\varphi \in C^\infty(\overline{U}; \mathbb{R}^d) \\ \varphi \neq 0}} \frac{|\oint_U F \cdot \varphi|}{[\varphi]_{\underline{H}^s(U)} + |U|^{-s/d} \|\varphi\|_{\underline{L}^2(U)}}.$$

Here and below, $\underline{H}^s = \underline{W}^{s,2}$. If $F \in (H^s(U; \mathbb{R}^d))^*$, then the numerator in (2.5) is interpreted as $|U|^{-1} \langle F, \varphi \rangle$; by density of smooth functions, the supremum is then the corresponding dual

norm. For every nonempty finite set A , write

$$(2.6) \quad \sum_{z \in A} c_z := \frac{1}{\#A} \sum_{z \in A} c_z.$$

In particular, if a cube U is the disjoint union of equal-volume cubes $(Q_z)_{z \in A}$, then

$$(2.7) \quad \sum_{z \in A} \|f\|_{\underline{L}^p(Q_z)}^p = \|f\|_{\underline{L}^p(U)}^p.$$

We work with the (volume-normalized) Besov-type seminorm defined in the cube $\square_m$ for exponents $s \in (0, 1)$ and $p, q, r \in [1, \infty)$ by

$$(2.8) \quad [u]_{\underline{B}_{p,q,r}^s(\square_m)} := \left(s \sum_{n=-\infty}^m 3^{-nsq} \left(\sum_{z \in 3^{n-1}\mathbb{Z}^d \cap \square_m, z+\square_n \subseteq \square_m} \|u - (u)_{z+\square_n}\|_{\underline{L}^p(z+\square_n)}^r \right)^{q/r} \right)^{1/q}.$$

For $r = p$, it reads

$$(2.9) \quad [u]_{\underline{B}_{p,q,p}^s(\square_m)} = \left(s \sum_{n=-\infty}^m 3^{-nsq} \left(\sum_{z \in 3^{n-1}\mathbb{Z}^d \cap \square_m, z+\square_n \subseteq \square_m} \|u - (u)_{z+\square_n}\|_{\underline{L}^p(z+\square_n)}^p \right)^{q/p} \right)^{1/q}.$$

The corresponding norm is $\|f\|_{\underline{B}_{p,q,r}^s(\square_m)} := [f]_{\underline{B}_{p,q,r}^s(\square_m)} + 3^{-sm}|(f)_{\square_m}|$. By Lemma C.2, there exists $C(d) < \infty$ such that, for every $s \in (0, 1)$ and $p \in [1, \infty)$,

$$(2.10) \quad C^{-1}[u]_{\underline{W}^{s,p}(\square_m)} \leq [u]_{\underline{B}_{p,p,p}^s(\square_m)} \leq C[u]_{\underline{W}^{s,p}(\square_m)}.$$

The corresponding norms are equivalent for each fixed s, p ; unlike the seminorm comparison, their constants need not be uniform in these parameters.

We allow each of p, q, r to equal ∞ . If $p = \infty$, the volume-normalized L^p norm is replaced by the L^∞ norm; if $r = \infty$, the average over cube centers is replaced by a supremum; and if $q = \infty$, the sum over scales is replaced by a supremum. We also allow $s = 1$ when $q = \infty$.

When $q = r = \infty$, the Besov seminorm is equivalent to the Hölder seminorm of order s for every p . More precisely,

$$(2.11) \quad \begin{aligned} C^{-1}s \sup_{x,y \in \square_m, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^s} &\leq [u]_{\underline{B}_{1,\infty,\infty}^s(\square_m)} \leq [u]_{\underline{B}_{\infty,\infty,\infty}^s(\square_m)} \\ &\leq C \sup_{x,y \in \square_m, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^s}. \end{aligned}$$

The first inequality of (2.11) is the only nontrivial one, and it is the Campanato characterization of Hölder spaces.

For $s \in (0, 1]$ and $q, r \in [1, \infty]$, define

$$(2.12) \quad \|f\|_{\dot{\underline{B}}_{r,q}^{-s}(\square_m)} := \left(s \sum_{n=-\infty}^m 3^{nsq} \left(\sum_{z \in 3^n\mathbb{Z}^d \cap \square_m} |(f)_{z+\square_n}|^r \right)^{q/r} \right)^{1/q}.$$

If $q^{-1} + (q')^{-1} = r^{-1} + (r')^{-1} = 1$ and either $s < 1$ or $(s, q) = (1, 1)$, then Lemma C.1 gives, for every f, g for which the right side is finite and $fg \in L^1(\square_m)$,

$$(2.13) \quad \left| \int_{\square_m} fg \right| \leq Cs^{-1} \|g\|_{\underline{B}_{1,q',r'}^s(\square_m)} \|f\|_{\dot{\underline{B}}_{r,q}^{-s}(\square_m)}.$$

For a bounded convex domain $U \subseteq \mathbb{R}^d$, set

$$(2.14) \quad \begin{cases} L_{\text{pot}}^2(U) := \nabla H^1(U), & L_{\text{pot},0}^2(U) := \nabla H_0^1(U), \\ L_{\text{sol}}^2(U) := L_{\text{pot},0}^2(U)^\perp, & L_{\text{sol},0}^2(U) := L_{\text{pot}}^2(U)^\perp. \end{cases}$$

All orthogonal complements in (2.14) are taken in $L^2(U; \mathbb{R}^d)$.

2.2. Coarse-grained matrices. We denote the linear space of solutions u of $-\nabla \cdot \mathbf{a} \nabla u = 0$ in an open set $U \subseteq \mathbb{R}^d$ by

$$(2.15) \quad \mathcal{A}(U; \mathbf{a}) := \{u \in H_{\text{loc}}^1(U) : -\nabla \cdot \mathbf{a} \nabla u = 0 \text{ in } U\}.$$

For every $p, q \in \mathbb{R}^d$ and bounded convex domain $U \subseteq \mathbb{R}^d$, we define the quantity

$$(2.16) \quad J(U, p, q; \mathbf{a}) := \sup_{v \in \mathcal{A}(U; \mathbf{a}) \cap H^1(U)} \int_U \left(-\frac{1}{2} \nabla v \cdot \mathbf{a} \nabla v - p \cdot \mathbf{a} \nabla v + q \cdot \nabla v \right).$$

The supremum is attained in $H^1(U)$, and the maximizer is unique up to an additive constant. We denote its mean-zero representative by $v(\cdot, U, p, q; \mathbf{a})$, so that

$$(v(\cdot, U, p, q; \mathbf{a}))_U = 0.$$

For a bounded convex domain $U \subseteq \mathbb{R}^d$, define the **coarse-grained matrices** $\mathbf{a}(U)$ and $\mathbf{a}_*(U)$ by the identity, valid for every $p, q \in \mathbb{R}^d$,

$$(2.17) \quad J(U, p, q; \mathbf{a}) =: \frac{1}{2} p \cdot \mathbf{a}(U) p + \frac{1}{2} q \cdot \mathbf{a}_*^{-1}(U) q - p \cdot q.$$

These matrices can also be defined variationally [Armstrong et al., 2019, Lemma 10.5]: for $p, q \in \mathbb{R}^d$,

$$(2.18) \quad \frac{1}{2} p \cdot \mathbf{a}(U) p = \min_{u \in p \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u = \max_{\mathbf{g} \in L_{\text{sol}}^2(U)} \int_U \left(-\frac{1}{2} \mathbf{g} \cdot \mathbf{a}^{-1} \mathbf{g} + p \cdot \mathbf{g} \right),$$

and

$$(2.19) \quad \frac{1}{2} q \cdot \mathbf{a}_*^{-1}(U) q = \max_{w \in H^1(U)} \int_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{a} \nabla w + q \cdot \nabla w \right) = \min_{\mathbf{g} \in q + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{g} \cdot \mathbf{a}^{-1} \mathbf{g}.$$

2.2.1. Properties of the coarse-grained matrices. We use the following identities and inequalities; see Armstrong and Kuusi [2024, 2025]. Throughout, $U \subseteq \mathbb{R}^d$ is a bounded convex domain.

The coarse-grained matrices are bounded by averages of the field, in the Loewner order:

$$(2.20) \quad \left(\int_U \mathbf{a}^{-1} \right)^{-1} \leq \mathbf{a}_*(U) \leq \mathbf{a}(U) \leq \int_U \mathbf{a}.$$

The coarse-grained matrices are **subadditive**: for every partition $\{U_i\}_{i=1}^N$ of U (up to a Lebesgue null set)

$$(2.21) \quad \mathbf{a}(U) \leq \sum_{i=1}^N \frac{|U_i|}{|U|} \mathbf{a}(U_i) \quad \text{and} \quad \mathbf{a}_*(U) \geq \left(\sum_{i=1}^N \frac{|U_i|}{|U|} \mathbf{a}_*^{-1}(U_i) \right)^{-1},$$

which generalizes the first and third inequalities of (2.20). Equivalently, for every $p, q \in \mathbb{R}^d$,

$$(2.22) \quad J(U, p, q; \mathbf{a}) \leq \sum_{i=1}^N \frac{|U_i|}{|U|} J(U_i, p, q; \mathbf{a}).$$

The spatial averages of the gradient and the flux of the maximizer can be written explicitly in terms of the coarse-grained matrices: for $p, q \in \mathbb{R}^d$,

$$(2.23) \quad \begin{cases} \int_U \nabla v(\cdot, U, p, q; \mathbf{a}) = -p + \mathbf{a}_*^{-1}(U) q \\ \int_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) = q - \mathbf{a}(U) p. \end{cases}$$

We also denote, for $e \in \mathbb{R}^d$,

$$(2.24) \quad v(\cdot, U, e; \mathbf{a}) := v(\cdot, U, 0, \mathbf{a}_*(U) e),$$

so that by (2.23) we have

$$(2.25) \quad \begin{pmatrix} (\nabla v(\cdot, U, e; \mathbf{a}))_U \\ (\mathbf{a} \nabla v(\cdot, U, e; \mathbf{a}))_U \end{pmatrix} = \begin{pmatrix} e \\ \mathbf{a}_*(U)e \end{pmatrix}.$$

The identity (2.35) below gives

$$(2.26) \quad J(U, e, \mathbf{a}_*(U)e; \mathbf{a}) = \frac{1}{2}e \cdot (\mathbf{a}(U) - \mathbf{a}_*(U))e.$$

For every $w \in \mathcal{A}(U; \mathbf{a}) \cap H^1(U)$ and $p, q \in \mathbb{R}^d$, the first variation of (2.16) gives

$$(2.27) \quad q \cdot \int_U \nabla w - p \cdot \int_U \mathbf{a} \nabla w = \int_U \nabla w \cdot \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}).$$

The second variation gives

$$(2.28) \quad \begin{aligned} J(U, p, q; \mathbf{a}) - \int_U \left(-\frac{1}{2} \nabla w \cdot \mathbf{a} \nabla w - p \cdot \mathbf{a} \nabla w + q \cdot \nabla w \right) \\ = \int_U \frac{1}{2} (\nabla v(\cdot, U, p, q; \mathbf{a}) - \nabla w) \cdot \mathbf{a} (\nabla v(\cdot, U, p, q; \mathbf{a}) - \nabla w). \end{aligned}$$

It follows that J can be written as the energy of the maximizer: for every $p, q \in \mathbb{R}^d$,

$$(2.29) \quad J(U, p, q; \mathbf{a}) = \int_U \frac{1}{2} \nabla v(\cdot, U, p, q; \mathbf{a}) \cdot \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}).$$

The Euler equation also gives

$$(2.30) \quad J(U, p, q; \mathbf{a}) = \frac{1}{2} \left(q \cdot \int_U \nabla v(\cdot, U, p, q; \mathbf{a}) - p \cdot \int_U \mathbf{a} \nabla v(\cdot, U, p, q; \mathbf{a}) \right).$$

The coarse-grained matrix $\mathbf{a}_*(U)$ bounds the energy of an arbitrary $H^1(U)$ function from below in terms of the spatial average of its gradient: for every $u \in H^1(U)$,

$$(2.31) \quad \frac{1}{2} (\nabla u)_U \cdot \mathbf{a}_*(U) (\nabla u)_U \leq \int_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u.$$

The dual inequality is, for every $\mathbf{h} \in L^2_{\text{sol}}(U)$,

$$(2.32) \quad \frac{1}{2} (\mathbf{h})_U \cdot \mathbf{a}^{-1}(U) (\mathbf{h})_U \leq \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{a}^{-1} \mathbf{h}.$$

For every constant positive-definite matrix $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$ and every $u \in \mathcal{A}(U; \mathbf{a}) \cap H^1(U)$,

$$(2.33) \quad |((\mathbf{a} - \mathbf{b}) \nabla u)_U| \leq 2 \left(\max_{|e|=1} J(U, e, \mathbf{b}e; \mathbf{a}) \right)^{1/2} \left(\int_U \frac{1}{2} \nabla u \cdot \mathbf{a} \nabla u \right)^{1/2},$$

in particular, with the coarse-graining error $\mathcal{E}_{s,\infty,1}$ of Definition 2.3 below, for every $m \in \mathbb{Z}$ and every $u \in \mathcal{A}(\square_m; \mathbf{a}) \cap H^1(\square_m)$,

$$(2.34) \quad 3^{-sm} \|(\mathbf{a} - \mathbf{b}) \nabla u\|_{\dot{B}_{2,1}^{-s}(\square_m)} \leq 4s^{-1} |\mathbf{b}|^{1/2} \|\mathbf{a}^{1/2} \nabla u\|_{L^2(\square_m)} \mathcal{E}_{s,\infty,1}(\square_m; \mathbf{a}, \mathbf{b}).$$

The bound (2.33) follows from (2.27), (2.29), and the Cauchy–Schwarz inequality.

The identity (2.17) can be rewritten for every symmetric matrix $\mathbf{b} \in \mathbb{R}_{\text{sym}}^{d \times d}$ as

$$(2.35) \quad J(U, e, \mathbf{b}e; \mathbf{a}) = \frac{1}{2}e \cdot (\mathbf{a}(U) - \mathbf{a}_*(U))e + \frac{1}{2}e \cdot (\mathbf{b} - \mathbf{a}_*(U))\mathbf{a}_*^{-1}(U)(\mathbf{b} - \mathbf{a}_*(U))e.$$

If $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$, then completing the square in the first argument of J instead of the second gives

$$(2.36) \quad J(U, \mathbf{b}^{-1}e, e; \mathbf{a}) = \frac{1}{2}e \cdot (\mathbf{a}_*^{-1}(U) - \mathbf{a}^{-1}(U))e + \frac{1}{2}e \cdot (\mathbf{b}^{-1} - \mathbf{a}^{-1}(U))\mathbf{a}(U)(\mathbf{b}^{-1} - \mathbf{a}^{-1}(U))e.$$

Applying (2.35) and (2.36) with $\mathbf{b}^{-1/2}e$ and $\mathbf{b}^{1/2}e$, respectively, in place of e yields

$$(2.37) \quad \begin{aligned} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) &= \frac{1}{2}e \cdot \mathbf{b}^{-1/2}(\mathbf{a}(U) - \mathbf{a}_*(U))\mathbf{b}^{-1/2}e \\ &\quad + \frac{1}{2}e \cdot (\mathbf{I}_d - \mathbf{b}^{-1/2}\mathbf{a}_*(U)\mathbf{b}^{-1/2})\mathbf{b}^{1/2}\mathbf{a}_*^{-1}(U)\mathbf{b}^{1/2}(\mathbf{I}_d - \mathbf{b}^{-1/2}\mathbf{a}_*(U)\mathbf{b}^{-1/2})e \end{aligned}$$

$$(2.38) \quad \begin{aligned} &= \frac{1}{2}e \cdot \mathbf{b}^{1/2}(\mathbf{a}_*^{-1}(U) - \mathbf{a}^{-1}(U))\mathbf{b}^{1/2}e \\ &\quad + \frac{1}{2}e \cdot (\mathbf{I}_d - \mathbf{b}^{1/2}\mathbf{a}^{-1}(U)\mathbf{b}^{1/2})\mathbf{b}^{-1/2}\mathbf{a}(U)\mathbf{b}^{-1/2}(\mathbf{I}_d - \mathbf{b}^{1/2}\mathbf{a}^{-1}(U)\mathbf{b}^{1/2})e. \end{aligned}$$

Lemma 2.1. *For every bounded convex domain $U \subseteq \mathbb{R}^d$ and $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$,*

$$(2.39) \quad \begin{aligned} \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) &\leq \frac{1}{2}|\mathbf{b}^{-1/2}(\mathbf{a}(U) - \mathbf{a}_*(U))\mathbf{b}^{-1/2}| + \frac{1}{2}|\mathbf{b}^{-1/2}\mathbf{a}_*^{1/2}(U) - \mathbf{b}^{1/2}\mathbf{a}_*^{-1/2}(U)|^2 \\ &\leq 2 \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}), \end{aligned}$$

and

$$(2.40) \quad \begin{aligned} \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) &\leq \frac{1}{2}|\mathbf{b}^{1/2}(\mathbf{a}^{-1}(U) - \mathbf{a}_*^{-1}(U))\mathbf{b}^{1/2}| + \frac{1}{2}|\mathbf{b}^{1/2}\mathbf{a}^{-1/2}(U) - \mathbf{b}^{-1/2}\mathbf{a}^{1/2}(U)|^2 \\ &\leq 2 \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}). \end{aligned}$$

In particular, there exists a universal constant $C < \infty$ such that

$$(2.41) \quad \begin{aligned} &|\mathbf{b}^{-1/2}(\mathbf{a}(U) - \mathbf{a}_*(U))\mathbf{b}^{-1/2}| + |\mathbf{b}^{1/2}\mathbf{a}_*^{-1}(U)\mathbf{b}^{1/2} - \mathbf{I}_d|^2 + |\mathbf{b}^{-1/2}\mathbf{a}(U)\mathbf{b}^{-1/2} - \mathbf{I}_d|^2 \\ &\leq C \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) \left(1 + \max_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a})\right). \end{aligned}$$

Proof. For positive-semidefinite matrices M and N ,

$$\max\{|M|, |N|\} \leq |M + N| \leq |M| + |N| \leq 2|M + N|.$$

Applying this inequality to the two summands in (2.37) proves (2.39); applying it to the two summands in (2.38) proves (2.40).

For every positive-definite matrix A and $J \geq 0$,

$$(2.42) \quad \frac{1}{2}|(\mathbf{I}_d - A)A^{-1}(\mathbf{I}_d - A)| \leq J \quad \implies \quad |\mathbf{I}_d - A|^2 \leq 4J(1 + J).$$

Indeed, the hypothesis implies

$$|\mathbf{I}_d - A|^2 \leq 2J|A| \leq 2J|\mathbf{I}_d - A| + 2J \leq \frac{1}{2}|\mathbf{I}_d - A|^2 + 2J^2 + 2J.$$

Since

$$(\mathbf{I}_d - A)A^{-1}(\mathbf{I}_d - A) = (\mathbf{I}_d - A^{-1})A(\mathbf{I}_d - A^{-1}),$$

apply (2.42) with $A = (\mathbf{b}^{-1/2}\mathbf{a}_*(U)\mathbf{b}^{-1/2})^{-1}$ in (2.37) and with $A = (\mathbf{b}^{1/2}\mathbf{a}^{-1}(U)\mathbf{b}^{1/2})^{-1}$ in (2.38). This gives

$$(2.43) \quad \begin{aligned} &\max\left\{|\mathbf{b}^{-1/2}\mathbf{a}(U)\mathbf{b}^{-1/2} - \mathbf{I}_d|^2, |\mathbf{b}^{1/2}\mathbf{a}_*^{-1}(U)\mathbf{b}^{1/2} - \mathbf{I}_d|^2\right\} \\ &\leq 10 \sup_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a}) \left(1 + \sup_{|e|=1} J(U, \mathbf{b}^{-1/2}e, \mathbf{b}^{1/2}e; \mathbf{a})\right). \end{aligned}$$

Combining (2.43) with (2.39) gives (2.41). $\square$

For every $\lambda > 0$, bounded convex domain $U \subseteq \mathbb{R}^d$ and $p, q \in \mathbb{R}^d$,

$$(2.44) \quad J(U, p, q; \mathbf{a}) = J(U, \lambda^{-1/2}p, \lambda^{1/2}q; \lambda\mathbf{a}).$$

2.3. Coarse-grained elliptic and functional inequalities. Following [Armstrong and Kuusi \[2025, \(2.39\)\]](#), we attach to each cube $\square_m$ the coarse-grained ellipticity constants $\lambda_{s,q}(\square_m; \mathbf{a})$ and $\Lambda_{s,q}(\square_m; \mathbf{a})$. The resulting Poincaré and Caccioppoli inequalities depend on these constants rather than on the pointwise ellipticity ratio.

The weights $(1 - 3^{-s})3^{-ns}$ sum to one over $n \geq 0$. For $0 < s \leq 2$, their normalizing factor satisfies $4/9s \leq 1 - 3^{-s} \leq (\log 3)s$.

Definition 2.2 (Coarse-grained ellipticity constants). For $s > 0$, $q \in [1, \infty)$, $m \in \mathbb{Z}$, and a uniformly elliptic field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, define the **coarse-grained ellipticity constants** by

$$(2.45) \quad \begin{cases} \Lambda_{s,q}(\square_m; \mathbf{a}) := \left((1 - 3^{-sq}) \sum_{k=-\infty}^m 3^{-sq(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}(z + \square_k; \mathbf{a})|^{q/2} \right)^{2/q}, \\ \lambda_{s,q}(\square_m; \mathbf{a}) := \left((1 - 3^{-sq}) \sum_{k=-\infty}^m 3^{-sq(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_k; \mathbf{a})|^{q/2} \right)^{-2/q}, \end{cases}$$

and, for $q = \infty$,

$$(2.46) \quad \begin{cases} \Lambda_{s,\infty}(\square_m; \mathbf{a}) := \sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}(z + \square_k; \mathbf{a})|, \\ \lambda_{s,\infty}(\square_m; \mathbf{a}) := \left(\sup_{k \in (-\infty, m] \cap \mathbb{Z}} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_k; \mathbf{a})| \right)^{-1}. \end{cases}$$

When the exponent q is omitted, it is understood that $q = 1$, that is,

$$\Lambda_s(\square_m; \mathbf{a}) = \Lambda_{s,1}(\square_m; \mathbf{a}) \quad \text{and} \quad \lambda_s(\square_m; \mathbf{a}) = \lambda_{s,1}(\square_m; \mathbf{a}).$$

Equation (2.46) is the limit of (2.45) as $q \rightarrow \infty$. On $y + \square_m$, every subcube $z + \square_k$ in these definitions is replaced by $y + z + \square_k$, with $z \in 3^k \mathbb{Z}^d \cap \square_m$.

For every $m \in \mathbb{Z}$, $q \in [1, \infty]$ and $t, s \in (0, \infty)$ with $t < s$, we have

$$(2.47) \quad \lambda_{t,q}(\square_m; \mathbf{a}) \leq \lambda_{s,q}(\square_m; \mathbf{a}) \leq |\mathbf{a}_*^{-1}(\square_m; \mathbf{a})|^{-1} \leq |\mathbf{a}(\square_m; \mathbf{a})| \leq \Lambda_{s,q}(\square_m; \mathbf{a}) \leq \Lambda_{t,q}(\square_m; \mathbf{a}).$$

There exists $C(d) < \infty$ such that, for every $s \in (0, 1]$ and $p, q \in [1, \infty]$ with $p \leq q$,

$$(2.48) \quad \begin{cases} \Lambda_{s,q}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \Lambda_{s,p}(\square_m; \mathbf{a}) \leq C s^{2/q-2} \Lambda_s(\square_m; \mathbf{a}), \\ \lambda_{s,q}^{-1}(\square_m; \mathbf{a}) \leq C s^{2/q-2/p} \lambda_{s,p}^{-1}(\square_m; \mathbf{a}) \leq C s^{2/q-2} \lambda_s^{-1}(\square_m; \mathbf{a}). \end{cases}$$

For every $k \in (-\infty, m] \cap \mathbb{Z}$, applying (2.47) on the subcubes and comparing the definitions gives

$$(2.49) \quad \begin{cases} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}(z + \square_k; \mathbf{a})| \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \Lambda_{s,q}(z + \square_k; \mathbf{a}) \leq 3^{2s(m-k)} \Lambda_{s,q}(\square_m; \mathbf{a}), \\ \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_k; \mathbf{a})| \leq \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \lambda_{s,q}^{-1}(z + \square_k; \mathbf{a}) \leq 3^{2s(m-k)} \lambda_{s,q}^{-1}(\square_m; \mathbf{a}). \end{cases}$$

We measure the error in replacing $\mathbf{a}$ by a constant matrix $\mathbf{a}_0$ through the quantities $J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})$ on the subcubes $z + \square_l$.

Definition 2.3 (Coarse-graining error). For $s \in (0, 1]$, $p, q \in [1, \infty]$, $m, n \in \mathbb{Z}$ with $n \leq m$, a uniformly elliptic field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, and a matrix $\mathbf{a}_0 \in \mathbb{R}_{\text{sym},+}^{d \times d}$, define the **coarse-graining**

error by

$$\mathcal{E}_{s,p,q}(\square_m, n; \mathbf{a}, \mathbf{a}_0) = \begin{cases} \left((1 - 3^{-sq}) \sum_{l=-\infty}^n 3^{-sq(n-l)} \left(\sum_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{p/2} \right)^{q/p} \right)^{1/q}, & p, q < \infty, \\ \left((1 - 3^{-sq}) \sum_{l=-\infty}^n 3^{-sq(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{q/2} \right)^{1/q}, & p = \infty, q < \infty, \\ \sup_{l \in (-\infty, n] \cap \mathbb{Z}} 3^{-s(n-l)} \left(\sum_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{p/2} \right)^{1/p}, & p < \infty, q = \infty, \\ \sup_{l \in (-\infty, n] \cap \mathbb{Z}} 3^{-s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap \square_m} \max_{|e|=1} (J(z + \square_l, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a}))^{1/2}, & p = q = \infty. \end{cases}$$

On $y + \square_m$, every subcube $z + \square_l$ in this definition is replaced by $y + z + \square_l$, with $z \in 3^l \mathbb{Z}^d \cap \square_m$. We use the abbreviation

$$\mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0) := \mathcal{E}_{s,p,q}(\square_m, m; \mathbf{a}, \mathbf{a}_0).$$

When the exponent q is omitted, it is understood that $q = 1$.

The definition gives, for $m \in \mathbb{Z}$, $p \in [2, \infty]$, $q \in [1, \infty]$, and $0 < t < s \leq 1$,

$$(2.50) \quad \max_{|e|=1} J(\square_m, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})^{1/2} \leq \mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \mathcal{E}_{t,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0),$$

and, for every $k \in (-\infty, m] \cap \mathbb{Z}$,

$$(2.51) \quad \begin{cases} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s,\infty,q}(z + \square_k; \mathbf{a}, \mathbf{a}_0) \leq 3^{s(m-k)} \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0), \\ \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s,p,q}(z + \square_k; \mathbf{a}, \mathbf{a}_0) \leq 3^{(s+d(1/p-1/q)_+)(m-k)} \mathcal{E}_{s,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0). \end{cases}$$

For a positive scalar $\mathbf{a}_0$, $s \in (0, 1]$ and $q \in [1, \infty]$,

$$(2.52) \quad \begin{aligned} \frac{1}{2} \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0)^2 &\leq \max\{|\mathbf{a}_0|^{-1} \Lambda_{s,q}(\square_m; \mathbf{a}), |\mathbf{a}_0| \lambda_{s,q}^{-1}(\square_m; \mathbf{a})\} \\ &\leq (1 + \sqrt{2} \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0))^2. \end{aligned}$$

To see the upper bound, put $J_U = \max_{|e|=1} J(U, \mathbf{a}_0^{-1/2} e, \mathbf{a}_0^{1/2} e; \mathbf{a})$. The ordering $\mathbf{a}_*(U) \leq \mathbf{a}(U)$ and (2.17) imply that each eigenvalue t of $\mathbf{a}_0^{-1} \mathbf{a}(U)$ or $\mathbf{a}_0 \mathbf{a}_*^{-1}(U)$ satisfies $t + t^{-1} - 2 \leq 2J_U$. Thus

$$\sqrt{t} \leq 1 + \sqrt{2J_U} \quad \text{and} \quad t \leq 1 + 2J_U + \sqrt{2J_U}.$$

The first inequality, followed by the weighted ℓ^q triangle inequality, gives the upper bound in (2.52). For the lower bound, use $\sqrt{J_U} \leq 2^{-1/2} (|\mathbf{a}_0^{-1} \mathbf{a}(U)|^{1/2} + |\mathbf{a}_0 \mathbf{a}_*^{-1}(U)|^{1/2})$ and the same triangle inequality. The second eigenvalue bound also gives

$$\max\{\mathbf{a}_0^{-1} \Lambda_{s,2}(\square_m; \mathbf{a}), \mathbf{a}_0 \lambda_{s,2}^{-1}(\square_m; \mathbf{a})\} \leq 1 + 2\mathcal{E}_{s,\infty,2}^2(\square_m; \mathbf{a}, \mathbf{a}_0) + 2\mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}, \mathbf{a}_0).$$

Moreover, in the case $q = 2$,

$$(2.53) \quad 1 + \frac{1}{2} \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}, \mathbf{a}_0)^2 \leq \max\{|\mathbf{a}_0|^{-1} \Lambda_{s,2}(\square_m; \mathbf{a}), |\mathbf{a}_0| \lambda_{s,2}^{-1}(\square_m; \mathbf{a})\}.$$

By Definition 2.3, for every $p, q \in [1, \infty]$ and $s \in (d/p, 1)$,

$$(2.54) \quad \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \left(\frac{(1 - 3^{-sq})}{(1 - 3^{-q(s-d/p)})} \right)^{1/q} \mathcal{E}_{s-d/p,p,q}(\square_m; \mathbf{a}, \mathbf{a}_0).$$

For $1 \leq q \leq p < \infty$ and $s > 0$,

$$(2.55) \quad \mathcal{E}_{s,\infty,q}(\square_m; \mathbf{a}, \mathbf{a}_0) \leq \mathcal{E}_{\frac{sq}{p},\infty,p}(\square_m; \mathbf{a}, \mathbf{a}_0).$$

The same definitions give

$$(2.56) \quad \Lambda_{s,q}(\square_m; \mathbf{a}) \leq \Lambda_{\frac{sq}{p},p}(\square_m; \mathbf{a}) \quad \text{and} \quad \lambda_{s,q}^{-1}(\square_m; \mathbf{a}) \leq \lambda_{\frac{sq}{p},p}^{-1}(\square_m; \mathbf{a}).$$

By Hölder's inequality, there exists a universal constant $C < \infty$ such that, for $0 < s < t \leq 1$,

$$(2.57) \quad \mathcal{E}_{t,\infty,1}(\square_m, n; \mathbf{a}, \mathbf{a}_0) \leq \frac{Ct}{(t-s)^{1/2} s^{1/2}} \mathcal{E}_{s,\infty,2}(\square_m, n; \mathbf{a}, \mathbf{a}_0).$$

The coarse-grained ellipticity constants satisfy the following stability estimate, which is [Armstrong et al. \[2026, Lemma 2.5\]](#).

Lemma 2.4 (Stability of the coarse-grained ellipticity constants). *There exists a constant $C(d) < \infty$ such that, for every $q \in [1, \infty]$, $s, t \in (0, 1/2]$ with $s < t$, and $x \in \square_0$ such that $x + \square_{-1} \subseteq \square_0$, we have the estimates*

$$(2.58) \quad \lambda_{t,q}^{-1}(x + \square_{-1}; \mathbf{a}) \leq \frac{C}{1-2s} \left(\frac{t}{t-s} \right)^{2/q} \lambda_{s,q}^{-1}(\square_0; \mathbf{a}),$$

$$(2.59) \quad \Lambda_{t,q}(x + \square_{-1}; \mathbf{a}) \leq \frac{C}{1-2s} \left(\frac{t}{t-s} \right)^{2/q} \Lambda_{s,q}(\square_0; \mathbf{a}),$$

and, for every $\mathbf{a}_0 \in \mathbb{R}_{\text{sym},+}^{d \times d}$ (see [Armstrong et al. \[2026, \(2.62\)\]](#)),

$$(2.60) \quad \mathcal{E}_{t,\infty,2}(x + \square_{-1}; \mathbf{a}, \mathbf{a}_0) \leq \frac{C}{(1-2s)^{1/2}} \left(\frac{t}{t-s} \right)^{1/2} \mathcal{E}_{s,\infty,2}(\square_0; \mathbf{a}, \mathbf{a}_0).$$

The next proposition is taken from [Armstrong and Kuusi \[2025, Lemma 2.3\]](#).

Proposition 2.5 (Coarse-grained Poincaré inequality). *For every $s \in (0, 1]$, $q \in [1, \infty]$, $m \in \mathbb{Z}$, every $u \in H^1(\square_m)$ and $\mathbf{h} \in L_{\text{sol}}^2(\square_m)$,*

$$(2.61) \quad 3^{-sm} [\nabla u]_{\dot{B}_{2,q}^{-s}(\square_m)} \leq (1 - 3^{-sq})^{-1/q} \lambda_{s,q}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)},$$

and

$$(2.62) \quad 3^{-sm} [\mathbf{h}]_{\dot{B}_{2,q}^{-s}(\square_m)} \leq (1 - 3^{-sq})^{-1/q} \Lambda_{s,q}^{1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{-1/2} \mathbf{h}\|_{\underline{L}^2(\square_m)}.$$

We combine the functional inequality

$$(2.63) \quad [u]_{\dot{B}_{2,\infty}^{1-s}(\square_0)} \leq C [\nabla u]_{\dot{B}_{2,1}^{-s}(\square_0)},$$

with (2.61) and the bound $\|u - (u)_{\square_0}\|_{\underline{L}^2(\square_0)} \leq C \liminf_{s \uparrow 1} [u]_{\dot{B}_{2,\infty}^{1-s}(\square_0)}$, and let $s \uparrow 1$. This gives, for some $C(d) < \infty$ and every $u \in H^1(\square_0)$,

$$(2.64) \quad \|u - (u)_{\square_0}\|_{\underline{L}^2(\square_0)} \leq C \lambda_{1,1}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}.$$

For zero boundary data we also have the following estimate. Let $w \in H_0^1(\square_0)$ be the **torsion function** of $\square_0$, that is, the solution of

$$\begin{cases} -\Delta w = 1 & \text{in } \square_0, \\ w = 0 & \text{on } \partial \square_0. \end{cases}$$

Then, for every $u \in H_0^1(\square_0)$,

$$|(u)_{\square_0}| = \left| \int_{\square_0} \nabla w \cdot \nabla u \right| \leq C [\nabla w]_{\dot{B}_{2,\infty}^1(\square_0)} [\nabla u]_{\dot{B}_{2,1}^{-1}(\square_0)} \leq C [\nabla u]_{\dot{B}_{2,1}^{-1}(\square_0)}.$$

Combining this with (2.64) yields, for some $C(d) < \infty$ and every $u \in H_0^1(\square_0)$,

$$(2.65) \quad \|u\|_{\underline{L}^2(\square_0)} \leq C \lambda_{1,1}^{-1/2}(\square_0; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_0)}.$$

The next proposition is taken from [Armstrong and Kuusi \[2025\]](#), Proposition 2.5].

Proposition 2.6 (Coarse-grained Caccioppoli inequality). *There exists a constant $C(d) < \infty$ such that, for every $s \in (0, 1)$, $t \in (0, 1/2)$ with $s + t < 1$, every coefficient field $\mathbf{a} \in L^\infty(\square_0; \mathbb{R}_{\text{sym},+}^{d \times d})$ and every $u \in H^1(\square_0)$ and $\mathbf{f} \in H^{2t}(\square_0; \mathbb{R}^d)$ satisfying*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{f} \quad \text{in } \square_0,$$

we have the estimate

$$(2.66) \quad \begin{aligned} \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_{-1})}^2 &\leq \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} \left(\frac{\Lambda_{s,1}(\square_0; \mathbf{a})}{\lambda_{t,1}(\square_0; \mathbf{a})} \right)^{\frac{s}{1-s-t}} \Lambda_{s,1}(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2 \\ &\quad + t^{-11} \left(\frac{C}{1-s-t} \right)^{2+\frac{4s}{1-s-t}} \left(\frac{\Lambda_{s,1}(\square_0; \mathbf{a})}{\lambda_{t,1}(\square_0; \mathbf{a})} \right)^{\frac{1-t}{1-s-t}} \lambda_{t,1}^{-1}(\square_0; \mathbf{a}) \|\mathbf{f}\|_{\underline{H}^{2t}(\square_0)}^2. \end{aligned}$$

Proof. Apply Lemma 2.10 with zero boundary datum on each cube of a fixed finite cover of $\square_{-1}$ by cubes of side 3^{-2} compactly contained in $\square_0$, and sum the resulting estimates. The forcing term has the displayed exponent since

$$\frac{1-t}{1-s-t} = 1 + \frac{s}{1-s-t}.$$

□

2.4. Sensitivity estimates. We quantify the dependence of J , $\mathbf{a}(U)$, and $\mathbf{a}_*(U)$ on a scalar coefficient field.

Lemma 2.7. *Let U be a bounded convex domain, let $\mathbf{a}, \mathbf{b} : U \rightarrow (0, \infty)$ be scalar, uniformly elliptic coefficient fields, and set $\varepsilon_U := \|\mathbf{a}\mathbf{b}^{-1} - 1\|_{L^\infty(U)} \vee \|\mathbf{a}^{-1}\mathbf{b} - 1\|_{L^\infty(U)}$. Then*

$$(2.67) \quad |\mathbf{b}_*^{-1/2}(U) \mathbf{a}_*(U) \mathbf{b}_*^{-1/2}(U) - \text{Id}| \leq \varepsilon_U \quad \text{and} \quad |\mathbf{b}^{-1/2}(U) \mathbf{a}(U) \mathbf{b}^{-1/2}(U) - \text{Id}| \leq \varepsilon_U.$$

Proof. The inequality (2.67) follows from the more precise bounds

$$(2.68) \quad \begin{cases} (\inf_U (\mathbf{a}\mathbf{b}^{-1})) \mathbf{b}(U) \leq \mathbf{a}(U) \leq (\sup_U (\mathbf{a}\mathbf{b}^{-1})) \mathbf{b}(U), \\ (\inf_U (\mathbf{a}^{-1}\mathbf{b})) \mathbf{b}_*^{-1}(U) \leq \mathbf{a}_*^{-1}(U) \leq (\sup_U (\mathbf{a}^{-1}\mathbf{b})) \mathbf{b}_*^{-1}(U). \end{cases}$$

For the upper bound in the first line of (2.68), the variational formula (2.18) gives, for every $e \in \mathbb{R}^d$,

$$\begin{aligned} \frac{1}{2} e \cdot \mathbf{a}(U) e &= \inf_{v \in e \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla v \cdot \mathbf{a} \nabla v \leq \sup_U (\mathbf{a}\mathbf{b}^{-1}) \inf_{v \in e \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla v \cdot \mathbf{b} \nabla v \\ &= \sup_U (\mathbf{a}\mathbf{b}^{-1}) \frac{1}{2} e \cdot \mathbf{b}(U) e. \end{aligned}$$

The dual variational formula (2.19) gives, for every $e \in \mathbb{R}^d$,

$$\begin{aligned} \frac{1}{2} e \cdot \mathbf{a}_*^{-1}(U) e &= \min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{a}^{-1} \mathbf{h} \leq \sup_U (\mathbf{a}^{-1} \mathbf{b}) \min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{b}^{-1} \mathbf{h} \\ &= \sup_U (\mathbf{a}^{-1} \mathbf{b}) \frac{1}{2} e \cdot \mathbf{b}_*^{-1}(U) e. \end{aligned}$$

Interchanging $\mathbf{a}$ and $\mathbf{b}$ gives the lower bounds in (2.68). □

Lemma 2.8. *Let $U \subseteq \mathbb{R}^d$ be a bounded convex domain and $\mathbf{a}, \mathbf{b} : U \rightarrow (0, \infty)$ be scalar, uniformly elliptic coefficient fields. Then, for every $\lambda \in (0, \infty)$, $p, q \in \mathbb{R}^d$ and $\delta \in (0, 1]$,*

$$(2.69) \quad \begin{aligned} & J(U, \lambda^{-1/2}p, \lambda^{1/2}q; \mathbf{b}) \\ & \leq (1 + \delta)J(U, p, q; \mathbf{a}) + \frac{3}{\delta} \left(\|\lambda \mathbf{b}^{-1} \mathbf{a} - 1\|_{L^\infty(U)}^2 + \|\lambda^{-1} \mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 \right) (J(U, p, q; \mathbf{a}) + p \cdot q). \end{aligned}$$

Proof. By (2.44), we may reduce to the case $\lambda = 1$. Let v and u be the maximizers of $J(U, p, 0; \mathbf{a})$ and $J(U, 0, q; \mathbf{a})$, respectively. Then $-v + c \in p \cdot x + H_0^1(U)$ for an additive constant c , and $\mathbf{a} \nabla u \in q + L_{\text{sol},0}^2(U)$. Therefore, by (2.18) and (2.19),

$$\begin{aligned} J(U, p, 0; \mathbf{b}) &= \min_{w \in p \cdot x + H_0^1(U)} \int_U \frac{1}{2} \nabla w \cdot \mathbf{b} \nabla w \leq \int_U \frac{1}{2} \nabla v \cdot \mathbf{b} \nabla v \\ &= J(U, p, 0; \mathbf{a}) + \int_U \frac{1}{2} \nabla v \cdot (\mathbf{b} \mathbf{a}^{-1} - 1) \mathbf{a} \nabla v, \end{aligned}$$

and

$$\begin{aligned} J(U, 0, q; \mathbf{b}) &= \min_{\mathbf{h} \in q + L_{\text{sol},0}^2(U)} \int_U \frac{1}{2} \mathbf{h} \cdot \mathbf{b}^{-1} \mathbf{h} \leq \int_U \frac{1}{2} \mathbf{a} \nabla u \cdot \mathbf{b}^{-1} \mathbf{a} \nabla u \\ &= J(U, 0, q; \mathbf{a}) + \int_U \frac{1}{2} \nabla u \cdot (\mathbf{a} \mathbf{b}^{-1} - 1) \mathbf{a} \nabla u. \end{aligned}$$

Hence,

$$J(U, p, q; \mathbf{b}) \leq J(U, p, q; \mathbf{a}) + \int_U \frac{1}{2} \nabla v \cdot (\mathbf{b} \mathbf{a}^{-1} - 1) \mathbf{a} \nabla v + \int_U \frac{1}{2} \nabla u \cdot (\mathbf{a} \mathbf{b}^{-1} - 1) \mathbf{a} \nabla u.$$

Since

$$\mathbf{a} \mathbf{b}^{-1} - 1 = -\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1) = -(\mathbf{b} \mathbf{a}^{-1} - 1) + \mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2,$$

we have

$$\int_U \frac{1}{2} \nabla u (\mathbf{a} \mathbf{b}^{-1} - 1) \cdot \mathbf{a} \nabla u \leq - \int_U \frac{1}{2} \nabla u (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a} \nabla u + \|\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2\|_{L^\infty(U)} J(U, 0, q; \mathbf{a}).$$

Pointwise,

$$\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2 \leq 2(\mathbf{a} \mathbf{b}^{-1} - 1)^2 + 2(\mathbf{b} \mathbf{a}^{-1} - 1)^2,$$

and, hence,

$$\|\mathbf{a} \mathbf{b}^{-1} (\mathbf{b} \mathbf{a}^{-1} - 1)^2\|_{L^\infty(U)} J(U, 0, q; \mathbf{a}) \leq 2(\|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 + \|\mathbf{b}^{-1} \mathbf{a} - 1\|_{L^\infty(U)}^2) (J(U, p, q; \mathbf{a}) + p \cdot q).$$

By Hölder's and Young's inequalities, for every $\delta > 0$,

$$\begin{aligned} & \int_U \frac{1}{2} \nabla v (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a} \nabla v - \int_U \frac{1}{2} \nabla u (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a} \nabla u \\ &= \int_U \frac{1}{2} \mathbf{a}^{1/2} \nabla (v + u) (\mathbf{b} \mathbf{a}^{-1} - 1) \cdot \mathbf{a}^{1/2} \nabla (v - u) \\ &\leq \|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)} (J(U, p, q; \mathbf{a}))^{1/2} 2^{-1/2} \|\mathbf{a}^{1/2} \nabla (v - u)\|_{\underline{L}^2(U)} \\ &\leq \delta J(U, p, q; \mathbf{a}) + \frac{1}{8\delta} \|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 \|\mathbf{a}^{1/2} \nabla (v - u)\|_{\underline{L}^2(U)}^2 \\ &\leq \delta J(U, p, q; \mathbf{a}) + \frac{1}{2\delta} \|\mathbf{b} \mathbf{a}^{-1} - 1\|_{L^\infty(U)}^2 (J(U, p, 0; \mathbf{a}) + J(U, 0, q; \mathbf{a})). \end{aligned}$$

Combining these displays gives (2.69). □

2.5. General coarse-graining estimates. Proposition 2.11 compares heterogeneous and constant-coefficient equations with the same data. We use the special case for symmetric coefficients of Armstrong et al. [2026, Lemmas 2.11–2.12 and Proposition 2.21] (version 2); for such coefficients, the boundary estimate below has a direct proof.

The first lemma is taken from Armstrong et al. [2026, Lemma 2.11].

Lemma 2.9 (Coarse-grained estimates with a divergence-form right-hand side). *There exists $C(d) < \infty$ such that the following holds. Fix $m \in \mathbb{N}$ and $s \in (0, 1)$. Let $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$ be uniformly elliptic and $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$. If $u \in H^1(\square_m)$ solves*

$$-\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} \quad \text{in } \square_m,$$

then we have the estimates

$$(2.70) \quad 3^{-sm} \|\nabla u\|_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C s^{-3/2} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + C s^{-3} \lambda_{s/2,2}^{-1}(\square_m; \mathbf{a}) 3^{sm} [\mathbf{g}]_{\underline{H}^s(\square_m)},$$

(2.71)

$$3^{-sm} \|\mathbf{a} \nabla u\|_{\dot{B}_{2,2}^{-s}(\square_m)} \leq C s^{-3/2} \Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + C s^{-9/2} \frac{\Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})}{\lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a})} 3^{sm} [\mathbf{g}]_{\underline{H}^s(\square_m)}.$$

Moreover, for every $h \in H^{1+s}(\square_m)$, the solutions $v, w \in H^1(\square_m)$ of the Dirichlet and Neumann problems

$$(2.72) \quad \begin{cases} -\nabla \cdot \mathbf{a} \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ v = h & \text{on } \partial \square_m, \end{cases} \quad \text{and} \quad \begin{cases} -\nabla \cdot \mathbf{a} \nabla w = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ \mathbf{n} \cdot (\mathbf{a} \nabla w + \mathbf{g} - (\mathbf{g})_{\square_m}) = 0 & \text{on } \partial \square_m \end{cases}$$

satisfy the estimate

$$(2.73) \quad \begin{aligned} & \|\mathbf{a}^{1/2} \nabla v\|_{\underline{L}^2(\square_m)} + \|\mathbf{a}^{1/2} \nabla w\|_{\underline{L}^2(\square_m)} \\ & \leq C s^{-3} \lambda_{s/2,2}^{-1/2}(\square_m; \mathbf{a}) 3^{sm} [\mathbf{g}]_{\underline{H}^s(\square_m)} + C s^{-3/2} \Lambda_{s/2,2}^{1/2}(\square_m; \mathbf{a}) 3^{sm} \|\nabla h\|_{\underline{H}^s(\square_m)}. \end{aligned}$$

The next lemma is the version for symmetric coefficients of the boundary Caccioppoli estimate in Armstrong et al. [2026, Lemma 2.12].

Lemma 2.10 (Coarse-grained Caccioppoli inequality with right-hand side). *There exists $C(d) < \infty$ such that, for every $s \in (0, 1)$, $t \in (0, 1/2)$ with $\sigma := 1 - s - t > 0$, every $x \in \square_0$, every $\mathbf{a} \in L^\infty(\square_0; \mathbb{R}_{\text{sym},+}^{d \times d})$, $u \in H^1(\square_0)$, $\mathbf{g} \in H^{2t}(\square_0; \mathbb{R}^d)$ and $h \in H^{1+2t}(\square_0)$ with $(h)_{\square_0} = 0$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_0, \\ u = h & \text{on } (\partial \square_0) \cap (x + \square_{-1}), \end{cases}$$

we have the estimate

$$(2.74) \quad \begin{aligned} & \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2((x+\square_{-2}) \cap \square_0)}^2 \\ & \leq \left(\frac{C}{\sigma} \right)^{2+4s/\sigma} \left(\frac{\Lambda_s(\square_0; \mathbf{a})}{\lambda_t(\square_0; \mathbf{a})} \right)^{\frac{s+\sigma}{\sigma}} \\ & \quad \times \left(\lambda_t(\square_0; \mathbf{a}) \|u\|_{\underline{L}^2(\square_0)}^2 + t^{-11} \lambda_t^{-1}(\square_0; \mathbf{a}) \|\mathbf{g}\|_{\underline{H}^{2t}(\square_0)}^2 + t^{-3} \Lambda_t(\square_0; \mathbf{a}) \|\nabla h\|_{\underline{H}^{2t}(\square_0)}^2 \right). \end{aligned}$$

Proof. In this proof, all norms and coarse-grained ellipticity constants are taken on $\square_0$ unless otherwise indicated, and $\|\cdot\|_2$ denotes $\|\cdot\|_{\underline{L}^2(\square_0)}$. Let v solve the Dirichlet problem in $\square_0$ with forcing $\nabla \cdot \mathbf{g}$ and boundary values h on all of $\partial \square_0$, and put $V = \|\mathbf{a}^{1/2} \nabla v\|_2$. By subadditivity, $\Lambda_{t,2} \leq \Lambda_t$ and $\lambda_{t,2}^{-1} \leq \lambda_t^{-1}$. Consequently, (2.73), applied with $2t < 1$ in place of s , gives

$$V \leq C t^{-3} \lambda_t^{-1/2} \|\mathbf{g}\|_{\underline{H}^{2t}} + C t^{-3/2} \Lambda_t^{1/2} \|\nabla h\|_{\underline{H}^{2t}}.$$

Since $v - h \in H_0^1(\square_0)$ and $(h)_{\square_0} = 0$, testing $v - h$ against the torsion function of $\square_0$, as in the argument preceding (2.65), controls $(v)_{\square_0}$ by the negative Besov seminorms of ∇v and ∇h . Combining this bound with (2.64) and with (2.61) for $s = q = 1$, we obtain

$$\|v\|_2 \leq C\lambda_t^{-1/2}V + C\|\nabla h\|_2.$$

Apply the homogeneous boundary Caccioppoli inequality [Armstrong et al., 2026, Proposition 2.9] to $u - v$, which has zero trace on the part of $\partial\square_0$ where the boundary condition is imposed. Put

$$B = \left(\frac{C}{\sigma}\right)^{2+4s/\sigma} \left(\frac{\Lambda_s}{\lambda_t}\right)^{s/\sigma} \Lambda_s.$$

By the triangle inequality, the left side of (2.74) is at most $CB(\|u\|_2^2 + \|v\|_2^2) + 2V^2$. Insert the preceding estimates and absorb $2V^2$ to obtain (2.74). $\square$

The next proposition is the general coarse-graining estimate from Armstrong et al. [2026, Proposition 2.21].

Proposition 2.11 (General coarse-graining estimate). *Let $m, n \in \mathbb{Z}$ with $n < m$, let $\mathbf{a} \in L^\infty(\square_m; \mathbb{R}_{\text{sym},+}^{d \times d})$ and $\mathbf{a}_0 \in (0, \infty)$. Fix $p \in [2, \infty)$, $s, s_1, s_2 \in (0, 1)$ with $s_1 < s < s_2 < 1$ and $sp' < 1$, where $p' = p/(p-1)$. There exists $C(p, d) < \infty$ such that, for every $\mathbf{g} \in W^{s_2, p}(\square_m; \mathbb{R}^d)$ and $u, v \in H^1(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a} \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ -\nabla \cdot \mathbf{a}_0 \nabla v = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u - v \in H_0^1(\square_m), \end{cases}$$

we have the estimate

$$\begin{aligned} & 3^{-ms} \mathbf{a}_0 \|\nabla u - \nabla v\|_{\widehat{W}^{-s, p}(\square_m)} + 3^{-ms} \|\mathbf{a} \nabla u - \mathbf{a}_0 \nabla v\|_{\widehat{W}^{-s, p}(\square_m)} \\ & \leq C s^{-1-1/p'} \mathbf{a}_0^{1/2} \mathcal{E}_{s_1, \infty, 1}(\square_m, n; \mathbf{a}, \mathbf{a}_0) \left(\sum_{k=-\infty}^n 3^{-(s-s_1)p(n-k)} \sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} \|\mathbf{a}^{1/2} \nabla u\|_{\underline{L}^2(z+\square_k)}^p \right)^{1/p} \\ (2.75) \quad & + C s^{-11/2} (s_2 - s)^{-1} (1 + \mathcal{E}_{s_1/2, \infty, 2}^2(\square_m, n; \mathbf{a}, \mathbf{a}_0)) 3^{s_2 n} [\mathbf{g}]_{W^{s_2, p}(\square_m)}. \end{aligned}$$

By Sobolev embedding, if $p - 2$ is small in terms of s , then the left side of (2.75) controls the quantity $\mathbf{a}_0 3^{-m} \|u - v\|_{\underline{L}^2(\square_m)}$.

3. ESTIMATES FOR THE CUTOFF COEFFICIENTS

In this section we prove the probabilistic estimates on the cutoff coefficients used in the rest of the paper. They are of two kinds. Sensitivity bounds (Lemma 3.3), on which the induction of Section 4 rests, measure how much the coarse-grained matrices change when the fields of a range of scales are added to the coefficient. Negative Besov bounds (Lemma 3.2) control the ratio $\mathbf{a}_m \mathbf{a}_n^{-1}$ of two cutoff coefficients. We also prove, in Proposition 3.5, the planar identity $\bar{\mathbf{a}}_m = e^{-\tau^2(m+1)}$, which gives the exact two-dimensional exponents.

Throughout this section, $\mathbb{P}$ is a probability measure satisfying Assumptions (g1)–(g4). On a given cube, a single field has little effect in two opposite situations. A field whose scale is much smaller than the cube oscillates on it, and is therefore small in a negative Besov norm on the cube. A field whose scale is much larger than the cube is nearly constant on it, and multiplying the coefficient by a constant does not change the solutions of the equation.

Stochastic bounds are written in the Orlicz notation: for $\sigma, A > 0$ we write $X \leq \mathcal{O}_{\Gamma_\sigma}(A)$ if $\mathbb{E}[\exp((A^{-1} \max\{X, 0\})^\sigma)] \leq 2$, and $X = \mathcal{O}_{\Gamma_\sigma}(A)$ if this holds for both X and $-X$; Appendix A collects the properties we use.

Recall from (1.7) that, for $L \in \mathbb{N}_0$, the cutoff coefficient $\mathbf{a}_L$ is

$$(3.1) \quad \mathbf{a}_L(x) = \prod_{k=0}^L \exp(\mathbf{g}_k(x) - \tau^2).$$

By (1.6), $\tau \leq C\bar{\delta}$. We also set $\mathbf{a}_{-1} \equiv 1$. For every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z}$ with $-1 \leq n < m$,

$$(3.2) \quad \mathbb{E}[(\mathbf{a}_m \mathbf{a}_n^{-1})(0)] = 1 \quad \text{and} \quad \mathbb{E}[(\mathbf{a}_m^{-1} \mathbf{a}_n)(0)] = \exp(2\tau^2(m-n)).$$

Moreover,

$$(3.3) \quad \text{the fields } \mathbf{a}_m \mathbf{a}_n^{-1} \text{ and } \exp(-2\tau^2(m-n)) \mathbf{a}_m^{-1} \mathbf{a}_n \text{ have the same law.}$$

3.1. Besov space estimates. We first estimate the oscillation of the ratios $\mathbf{a}_m \mathbf{a}_n^{-1}$.

Lemma 3.1. *There exists a constant $C(d) < \infty$ such that, for every $n \in \mathbb{N}_0$ and $k \in \mathbb{Z}$,*

$$(3.4) \quad \sup_{m \geq n} \sup_{x, y \in \square_k} \log \left(\frac{(\mathbf{a}_m \mathbf{a}_n^{-1})(x)}{(\mathbf{a}_m \mathbf{a}_n^{-1})(y)} \right) \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta} 3^{k-n}),$$

and, for every $\xi \in [1, \infty)$,

$$(3.5) \quad \mathbb{E} \left[\left| \sup_{m \geq n} \sup_{x, y \in \square_k} \frac{(\mathbf{a}_m \mathbf{a}_n^{-1})(x)}{(\mathbf{a}_m \mathbf{a}_n^{-1})(y)} - 1 \right|^\xi \right]^{1/\xi} \leq C\xi^{1/2} \bar{\delta} 3^{k-n} \exp(C\xi \bar{\delta}^2 3^{2(k-n)}).$$

Moreover, for every $m \in \mathbb{N}_0$ with $m > n$,

$$(3.6) \quad \|\log(\mathbf{a}_m \mathbf{a}_n^{-1}) + (m-n)\tau^2\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(|m-n|^{1/2} + (k-n)_+)),$$

and, for every $\xi \in [1, \infty)$,

$$(3.7) \quad \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{L^\infty(\square_k)}^\xi \right]^{1/\xi} + \mathbb{E} \left[\|\mathbf{a}_m^{-1} \mathbf{a}_n - 1\|_{L^\infty(\square_k)}^\xi \right]^{1/\xi} \leq (C\xi^{1/2} \bar{\delta}((m-n)^{1/2} + (k-n)_+)) \exp(C\xi \bar{\delta}^2(m-n + (k-n)_+^2)).$$

Finally, for every bounded domain $U \subseteq \mathbb{R}^d$, every $p \in [1, \infty)$, and every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z}$ with $-1 \leq n < m$,

$$(3.8) \quad \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - \text{Id}\|_{L^p(U)}^p \right]^{1/p} \leq (Cp^{1/2} \bar{\delta}(m-n)^{1/2}) \exp(Cp \bar{\delta}^2(m-n)),$$

and

$$(3.9) \quad \mathbb{E} \left[\|\mathbf{a}_m^{-1} \mathbf{a}_n - \text{Id}\|_{L^p(U)}^p \right]^{1/p} \leq (Cp^{1/2} \bar{\delta}(m-n)^{1/2}) \exp(Cp \bar{\delta}^2(m-n)).$$

Proof. We first prove (3.4). By (1.5),

$$\begin{aligned} \sup_{m \geq n} \sup_{x, y \in \square_k} \log \frac{(\mathbf{a}_m \mathbf{a}_n^{-1})(x)}{(\mathbf{a}_m \mathbf{a}_n^{-1})(y)} &\leq \sum_{j=n+1}^{\infty} \sup_{x, y \in \square_k} |\mathbf{g}_j(x) - \mathbf{g}_j(y)| \\ &\leq \mathcal{O}_{\Gamma_2}(C3^{k-n} \bar{\delta}). \end{aligned}$$

Indeed, for $j \geq k$ the mean value theorem bounds the oscillation by $C3^k \|\nabla \mathbf{g}_j\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C3^{k-j} \bar{\delta})$. For $j < k$, cover $\square_k$ by the $3^{d(k-j)}$ cubes $z + \square_j$ with $z \in 3^j \mathbb{Z}^d \cap \square_k$ and use (A.4) to bound the oscillation by $2\|\mathbf{g}_j\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}\sqrt{1+k-j}) \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}3^{k-j})$. Then (3.5) follows from (A.8), applied to the supremum in (3.4). Since $\log(\mathbf{a}_m \mathbf{a}_n^{-1}) + (m-n)\tau^2 = \sum_{j=n+1}^m \mathbf{g}_j$, the bound (3.6) follows once we prove, for a fixed $m \in \mathbb{N}_0$ with $m > n$, that

$$(3.10) \quad \sup_{x \in \square_k} \left| \sum_{j=n+1}^m \mathbf{g}_j(x) \right| \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}((m-n)^{1/2} + (k-n)_+)).$$

By (1.5), (A.5), and (A.3),

$$(3.11) \quad \sup_{x \in \square_k} \left| \sum_{j=k \vee (n+1)}^m \mathbf{g}_j(x) \right| \leq \underbrace{\left| \sum_{j=k \vee (n+1)}^m \mathbf{g}_j(0) \right|}_{\leq \mathcal{O}_{\Gamma_2}(\bar{\delta}(m-n)^{1/2})} + \sum_{j=k \vee (n+1)}^m \underbrace{\|\mathbf{g}_j - \mathbf{g}_j(0)\|_{L^\infty(\square_k)}}_{\leq \mathcal{O}_{\Gamma_2}(C3^{k-j}\bar{\delta})} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(m-n)^{1/2}).$$

This proves (3.10) when $k \leq n$. Suppose that $k > n$. The same three estimates, now on $\square_n$ and with the sums stopped at $m \wedge k$, give

$$(3.12) \quad \sup_{x \in \square_n} \left| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j(x) \right| \leq \underbrace{\left| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j(0) \right|}_{\mathcal{O}_{\Gamma_2}(\bar{\delta}((m \wedge k) - n)^{1/2})} + \sum_{j=n+1}^{m \wedge k} \underbrace{\|\mathbf{g}_j - \mathbf{g}_j(0)\|_{L^\infty(\square_n)}}_{\mathcal{O}_{\Gamma_2}(C3^{n-j}\bar{\delta})} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}((m \wedge k) - n)^{1/2}).$$

The maximum of $N \geq 2$ random variables, each of which is $\mathcal{O}_{\Gamma_2}(A)$, is $\mathcal{O}_{\Gamma_2}((3 \log N)^{1/2} A)$; this is [Armstrong et al. \[2024, Lemma 2.3\]](#). Applying it with $N = 3^{d(k-n)}$ gives

$$(3.13) \quad \sup_{x \in \square_k} \left| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j(x) \right| = \max_{z \in 3^n \mathbb{Z}^d \cap \square_k} \left\| \sum_{j=n+1}^{m \wedge k} \mathbf{g}_j \right\|_{L^\infty(z + \square_n)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(k-n)).$$

If $m > k$, then, by (3.11), the remaining sum $\sum_{j=k+1}^m \mathbf{g}_j$ contributes at most $\mathcal{O}_{\Gamma_2}(C\bar{\delta}(m-n)^{1/2})$ on $\square_k$. This proves (3.10).

To prove (3.7), we use

$$\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{L^\infty(\square_k)} \vee \|\mathbf{a}_m^{-1} \mathbf{a}_n - 1\|_{L^\infty(\square_k)} \leq \|\log(\mathbf{a}_m \mathbf{a}_n^{-1})\|_{L^\infty(\square_k)} \exp(\|\log(\mathbf{a}_m \mathbf{a}_n^{-1})\|_{L^\infty(\square_k)}).$$

Equations (1.6) and (3.6), together with (A.8) and (A.6) at exponent 2ξ , bound the $L^{2\xi}(\mathbb{P})$ norms of the random variable $\|\log(\mathbf{a}_m \mathbf{a}_n^{-1})\|_{L^\infty(\square_k)}$ and of its exponential. Hölder's inequality then gives (3.7).

At a fixed point, the pointwise version of (3.6) and the exponential moment bound used above give

$$\|\mathbf{a}_m(x) \mathbf{a}_n^{-1}(x) - 1\|_{L^p(\mathbb{P})} \leq Cp^{1/2} \bar{\delta}(m-n)^{1/2} \exp(Cp\bar{\delta}^2(m-n)).$$

For $n = -1$, separate the independent factor $e^{\mathbf{g}_0 - \tau^2}$ and apply the estimate with $n = 0$. Stationarity now gives (3.8). Finally, (3.3) and (1.6) give (3.9). $\square$

Lemma 3.2. *There exist $C(d) < \infty$ and $c(d) > 0$ such that, for every $s \in (0, 1]$ and $p \in [2, \infty)$ satisfying*

$$(3.14) \quad s \geq Cp\bar{\delta}^2 \quad \text{and} \quad p\bar{\delta}^2 \log(2+p) \leq c,$$

and every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z}$ with $-1 \leq n < m$, we have the estimates

$$(3.15) \quad 3^{-sm} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{\underline{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} \leq Cs^{-1} \bar{\delta} p^{1/2} (\log p),$$

and

$$(3.16) \quad 3^{-sm} \mathbb{E} \left[\|\exp(-2\tau^2(m-n)) \mathbf{a}_m^{-1} \mathbf{a}_n - 1\|_{\underline{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} \leq Cs^{-1} \bar{\delta} p^{1/2} (\log p).$$

Proof. The two estimates are equivalent by (3.3); we prove (3.15).

When $n = -1$, the summand with $k = -1$ in the decomposition (3.18) below contains $\mathbf{g}_0$ and $\mathbf{a}_m \mathbf{a}_0^{-1}$. Assumptions (g1)–(g2), independence, and Lemma 3.1 with $n = 0$ give the bound required for this summand. Every remaining summand has $k \geq 0$.

By (2.12),

$$(3.17) \quad 3^{-sm} \mathbb{E} \left[\left\| \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right\|_{\dot{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} = \mathbb{E} \left[\sum_{j=-\infty}^m s 3^{sp(j-m)} \sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\mathbf{a}_m \mathbf{a}_n^{-1})_{z+\square_j} - 1|^p \right]^{1/p}.$$

For each $j \leq m$, we write

$$(3.18) \quad \oint_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 = \sum_{k=n}^{m-1} \oint_{\square_j} (\mathbf{a}_m \mathbf{a}_k^{-1} - \mathbf{a}_m \mathbf{a}_{k+1}^{-1}) = \sum_{k=n}^{m-1} \oint_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1).$$

Step 1. We show the existence of $C(d) < \infty$ such that, for every $j \in \{k, \dots, m\}$,

$$(3.19) \quad \mathbb{E} \left[\left| \oint_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \leq C \bar{\delta} p^{1/2} \left(3^{-d/2(j-k)} + \min\{p 3^{-d(1-1/p)(j-k)}, 1\} \right) \exp(C p \bar{\delta}^2 (m-k)).$$

For each $k \leq j$, we have

$$(3.20) \quad \oint_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) = \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \oint_{z+\square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1).$$

Condition on $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$, which is independent of the field $\mathbf{g}_{k+1}$, and apply Lemma B.1. By (3.7), (3.14), and (1.6),

$$(3.21) \quad \mathbb{E} \left[\left| \oint_{\square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \leq \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_k)} \mathbb{E} \left[\left(\oint_{\square_k} |\exp(\mathbf{g}_{k+1} - \tau^2) - 1|^2 \right)^{p/2} \right]^{1/p} \leq C p^{1/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_k)}.$$

In particular,

$$\begin{aligned} & \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \mathbb{E} \left[\left| \oint_{z+\square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^2 \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right] \right)^{1/2} \\ & \leq C \bar{\delta} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(z+\square_k)}^2 \right)^{1/2} = C \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)}, \end{aligned}$$

and

$$(3.22) \quad \begin{aligned} & \mathbb{E} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_j} \left| \oint_{z+\square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\ & \leq \mathbb{E} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \left| \oint_{z+\square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\ & \leq C p^{1/2} \bar{\delta} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(z+\square_k)}^p \right)^{1/p} \leq C 3^{d/p(j-k)} p^{1/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}. \end{aligned}$$

Conditionally on $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$, the field $\mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1)$ has range of dependence at most $\sqrt{d} 3^{k+1}$. The summands indexed by $z \in 3^k \mathbb{Z}^d \cap \square_j$ can therefore be partitioned into $C(d)$

classes within each of which they are independent. Applying Lemma B.1 to each class, we obtain

$$\begin{aligned}
& \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq C p^{1/2} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \mathbb{E} \left[\left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^2 \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right] \right)^{1/2} \\
& \quad + C p \mathbb{E} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_j} \left| \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq C 3^{d/2(j-k)} \bar{\delta} p^{1/2} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)} + C p^{3/2} \bar{\delta} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(z + \square_k)}^p \right)^{1/p}.
\end{aligned}$$

Dividing by $3^{d(j-k)}$ gives

$$\begin{aligned}
& \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
& \leq C 3^{-d/2(j-k)} \bar{\delta} p^{1/2} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)} + C 3^{-d(1-1/p)(j-k)} p^{3/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}.
\end{aligned}$$

Separately, the triangle inequality and (3.21) give the cruder bound

$$(3.23) \quad \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_j} \int_{z + \square_k} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \leq C p^{1/2} \bar{\delta} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}.$$

Combining the previous two displays yields

$$\begin{aligned}
& \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \left| \mathbf{a}_m \mathbf{a}_{k+1}^{-1} \right| \right]^{1/p} \\
(3.24) \quad & \leq C \bar{\delta} p^{1/2} 3^{-d/2(j-k)} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^2(\square_j)} + C \bar{\delta} p^{1/2} \min\{p 3^{-d(1-1/p)(j-k)}, 1\} \|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}.
\end{aligned}$$

Taking $L^p(\mathbb{P})$ norms and using (3.8) gives

$$\begin{aligned}
& \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \\
& \leq C \bar{\delta} p^{1/2} \left(3^{-d/2(j-k)} + \min\{p 3^{-d(1-1/p)(j-k)}, 1\} \right) \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{\underline{L}^p(\square_j)}^p \right]^{1/p} \\
& \leq C \bar{\delta} p^{1/2} \left(3^{-d/2(j-k)} + \min\{p 3^{-d(1-1/p)(j-k)}, 1\} \right) \exp(C p \bar{\delta}^2 (m - k)).
\end{aligned}$$

This is (3.19).

Step 2. We show that, for every $j \leq m$,

$$(3.25) \quad \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right|^p \right]^{1/p} \leq C \bar{\delta} p^{1/2} (\log p) (1 + m - j) \exp(C p \bar{\delta}^2 (m - j)).$$

By the triangle inequality and (3.18) we have

$$\begin{aligned} & \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right| \\ & \leq \sum_{k=n}^{j-1} \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right| + \sum_{k=j}^{m-1} \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|. \end{aligned}$$

We bound the first term on the right side by summing (3.19) over $k \in \{n, \dots, j-1\}$. The resulting sum, after factoring out $\exp(Cp\bar{\delta}^2(m-j))$, is at most $C \log(2+p)$ by (3.14). Thus

$$\begin{aligned} \mathbb{E} \left[\left| \sum_{k=n}^{j-1} \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} & \leq \sum_{k=n}^{j-1} \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \\ & \leq C\bar{\delta}p^{1/2}(\log p) \exp(Cp\bar{\delta}^2(m-j)). \end{aligned}$$

For the second sum over $k \geq j$ we use the cruder bound coming from (3.7) and $\tau \leq C\bar{\delta}$, to obtain

$$\begin{aligned} & \mathbb{E} \left[\left(\sum_{k=j}^{m-1} \left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right| \right)^p \right]^{1/p} \\ & \leq \sum_{k=j}^{m-1} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_{k+1}^{-1}\|_{L^\infty(\square_j)}^{2p} \right]^{1/2p} \mathbb{E} \left[\|(\exp(\mathbf{g}_{k+1} - \tau^2) - 1)\|_{L^\infty(\square_j)}^{2p} \right]^{1/2p} \\ & \leq \sum_{k=j}^{m-1} C\bar{\delta}p^{1/2} \exp(Cp\bar{\delta}^2(m-k)) \leq C\bar{\delta}p^{1/2}(m-j) \exp(Cp\bar{\delta}^2(m-j)). \end{aligned}$$

Combining these yields (3.25) when $j \geq n$.

Suppose now that $j < n$. By (3.18), the triangle inequality, independence of $\mathbf{g}_{k+1}$ from $\mathbf{a}_m \mathbf{a}_{k+1}^{-1}$, Assumption (g2), and Lemma 3.1,

$$\begin{aligned} \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right|^p \right]^{1/p} & \leq \sum_{k=n}^{m-1} \mathbb{E} \left[\left| \int_{\square_j} \mathbf{a}_m \mathbf{a}_{k+1}^{-1} (\exp(\mathbf{g}_{k+1} - \tau^2) - 1) \right|^p \right]^{1/p} \\ & \leq C\bar{\delta}p^{1/2}(m-n) \exp(Cp\bar{\delta}^2(m-n)) \\ & \leq C\bar{\delta}p^{1/2}(1+m-j) \exp(Cp\bar{\delta}^2(m-j)). \end{aligned}$$

This proves (3.25) for every $j \leq m$.

Step 3. We sum over the scales to prove (3.15). Returning to (3.17) and applying (3.25) and the triangle inequality, we obtain

$$\begin{aligned} & 3^{-spm} \mathbb{E} \left[\|\mathbf{a}_m \mathbf{a}_n^{-1} - 1\|_{\underline{B}_{p,p}^{-s}(\square_m)}^p \right] \\ & \leq C \sum_{j=-\infty}^m s 3^{sp(j-m)} \mathbb{E} \left[\sum_{z \in 3^j \mathbb{Z}^d \cap \square_m} |(\mathbf{a}_m \mathbf{a}_n^{-1})_{z+\square_j} - 1|^p \right] \\ (3.26) \quad & \leq s(C\bar{\delta}p^{1/2}(\log p))^p \sum_{j=-\infty}^m (1+m-j)^p \exp((\log 3)sp - Cp^2\bar{\delta}^2)(j-m). \end{aligned}$$

By (3.14), after increasing the constant there, the exponent in the last line of (3.26) satisfies $(\log 3)sp - Cp^2\bar{\delta}^2 \geq 1/2(\log 3)sp$, so we obtain

$$\begin{aligned} 3^{-sm} \mathbb{E} \left[\left\| \mathbf{a}_m \mathbf{a}_n^{-1} - 1 \right\|_{\dot{B}_{p,p}^{-s}(\square_m)}^p \right]^{1/p} &\leq Cs^{1/p} \bar{\delta} p^{1/2} (\log p) \left(\sum_{j=-\infty}^m (1+m-j)^p \exp\left(\frac{1}{2}sp(j-m)\right) \right)^{1/p} \\ &\leq Cs^{-1} \bar{\delta} p^{1/2} (\log p). \end{aligned}$$

□

3.2. Sensitivity estimates for coarse-grained matrices.

Lemma 3.3. *There exists a constant $C(d) < \infty$ such that, for every $n \in \mathbb{N}_0$ and $k \in \mathbb{Z}$, every $\xi \in [1, \infty)$, and every convex domain $U \subseteq \square_k$,*

$$\begin{aligned} (3.27) \quad &\mathbb{E} \left[\sup_{L \geq n} \left(\left| \frac{\mathbf{a}_{L,*}(U)^{-1/2} \mathbf{a}_{n,*}(U) \mathbf{a}_{L,*}(U)^{-1/2}}{(\mathbf{a}_n \mathbf{a}_L^{-1})(0)} - \mathbf{I}_d \right| + \left| \frac{\mathbf{a}_L(U)^{-1/2} \mathbf{a}_n(U) \mathbf{a}_L(U)^{-1/2}}{(\mathbf{a}_n \mathbf{a}_L^{-1})(0)} - \mathbf{I}_d \right| \right)^\xi \right]^{1/\xi} \\ &+ \mathbb{E} \left[\sup_{L \geq n} \left(\left| \frac{\mathbf{a}_{n,*}(U)^{-1/2} \mathbf{a}_{L,*}(U) \mathbf{a}_{n,*}(U)^{-1/2}}{(\mathbf{a}_L \mathbf{a}_n^{-1})(0)} - \mathbf{I}_d \right| + \left| \frac{\mathbf{a}_n(U)^{-1/2} \mathbf{a}_L(U) \mathbf{a}_n(U)^{-1/2}}{(\mathbf{a}_L \mathbf{a}_n^{-1})(0)} - \mathbf{I}_d \right| \right)^\xi \right]^{1/\xi} \\ &\leq C\xi^{1/2} \bar{\delta} 3^{k-n} \exp(C\xi \bar{\delta}^2 3^{2(k-n)}), \end{aligned}$$

and, for every $m \in \mathbb{N}_0$ with $m > n$,

$$\begin{aligned} (3.28) \quad &\mathbb{E} \left[\left| \mathbf{a}_{m,*}(U)^{-1/2} \mathbf{a}_{n,*}(U) \mathbf{a}_{m,*}(U)^{-1/2} - \mathbf{I}_d \right|^\xi + \left| \mathbf{a}_m(U)^{-1/2} \mathbf{a}_n(U) \mathbf{a}_m(U)^{-1/2} - \mathbf{I}_d \right|^\xi \right]^{1/\xi} \\ &+ \mathbb{E} \left[\left| \mathbf{a}_{n,*}(U)^{-1/2} \mathbf{a}_{m,*}(U) \mathbf{a}_{n,*}(U)^{-1/2} - \mathbf{I}_d \right|^\xi + \left| \mathbf{a}_n(U)^{-1/2} \mathbf{a}_m(U) \mathbf{a}_n(U)^{-1/2} - \mathbf{I}_d \right|^\xi \right]^{1/\xi} \\ &\leq (C\xi^{1/2} \bar{\delta} ((m-n)^{1/2} + (k-n)_+)) \exp(C\xi \bar{\delta}^2 (m-n + (k-n)_+^2)). \end{aligned}$$

Proof. Apply Lemmas 3.1 and 2.7 on $U \subseteq \square_k$, first to $(\mathbf{a}_n, (\mathbf{a}_n \mathbf{a}_L^{-1})(0) \mathbf{a}_L)$ and then with the two coefficients interchanged. Taking the supremum over $L \geq n$ and using (3.5) gives (3.27). For (3.28), apply Lemma 2.7 directly to the unnormalized pair $(\mathbf{a}_n, \mathbf{a}_m)$, in both orders, and use (3.7). □

For $m \in \mathbb{N}_0$, take the expectation of $\mathbf{a}_m(U)$ and the inverse of the expectation of $\mathbf{a}_{m,*}^{-1}(U)$:

$$(3.29) \quad \bar{\mathbf{a}}_m(U) := \mathbb{E}[\mathbf{a}_m(U)] \quad \text{and} \quad \bar{\mathbf{a}}_{m,*}(U) := \mathbb{E}[\mathbf{a}_{m,*}^{-1}(U)]^{-1}.$$

By subadditivity, the matrices $\bar{\mathbf{a}}_m(\square_n)$ and $\bar{\mathbf{a}}_{m,*}^{-1}(\square_n)$ are nonincreasing in n in the Loewner order. Using (2.20) and (2.22), the limits of $\bar{\mathbf{a}}_m(\square_n)$ and $\bar{\mathbf{a}}_{m,*}(\square_n)$ as $n \rightarrow \infty$ exist. We define

$$(3.30) \quad \bar{\mathbf{a}}_m := \lim_{n \rightarrow \infty} \bar{\mathbf{a}}_m(\square_n).$$

For each fixed m , the field $\mathbf{a}_m$ and its inverse have moments of every finite order by Lemma 3.1, so they satisfy the integrability hypothesis of the qualitative corrector construction [Bella and Kniely, 2024, Remark 1.5]. The stationary primal and dual variational formulas of that construction, together with the subadditive ergodic theorem for the energies on the cubes $\square_n$, identify the two limits with the same deterministic homogenized matrix; in particular $\bar{\mathbf{a}}_{m,*}(\square_n) \rightarrow \bar{\mathbf{a}}_m$. By Assumption (g3), $\bar{\mathbf{a}}_m(\square_n)$ and $\bar{\mathbf{a}}_{m,*}(\square_n)$ are scalar matrices for every n ; hence so is $\bar{\mathbf{a}}_m$.

Lemma 3.4. *For every bounded convex domain $U \subseteq \mathbb{R}^d$ and $m, n \in \mathbb{N}_0$ with $n < m$, we have*

$$(3.31) \quad \begin{cases} \bar{\mathbf{a}}_m(U) \leq \bar{\mathbf{a}}_n(U) \leq \mathbb{E} \left[\sup_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \right] \bar{\mathbf{a}}_m(U), \\ \exp(-2\tau^2(m-n)) \bar{\mathbf{a}}_{m,*}^{-1}(U) \leq \bar{\mathbf{a}}_{n,*}^{-1}(U) \leq \mathbb{E} \left[\sup_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \right] \bar{\mathbf{a}}_{m,*}^{-1}(U). \end{cases}$$

In particular, for every $m, n \in \mathbb{N}_0$ with $n < m$, we have

$$(3.32) \quad \bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_n \leq \exp(2\tau^2(m-n))\bar{\mathbf{a}}_m.$$

Proof. Since $\mathbf{a}_m \mathbf{a}_n^{-1}$ is independent of $\mathbf{a}_n$, it is independent of $v(\cdot, U, e, 0; \mathbf{a}_n)$. Moreover, for a constant c , the function $-v(\cdot, U, e, 0; \mathbf{a}_n) + c$ belongs to $e \cdot x + H_0^1(U)$. Hence

$$\begin{aligned} e \cdot \bar{\mathbf{a}}_m(U)e &= \mathbb{E} \left[\min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_m \nabla w \right] \\ &\leq \mathbb{E} \left[\int_U \nabla v(\cdot, U, e, 0; \mathbf{a}_n) \cdot \mathbf{a}_m \nabla v(\cdot, U, e, 0; \mathbf{a}_n) \right] \\ &= \int_U \mathbb{E} [\nabla v(x, U, e, 0; \mathbf{a}_n) \cdot \mathbf{a}_m(x) \nabla v(x, U, e, 0; \mathbf{a}_n)] dx \\ &= \int_U \underbrace{\mathbb{E} [\mathbf{a}_m(x) \mathbf{a}_n^{-1}(x)]}_{=1} \mathbb{E} [\nabla v(x, U, e, 0; \mathbf{a}_n) \cdot \mathbf{a}_n(x) \nabla v(x, U, e, 0; \mathbf{a}_n)] dx = e \cdot \bar{\mathbf{a}}_n(U)e. \end{aligned}$$

For the other direction, we have

$$\begin{aligned} e \cdot \bar{\mathbf{a}}_m(U)e &= \mathbb{E} \left[\min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_m \nabla w \right] \\ &\geq \mathbb{E} \left[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_n \nabla w \right] \\ &= \mathbb{E} \left[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \right] \mathbb{E} \left[\min_{w \in e \cdot x + H_0^1(U)} \int_U \nabla w \cdot \mathbf{a}_n \nabla w \right] = \mathbb{E} \left[\inf_U (\mathbf{a}_m \mathbf{a}_n^{-1}) \right] (e \cdot \bar{\mathbf{a}}_n(U)e). \end{aligned}$$

For the second line of (3.31), use the dual variational formula with the minimizing flux for $\mathbf{a}_n$:

$$\begin{aligned} e \cdot \bar{\mathbf{a}}_{m,*}^{-1}(U)e &= \mathbb{E} \left[\min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_m^{-1} \mathbf{h} \right] \\ &\leq \mathbb{E} \left[\int_U (\mathbf{a}_n \nabla v(\cdot, U, 0, e; \mathbf{a}_n)) \cdot \mathbf{a}_m^{-1} (\mathbf{a}_n \nabla v(\cdot, U, 0, e; \mathbf{a}_n)) \right] \\ &= \int_U \mathbb{E} [(\mathbf{a}_n(x) \nabla v(x, U, 0, e; \mathbf{a}_n)) \cdot \mathbf{a}_m^{-1}(x) (\mathbf{a}_n(x) \nabla v(x, U, 0, e; \mathbf{a}_n))] dx \\ &= \int_U \underbrace{\mathbb{E} [\mathbf{a}_m^{-1}(x) \mathbf{a}_n(x)]}_{=\exp(2\tau^2(m-n))} \mathbb{E} [\nabla v(x, U, 0, e; \mathbf{a}_n) \cdot \mathbf{a}_n(x) \nabla v(x, U, 0, e; \mathbf{a}_n)] dx \\ &= \exp(2\tau^2(m-n)) (e \cdot \bar{\mathbf{a}}_{n,*}^{-1}(U)e), \end{aligned}$$

and

$$\begin{aligned} e \cdot \bar{\mathbf{a}}_{m,*}^{-1}(U)e &= \mathbb{E} \left[\min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_m^{-1} \mathbf{h} \right] \\ &\geq \mathbb{E} \left[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_n^{-1} \mathbf{h} \right] \\ &= \mathbb{E} \left[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \right] \mathbb{E} \left[\min_{\mathbf{h} \in e + L_{\text{sol},0}^2(U)} \int_U \mathbf{h} \cdot \mathbf{a}_n^{-1} \mathbf{h} \right] = \mathbb{E} \left[\inf_U (\mathbf{a}_m^{-1} \mathbf{a}_n) \right] (e \cdot \bar{\mathbf{a}}_{n,*}^{-1}(U)e). \end{aligned}$$

□

By (3.32) and (1.6), there exists $C(d) < \infty$ such that, for every $m, n \in \mathbb{N}_0$ with $n \leq m$,

$$(3.33) \quad |\bar{\mathbf{a}}_m \bar{\mathbf{a}}_n^{-1} - \text{Id}| + |\bar{\mathbf{a}}_m^{-1} \bar{\mathbf{a}}_n - \text{Id}| \leq C \bar{\delta}^2(m-n) \exp(C \bar{\delta}^2(m-n)).$$

In dimension two, planar duality determines $\bar{\mathbf{a}}_m$ exactly.

Proposition 3.5. *In dimension $d = 2$, for every $m \in \mathbb{N}_0$, we have $\bar{\mathbf{a}}_m = \exp(-\tau^2(m+1))$.*

Proof. Let $R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. On the simply connected square $\square_n$, the divergence-free fields with zero normal component on the boundary are exactly the rotated gradients of functions vanishing on the boundary: $L_{\text{sol},0}^2(\square_n) = R\nabla H_0^1(\square_n)$. Substituting this into the dual minimum in (2.19) therefore gives, for every scalar coefficient a bounded above and away from zero on $\square_n$,

$$a_*^{-1}(\square_n; a) = R a(\square_n; a^{-1}) R^\top.$$

By the invariance of the law of $\mathbf{g}_0$ under $\mathbf{g}_0 \mapsto -\mathbf{g}_0$ in (g3), $\mathbf{a}_m^{-1}$ has the same law as $e^{2(m+1)\tau^2} \mathbf{a}_m$. Taking expectations in the preceding identity and using homogeneity and the invariance of the law under signed permutations, we obtain

$$\bar{\mathbf{a}}_{m,*}^{-1}(\square_n) = e^{2(m+1)\tau^2} \bar{\mathbf{a}}_m(\square_n).$$

Let $n \rightarrow \infty$ and use that $\bar{\mathbf{a}}_{m,*}(\square_n) \rightarrow \bar{\mathbf{a}}_m$, as shown after (3.30). The positive scalar $\bar{\mathbf{a}}_m$ satisfies $\bar{\mathbf{a}}_m^{-1} = e^{2(m+1)\tau^2} \bar{\mathbf{a}}_m$, hence $\bar{\mathbf{a}}_m = e^{-(m+1)\tau^2}$. This argument is a finite-volume variational form of planar duality [Dykhne, 1971]. $\square$

Lemma 3.6. *There exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{N}_0$, every $\xi \in [1, \infty)$, and every convex domain $U \subseteq \square_m$,*

$$(3.34) \quad \mathbb{E} \left[\max_{|e|=1} J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m)^\xi \right]^{1/\xi} \leq C \xi (m+1) \bar{\delta}^2 \exp(C \xi (m+1) \bar{\delta}^2).$$

Proof. By (2.17) and (2.20),

$$\max_{|e|=1} J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) \leq \frac{1}{2} \int_U \left(\frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} + \frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} - 2 \right) = \frac{1}{2} \int_U \left(\frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} - 1 \right) \left(1 - \frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} \right) \leq \frac{1}{4} \int_U \left(\left(\frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} - 1 \right)^2 + \left(\frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} - 1 \right)^2 \right).$$

Taking expectations in (2.20) also gives $e^{-2(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq 1$. Consequently, Lemma 3.1, with $n = -1$ and exponent 2ξ , and (1.6) bound the $L^\xi(\mathbb{P})$ norm of the right side of the last display by the right side of (3.34). $\square$

4. INDUCTIVE COARSE-GRAINING UP THE SCALES

In this section we prove the coarse-graining estimate at the cutoff scale, stated below in Proposition 4.1. For a bounded convex domain U and $m \in \mathbb{N}_0$, abbreviate

$$(4.1) \quad \mathcal{J}_m(U) := \max_{|e|=1} J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m).$$

On the cube $\square_m$, whose size matches the cutoff scale, the quantity $\mathcal{J}_m(\square_m)$ is bounded in $L^\xi(\mathbb{P})$ by $C\xi \log(2 + \xi) \bar{\delta}^2$ whenever $\xi \leq C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$. In particular, the bound does not deteriorate as further scales are added.

The quantity $\mathcal{J}_m(\square_m)$ is the gap between the two variational principles that define the coarse-grained matrices. It measures the error in replacing the coefficient on a cube by a constant matrix: by Lemma 2.1, it controls $\mathbf{a}_m(\square_m) - \mathbf{a}_{m,*}(\square_m)$ and the distance of these matrices from $\bar{\mathbf{a}}_m$, and it vanishes if and only if $\mathbf{a}_m(\square_m) = \mathbf{a}_{m,*}(\square_m) = \bar{\mathbf{a}}_m \mathbf{I}_d$. We argue by induction up the scales: the inductive hypothesis of Definition 4.2 bounds $\mathcal{J}_m(\square_m)$ in $L^\xi(\mathbb{P})$ for every m up to some m_0 , and the induction step extends this bound to the next scale. The field added at each step has a scale larger than the cubes already treated, so, by Section 3, it acts on them almost as a constant factor; the gain comes from estimating its effect on the larger cube in negative norms.

Proposition 4.1 (Coarse-graining estimate at the cutoff scale). *There are constants $c(d) > 0$ and $C(d) < \infty$ such that, for every $\xi \in [1, \infty)$, if the parameter $\bar{\delta}$ in Assumption (g2) satisfies*

$$(4.2) \quad \xi \leq C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1},$$

then, for every $m \in \mathbb{N}_0$, we have

$$(4.3) \quad \mathbb{E} \left[\max_{|e|=1} \left(J(\square_m, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) \right)^\xi \right]^{1/\xi} \leq C \xi \log(2 + \xi) \bar{\delta}^2.$$

Moreover, for every $s \in (0, 1]$ satisfying

$$4ds^{-1} \leq \xi, \quad C\xi \log(2 + \xi) \bar{\delta}^2 \leq cs \quad \text{and} \quad s \log(2 + \xi) \geq c,$$

we have

$$(4.4) \quad \mathbb{E} [\mathcal{E}_{s,\infty}(\square_m; \mathbf{a}_m, \bar{\mathbf{a}}_m)^\xi]^{1/\xi} \leq Cs^{-1} (\xi \log(2 + \xi) \bar{\delta}^2)^{1/2}.$$

After reducing the constant in (4.2), apply (4.3) with moment 2ξ and use Lemma 2.1 to obtain

$$(4.5) \quad \mathbb{E} [|\bar{\mathbf{a}}_m^{-1} \mathbf{a}_m(\square_m) - \mathbf{I}_d|^{2\xi}]^{1/\xi} + \mathbb{E} [|\bar{\mathbf{a}}_m \mathbf{a}_{m,*}^{-1}(\square_m) - \mathbf{I}_d|^{2\xi}]^{1/\xi} \leq C \xi \log(2 + \xi) \bar{\delta}^2.$$

The dependence on $\bar{\delta}$ is optimal: the scale- m field alone produces fluctuations of order $\bar{\delta}$ in the coarse-grained matrices.

Definition 4.2 (Inductive hypothesis $\mathcal{S}(m_0, \xi, \delta_1)$). Given $m_0 \in \mathbb{N}_0$, an exponent $\xi \in [1, \infty)$, and $\delta_1 \in (0, 1)$, let $\mathcal{S}(m_0, \xi, \delta_1)$ denote the statement

$$(4.6) \quad \max_{m \in \{0, \dots, m_0\}} \left\| \max_{|e|=1} J(\square_m, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) \right\|_{L^\xi(\mathbb{P})} \leq \delta_1.$$

Let $\mathcal{S}(\infty, \xi, \delta_1)$ denote the statement

$$\sup_{m \in \mathbb{N}_0} \left\| \max_{|e|=1} J(\square_m, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) \right\|_{L^\xi(\mathbb{P})} \leq \delta_1.$$

The bound (4.3) is the assertion that $\mathcal{S}(\infty, \xi, C\xi \log(2 + \xi) \bar{\delta}^2)$ holds throughout the stated range of ξ . The bound (4.4) on $\mathcal{E}_{s,\infty}$ then follows from Lemma 4.4 with $\delta_1 = C\xi \log(2 + \xi) \bar{\delta}^2$.

Proposition 4.1 is used in every later section: in Section 5 to derive the decay of $\bar{\mathbf{a}}_m$, in Section 6 to show that the scales with a large homogenization error are rare, and in Sections 8–10 to bound the coarse-grained ellipticity of $\mathbf{a}_L$. The proof has three parts and uses two inductions, the first of which supplies the low-moment estimate used in the second.

- In Section 4.1 we use $\mathcal{S}(m_0, \xi, \delta_1)$ to bound the coarse-graining error $\mathcal{E}_{s,\infty}$ of $\mathbf{a}_L$ on cubes below the cutoff scale (Lemma 4.4) and above it (Lemma 4.5).
- In Section 4.2 we prove the contraction of Proposition 4.10: under $\mathcal{S}(m_0, \xi, \delta_1)$, the quantity $\mathcal{J}_L(\square_m)$ with $m \geq L + C \log(2 + \xi)$ is at most $1/4\delta_1$ in $L^\xi(\mathbb{P})$.
- At the end of Section 4.2 we add the omitted fields $\mathbf{g}_{L+1}, \dots, \mathbf{g}_m$ using Lemma 2.8 and close the inductions.

4.1. A multiscale estimate. The statement $\mathcal{S}(m_0, \xi, \delta_1)$ bounds, for every $m \leq m_0$, the $L^\xi(\mathbb{P})$ norm of the maximum in (4.6). We now deduce multiscale bounds from this estimate.

Definition 4.3 (The coefficient $\mathbf{b}_{L,m}$). For $L, m \in \mathbb{N}_0$, define

$$(4.7) \quad \mathbf{b}_{L,m} := \bar{\mathbf{a}}_{m \wedge L} \frac{\mathbf{a}_L}{\mathbf{a}_{m \wedge L}}.$$

Thus, if $m \leq L$, then

$$\mathbf{b}_{L,m} = \bar{\mathbf{a}}_m \exp \left(\sum_{j=m+1}^L (\mathbf{g}_j - \tau^2) \right),$$

whereas $\mathbf{b}_{L,m} = \bar{\mathbf{a}}_L$ if $m \geq L$.

Lemma 4.4. *Let $m_0 \in \mathbb{N}_0$, $\xi \in [6, \infty)$ and $\delta_1 \in [\bar{\delta}^2, 1)$, and assume that $\mathcal{S}(m_0, \xi, \delta_1)$ holds. There exist constants $c(d) \in (0, 1]$ and $C(d) \in [1, \infty)$ such that, for every $s \in (0, 1]$ satisfying*

$$(4.8) \quad 4ds^{-1} \leq \xi \leq cs\bar{\delta}^{-2}\delta_1,$$

and every $m \in \mathbb{N}_0$ and $n \in \mathbb{Z} \cap (-\infty, m_0]$ with $m \geq n$, we have

$$(4.9) \quad \mathbb{E} \left[\sup_{L \geq m} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} (\mathcal{E}_{s,\infty}(z + \square_n; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}))^\xi \right]^{1/\xi} \leq Cs^{-1} \delta_1^{1/2} 3^{(d/\xi + s/2)(m-n)},$$

and, for every $L \in \mathbb{N}_0$ with $L \leq m_0$,

$$(4.10) \quad \mathbb{E} [\mathcal{E}_{s,\infty}(\square_L; \mathbf{a}_L, \bar{\mathbf{a}}_L)^\xi]^{1/\xi} \leq Cs^{-1} \delta_1^{1/2}.$$

Proof. Let $s \in (0, 1]$ satisfy (4.8).

Step 1. We first treat the scales at which the induction hypothesis applies. We show the existence of $C(d) < \infty$ such that for every $m, k \in \mathbb{N}_0$ with $k \leq m \wedge m_0$ and every $z \in 3^k \mathbb{Z}^d \cap \square_m$,

$$(4.11) \quad \mathbb{E} \left[\sup_{L \geq m} \max_{|e|=1} \left(J(z + \square_k, (\mathbf{b}_{L,m})_{\square_m}^{-1/2} e, (\mathbf{b}_{L,m})_{\square_m}^{1/2} e; \mathbf{a}_L) \right)^\xi \right]^{1/\xi} \leq C \delta_1 3^{s(m-k)}.$$

We apply Lemma 2.8 with $\delta = 1$ and $\lambda = (\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m} \bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1}$, and with $\mathbf{b} = \mathbf{a}_L$ and $\mathbf{a} = \mathbf{a}_k$. We obtain, for every $L, m, k \in \mathbb{N}_0$ and $z \in 3^k \mathbb{Z}^d \cap \square_m$ with $L \geq m \geq k$, and $e \in \mathbb{R}^d$ with $|e| = 1$,

$$(4.12) \quad \begin{aligned} & J(z + \square_k, (\mathbf{b}_{L,m})_{\square_m}^{-1/2} e, (\mathbf{b}_{L,m})_{\square_m}^{1/2} e; \mathbf{a}_L) \\ & \leq 2J(z + \square_k, \bar{\mathbf{a}}_k^{-1/2} e, \bar{\mathbf{a}}_k^{1/2} e; \mathbf{a}_k) \\ & \quad + 3 \left(\sup_{L \geq m} \sup_{x \in \square_m, y \in z + \square_k} (|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} (\mathbf{a}_L \mathbf{a}_m^{-1})(x) (\mathbf{a}_L^{-1} \mathbf{a}_k)(y) - 1| \right. \\ & \quad \left. + |\bar{\mathbf{a}}_m^{-1} \bar{\mathbf{a}}_k (\mathbf{a}_L^{-1} \mathbf{a}_m)(x) (\mathbf{a}_L \mathbf{a}_k^{-1})(y) - 1|) \right)^2 (J(z + \square_k, \bar{\mathbf{a}}_k^{-1/2} e, \bar{\mathbf{a}}_k^{1/2} e; \mathbf{a}_k) + 1). \end{aligned}$$

We first bound the moments of the supremum in this display:

$$(4.13) \quad \mathbb{E} \left[\left(\sup_{L \geq m} \sup_{x \in \square_m, y \in z + \square_k} (|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} (\mathbf{a}_L \mathbf{a}_m^{-1})(x) (\mathbf{a}_L^{-1} \mathbf{a}_k)(y) - 1| + |\bar{\mathbf{a}}_m^{-1} \bar{\mathbf{a}}_k (\mathbf{a}_L^{-1} \mathbf{a}_m)(x) (\mathbf{a}_L \mathbf{a}_k^{-1})(y) - 1|) \right)^{2\xi} \right]^{1/\xi} \leq C \delta_1 3^{s(m-k)}.$$

Given (4.13), the induction hypothesis $\mathcal{S}(m_0, \xi, \delta_1)$ and the independence of this supremum from the field $\mathbf{a}_k$ give (4.11).

To prove (4.13), we write

$$|\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} (\mathbf{a}_L \mathbf{a}_m^{-1})(x) (\mathbf{a}_L^{-1} \mathbf{a}_k)(y) - 1| = \left| \bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} \frac{(\mathbf{a}_L \mathbf{a}_m^{-1})(x)}{(\mathbf{a}_L \mathbf{a}_m^{-1})(y)} (\mathbf{a}_m^{-1} \mathbf{a}_k)(y) - 1 \right|,$$

and use the general bound

$$\begin{aligned} |uvw - 1| & \leq |u - 1| + |v - 1| + |w - 1| + |(u - 1)(v - 1)| \\ & \quad + |(u - 1)(w - 1)| + |(v - 1)(w - 1)| + |(u - 1)(v - 1)(w - 1)|, \end{aligned}$$

together with the independence of the fields $\mathbf{a}_L \mathbf{a}_m^{-1}$ and $\mathbf{a}_m^{-1} \mathbf{a}_k$ and the following inequalities, invoked at moment 2ξ , which follow from (3.33), (3.5), (3.7) and (4.8):

$$(4.14) \quad |\bar{\mathbf{a}}_m \bar{\mathbf{a}}_k^{-1} - 1| \leq C \bar{\delta}^2(m-k) \exp(C \bar{\delta}^2(m-k)) \leq C \delta_1^{1/2} 3^{s/6(m-k)},$$

$$(4.15) \quad \mathbb{E} \left[\left(\sup_{L \geq m} \sup_{x, y \in \square_m} \frac{(\mathbf{a}_L \mathbf{a}_m^{-1})(x)}{(\mathbf{a}_L \mathbf{a}_m^{-1})(y)} - 1 \right)^{2\xi} \right]^{1/\xi} \leq C \xi \bar{\delta}^2 \exp(C \xi \bar{\delta}^2(m-k)) \leq C \delta_1 3^{s/3(m-k)},$$

$$(4.16) \quad \mathbb{E} \left[\|\mathbf{a}_m^{-1} \mathbf{a}_k - 1\|_{L^\infty(\square_k)}^{2\xi} \right]^{1/\xi} \leq C \xi \bar{\delta}^2(m-k) \exp(C \xi \bar{\delta}^2(m-k)) \leq C \delta_1 3^{s/3(m-k)}.$$

Each of the three deviations thus has $L^{2\xi}$ norm at most $C \delta_1^{1/2} 3^{s(m-k)/6}$. The two random factors depend on disjoint sets of fields, and the remaining factor is deterministic, so the $L^{2\xi}$ norm of a product of deviations is the product of their $L^{2\xi}$ norms. Since $\delta_1 < 1$, even the product of all three is bounded by $C \delta_1^{1/2} 3^{s(m-k)/2}$. The bound on $|XYZ - 1|$ above therefore proves (4.13), and hence (4.11).

We next estimate the terms in Definition 2.3 with cube size smaller than one. Equations (2.17) and (2.20) imply that, for every positive scalar coefficient $\mathbf{a}$, positive scalar b , bounded convex domain V , and $|e| = 1$,

$$(4.17) \quad J(V, b^{-1/2}e, b^{1/2}e; \mathbf{a}) \leq \frac{1}{2} \int_V \left(\frac{\mathbf{a}}{b} + \frac{b}{\mathbf{a}} - 2 \right).$$

Indeed, (2.20) bounds the two quadratic terms in (2.17) by the corresponding averages of $\mathbf{a}$ and $\mathbf{a}^{-1}$. To estimate the right side with $\mathbf{a} = \mathbf{a}_L$ and $b = (\mathbf{b}_{L,m})_{\square_m}$, we factor

$$\frac{\mathbf{a}_L}{(\mathbf{b}_{L,m})_{\square_m}} = \frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} \cdot \frac{\mathbf{a}_L \mathbf{a}_m^{-1}}{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}}.$$

The elementary inequality

$$r + r^{-1} - 2 \leq (r - 1)^2 + (r^{-1} - 1)^2,$$

valid for every $r > 0$, allows us to estimate the two factors separately. We first estimate $\mathbf{a}_m / \bar{\mathbf{a}}_m$. For each unit cube $z + \square_0$ meeting $\square_m$, independence and Assumptions (g1)–(g2) give, for every $p \geq 1$,

$$\left\| \sum_{i=0}^m \mathbf{g}_i(z) \right\|_{L^p(\mathbb{P})} \leq C \sqrt{p} \bar{\delta} \sqrt{m+1} \quad \text{and} \quad \left\| \sum_{i=0}^m \text{osc}_{z+\square_0} \mathbf{g}_i \right\|_{L^p(\mathbb{P})} \leq C \sqrt{p} \bar{\delta}.$$

There are at most $C 3^{dm}$ such unit cubes. We apply (A.8) on each cube and then take a union bound over these cubes. The variational bounds (2.20), together with $\mathbb{E}[\mathbf{a}_0(0)] = 1$ and $\mathbb{E}[\mathbf{a}_0(0)^{-1}] = e^{2\tau^2}$, give $e^{-2\tau^2} \leq \bar{\mathbf{a}}_0 \leq 1$. Combining this with (3.33) controls the deterministic factor $\bar{\mathbf{a}}_m$. This yields

$$\begin{aligned} & \mathbb{E} \left[\left\| \frac{\mathbf{a}_m}{\bar{\mathbf{a}}_m} + \frac{\bar{\mathbf{a}}_m}{\mathbf{a}_m} - 2 \right\|_{L^\infty(\square_m)}^\xi \right]^{1/\xi} \\ & \leq C \xi \bar{\delta}^2(m+1) \exp \left(\frac{d(\log 3)m + C}{\xi} + C \xi \bar{\delta}^2(m+1) + C \bar{\delta}^2 m \right). \end{aligned}$$

For the second factor, the average in the denominator lies between the minimum and the maximum of $\mathbf{a}_L \mathbf{a}_m^{-1}$ on $\square_m$. Hence (3.5), applied with $n = k = m$, gives the same bound with $m+1$ replaced by a dimensional constant. The two factors are independent. Hölder's inequality and $|uv - 1| \leq$

$|u - 1||v| + |v - 1|$, applied also to the inverses, therefore give

$$(4.18) \quad \mathbb{E} \left[\sup_{L \geq m} \left\| \frac{\mathbf{a}_L}{(\mathbf{b}_{L,m})_{\square_m}} + \frac{(\mathbf{b}_{L,m})_{\square_m}}{\mathbf{a}_L} - 2 \right\|_{L^\infty(\square_m)}^\xi \right]^{1/\xi} \\ \leq C \xi \bar{\delta}^2 (m+1) \exp \left(\frac{d(\log 3)m + C}{\xi} + C \xi \bar{\delta}^2 (m+1) + C \bar{\delta}^2 m \right) \leq C \delta_1 3^{sm}.$$

The last inequality follows from (4.8). Thus, by (4.17), for every $k < 0$,

$$(4.19) \quad \mathbb{E} \left[\sup_{L \geq m} \max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \max_{|e|=1} J(z + \square_k, (\mathbf{b}_{L,m})_{\square_m}^{-1/2} e, (\mathbf{b}_{L,m})_{\square_m}^{1/2} e; \mathbf{a}_L)^\xi \right]^{1/\xi} \leq C \delta_1 3^{sm}.$$

Step 2. We prove (4.9). Split the sum in Definition 2.3 into $k \geq 0$ and $k < 0$. Minkowski's inequality, (4.11), (4.19), and the union bound over the $O(3^{d(m-k)})$ cubes of scale k give

$$\left\| \sup_{L \geq m} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \mathcal{E}_{s,\infty}(z + \square_n; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}) \right\|_{L^\xi(\mathbb{P})} \leq C \delta_1^{1/2} 3^{d/\xi(m-n)} \sum_{k=0}^n 3^{-(s/2-d/\xi)(n-k)+s/2(m-n)} \mathbb{1}_{\{n \geq 0\}} \\ + C \delta_1^{1/2} 3^{s/2m} \sum_{k=-\infty}^{n \wedge (-1)} 3^{-s(n-k)} \\ \leq C s^{-1} \delta_1^{1/2} 3^{(d/\xi+s/2)(m-n)}.$$

Here we used $\xi \geq 4ds^{-1}$ from (4.8).

Step 3. We prove (4.10). Specializing (4.9) to $m = n = L$ and using $\mathbf{b}_{L,L} = \bar{\mathbf{a}}_L$ gives (4.10). $\square$

Lemma 4.5 (Coarse-graining error on cubes above the cutoff scale). *There exist $c(d) > 0$ and $C(d) < \infty$ such that the following holds. Let $L, m_0 \in \mathbb{N}_0$ satisfy $L \leq m_0$, let $s \in (0, 1]$ and $\delta_1 \in [\bar{\delta}^2, 1)$, and let ξ satisfy*

$$(4.20) \quad 4ds^{-1} \leq \xi \leq cs \bar{\delta}^{-2} \delta_1.$$

If $\mathcal{S}(m_0, \xi, \delta_1)$ is valid, then the following estimate holds for every $M \geq L$, every translate U of $\square_M$, and $r \in \{1, 2\}$:

$$(4.21) \quad \|\mathcal{E}_{s,\infty,r}(U; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^\xi(\mathbb{P})} \leq C s^{-1/r} \delta_1^{1/2}.$$

Proof. It is enough, by stationarity, to take $U = \square_M$. Recall the notation $\mathcal{J}_L(U)$ of (4.1). Definition 2.3 and Minkowski's inequality give

$$(4.22) \quad \|\mathcal{E}_{s,\infty,r}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^\xi}^r \leq C \sum_{k=-\infty}^M 3^{-sr(M-k)} \left\| \max_{z \in 3^k \mathbb{Z}^d \cap \square_M} \mathcal{J}_L(z + \square_k)^{1/2} \right\|_{L^\xi}^r.$$

We split the sum into three ranges.

If $L \leq k \leq M$, then subadditivity on the partition into cubes of scale L , the induction hypothesis, and Jensen's inequality yield

$$\|\mathcal{J}_L(z + \square_k)\|_{L^\xi} \leq C \delta_1.$$

Taking the maximum over at most $C 3^{d(M-k)}$ cubes gives a contribution to (4.22) bounded by

$$C \delta_1^{r/2} \sum_{j=0}^{M-L} 3^{-r(s-d/\xi)j} \leq C s^{-1} \delta_1^{r/2}.$$

If $0 \leq k < L$, then (4.11) with $m = L$ and cutoff L gives

$$\|\mathcal{J}_L(z + \square_k)\|_{L^\xi} \leq C \delta_1 3^{s(L-k)}.$$

After taking the maximum over the cubes of scale k in $\square_M$, we bound the corresponding summand by

$$C\delta_1^{r/2}3^{-r(s-d/\xi)(M-L)}3^{-r(s/2-d/\xi)(L-k)}.$$

The sum of these bounds over k is at most $Cs^{-1}\delta_1^{r/2}$.

Finally, if $k < 0$, apply (4.19) with $m = L$ on each scale- L parent cube. The maximum over the at most $C3^{d(M-L)}$ such cubes gives

$$\begin{aligned} & \sum_{k=-\infty}^{-1} 3^{-sr(M-k)} \left\| \max_{z \in 3^k \mathbb{Z}^d \cap \square_M} \mathcal{J}_L(z + \square_k)^{1/2} \right\|_{L^\xi}^r \\ & \leq C\delta_1^{r/2} 3^{-r(s-d/\xi)(M-L)} 3^{-r(s/2-d/\xi)L} \sum_{j=1}^{\infty} 3^{-srj} \leq Cs^{-1}\delta_1^{r/2}. \end{aligned}$$

Inserting the estimates into (4.22) and taking the r th root proves the lemma. $\square$

4.2. The contraction step. We now prove the contraction step: if $\mathcal{J}_m(\square_m)$ is at most δ_1 in $L^\xi(\mathbb{P})$ for every scale m up to m_0 , then, for $L \leq m_0$ and a cube $\square_m$ of scale $m \geq L + C \log(2 + \xi)$, the quantity $\mathcal{J}_L(\square_m)$, which measures $\mathbf{a}_L$ against $\bar{\mathbf{a}}_L$, is at most $1/4\delta_1$ in $L^\xi(\mathbb{P})$. This is Proposition 4.10, and the factor $1/4$ compensates for the losses incurred elsewhere in the induction. We prove it by the multiscale variational argument of Armstrong et al. [2019], using the estimates of Armstrong and Kuusi [2025], whose constants depend on coarse-grained ellipticity constants rather than on pointwise bounds on the coefficient. We first bound J by a negative Sobolev norm of the gradient of its maximizer and by the amount by which the values of J on the subcubes exceed its value on the cube (Lemma 4.6). Throughout this subsection, for $p, q \in \mathbb{R}^d$, a coefficient field $\mathbf{a}$, $n \in \mathbb{Z}$ and $z \in 3^n \mathbb{Z}^d$, we write $v_{n,z} := v(\cdot, z + \square_n, p, q; \mathbf{a})$ for the maximizer of $J(z + \square_n, p, q; \mathbf{a})$, and we abbreviate $v_n := v_{n,0}$.

Lemma 4.6 (Estimate of J by a negative Sobolev norm). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{N}$, coefficient field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$ and $p, q \in \mathbb{R}^d$,*

$$\begin{aligned} J(\square_{m-1}, p, q; \mathbf{a}) & \leq C\Lambda_{1/4}(\square_m; \mathbf{a}) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a})}{\lambda_{1/4}(\square_m; \mathbf{a})} \right)^{1/2} 3^{-2m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a})\|_{\underline{H}^{-1}(\square_m)}^2 \\ (4.23) \quad & + \sum_{z \in 3^{m-1} \mathbb{Z}^d \cap \square_m} 2(J(z + \square_{m-1}, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a})). \end{aligned}$$

Proof. We compute, using the triangle inequality, (2.28) and (2.29),

$$\begin{aligned} J(\square_{m-1}, p, q; \mathbf{a}) & = \int_{\square_{m-1}} \frac{1}{2} \nabla v_{m-1} \cdot \mathbf{a} \nabla v_{m-1} \\ & \leq \int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m + \int_{\square_{m-1}} (\nabla v_m - \nabla v_{m-1}) \cdot \mathbf{a} (\nabla v_m - \nabla v_{m-1}) \\ & \leq \int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m + \sum_{z \in 3^{m-1} \mathbb{Z}^d \cap \square_m} \int_{z + \square_{m-1}} (\nabla v_m - \nabla v_{m-1,z}) \cdot \mathbf{a} (\nabla v_m - \nabla v_{m-1,z}) \\ & = \int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m + \sum_{z \in 3^{m-1} \mathbb{Z}^d \cap \square_m} 2(J(z + \square_{m-1}, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a})). \end{aligned}$$

The coarse-grained Caccioppoli inequality (Proposition 2.6) yields

$$\int_{\square_{m-1}} \nabla v_m \cdot \mathbf{a} \nabla v_m \leq C\Lambda_{1/4}(\square_m; \mathbf{a}) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a})}{\lambda_{1/4}(\square_m; \mathbf{a})} \right)^{1/2} 3^{-2m} \int_{\square_m} |v_m - (v_m)_{\square_m}|^2.$$

Combining these and using $\|v_m - (v_m)_{\square_m}\|_{L^2(\square_m)} \leq C\|\nabla v_m\|_{\underline{H}^{-1}(\square_m)}$, we obtain (4.23). $\square$

Lemma 4.7 (Negative-norm estimates). *There exists $C(d) < \infty$ such that, for every $m \in \mathbb{N}$, coefficient field $\mathbf{a} : \square_m \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, exponent $s \in (0, 1)$, integer $L_0 \leq m$, and $p, q \in \mathbb{R}^d$, we have*

$$\begin{aligned}
& 3^{-m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a}) - (\mathbf{a}_*^{-1}(\square_m)q - p)\|_{\hat{H}^{-1}(\square_m)} \\
& \leq C \lambda_s^{-1/2}(\square_m; \mathbf{a}) \sum_{k=L_0}^m 3^{-(1-s)(m-k)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2} \\
& \quad + C \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(\square_m)q - \mathbf{a}_*^{-1}(z + \square_n)q|^2 \right)^{1/2} \\
& \quad + \frac{C \lambda_s^{-1/2}(\square_m; \mathbf{a})}{1-s} 3^{-(1-s)(m-L_0)} \|\mathbf{a}^{1/2} \nabla v(\cdot, \square_m, p, q; \mathbf{a})\|_{\underline{L}^2(\square_m)} \\
& \quad + C 3^{-(m-L_0)} |\mathbf{a}_*^{-1}(\square_m)q - p|.
\end{aligned} \tag{4.24}$$

Proof. The duality inequality (2.13), applied to $\nabla v_m - (\nabla v_m)_{\square_m}$ and an H^1 test function, together with the Poincaré inequality on cubes, gives the first inequality below. For the second, we split the sum at L_0 and use the triangle inequality:

$$\begin{aligned}
& 3^{-m} \|\nabla v_m - (\mathbf{a}_*^{-1}(\square_m)q - p)\|_{\hat{H}^{-1}(\square_m)} \\
& \leq C \sum_{n=-\infty}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_n} - (\mathbf{a}_*^{-1}(\square_m)q - p)|^2 \right)^{1/2} \\
& \leq C \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p)|^2 \right)^{1/2} \\
& \quad + C \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(\square_m)q - \mathbf{a}_*^{-1}(z + \square_n)q|^2 \right)^{1/2} \\
& \quad + C \sum_{n=-\infty}^{L_0-1} 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(|(\nabla v_m)_{z+\square_n}|^2 + |\mathbf{a}_*^{-1}(\square_m)q - p|^2 \right) \right)^{1/2}.
\end{aligned} \tag{4.25}$$

For $n \in \mathbb{Z}$ with $n \leq L_0 - 1$, we have, by (2.31) and (2.49),

$$\begin{aligned}
\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z+\square_n}|^2 & \leq \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_*^{-1}(z + \square_n)| \|\mathbf{a}^{1/2} \nabla v_m\|_{\underline{L}^2(\square_m)}^2 \\
& \leq 3^{2s(m-n)} \lambda_s^{-1}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla v_m\|_{\underline{L}^2(\square_m)}^2.
\end{aligned}$$

This gives the following bound for the third term on the right side of (4.25):

$$\begin{aligned}
& \sum_{n=-\infty}^{L_0-1} 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left(|(\nabla v_m)_{z+\square_n}|^2 + |\mathbf{a}_*^{-1}(\square_m)q - p|^2 \right) \right)^{1/2} \\
& \leq \frac{C}{1-s} 3^{-(1-s)(m-L_0)} \lambda_s^{-1/2}(\square_m; \mathbf{a}) \|\mathbf{a}^{1/2} \nabla v_m\|_{\underline{L}^2(\square_m)} + C 3^{-(m-L_0)} |\mathbf{a}_*^{-1}(\square_m)q - p|.
\end{aligned} \tag{4.26}$$

Next we use (2.23) and (2.31) to obtain, for each $n \leq m$,

$$\begin{aligned}
|\mathbf{a}_*^{1/2}(z + \square_n)((\nabla v_m)_{z+\square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p))|^2 & = |\mathbf{a}_*^{1/2}(z + \square_n)(\nabla v_m - \nabla v_{n,z})_{z+\square_n}|^2 \\
& \leq \int_{z+\square_n} (\nabla v_m - \nabla v_{n,z}) \cdot \mathbf{a}(\nabla v_m - \nabla v_{n,z}).
\end{aligned}$$

Summing this over $z \in 3^n \mathbb{Z}^d \cap \square_m$ and applying (2.28) yields, for each $n \leq m$,

$$\begin{aligned} & \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \left| \mathbf{a}_*^{1/2}(z + \square_n) ((\nabla v_m)_{z + \square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p)) \right|^2 \\ & \leq \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} \int_{z + \square_n} (\nabla v_m - \nabla v_{n,z}) \cdot \mathbf{a}(\nabla v_m - \nabla v_{n,z}) \\ & = 2 \sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (J(z + \square_n, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a})). \end{aligned}$$

Using that, by (2.49), for every $P \in \mathbb{R}^d$, every $n \leq m$ and every $z \in 3^n \mathbb{Z}^d \cap \square_m$,

$$(4.27) \quad |P|^2 \leq |\mathbf{a}_*^{-1}(z + \square_n)| |\mathbf{a}_*^{1/2}(z + \square_n)P|^2 \leq 3^{2s(m-n)} \lambda_s^{-1}(\square_m; \mathbf{a}) |\mathbf{a}_*^{1/2}(z + \square_n)P|^2,$$

we obtain the following bound for the first term on the right side of (4.25):

$$\begin{aligned} & \sum_{n=L_0}^m 3^{-(m-n)} \left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |(\nabla v_m)_{z + \square_n} - (\mathbf{a}_*^{-1}(z + \square_n)q - p)|^2 \right)^{1/2} \\ (4.28) \quad & \leq C \lambda_s^{-1/2}(\square_m; \mathbf{a}) \sum_{k=L_0}^m 3^{-(1-s)(m-k)} \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_m} J(z + \square_k, p, q; \mathbf{a}) - J(\square_m, p, q; \mathbf{a}) \right)^{1/2}. \end{aligned}$$

Combining (4.25), (4.26) and (4.28) yields (4.24). $\square$

For a random matrix M , we write $\text{var}[M] := \mathbb{E}[|M - \mathbb{E}M|^2]$, with the matrix norm fixed in Section 2.1.

Lemma 4.8 (Variance estimate). *For every $\mathbb{Z}^d$ -stationary random coefficient field $\mathbf{a} : \mathbb{R}^d \rightarrow \mathbb{R}_{\text{sym},+}^{d \times d}$, every $n, k \in \mathbb{Z}$ with $k < n$, and every positive-definite matrix $\mathbf{b} \in \mathbb{R}_{\text{sym},+}^{d \times d}$, we have*

$$(4.29) \quad \text{var}[\mathbf{a}_*^{-1}(\square_n)] \leq 2 \text{var} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) \right] + 32 \mathbb{E} \left[\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1}e_i, e_i; \mathbf{a}) \right)^2 \right].$$

Proof. For the operator norm, the triangle inequality and Jensen's inequality give $\text{var}[M] \leq 4\mathbb{E}[|M|^2]$ for every random matrix M . We apply this to the difference between $\mathbf{a}_*^{-1}(\square_n)$ and the average of $\mathbf{a}_*^{-1}(z + \square_k)$ over $z \in 3^k \mathbb{Z}^d \cap \square_n$, and obtain, for every $k < n$,

$$(4.30) \quad \text{var}[\mathbf{a}_*^{-1}(\square_n)] \leq 2 \text{var} \left[\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) \right] + 8 \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \mathbf{a}_*^{-1}(\square_n) \right|^2 \right].$$

We compare the arithmetic and harmonic means. Let $\mathbf{b}_1, \dots, \mathbf{b}_N$ be symmetric positive-definite matrices, and write $\mathbf{h} := (\sum_{i=1}^N \mathbf{b}_i^{-1})^{-1}$ for their harmonic mean. For every positive-definite matrix $\mathbf{b}$,

$$(4.31) \quad \sum_{i=1}^N \mathbf{b}_i = \mathbf{h} + \sum_{i=1}^N (\mathbf{b}_i - \mathbf{b}) \mathbf{b}_i^{-1} (\mathbf{b}_i - \mathbf{b}) - \underbrace{(\mathbf{h} - \mathbf{b}) \mathbf{h}^{-1} (\mathbf{h} - \mathbf{b})}_{\geq 0} \leq \mathbf{h} + \sum_{i=1}^N (\mathbf{b}_i - \mathbf{b}) \mathbf{b}_i^{-1} (\mathbf{b}_i - \mathbf{b}).$$

Apply (4.31) with $\{\mathbf{a}^{-1}(z + \square_k) : z \in 3^k \mathbb{Z}^d \cap \square_n\}$ in place of the $\mathbf{b}_i$ and combine this with (2.36) and (2.20) to obtain, for every positive-definite matrix $\mathbf{b}$,

$$\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}(z + \square_k) \right)^{-1} \leq 2 \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1}e_i, e_i; \mathbf{a}) \mathbf{I}_d.$$

Using the previous display and the subadditivity of the coarse-grained matrices $\mathbf{a}_*^{-1}$ and $\mathbf{a}$, we find

$$\begin{aligned}
 \mathbf{a}^{-1}(\square_n) &\leq \mathbf{a}_*^{-1}(\square_n) \leq \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) \\
 &\leq \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}(z + \square_k) \right)^{-1} + 2 \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right) \mathbf{I}_d \\
 (4.32) \quad &\leq \mathbf{a}^{-1}(\square_n) + 2 \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right) \mathbf{I}_d.
 \end{aligned}$$

We deduce from (4.32) that

$$0 \leq \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \mathbf{a}_*^{-1}(\square_n) \leq 2 \left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right) \mathbf{I}_d,$$

and therefore

$$(4.33) \quad \mathbb{E} \left[\left| \sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \mathbf{a}_*^{-1}(z + \square_k) - \mathbf{a}_*^{-1}(\square_n) \right|^2 \right] \leq 4 \mathbb{E} \left[\left(\sum_{z \in 3^k \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_k, \mathbf{b}^{-1} e_i, e_i; \mathbf{a}) \right)^2 \right].$$

Combining this estimate with (4.30) yields (4.29). $\square$

We next combine the previous four lemmas into an upper bound on $\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$.

Proposition 4.9. *There exist constants $c(d) > 0$ and $C(d) < \infty$ such that the following holds. Let $L \in \mathbb{N}_0$, $\xi \in [16d, \infty)$ and $\delta_1 \in [C\xi\bar{\delta}^2, c]$ be such that $\mathcal{S}(L, \xi, \delta_1)$ is valid. For every $m \in \mathbb{N}$ with $m \geq L$ and $C3^{-(m-L)} \leq 1/4$, and every $p, q \in \mathbb{R}^d$ satisfying*

$$(4.34) \quad q = \bar{\mathbf{a}}_L p \quad \text{and} \quad |q|^2 \leq \bar{\mathbf{a}}_L,$$

we have the estimate

$$\begin{aligned}
 &\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)] \\
 (4.35) \quad &\leq C\delta_1(\delta_1 + 3^{-(m-L)}) + C \sum_{n=L}^m 3^{-(m-n)} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)].
 \end{aligned}$$

Proof. Define the event $\mathbf{G}$ by

$$(4.36) \quad \mathbf{G} := \left\{ \frac{1}{2} \bar{\mathbf{a}}_L \leq \lambda_{1/4}(\square_m; \mathbf{a}_L) \leq \Lambda_{1/4}(\square_m; \mathbf{a}_L) \leq 2\bar{\mathbf{a}}_L \right\}.$$

Step 1. We combine the previous lemmas to obtain an upper bound for $\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$. We show that there exists $C(d) < \infty$ such that, if $3^{-3/2(m-L)}C \leq 1/4$, then

$$\begin{aligned}
 \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)] &\leq C \sum_{n=L}^m 3^{-(m-n)} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)] \\
 &\quad + C\bar{\mathbf{a}}_L^2 \sum_{n=L}^m 3^{-(m-n)} \left(\text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] + |\bar{\mathbf{a}}_{L,*}^{-1}(\square_n) - \bar{\mathbf{a}}_L^{-1}|^2 \right) \\
 (4.37) \quad &\quad + C\mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbf{1}_{\mathbf{G}^c}].
 \end{aligned}$$

Taking expectations in (4.23) and using the subadditivity of J and stationarity, we obtain

$$\begin{aligned} \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)] &\leq \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L)] \\ &= \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_G] + \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_{G^c}] \\ &\leq C \mathbb{E} \left[\Lambda_{1/4}(\square_m; \mathbf{a}_L) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a}_L)}{\lambda_{1/4}(\square_m; \mathbf{a}_L)} \right)^{1/2} 3^{-2m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a}_L)\|_{\underline{H}^{-1}(\square_m)}^2 \mathbb{1}_G \right] \\ &\quad + C \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)] + C \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_{G^c}]. \end{aligned}$$

Applying Lemma 4.7, squaring both sides of (4.24), taking expectations and using that

$$\left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a}_L)}{\lambda_{1/4}(\square_m; \mathbf{a}_L)} \right)^{3/2} \mathbb{1}_G \leq 8 \quad \text{and} \quad \Lambda_{1/4}(\square_m; \mathbf{a}_L) \mathbb{1}_G \leq 2\bar{\mathbf{a}}_L,$$

we obtain

$$\begin{aligned} &\mathbb{E} \left[\Lambda_{1/4}(\square_m; \mathbf{a}_L) \left(\frac{\Lambda_{1/4}(\square_m; \mathbf{a}_L)}{\lambda_{1/4}(\square_m; \mathbf{a}_L)} \right)^{1/2} 3^{-2m} \|\nabla v(\cdot, \square_m, p, q; \mathbf{a}_L)\|_{\underline{H}^{-1}(\square_m)}^2 \mathbb{1}_G \right] \\ &\leq C \sum_{n=L}^m 3^{-(m-n)} \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L) - J(\square_m, p, q; \mathbf{a}_L)] \\ &\quad + C\bar{\mathbf{a}}_L \sum_{n=L}^m 3^{-(m-n)} \mathbb{E} \left[\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_{L,*}^{-1}(\square_m)q - \mathbf{a}_{L,*}^{-1}(z + \square_n)q|^2 \right] \\ &\quad + C\bar{\mathbf{a}}_L \mathbb{E}[|\mathbf{a}_{L,*}^{-1}(\square_m)q - p|^2] + C3^{-3/2(m-L)} \mathbb{E}[\|\mathbf{a}_L^{1/2} \nabla v\|_{\underline{L}^2(\square_m)}^2]. \end{aligned}$$

Under the condition $C3^{-3(m-L)/2} \leq 1/4$, we use (2.29) to bound the last term by $1/2 \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)]$, which we absorb into the left side. By stationarity and the triangle inequality,

$$\begin{aligned} &\mathbb{E} \left[\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} |\mathbf{a}_{L,*}^{-1}(\square_m) - \mathbf{a}_{L,*}^{-1}(z + \square_n)|^2 \right] \\ &\leq 3 \text{var}[\mathbf{a}_{L,*}^{-1}(\square_m)] + 3 \text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] + 3|\bar{\mathbf{a}}_{L,*}^{-1}(\square_m) - \bar{\mathbf{a}}_{L,*}^{-1}(\square_n)|^2. \end{aligned}$$

Combining the previous two displays and using (4.34), we obtain (4.37).

Step 2. Assuming $\mathcal{S}(L, \xi, \delta_1)$, we prove that, for every $n \geq L$,

$$(4.38) \quad |\bar{\mathbf{a}}_{L,*}^{-1}(\square_n) - \bar{\mathbf{a}}_L^{-1}|^2 \leq C\bar{\mathbf{a}}_L^{-2}\delta_1^2.$$

Indeed, subadditivity on the partition into cubes of scale L and the induction hypothesis give, for every $|e| = 1$,

$$\mathbb{E}[J(\square_n, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)] \leq \delta_1.$$

On the other hand, (2.17), the definition of the expected coarse-grained matrices, and their monotonicity imply

$$2\mathbb{E}[J(\square_n, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)] \geq \bar{\mathbf{a}}_L \bar{\mathbf{a}}_{L,*}^{-1}(\square_n) - 1 \geq 0.$$

This proves (4.38).

Step 3. Using the variance estimates in Lemma 4.8, the independence assumption, and the hypothesis $\mathcal{S}(L, \xi, \delta_1)$, we show that, for every $n \geq L$,

$$(4.39) \quad \text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] \leq C\bar{\mathbf{a}}_L^{-2}\delta_1(\delta_1 + 3^{-d(n-L)}).$$

For $n = L$, (4.39) follows directly from $\mathcal{S}(L, \xi, \delta_1)$ and (2.41). By Lemma 4.8, for every $n > L$ we have

$$\text{var}[\mathbf{a}_{L,*}^{-1}(\square_n)] \leq 2 \text{var} \left[\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \mathbf{a}_{L,*}^{-1}(z + \square_L) \right] + 32 \mathbb{E} \left[\left(\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_L, \bar{\mathbf{a}}_L^{-1} e_i, e_i; \mathbf{a}_L) \right)^2 \right].$$

By assumption (g1) and the definition of $\mathbf{a}_L$, for every $y, z \in 3^L \mathbb{Z}^d$ with $|y - z| > d3^{L+1}$, the random variables $\mathbf{a}_{L,*}^{-1}(y + \square_L)$ and $\mathbf{a}_{L,*}^{-1}(z + \square_L)$ are independent. Consequently, by $\mathcal{S}(L, \xi, \delta_1)$ and (2.41),

$$\begin{aligned} \text{var} \left[\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \mathbf{a}_{L,*}^{-1}(z + \square_L) \right] &\leq C 3^{-d(n-L)} \text{var}[\mathbf{a}_{L,*}^{-1}(\square_L)] \leq C 3^{-d(n-L)} \mathbb{E} \left[|\mathbf{a}_{L,*}^{-1}(\square_L) - \bar{\mathbf{a}}_L^{-1}|^2 \right] \\ &\leq C \bar{\mathbf{a}}_L^{-2} 3^{-d(n-L)} \delta_1. \end{aligned}$$

The identical calculation for the averages over z of $\sum_{i=1}^d J(z + \square_L, \bar{\mathbf{a}}_L^{-1} e_i, e_i; \mathbf{a}_L)$ gives

$$\text{var} \left[\sum_{z \in 3^L \mathbb{Z}^d \cap \square_n} \sum_{i=1}^d J(z + \square_L, \bar{\mathbf{a}}_L^{-1} e_i, e_i; \mathbf{a}_L) \right] \leq C \bar{\mathbf{a}}_L^{-2} 3^{-d(n-L)} \delta_1^2.$$

Finally, using $\mathcal{S}(L, \xi, \delta_1)$ again, we have

$$(4.40) \quad \mathbb{E}[J(z + \square_L, \bar{\mathbf{a}}_L^{-1} e_i, e_i; \mathbf{a}_L)]^2 = \bar{\mathbf{a}}_L^{-2} \mathbb{E}[J(\square_L, \bar{\mathbf{a}}_L^{-1/2} e_i, \bar{\mathbf{a}}_L^{1/2} e_i; \mathbf{a}_L)]^2 \leq \bar{\mathbf{a}}_L^{-2} \delta_1^2.$$

Combining the previous three displays yields (4.39).

Step 4. We prove that

$$(4.41) \quad \mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_{G^c}] \leq C \delta_1^2.$$

By (4.36) and (2.52),

$$G^c \subseteq \{\mathcal{E}_{1/4, \infty}(\square_m; \mathbf{a}_L, \bar{\mathbf{a}}_L) \geq 1/5\}.$$

Apply Lemma 4.5 with $s = 1/4$ and the same exponent ξ . This gives, for a constant $C_B(d) < \infty$,

$$(4.42) \quad \mathbb{P}[G^c] \leq \mathbb{P}[\mathcal{E}_{1/4, \infty}(\square_m; \mathbf{a}_L, \bar{\mathbf{a}}_L) \geq 1/5] \leq (C_B \delta_1)^{\xi/2}.$$

By subadditivity down to scale L , the induction hypothesis, and (4.34),

$$\mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L)^2] \leq C \delta_1^2.$$

After decreasing $c(d)$, we may suppose that $C_B \delta_1 \leq 1$. The Cauchy–Schwarz inequality, (4.42), and $\xi \geq 4$ give

$$\mathbb{E}[J(\square_{m-1}, p, q; \mathbf{a}_L) \mathbb{1}_{G^c}] \leq C \delta_1 (C_B \delta_1)^{\xi/4} \leq C \delta_1^2.$$

Step 5. We conclude. Combining (4.37), (4.38), (4.39), and (4.41) gives (4.35). $\square$

Proposition 4.10 (Contraction above the cutoff scale). *There exist constants $c(d) > 0$ and $C(d) < \infty$ such that the following holds. Let $m_0 \in \mathbb{N}$, $\xi \in [16d, \infty)$ and $\delta_1 \in [C \xi \bar{\delta}^2, c]$ be such that $\mathcal{S}(m_0, \xi, \delta_1)$ is valid. Then, for every $m \in \mathbb{N}$ and $L \in \mathbb{N}_0$ with $L \leq m_0$ and $m \geq L + C \log(2 + \xi)$, we have*

$$(4.43) \quad \mathbb{E} \left[\max_{|e|=1} (J(\square_m, \bar{\mathbf{a}}_L^{-1/2} e, \bar{\mathbf{a}}_L^{1/2} e; \mathbf{a}_L))^\xi \right]^{1/\xi} \leq \frac{1}{4} \delta_1.$$

Proof. Fix $p, q \in \mathbb{R}^d$ satisfying (4.34). By subadditivity on the partition into cubes of scale L , the induction hypothesis and (4.34), for every $n \geq L$,

$$(4.44) \quad \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \leq \delta_1.$$

Iterating (4.35), starting from (4.44), we obtain an exponent $\alpha(d) > 0$ and a constant $C(d) < \infty$ such that, for every $n \in \mathbb{N}$ with $n \geq L$,

$$(4.45) \quad \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)] \leq C \delta_1 (\delta_1 + 3^{-\alpha(n-L)}).$$

Using subadditivity and the triangle inequality, we have, for every $m \geq n \geq L$,

$$\begin{aligned}
& \mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \\
& \leq \mathbb{E}\left[\left(\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n, p, q; \mathbf{a}_L)\right)^\xi\right]^{1/\xi} \\
(4.46) \quad & \leq \mathbb{E}\left[\left|\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (J(z + \square_n, p, q; \mathbf{a}_L) - \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)])\right|^\xi\right]^{1/\xi} + \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)].
\end{aligned}$$

Since $J(z + \square_n, p, q; \mathbf{a}_L)$ depends only on $\mathbf{a}_L|_{z + \square_n}$ and $\mathbf{a}_L$ has range of dependence at most $\sqrt{d}3^n$, we may group the summands into $C(d)$ families of independent random variables. Applying Rosenthal's inequality (Lemma B.1) to each family and using (4.44), we find that

$$\begin{aligned}
& \mathbb{E}\left[\left|\sum_{z \in 3^n \mathbb{Z}^d \cap \square_m} (J(z + \square_n, p, q; \mathbf{a}_L) - \mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)])\right|^\xi\right]^{1/\xi} \\
& \leq C3^{-d/2(m-n)}\xi^{1/2}\mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)^2]^{1/2} + C3^{-\frac{d(\xi-1)}{\xi}(m-n)}\xi\mathbb{E}[J(\square_n, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \\
& \leq C3^{-d/2(m-n)}\xi^{1/2}\delta_1 + C3^{-\frac{d(\xi-1)}{\xi}(m-n)}\xi\delta_1 \\
(4.47) \quad & \leq C3^{-d/2(m-n)}\xi\delta_1.
\end{aligned}$$

Combining the previous two displays with (4.45), we obtain

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \leq C3^{-d/2(m-n)}\xi\delta_1 + C\delta_1(\delta_1 + 3^{-\alpha(n-L)}).$$

Selecting $n := L + \lceil 1/2(m-L) \rceil$ yields

$$\mathbb{E}[J(\square_m, p, q; \mathbf{a}_L)^\xi]^{1/\xi} \leq C3^{-d/4(m-L)}\xi\delta_1 + C\delta_1(\delta_1 + 3^{-1/2\alpha(m-L)}).$$

If $m \geq L + C \log(2 + \xi)$, then the right side is at most $c(d)\delta_1$. To pass from a fixed direction to the supremum in (4.43), we use a fixed $1/4$ -net of the unit sphere. Since the map

$$e \mapsto J(\square_m, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)$$

is a nonnegative quadratic form, its supremum on the sphere is at most twice its maximum on the net. Bounding the maximum over the net by the sum over its points, whose number depends only on d , we obtain

$$\mathbb{E}\left[\max_{|e|=1} J(\square_m, \bar{\mathbf{a}}_L^{-1/2}e, \bar{\mathbf{a}}_L^{1/2}e; \mathbf{a}_L)^\xi\right]^{1/\xi} \leq \frac{1}{4}\delta_1.$$

This completes the proof. $\square$

To complete the proof of Proposition 4.1, we compare the coefficient at each scale with the coefficient h scales below it using Lemma 2.8, and choose h so that the resulting error is absorbed by the gain in Proposition 4.10.

Proof of Proposition 4.1. We use two inductions. The first proves $\mathcal{S}(\infty, \xi, C\xi \log^2(2 + \xi)\bar{\delta}^2)$; the second removes one factor of $\log(2 + \xi)$.

Step 1. The base case of both inductions is the assertion that, for some constant $C(d) < \infty$ and every $\xi \in [1, \infty)$ and $h \in \mathbb{N}$ satisfying $C\xi\bar{\delta}^2h \leq 1$,

$$(4.48) \quad \mathcal{S}(3h, \xi, C\xi\bar{\delta}^2h) \text{ is valid.}$$

Lemma 3.6 implies that, for every $n \in \{0, \dots, 3h\}$,

$$\|\mathcal{J}_n(\square_n)\|_{L^\xi(\mathbb{P})} \leq C\xi\bar{\delta}^2h \exp(C\xi\bar{\delta}^2h).$$

This implies (4.48).

Step 2. First induction. We prove that there exists $C(d) < \infty$ such that $\mathcal{S}(\infty, \xi, C\xi \log^2(2+\xi)\bar{\delta}^2)$ is valid for every $\xi \in [16d, C^{-1}\bar{\delta}^{-2}|\log \bar{\delta}|^{-2}]$. We assume that

$$(4.49) \quad h \geq K \log(2+\xi) \quad \text{and} \quad \xi \bar{\delta}^2 h^2 \leq K^{-1} \delta_1 \leq K^{-2},$$

where $K(d)$ is a large constant to be determined below. We iterate in blocks of h scales. For parameters satisfying (4.49) and every $m \in \mathbb{N} \cap [3h, \infty)$, we prove

$$(4.50) \quad \mathcal{S}(m, \xi, \delta_1) \implies \mathcal{S}(m+h, \xi, \delta_1).$$

Equations (4.48) and (4.50) imply by induction that $\mathcal{S}(\infty, \xi, \delta_1)$ holds for every choice of parameters satisfying (4.49), after enlarging K .

To prove (4.50), we fix $m \in \mathbb{N}$ with $m \geq 3h$ such that $\mathcal{S}(m, \xi, \delta_1)$ is valid. By Proposition 4.10, if K is taken sufficiently large, then

$$(4.51) \quad \mathbb{E} \left[\max_{|e|=1} J(\square_{m+h}, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m)^\xi \right]^{1/\xi} \leq \frac{1}{4} \delta_1.$$

We apply Lemma 2.8 with $\delta = 1$ and $\lambda = \bar{\mathbf{a}}_{m+h} \bar{\mathbf{a}}_m^{-1}$, and with $\mathbf{b} = \mathbf{a}_{m+h}$ and $\mathbf{a} = \mathbf{a}_m$. This gives, for every bounded convex domain $U \subseteq \mathbb{R}^d$ and $|e| = 1$,

$$(4.52) \quad J(U, \bar{\mathbf{a}}_{m+h}^{-1/2} e, \bar{\mathbf{a}}_{m+h}^{1/2} e; \mathbf{a}_{m+h}) \leq 2J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) + 3\mathcal{X}_{m,h}(U)^2 (J(U, \bar{\mathbf{a}}_m^{-1/2} e, \bar{\mathbf{a}}_m^{1/2} e; \mathbf{a}_m) + 1),$$

where we have defined

$$\mathcal{X}_{m,h}(U) := \|\bar{\mathbf{a}}_{m+h} \bar{\mathbf{a}}_m^{-1} (\mathbf{a}_{m+h}^{-1} \mathbf{a}_m) - 1\|_{L^\infty(U)} + \|\bar{\mathbf{a}}_{m+h}^{-1} \bar{\mathbf{a}}_m (\mathbf{a}_{m+h} \mathbf{a}_m^{-1}) - 1\|_{L^\infty(U)}.$$

Taking the maximum over $|e| = 1$ in (4.52) gives

$$(4.53) \quad \mathcal{J}_{m+h}(U) \leq 2\mathcal{J}_m(U) + 3\mathcal{X}_{m,h}(U)^2 (\mathcal{J}_m(U) + 1).$$

The fields $\bar{\mathbf{a}}_{m+h}^{-1} \mathbf{a}_m$ and $\mathbf{a}_m$ are independent; hence so are $\mathcal{X}_{m,h}(\square_{m+h})$ and $\mathcal{J}_m(\square_{m+h})$. The triangle inequality therefore gives

$$(4.54) \quad \begin{aligned} \|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} &\leq 2\|\mathcal{J}_m(\square_{m+h})\|_{L^\xi(\mathbb{P})} \\ &\quad + 3\|\mathcal{X}_{m,h}(\square_{m+h})\|_{L^\xi(\mathbb{P})}^2 (1 + \|\mathcal{J}_m(\square_{m+h})\|_{L^\xi(\mathbb{P})}). \end{aligned}$$

By (3.33) and (3.7), applied with moment 2ξ , together with (4.49), we have the estimate

$$\mathbb{E}[\mathcal{X}_{m,h}(\square_{m+h})^{2\xi}]^{1/\xi} \leq C\xi \bar{\delta}^2 h^2 \leq \frac{1}{12} \delta_1.$$

Inserting this and (4.51) into (4.54) gives

$$\|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} \leq \frac{1}{2} \delta_1 + C\xi \bar{\delta}^2 h^2 \leq \delta_1.$$

The same argument, with $(m, m+h)$ replaced by $(r-h, r)$, gives the same bound on $\square_r$ for each $r \in \{m+1, \dots, m+h\}$; together with $\mathcal{S}(m, \xi, \delta_1)$, this proves (4.50). Hence $\mathcal{S}(\infty, \xi, C\xi \log^2(2+\xi)\bar{\delta}^2)$ holds for all $\xi \geq 16d$ in the stated range. The remaining range $1 \leq \xi < 16d$ follows from monotonicity of $L^\xi(\mathbb{P})$ norms.

Step 3. Second induction. We prove (4.3) and (4.4). We now consider parameters ξ, δ_1 and h satisfying

$$(4.55) \quad h \geq K \log(2+\xi) \quad \text{and} \quad \xi \bar{\delta}^2 h \leq K^{-1} \delta_1 \leq K^{-2},$$

where $K(d)$ is a large constant to be determined below. We also assume $\xi \geq 4$.

For every $m \in \mathbb{N} \cap [3h, \infty)$, we prove the implication

$$(4.56) \quad \mathcal{S}(m, \xi, \delta_1) \implies \mathcal{S}(m+h, \xi, \delta_1).$$

By subadditivity, (4.53), Hölder's inequality, and the independence of the fields defining $\mathcal{X}_{m,h}$ and $\mathcal{J}_m$, we have

$$\begin{aligned}
\|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} &\leq \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_{m+h}(z + \square_m) \right)^\xi \right]^{1/\xi} \\
&\leq 3 \mathbb{E} \left[\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} (\mathcal{X}_{m,h}(z + \square_m))^{2\xi} \right]^{1/\xi} \\
&\quad + 3 \mathbb{E} \left[\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} (1 + \mathcal{X}_{m,h}(z + \square_m)^2)^\xi \right]^{1/\xi} \\
&\quad \times \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_m(z + \square_m)^2 \right)^{\xi/2} \right]^{1/\xi} \\
&\leq C\xi \bar{\delta}^2 h + 3(1 + C\xi \bar{\delta}^2 h) \mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_m(z + \square_m)^2 \right)^{\xi/2} \right]^{1/\xi}.
\end{aligned}$$

Each summand depends only on the restriction of $\mathbf{a}_m$ to its own cube, and $\mathbf{a}_m$ has range of dependence at most $\sqrt{d}3^m$. We can therefore color the cubes with $C(d)$ colors so that the summands of each color are independent. Applying Rosenthal's inequality (Lemma B.1) to each color as in (4.47), with $\mathcal{J}_m^2$ in place of J and $\xi/2$ in place of ξ , and using $\mathcal{S}(m, \xi, \delta_1)$ and the estimate $\mathcal{S}(\infty, 2, C\bar{\delta}^2)$ proved in Step 2, we obtain

$$\begin{aligned}
&\mathbb{E} \left[\left(\sum_{z \in 3^m \mathbb{Z}^d \cap \square_{m+h}} \mathcal{J}_m(z + \square_m)^2 \right)^{\xi/2} \right]^{2/\xi} \\
&\leq C3^{-d/2h} \xi^{1/2} \mathbb{E}[\mathcal{J}_m(\square_m)^4]^{1/2} + C3^{-\frac{d(\xi-2)}{\xi}h} \xi \mathbb{E}[\mathcal{J}_m(\square_m)^\xi]^{2/\xi} \\
&\quad + \mathbb{E}[\mathcal{J}_m(\square_m)^2] \\
&\leq C3^{-d/2h} \xi^{1/2} \delta_1^2 + C3^{-d/2h} \xi \delta_1^2 + C\bar{\delta}^4 \\
&\leq \frac{1}{36} \delta_1^2.
\end{aligned}$$

Combining the previous two displays and using $\delta_1 \leq 1$, we obtain

$$\|\mathcal{J}_{m+h}(\square_{m+h})\|_{L^\xi(\mathbb{P})} \leq C\xi \bar{\delta}^2 h + \frac{1}{2}(1 + C\xi \bar{\delta}^2 h) \delta_1 \leq \frac{1}{2} \delta_1 + C\xi \bar{\delta}^2 h \leq \delta_1.$$

The last inequality follows from (4.55) after fixing $K(d)$ sufficiently large. The same calculation, with $(m, m+h)$ replaced by $(r-h, r)$, gives the same bound on $\square_r$ for each $r \in \{m+1, \dots, m+h\}$; together with $\mathcal{S}(m, \xi, \delta_1)$, this proves (4.56).

The base case (4.48) and (4.56) imply that $\mathcal{S}(\infty, \xi, C\xi \log(2 + \xi) \bar{\delta}^2)$ holds for $\xi \geq 4$ whenever $\xi \log(2 + \xi) \bar{\delta}^2 \leq C^{-1}$. The range $1 \leq \xi < 4$ follows by monotonicity of $L^\xi(\mathbb{P})$ norms. This proves (4.3). The bound (4.4) follows from (4.10) with $\delta_1 = C\xi \log(2 + \xi) \bar{\delta}^2$. $\square$

5. EFFECTIVE DIFFUSIVITIES

In this section we determine how fast the effective diffusivities $\bar{\mathbf{a}}_m$ decay. We prove that they are nonincreasing in m , that

$$e^{-(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)} \quad (m \in \mathbb{N}_0)$$

for an exponent $\eta_{\text{sub}} > 0$ depending only on d and the law of $\mathbf{g}_0$, and that, at weak disorder,

$$\left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

In dimension two, planar duality gives the exact identity $\bar{\mathbf{a}}_m = e^{-\tau^2(m+1)}$. These statements are collected in Theorem 5.1. The decay of $\bar{\mathbf{a}}_m$ gives the time scale $3^{2N}/\bar{\mathbf{a}}_N$ in Theorem A and the exponent η in Theorem A(iii).

We prove the asymptotics at weak disorder through matching upper and lower bounds on $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$ for integers $h \leq \bar{\delta}^{-1}$. We compare the corrector of $\mathbf{a}_m$ with the solution of the linearized corrector equation. The corrector is the function added to an affine function to make it solve the equation with coefficient $\mathbf{a}_m$. The linearized corrector solves a Poisson equation forced by the added fields, and we compute its energy exactly in Lemma 5.3. The error term in these asymptotics exceeds the main term unless τ^2 is bounded below by a power of $\bar{\delta}$. We therefore prove the two-sided bound $e^{-(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}$ separately, by a nonperturbative argument which assumes no smallness of $\bar{\delta}$. Its upper bound makes the process subdiffusive in every dimension, and the asymptotics identify the exponent at weak disorder.

Theorem 5.1 (Bounds on the effective diffusivities). *There exists $\eta_{\text{sub}} = \eta_{\text{sub}}(d, \text{law}(\mathbf{g}_0)) > 0$ such that, for every $0 \leq n \leq m$,*

$$(5.1) \quad e^{-(m+1)\tau^2} \leq \bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_n \leq \min\{1, e^{2\tau^2(m-n)}\bar{\mathbf{a}}_m\}, \quad \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}.$$

In dimension two, for every $m \in \mathbb{N}_0$,

$$(5.2) \quad \bar{\mathbf{a}}_m = e^{-(m+1)\tau^2}.$$

Moreover, there exist $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0(d)$, then, for every $m \in \mathbb{N}_0$,

$$(5.3) \quad \left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

Proof. The monotonicity in m and the comparison between two scales are (3.32), and the affine competitor gives $\bar{\mathbf{a}}_0 \leq 1$. The remaining assertions are Propositions 3.5, 5.2, 5.6, and 5.7. $\square$

Proposition 5.2 (Asymptotics of the effective diffusivities at weak disorder). *There exist constants $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0(d)$, then, for every $m \in \mathbb{N}_0$,*

$$(5.4) \quad \left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1 + m\bar{\delta}).$$

Lemma 5.3 (Energy of the linearized corrector). *Let $m \in \mathbb{N}_0$ and $h \in \mathbb{N}$ satisfy $h \leq m+1$ and $h \leq \bar{\delta}^{-1}$, let $X := \prod_{k=m-h+1}^m \exp(\mathbf{g}_k - \tau^2) - 1$, and let $p \in \mathbb{R}^d$ be a unit vector. If $w_p^{(K)} \in H_0^1(\square_K)$ solves*

$$-\Delta w_p^{(K)} = \nabla \cdot (Xp) \quad \text{in } \square_K,$$

then

$$(5.5) \quad \lim_{K \rightarrow \infty} \mathbb{E}[\|\nabla w_p^{(K)}\|_{L^2(\square_K)}^2] = \frac{2\tau^2 h}{d} + O(\bar{\delta}^4 h^2).$$

The same conclusion holds if $w_p^{(K)}$ is the mean-zero solution of

$$-\Delta w_p^{(K)} = -\nabla \cdot (Xp) \quad \text{in } \square_K, \quad \partial_{\mathbf{n}} w_p^{(K)} = Xp \cdot \mathbf{n} \quad \text{on } \partial \square_K.$$

Proof. The stationary Weyl decomposition of Xp in $L^2(\Omega; \mathbb{R}^d)$ provides potentials ϕ and $\mathbf{S} = -\mathbf{S}^t$ such that $Xp = \nabla \phi + \nabla \cdot \mathbf{S}$. Here $\nabla \phi$ is the orthogonal projection of Xp onto the closure of the stationary potential fields. After fixing the additive constants, the potentials also satisfy

$$(5.6) \quad \lim_{K \rightarrow \infty} 3^{-2K} \mathbb{E} \left[\int_{\square_K} (|\phi - (\phi)_{\square_K}|^2 + |\mathbf{S} - (\mathbf{S})_{\square_K}|^2) \right] = 0.$$

Fix $L \in \mathbb{N}$. Choose $\eta_{K,L} \in C_c^\infty(\square_K)$ equal to one at distance at least 3^{K-L} from $\partial\square_K$ and satisfying $0 \leq \eta_{K,L} \leq 1$ and $|\nabla \eta_{K,L}| \leq C3^{L-K}$. The support of $1 - \eta_{K,L}$ has relative volume at most $C3^{-L}$. Consequently, the function

$$v_{K,L} := -\eta_{K,L}(\phi - (\phi)_{\square_K}) \in H_0^1(\square_K)$$

and the field

$$\mathbf{j}_{K,L} := \nabla \cdot (\eta_{K,L}(\mathbf{S} - (\mathbf{S})_{\square_K})) \in L_{\text{sol},0}^2(\square_K)$$

satisfy, by stationarity and (5.6),

$$(5.7) \quad \limsup_{K \rightarrow \infty} \mathbb{E}[\|\nabla v_{K,L} + \nabla \phi\|_{\underline{L}^2(\square_K)}^2] \leq C3^{-L} \mathbb{E}[|\nabla \phi(0)|^2],$$

$$(5.8) \quad \limsup_{K \rightarrow \infty} \mathbb{E}[\|\mathbf{j}_{K,L} - \nabla \cdot \mathbf{S}\|_{\underline{L}^2(\square_K)}^2] \leq C3^{-L} \mathbb{E}[|\nabla \cdot \mathbf{S}(0)|^2].$$

For the Dirichlet problem, the variational identity

$$\frac{1}{2} \|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2 = \sup_{\varphi \in H_0^1(\square_K)} \int_{\square_K} \left(-Xp \cdot \nabla \varphi - \frac{1}{2} |\nabla \varphi|^2 \right)$$

and the orthogonality of $\nabla \cdot \mathbf{S}$ to compactly supported gradients give

$$\mathbb{E}[\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2] \leq \mathbb{E}[|\nabla \phi(0)|^2].$$

Using $v_{K,L}$ in the variational identity, then sending $K \rightarrow \infty$ and $L \rightarrow \infty$ in this order, proves the reverse inequality.

For the Neumann problem, let $w_p^{(K)}$ now denote the mean-zero solution. Its weak formulation says that $\nabla w_p^{(K)}$ is the orthogonal projection of Xp onto $\nabla H^1(\square_K)$. Using the restriction of ϕ to $\square_K$ as a competitor in the variational formula for the norm of this projection, and using $\mathbb{E}[(Xp)(0) \cdot \nabla \phi(0)] = \mathbb{E}[|\nabla \phi(0)|^2]$, we obtain

$$\mathbb{E}[\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2] \geq \mathbb{E}[|\nabla \phi(0)|^2].$$

On the other hand, the field $\mathbf{j}_{K,L} \in L_{\text{sol},0}^2(\square_K)$ is orthogonal to $\nabla w_p^{(K)}$, and hence

$$\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2 \leq \|Xp - \mathbf{j}_{K,L}\|_{\underline{L}^2(\square_K)}^2.$$

Sending first $K \rightarrow \infty$ and then $L \rightarrow \infty$ and using (5.8) proves the reverse inequality. Consequently, in both cases,

$$\lim_{K \rightarrow \infty} \mathbb{E}[\|\nabla w_p^{(K)}\|_{\underline{L}^2(\square_K)}^2] = \mathbb{E}[|\nabla \phi(0)|^2].$$

The spectral measure of the scalar stationary field X is invariant under coordinate reflections and permutations. The off-diagonal entries of the Fourier multiplier of the projection therefore integrate to zero against this measure, while the diagonal entries have equal integrals, which sum to $\mathbb{E}[X(0)^2]$. Hence

$$\mathbb{E}[|\nabla \phi(0)|^2] = \frac{1}{d} \mathbb{E}[X(0)^2].$$

Independence of the fields and the definition $\tau^2 = \log \mathbb{E}[\exp(\mathbf{g}_0(0))]$ yield

$$\mathbb{E}[X(0)^2] = \exp(h(\log \mathbb{E}[\exp(2\mathbf{g}_0(0))] - 2\tau^2)) - 1.$$

The symmetry and sub-Gaussian bound in Assumptions (g2)–(g3) give

$$\log \mathbb{E}[\exp(2\mathbf{g}_0(0))] - 2\tau^2 = 2\tau^2 + O(\bar{\delta}^4).$$

Since $h\bar{\delta}^2 \leq \bar{\delta}$, expansion of the exponential proves (5.5). $\square$

Lemma 5.4 (Upper bound on $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$). *There exists $C(d) < \infty$ such that, for every $m, h \in \mathbb{N}$ with*

$$h \leq \bar{\delta}^{-1} \quad \text{and} \quad m - h \geq 16\lceil \log_3 \bar{\delta} \rceil,$$

we have

$$(5.9) \quad \bar{\mathbf{a}}_m \leq \bar{\mathbf{a}}_{m-h} \left(1 - \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2 + C\bar{\delta}^2 |\log \bar{\delta}| \right).$$

Proof. Step 1. We set up the Dirichlet problem and record the interior estimates on the **cells** of scale 3^n , the cubes $z + \square_n$ with $z \in 3^n \mathbb{Z}^d$. Fix $p \in \mathbb{R}^d$ with $|p| = 1$, and set

$$(5.10) \quad \mathbf{h} := \sum_{k=m-h+1}^m (\mathbf{g}_k - \tau^2).$$

Choose

$$(5.11) \quad n := m - h - 16\lceil \log_3 \bar{\delta} \rceil,$$

so that $3^{-(m-h-n)} \leq \bar{\delta}^{16}$. For $K > m$, let $w \in H_0^1(\square_K)$ solve

$$-\Delta w = \nabla \cdot ((e^{\mathbf{h}} - 1)p) \quad \text{in } \square_K.$$

Testing the equation with w and applying the Calderón–Zygmund estimate give, for $1 \leq \xi \leq 8$,

$$\|\nabla w\|_{L^\xi(\mathbb{P}; \underline{L}^\xi(\square_K))} \leq C\bar{\delta} h^{1/2}, \quad \|\nabla w\|_{\underline{L}^2(\square_K)}^2 = - \int_{\square_K} (e^{\mathbf{h}} - 1)p \cdot \nabla w.$$

Lemma 5.3 yields

$$\lim_{K \rightarrow \infty} \mathbb{E} \|\nabla w\|_{\underline{L}^2(\square_K)}^2 = \frac{2\tau^2 h}{d} + O(\bar{\delta}^4 h^2).$$

For every scale- 3^n cell $Q_z = z + \square_n$ satisfying $z + \square_{m-h} \subseteq \square_K$, define

$$p_z = (p + \nabla w)_{Q_z}, \quad F_z = \nabla w - (\nabla w)_{Q_z}, \quad A_z = \|F_z\|_{\underline{L}^2(Q_z)}, \quad B_z = 3^n \|\nabla F_z\|_{\underline{L}^2(Q_z)}.$$

The interior estimates for the Laplacian give

$$A_z \leq C3^{-(m-h-n)} \|\nabla w\|_{\underline{L}^2(z + \square_{m-h})} + C3^n \|\nabla(e^{\mathbf{h}} - 1)\|_{L^\infty(z + \square_{m-h})}.$$

The corresponding interior $W^{2,4}$ estimate controls B_z by the same two terms, with the local L^4 norm of ∇w in place of its L^2 norm. Assumption (g2), the scaling relation (1.4), and the preceding moment bound therefore give

$$\limsup_{K \rightarrow \infty} \left(\sum_z \mathbb{E}[A_z^4] \right)^{1/4} \leq C\bar{\delta}^{15}, \quad \limsup_{K \rightarrow \infty} \left(\sum_z \mathbb{E}[B_z^4] \right)^{1/4} \leq C\bar{\delta}^{15}.$$

Here and below, the averages run over these interior cells. The omitted cells form a boundary layer whose relative volume tends to zero as $K \rightarrow \infty$.

Step 2. We solve cell problems on the interior cells, patch their solutions into a competitor for the Dirichlet principle, and bound the energy by a sum over cells. On Q_z , let $\chi_z, \phi_z \in H_0^1(Q_z)$ solve

$$-\nabla \cdot (\mathbf{a}_m \nabla \chi_z) = \nabla \cdot (\mathbf{a}_m p_z), \quad -\nabla \cdot (\mathbf{a}_m \nabla \phi_z) = \nabla \cdot (\mathbf{a}_m F_z),$$

and set

$$S_z = p_z + \nabla \chi_z, \quad \tilde{S}_z = F_z + \nabla \phi_z.$$

Then $\nabla \cdot (\mathbf{a}_m S_z) = \nabla \cdot (\mathbf{a}_m \tilde{S}_z) = 0$, $(\tilde{S}_z)_{Q_z} = 0$, and

$$\int_{Q_z} |S_z|^2 \mathbf{a}_m = p_z \cdot \mathbf{a}_m(Q_z) p_z.$$

Testing the Dirichlet principle with the function equal to $p \cdot x + w + \chi_z + \phi_z$ on each interior cell Q_z and to $p \cdot x + w$ in the boundary layer gives

$$\frac{1}{2} \mathbb{E}[p \cdot \mathbf{a}_m(\square_K)p] \leq \sum_z \mathbb{E} \left[\frac{1}{2} p_z \cdot \mathbf{a}_m(Q_z) p_z + \int_{Q_z} S_z \cdot \mathbf{a}_m \tilde{S}_z + \frac{1}{2} \int_{Q_z} |\tilde{S}_z|^2 \mathbf{a}_m \right] + o_K(1).$$

Since ϕ_z minimizes the energy of $F_z + \nabla \phi$ and $(F_z)_{Q_z} = 0$,

$$\int_{Q_z} |\tilde{S}_z|^2 \mathbf{a}_m = \int_{Q_z} F_z \cdot \mathbf{a}_m \tilde{S}_z.$$

Apply (2.13), (2.62), and

$$3^{n/4} [F_z]_{B_{2,\infty}^{1/4}(Q_z)} \leq C(A_z + A_z^{3/4} B_z^{1/4})$$

to obtain

$$\int_{Q_z} |\tilde{S}_z|^2 \mathbf{a}_m \leq C \Lambda_{1/4,1}(Q_z; \mathbf{a}_m) (A_z^2 + A_z^{3/2} B_z^{1/2}).$$

Proposition 4.1, (2.52), (3.28), (3.27), and Lemma 3.4 give

$$\|\Lambda_{1/4,1}(Q_z; \mathbf{a}_m)\|_{L^2(\mathbb{P})} \leq C \bar{\mathbf{a}}_{m-h}.$$

Hence the estimates of Step 1 and Hölder's inequality imply

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \int_{Q_z} |\tilde{S}_z|^2 \mathbf{a}_m \leq C \bar{\delta}^{30} \bar{\mathbf{a}}_{m-h}.$$

The same coefficient bounds give

$$\sum_z \mathbb{E} \int_{Q_z} |S_z|^2 \mathbf{a}_m \leq C \bar{\mathbf{a}}_{m-h}.$$

Cauchy–Schwarz therefore yields

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \left| \int_{Q_z} S_z \cdot \mathbf{a}_m \tilde{S}_z \right| \leq C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}.$$

Step 3. We estimate the sum of the cell energies $p_z \cdot \mathbf{a}_m(Q_z) p_z$. Since $\mathbf{a}_m = e^{\mathbf{h}} \mathbf{a}_{m-h}$,

$$p_z \cdot \mathbf{a}_m(Q_z) p_z \leq \left(\sup_{Q_z} e^{\mathbf{h}} \right) p_z \cdot \mathbf{a}_{m-h}(Q_z) p_z.$$

We have the deterministic estimate

$$\left| \left(\sup_{Q_z} e^{\mathbf{h}} \right) |p_z|^2 - \int_{Q_z} e^{\mathbf{h}} |p + \nabla w|^2 \right| \leq C e^{\|\mathbf{h}\|_{L^\infty(Q_z)}} \left(\text{osc}_{Q_z} \mathbf{h} |p_z|^2 + (|p_z| + A_z) A_z \right).$$

The choice of n gives $\|\text{osc}_{Q_z} \mathbf{h}\|_{L^8(\mathbb{P})} \leq C \bar{\delta}^{17}$. The exponential factor and p_z have bounded eighth moments, while Step 1 controls A_z . Combining these bounds with the independence of $\mathbf{h}$ and $\mathbf{a}_{m-h}$, we obtain

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E}[p_z \cdot \mathbf{a}_m(Q_z) p_z] \leq \bar{\mathbf{a}}_{m-h}(\square_n) \limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} e^{\mathbf{h}} |p + \nabla w|^2 + C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}.$$

It remains to evaluate the expectation in the last display. The equation for w , the zero trace of w , and $\mathbb{E}[e^{\mathbf{h}}] = 1$ give

$$\mathbb{E} \int_{\square_K} e^{\mathbf{h}} |p + \nabla w|^2 = 1 - \mathbb{E} \|\nabla w\|_{\underline{L}^2(\square_K)}^2 + \mathbb{E} \int_{\square_K} |\nabla w|^2 (e^{\mathbf{h}} - 1).$$

We now show that the last term is $O(\bar{\delta}^4 h^2)$, uniformly in K . Decompose $w = v + \bar{w}$, where

$$-\Delta v = \nabla \cdot (\mathbf{h} p), \quad -\Delta \bar{w} = \nabla \cdot ((e^{\mathbf{h}} - 1 - \mathbf{h}) p), \quad v, \bar{w} \in H_0^1(\square_K).$$

The Calderón–Zygmund and exponential-moment estimates give

$$\|\nabla v\|_{L^4(\mathbb{P}; \underline{L}^4)} \leq C\bar{\delta}h^{1/2}, \quad \|\nabla \bar{w}\|_{L^4(\mathbb{P}; \underline{L}^4)} \leq C\bar{\delta}^2h.$$

Since constants have zero divergence, v is linear in the sum $\mathbf{h} + h\tau^2 = \sum_{k=m-h+1}^m \mathbf{g}_k$ of the last h fields. Reversing the signs of these fields simultaneously therefore gives

$$\mathbb{E} \int_{\square_K} |\nabla v|^2(\mathbf{h} + h\tau^2) = 0, \quad \mathbb{E} \int_{\square_K} |\nabla v|^2 \mathbf{h} = -h\tau^2 \mathbb{E} \|\nabla v\|_{\underline{L}^2}^2 = O(\bar{\delta}^4 h^2).$$

The preceding moment bounds control the terms containing $\bar{w}$, and $|e^s - 1 - s| \leq 1/2 s^2 e^{|s|}$ controls the nonlinear remainder. Hölder's inequality and $\bar{\delta}h \leq 1$ thus give

$$\sup_K \left| \mathbb{E} \int_{\square_K} |\nabla w|^2 (e^{\mathbf{h}} - 1) \right| \leq C\bar{\delta}^4 h^2.$$

Lemma 5.3 now yields

$$\limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} e^{\mathbf{h}} |p + \nabla w|^2 \leq 1 - \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2.$$

Step 4. We conclude the proof of (5.9). Combining Steps 2 and 3 and passing to the thermodynamic limit gives

$$\frac{1}{2} \bar{\mathbf{a}}_m \leq \left(\frac{1}{2} - \frac{\tau^2 h}{d} + C\bar{\delta}^4 h^2 \right) \bar{\mathbf{a}}_{m-h}(\square_n) + C\bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}.$$

By (3.32) and (4.3),

$$\bar{\mathbf{a}}_{m-h} \leq \bar{\mathbf{a}}_{m-h}(\square_n) \leq (1 + C\bar{\delta}^2 |\log \bar{\delta}|) \bar{\mathbf{a}}_{m-h}.$$

Substituting this into the previous display and using $\tau^2 \leq C\bar{\delta}^2$ and $\bar{\delta}h \leq 1$ proves (5.9). $\square$

Lemma 5.5 (Lower bound on $\bar{\mathbf{a}}_m/\bar{\mathbf{a}}_{m-h}$). *There exists $C(d) < \infty$ such that, for every $m, h \in \mathbb{N}$ with*

$$h \leq \bar{\delta}^{-1} \quad \text{and} \quad m - h \geq 16[|\log_3 \bar{\delta}|],$$

we have

$$(5.12) \quad \bar{\mathbf{a}}_m^{-1} \leq \bar{\mathbf{a}}_{m-h}^{-1} \left(1 + \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2 + C\bar{\delta}^2 |\log \bar{\delta}| \right).$$

Proof. Step 1. We set up the Neumann problem and record the interior estimates on the scale- 3^n cells. Fix $q \in \mathbb{R}^d$ with $|q| = 1$, and define $\mathbf{h}$ and n by (5.10) and (5.11). For $K > m$, let W be the mean-zero solution of

$$-\Delta W = -\nabla \cdot ((e^{\mathbf{h}} - 1)q) \quad \text{in } \square_K, \quad \partial_{\mathbf{n}} W = (e^{\mathbf{h}} - 1)q \cdot \mathbf{n} \quad \text{on } \partial \square_K,$$

and set $\mathbf{z} = e^{\mathbf{h}} q - \nabla W$. Then $\nabla \cdot \mathbf{z} = 0$ and $\mathbf{z} \cdot \mathbf{n} = q \cdot \mathbf{n}$ on $\partial \square_K$. The Neumann Calderón–Zygmund estimate and Lemma 5.3 give, for $1 \leq \xi \leq 8$,

$$\mathbb{E} [\|\nabla W\|_{\underline{L}^\xi(\square_K)}^\xi]^{1/\xi} \leq C\bar{\delta}h^{1/2}, \quad \lim_{K \rightarrow \infty} \mathbb{E} [\|\nabla W\|_{\underline{L}^2(\square_K)}^2] = \frac{2\tau^2 h}{d} + O(\bar{\delta}^4 h^2).$$

For each cell $Q_z = z + \square_n$, put

$$q_z = (\mathbf{z})_{Q_z}, \quad G_z = \mathbf{z} - q_z, \quad A_z = \|G_z\|_{\underline{L}^2(Q_z)}, \quad B_z = 3^n \|\nabla G_z\|_{\underline{L}^2(Q_z)}.$$

With the Neumann estimate in place of the Dirichlet estimate, and with the additional term $e^{\mathbf{h}} q$ controlled by the same bound on the gradient of $e^{\mathbf{h}}$, the oscillation and interior estimates in Step 1 of the proof of Lemma 5.4 yield

$$\limsup_{K \rightarrow \infty} \left(\sum_z \mathbb{E}[A_z^4] \right)^{1/4} \leq C\bar{\delta}^{15}, \quad \limsup_{K \rightarrow \infty} \left(\sum_z \mathbb{E}[B_z^4] \right)^{1/4} \leq C\bar{\delta}^{15}.$$

As in the proof of Lemma 5.4, the averages run over the interior cells, and the omitted cells form a boundary layer of vanishing relative volume.

Step 2. We solve Neumann problems on the interior cells, patch the resulting fluxes into an admissible flux, and bound the dual energy by a sum over cells. On every interior cell, let ψ_z, η_z be the mean-zero solutions of

$$-\nabla \cdot (\mathbf{a}_m \nabla \psi_z) = 0, \quad \mathbf{a}_m \nabla \psi_z \cdot \mathbf{n} = q_z \cdot \mathbf{n},$$

and

$$-\nabla \cdot (\mathbf{a}_m \nabla \eta_z) = 0, \quad \mathbf{a}_m \nabla \eta_z \cdot \mathbf{n} = G_z \cdot \mathbf{n},$$

respectively. These problems are compatible because G_z is divergence free. The normal trace of the flux $\mathbf{a}_m \nabla(\psi_z + \eta_z)$ agrees with that of $\mathbf{z}$ on every cell boundary. Therefore the field equal to $\Sigma_z + \tilde{\Sigma}_z$ on each interior cell and to $\mathbf{z}$ in the boundary layer belongs to $q + L^2_{\text{sol},0}(\square_K)$, and the contribution of the boundary layer to its dual energy vanishes in expectation as $K \rightarrow \infty$. Hence

$$\frac{1}{2} q \cdot \mathbf{a}_{m,*}^{-1}(\square_K) q \leq \sum_z \left(\frac{1}{2} q_z \cdot \mathbf{a}_{m,*}^{-1}(Q_z) q_z + \int_{Q_z} \mathbf{a}_m \nabla \psi_z \cdot \nabla \eta_z + \frac{1}{2} \int_{Q_z} \mathbf{a}_m |\nabla \eta_z|^2 \right) + o_K(1).$$

The argument of Step 2 of the proof of Lemma 5.4, applied to fluxes instead of gradients and with (2.61) in place of (2.62), gives

$$\int_{Q_z} \mathbf{a}_m |\nabla \eta_z|^2 \leq C \lambda_{1/4,1}^{-1}(Q_z; \mathbf{a}_m) (A_z^2 + A_z^{3/2} B_z^{1/2}).$$

Proposition 4.1, (2.52), (3.28), and (3.27) imply

$$\mathbb{E}[\lambda_{1/4,1}^{-2}(\square_n; \mathbf{a}_m)]^{1/2} \leq C \bar{\mathbf{a}}_{m-h}^{-1}.$$

The preceding three displays and Cauchy–Schwarz thus yield

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \int_{Q_z} \mathbf{a}_m |\nabla \eta_z|^2 \leq C \bar{\delta}^{30} \bar{\mathbf{a}}_{m-h}^{-1},$$

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E} \left| \int_{Q_z} \mathbf{a}_m \nabla \psi_z \cdot \nabla \eta_z \right| \leq C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}.$$

Step 3. We estimate the sum of the cell energies $q_z \cdot \mathbf{a}_{m,*}^{-1}(Q_z) q_z$. The pair (q_z, G_z) is a function of the last h fields and is independent of $\mathbf{a}_{m-h}$. Since $\mathbf{a}_m^{-1} = e^{-\mathbf{h}} \mathbf{a}_{m-h}^{-1}$, the deterministic estimate in Step 3 of the proof of Lemma 5.4, with $\mathbf{h}$ replaced by $-\mathbf{h}$, $p + \nabla w$ by $\mathbf{z}$, and p_z by q_z , gives

$$\limsup_{K \rightarrow \infty} \sum_z \mathbb{E}[q_z \cdot \mathbf{a}_{m,*}^{-1}(Q_z) q_z] \leq \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) \limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} |\mathbf{z}|^2 e^{-\mathbf{h}} + C \bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}.$$

We next show that

$$(5.13) \quad \limsup_{K \rightarrow \infty} \left| \mathbb{E} \int_{\square_K} |\nabla W|^2 (e^{-\mathbf{h}} - 1) \right| \leq C \bar{\delta}^4 h^2.$$

Indeed, decompose $W = V + \bar{W}$, where V solves the Neumann problem defining W with $e^{\mathbf{h}} - 1$ replaced by its linearization $\mathbf{h}$. If V_0 is the solution of the same problem with $\mathbf{h}$ replaced by the sum $\mathbf{h} + h\tau^2 = \sum_{k=m-h+1}^m \mathbf{g}_k$ of the last h fields, then

$$\nabla V = \nabla V_0 - h\tau^2 q, \quad \mathbb{E} \|\nabla V_0\|_{\underline{L}^2(\square_K)}^2 \leq C \bar{\delta}^2 h.$$

Reversing the signs of the last h fields simultaneously gives

$$\mathbb{E} \int_{\square_K} |\nabla V_0|^2 (\mathbf{h} + h\tau^2) = 0, \quad \mathbb{E} \int_{\square_K} q \cdot \nabla V_0 = 0,$$

while Cauchy–Schwarz gives

$$\left| \mathbb{E} \int_{\square_K} (q \cdot \nabla V_0)(\mathbf{h} + h\tau^2) \right| \leq C\bar{\delta}^2 h.$$

Expanding $|\nabla V|^2 \mathbf{h}$ with the identities for ∇V and $\mathbf{h}$ therefore yields

$$\mathbb{E} \int_{\square_K} |\nabla V|^2 \mathbf{h} = O(\bar{\delta}^4 h^2).$$

The Neumann Calderón–Zygmund estimate gives $\mathbb{E}[\|\nabla \bar{W}\|_{\underline{L}^4}^4]^{1/4} \leq C\bar{\delta}^2 h$. The Hölder estimates used in Step 3 of the upper-bound proof, together with $|e^{-s} - 1 + s| \leq 1/2 s^2 e^{|s|}$, control every term containing $\bar{W}$ and the nonlinear remainder by $C\bar{\delta}^4 h^2$. This proves (5.13).

Since $\mathbf{z} = e^{\mathbf{h}} q - \nabla W$,

$$\int_{\square_K} |\mathbf{z}|^2 e^{-\mathbf{h}} = \int_{\square_K} e^{\mathbf{h}} - 2 \int_{\square_K} q \cdot \nabla W + \int_{\square_K} e^{-\mathbf{h}} |\nabla W|^2.$$

The expectation of the first term is one. Taking expectations in the mean-zero Neumann problem gives $\mathbb{E}W = 0$, so the expectation of the second term vanishes. Lemma 5.3 and (5.13) consequently give

$$\limsup_{K \rightarrow \infty} \mathbb{E} \int_{\square_K} |\mathbf{z}|^2 e^{-\mathbf{h}} \leq 1 + \frac{2\tau^2 h}{d} + C\bar{\delta}^4 h^2.$$

Step 4. We conclude the proof of (5.12). Combining Steps 2 and 3, letting $K \rightarrow \infty$, and recalling that $\bar{\mathbf{a}}_{m,*}(\square_K) \rightarrow \bar{\mathbf{a}}_m$, we obtain

$$\frac{1}{2} \bar{\mathbf{a}}_m^{-1} \leq \left(\frac{1}{2} + \frac{\tau^2 h}{d} + C\bar{\delta}^4 h^2 \right) \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) + C\bar{\delta}^{15} \bar{\mathbf{a}}_{m-h}^{-1}.$$

Finally, (3.32) and (4.3) give

$$\bar{\mathbf{a}}_{m-h}^{-1} \leq \bar{\mathbf{a}}_{m-h,*}^{-1}(\square_n) \leq (1 + C\bar{\delta}^2 |\log \bar{\delta}|) \bar{\mathbf{a}}_{m-h}^{-1}.$$

Using $\tau^2 \leq 4\bar{\delta}^2$ and $\bar{\delta} h \leq 1$ proves (5.12). $\square$

Proof of Proposition 5.2. It suffices to consider $\bar{\delta} \leq c(d)$. Set

$$J := 16 \lceil |\log_3 \bar{\delta}| \rceil.$$

The ordering (3.32) gives

$$(5.14) \quad \left| \log \bar{\mathbf{a}}_J + \frac{2\tau^2(J+1)}{d} \right| \leq C\bar{\delta}^2(J+1) \leq C\bar{\delta}^2 |\log \bar{\delta}|.$$

By Lemmas 5.4 and 5.5 and the inequality $|\log(1+x)| \leq 2|x|$ for $|x| \leq 1/2$, whenever $1 \leq h \leq \bar{\delta}^{-1}$ and $m-h \geq J$,

$$(5.15) \quad \left| \log \bar{\mathbf{a}}_m - \log \bar{\mathbf{a}}_{m-h} + \frac{2\tau^2 h}{d} \right| \leq C\bar{\delta}^4 h^2 + C\bar{\delta}^2 |\log \bar{\delta}|.$$

We may assume $m > J$. Starting at m , use (5.15) repeatedly with $h = \lfloor \bar{\delta}^{-1} \rfloor$, each time lowering the scale by h , for as long as the new scale stays at least J , and then once more with the remaining $h < \bar{\delta}^{-1}$, if any, to arrive at J . This takes at most $1 + (m-J)/\lfloor \bar{\delta}^{-1} \rfloor \leq C(1+m\bar{\delta})$ steps, and each step has $h \leq \bar{\delta}^{-1}$ and therefore contributes an error at most $C\bar{\delta}^2 + C\bar{\delta}^2 |\log \bar{\delta}|$, because $\bar{\delta}^4 h^2 \leq \bar{\delta}^2$. Together with (5.14), this gives

$$\left| \log \bar{\mathbf{a}}_m + \frac{2\tau^2(m+1)}{d} \right| \leq C\bar{\delta}^2 |\log \bar{\delta}| (1+m\bar{\delta}),$$

which is (5.4). $\square$

5.1. Further bounds on the effective diffusivities. The error term in Proposition 5.2 dominates $2\tau^2 m/d$ unless τ^2 is bounded below by a multiple of $\bar{\delta}^3 |\log \bar{\delta}|$, and at $m = 0$ the proposition gives no upper bound. The following decay estimate assumes no such lower bound.

Proposition 5.6 (Power-law decay of the effective diffusivities). *There exists $\eta_{\text{sub}}(d, \text{law}(\mathbf{g}_0)) > 0$ satisfying*

$$(5.16) \quad \eta_{\text{sub}} = \frac{\tau^2}{\log 3} \quad \text{if } d = 2, \quad 0 < \eta_{\text{sub}} \leq \frac{\tau^2}{d \log 3} \quad \text{if } d \geq 3,$$

and, for every $m \in \mathbb{N}_0$,

$$(5.17) \quad \bar{\mathbf{a}}_m \leq 3^{-\eta_{\text{sub}}(m+1)}.$$

Proof. In dimension $d = 2$, Proposition 3.5 proves the result with $\eta_{\text{sub}} = \tau^2/\log 3$. We may therefore suppose that $d \geq 3$.

Put

$$B_k := \exp(\mathbf{g}_k - \tau^2), \quad A_N^{(R)} := \prod_{j=0}^N B_{jR} \quad (N \in \mathbb{N}_0),$$

where the integer $R \geq 1$ will be fixed below. Thus $\mathbb{E}[B_k(x)] = 1$ for every k and x . Define, with the maximum over all permutations π of $\{1, \dots, d\}$,

$$(5.18) \quad q_* := \max_{\pi} \sup_{|p|=1} \mathbb{E} \left[\min_{u \in p \cdot x + H_0^1(\Delta_0^\pi)} \int_{\Delta_0^\pi} B_0 |\nabla u|^2 \right].$$

Step 1. We prove that $q_* < 1$. The affine competitor gives $q_* \leq 1$. If equality held, there would be a permutation π and $|p| = 1$ for which the expected minimum in (5.18) equals $\mathbb{E}[\int_{\Delta_0^\pi} B_0 |p|^2] = 1$. The energy of $p \cdot x$ minus the minimum is nonnegative and would then have zero expectation, so it would vanish almost surely. Hence $p \cdot x$ would be a minimizer, and the Euler–Lagrange equation would give

$$p \cdot \nabla B_0 = 0 \quad \text{in } \Delta_0^\pi$$

almost surely. By stationarity and the almost-sure C^1 regularity of B_0 , $p \cdot \nabla B_0 = 0$ on $\mathbb{R}^d$. Consequently, $B_0(0) = B_0(tp)$ for every t . Choosing t with $|tp| > \sqrt{d}$, we see that the independent random variables $B_0(0)$ and $B_0(tp)$ coincide almost surely; hence each is deterministic. It follows that $\mathbf{g}_0(0)$ is deterministic and, since it has mean zero, that $\tau^2 = 0$, contrary to Assumption (g4). Thus

$$(5.19) \quad q_* < 1.$$

Step 2. We show that q_R converges to q_* , and fix R with $q_R < 1$. For a permutation π and $p \in \mathbb{R}^d$, let T run over the Kuhn simplices of the cubes of side 3^{-R} contained in Δ_0^π , and let v run over the continuous functions on $\bar{\Delta}_0^\pi$ which are affine on every such T and equal to $p \cdot x$ on $\partial \Delta_0^\pi$. Set

$$(5.20) \quad Q_{R,\pi}(p) := \min_v \frac{1}{|\Delta_0^\pi|} \sum_T |T| \left(\sup_{\bar{T}} B_0 \right) |\nabla v|_T|^2, \\ q_R := \max_{\pi} \sup_{|p|=1} \mathbb{E}[Q_{R,\pi}(p)].$$

Since B_0 is almost surely continuous and conforming piecewise-affine functions are dense in $H_0^1(\Delta_0^\pi)$, we have, almost surely as $R \rightarrow \infty$,

$$Q_{R,\pi}(p) \longrightarrow \min_{u \in p \cdot x + H_0^1(\Delta_0^\pi)} \int_{\Delta_0^\pi} B_0 |\nabla u|^2.$$

The affine competitor bounds $Q_{R,\pi}(p)$ by $|p|^2 \sup_{\Delta_0^\pi} B_0$, whose expectation is finite by Assumption (g2). Dominated convergence and (5.19) show that $q_R \rightarrow q_* < 1$. Fix R such that

$$(5.21) \quad q_R < 1.$$

Step 3. We iterate (5.21) over scales separated by R . We show by induction on N that, for every $N \in \mathbb{N}_0$, every permutation π and every $p \in \mathbb{R}^d$,

$$(5.22) \quad \mathbb{E} \left[p \cdot A_N^{(R)}(\Delta_{NR}^\pi) p \right] \leq q_R^{N+1} |p|^2.$$

For $N = 0$, this follows from (5.20), because restricting the minimization to piecewise-affine functions and replacing B_0 by its supremum on each simplex can only increase the energy. Suppose that (5.22) holds with $N-1$ in place of N . Fix a parent simplex $U := \Delta_{NR}^\pi$ and a vector p . The Kuhn simplices T of the cubes of side $3^{(N-1)R}$ contained in U form, after rescaling, the mesh used in (5.20). On this mesh, we consider the continuous function with boundary values $p \cdot x$ that is affine on each simplex and minimizes the energy in (5.20) with the weights $\sup_{\bar{T}} B_{NR}$ in place of $\sup_{\bar{T}} B_0$, and let p_T be its slope on T . On every T , we then take the minimizer of $\int_T A_{N-1}^{(R)} |\nabla u|^2$ among the functions whose boundary values on ∂T agree with this piecewise-affine function. These minimizers glue to a competitor in $p \cdot x + H_0^1(U)$, and the variational formula gives

$$p \cdot A_N^{(R)}(U) p \leq \sum_T \frac{|T|}{|U|} \left(\sup_{\bar{T}} B_{NR} \right) p_T \cdot A_{N-1}^{(R)}(T) p_T.$$

Condition on B_{NR} . The slopes p_T and the displayed weights are then fixed, whereas $A_{N-1}^{(R)}$ is independent of B_{NR} . Stationarity and the induction hypothesis bound the conditional expectation of the right side by

$$q_R^N \sum_T \frac{|T|}{|U|} \left(\sup_{\bar{T}} B_{NR} \right) |p_T|^2.$$

Taking the remaining expectation and using the scaling relation $\mathbf{g}_{NR} \stackrel{\text{law}}{=} \mathbf{g}_0(3^{-NR} \cdot)$ and (5.20), we obtain

$$\mathbb{E} \left[p \cdot A_N^{(R)}(U) p \right] \leq q_R^{N+1} |p|^2,$$

which proves (5.22).

Step 4. We pass from the products $A_N^{(R)}$, which contain only every R -th field, to $\bar{\mathbf{a}}_m$ and conclude. Fix $m \in \mathbb{N}_0$ and put $N := \lfloor m/R \rfloor$. The field $\mathbf{a}_m/A_N^{(R)}$ is the product of the factors B_k with $0 \leq k \leq m$ and $R \nmid k$. It is independent of $A_N^{(R)}$ and satisfies $\mathbb{E}[(\mathbf{a}_m/A_N^{(R)})(x)] = 1$. If u minimizes $\int_V A_N^{(R)} |\nabla u|^2$ among the functions on a bounded convex domain V with boundary values $p \cdot x$, then u is a function of $A_N^{(R)}$, so conditioning on $A_N^{(R)}$ and using its independence from $\mathbf{a}_m/A_N^{(R)}$ give

$$(5.23) \quad \mathbb{E}[p \cdot \mathbf{a}_m(V) p] \leq \mathbb{E} \left[\int_V \mathbf{a}_m |\nabla u|^2 \right] = \mathbb{E} \left[p \cdot A_N^{(R)}(V) p \right].$$

For $K \geq NR$, partition $\square_K$ into the Kuhn simplices of the cubes of side 3^{NR} . Using on each simplex the minimizer for $A_N^{(R)}$ with boundary values $p \cdot x$ as a competitor, we deduce from (5.22) that

$$\mathbb{E} \left[p \cdot A_N^{(R)}(\square_K) p \right] \leq q_R^{N+1} |p|^2.$$

Letting $K \rightarrow \infty$ in (5.23) and using (3.30), we obtain

$$\bar{\mathbf{a}}_m \leq q_R^{\lfloor m/R \rfloor + 1}.$$

Since $R(\lfloor m/R \rfloor + 1) \geq m + 1$, the choice

$$\eta_{\text{sub}} := -\frac{\log q_R}{R \log 3} > 0$$

proves (5.17). Replacing this value by its minimum with $\tau^2/(d \log 3)$ proves (5.16) as well. $\square$

Proposition 5.2 controls $\bar{\mathbf{a}}_m$ only up to its error term. The following lower bound has no error term and holds for every δ .

Proposition 5.7 (Lower bound on the effective diffusivities). *For every $m \in \mathbb{N}_0$,*

$$(5.24) \quad \bar{\mathbf{a}}_m \geq e^{-(m+1)\tau^2}.$$

Proof. Fix $m, n \in \mathbb{N}_0$ and put

$$c_m := e^{(m+1)\tau^2} \mathbf{a}_m = \exp \left(\sum_{k=0}^m \mathbf{g}_k \right).$$

Assumption (g3) gives $c_m^{-1} \stackrel{\text{law}}{=} c_m$. Since coarse-grained matrices are homogeneous of degree one in the coefficient field, it follows that

$$(5.25) \quad \mathbb{E}[c_m(\square_n)] = \mathbb{E}[(c_m^{-1})(\square_n)] = e^{(m+1)\tau^2} \bar{\mathbf{a}}_m(\square_n).$$

Here $(c_m^{-1})(\square_n)$ denotes the coarse-grained matrix associated with the reciprocal coefficient field $x \mapsto c_m(x)^{-1}$, and not the inverse of the matrix $c_m(\square_n)$.

For $i \in \{1, \dots, d\}$, let u_i and v_i be the minimizers in (2.18), with $p = e_i$, for the respective coefficients c_m and c_m^{-1} on $\square_n$. Let E and F be the Jacobian matrices of $x \mapsto (u_i(x))_i$ and $x \mapsto (v_i(x))_i$. Since u_i and v_i have boundary values $e_i \cdot x$, both maps have trace x on $\partial \square_n$. For two d -by- d matrices, set

$$Q(E, F) := \text{tr}(E) \text{tr}(F) - \text{tr}(EF), \quad |M|_{\mathbb{F}} := \text{tr}(M^{\top} M)^{1/2},$$

where $|\cdot|_{\mathbb{F}}$ is the Frobenius norm. The bilinear form Q is the polarization of the sum of the second-order principal minors. These minors are null Lagrangians, and $(u_i)_i$ and $(v_i)_i$ have the same trace, so

$$(5.26) \quad \int_{\square_n} Q(E, F) = Q(\text{Id}, \text{Id}) = d(d-1).$$

Equivalently, this follows by integration by parts from

$$Q(E, F) = \sum_{i,j=1}^d (\partial_i(u_i \partial_j v_j) - \partial_j(u_i \partial_i v_j)).$$

For arbitrary matrices E, F , Cauchy–Schwarz and $\text{tr}(F)^2 \leq d|F|_{\mathbb{F}}^2$ give

$$\begin{aligned} Q(E, F) &= \text{tr}(E^{\top}(\text{tr}(F)\text{Id} - F^{\top})) \leq (d-1)|E|_{\mathbb{F}}|F|_{\mathbb{F}}, \\ \left| \text{tr}(F)\text{Id} - F^{\top} \right|_{\mathbb{F}}^2 &= |F|_{\mathbb{F}}^2 + (d-2)\text{tr}(F)^2 \leq (d-1)^2|F|_{\mathbb{F}}^2. \end{aligned}$$

Consequently, pointwise on $\square_n$,

$$c_m|E|_{\mathbb{F}}^2 + c_m^{-1}|F|_{\mathbb{F}}^2 \geq 2|E|_{\mathbb{F}}|F|_{\mathbb{F}} \geq \frac{2}{d-1}Q(E, F).$$

Integrating, taking expectations, and using (5.25), (2.18), and (5.26), we obtain

$$2e^{(m+1)\tau^2} \text{tr} \bar{\mathbf{a}}_m(\square_n) = \mathbb{E} \left[\int_{\square_n} (c_m|E|_{\mathbb{F}}^2 + c_m^{-1}|F|_{\mathbb{F}}^2) \right] \geq 2d.$$

The matrix $\bar{\mathbf{a}}_m(\square_n)$ is scalar by Assumption (g3). Dividing by $2d$ and letting $n \rightarrow \infty$ proves (5.24). $\square$

6. LARGE-SCALE REGULARITY AND DIRICHLET HOMOGENIZATION

In this section we prove the large-scale Hölder estimate (Theorem C) and the homogenization error bound for the Dirichlet problem (Theorem B). The regularity estimates are used in Sections 8 and 9.

A solution of an equation with a rapidly oscillating coefficient cannot be regular on small scales, but it can be regular on large ones: if on every scale it is close to a solution of the homogenized equation, then it inherits the regularity of that solution down to the scale at which the approximation fails. The quantity we iterate is the excess, the L^2 distance from a solution to the affine functions on a cube. On a scale where the coarse-graining error is small, the excess contracts by a fixed factor; if such scales are frequent enough, the excess therefore decays geometrically and a Hölder estimate follows.

The scales on which the error fails to be small occur in a proportion that we can make as small as we wish by lowering the Hölder exponent; here randomness and independence between the scales enter through a concentration inequality. The cutoff coefficient $\mathbf{a}_L$ contains no fields of scale larger than the cutoff scale 3^L , so averaging over cubes larger than the cutoff scale improves its comparison with a constant coefficient. For this reason no relation between the size of the domain and the cutoff scale is needed, and the estimates are uniform in L and pass to the limiting coefficient.

The section is organized as follows.

- In Section 6.1 we define the good event, on which the coarse-graining error at a given scale is deterministically bounded (Proposition 6.3), and we show that, outside an event of exponentially small probability, the proportion of scales on which the good event fails is small (Proposition 6.4).
- In Section 6.2 we show that, on a good event, a solution is close to a harmonic function and its excess contracts by a fixed factor (Lemmas 6.8 and 6.9). The iteration of Lemma 6.10 combines this contraction on the good scales with a loss of at most a bounded factor on each of the other scales.
- In Section 6.3 we define the minimal scale and prove Proposition 6.1, the large-scale Hölder estimate in $\square_m$ for $L \geq m$.
- In Section 6.4 we remove the restriction $L \geq m$ (Proposition 6.16) and show that the estimates are stable when the coefficient is multiplied by a positive continuous function that is uniformly close to a constant (Proposition 6.17).
- Theorems 6.18 and 6.19 are the two parts of Theorem B: they bound the homogenization error of the Dirichlet problem uniformly in the cutoff and, at a fixed cutoff, with an algebraic rate. They follow from the general coarse-graining estimate of Proposition 2.11, applied in the whole domain, and do not use the regularity estimates.
- In the last subsection we extend the regularity estimates to the limiting coefficient $\mathbf{a}$, which proves Theorem C in the form of Theorem 6.21.

In (6.1) and (6.2) below, ℓ denotes an affine function on $\mathbb{R}^d$, and the minimum and the minimizer are taken over all affine functions. For $m, n \in \mathbb{Z}$ with $n \leq m$ and $x \in \square_m$, we define the **excess** of a function $u \in L^2((x + \square_n) \cap \square_m)$ by

$$(6.1) \quad E(u, (x + \square_n) \cap \square_m) := 3^{-n} \min_{\ell} \|u - \ell\|_{\underline{L}^2((x + \square_n) \cap \square_m)}$$

and its best affine approximation by

$$(6.2) \quad \ell(u, (x + \square_n) \cap \square_m) := \operatorname{argmin}_{\ell} \|u - \ell\|_{\underline{L}^2((x + \square_n) \cap \square_m)}.$$

Recall that $\mathbf{b}_{L,m}$ is defined in Definition 4.3.

Proposition 6.1 (Large-scale Hölder estimate for $L \geq m$). *There exists a constant $C(d) < \infty$ such that, if $\bar{\delta} \leq C^{-1}$, then, for every $\gamma \in [1/2, 1 - C\bar{\delta}|\log \bar{\delta}|^{1/2}]$ and $m \in \mathbb{N}_0$, there exists an N -valued*

random variable $\mathcal{L}_L(\gamma, m)$ (the same for every $L \geq m$) satisfying

$$(6.3) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C\bar{\delta}^2|\log \bar{\delta}|}\right) \quad (k \in \mathbb{N})$$

such that, for every $L \in \mathbb{N}_0$ with $L \geq m$ and every $\mathbf{g} \in C^{0,1/2}(\square_m; \mathbb{R}^d)$, $h \in C^{1,1/2}(\square_m)$ and $u \in H^1(\square_m)$ solving

$$(6.4) \quad \begin{cases} -\nabla \cdot \mathbf{a}_L \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = h & \text{on } \partial \square_m, \end{cases}$$

we have the following estimates:

- Large-scale Hölder estimate: for every $n \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $x \in \square_m$,

$$(6.5) \quad \begin{aligned} & \sup_{\substack{j \in \mathbb{N}_0 \\ j \leq n}} \max_{y \in 3^j \mathbb{Z}^d \cap ((x + \square_n) \cap \square_m)} 3^{\gamma(n-j)} \|u - (u)_{(y + \square_j) \cap \square_m}\|_{\underline{L}^2((y + \square_j) \cap \square_m)} \\ & \leq C 3^{-\gamma(m-n)} \left(\|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)} + (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{3/2m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right) \\ & \quad + C 3^{-\gamma(m-n)} 3^{3/2m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{x \notin \square_{m-1}\}}; \end{aligned}$$

- Energy estimate: for every $n \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $x \in \square_m$,

$$(6.6) \quad \begin{aligned} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2((x + \square_n) \cap \square_m)} & \leq C 3^{(1-\gamma)(m-n)} \left(\|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + (\mathbf{b}_{L,m})_{\square_m}^{-1/2} 3^{1/2m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right) \\ & \quad + C 3^{(1-\gamma)(m-n)} (\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{1/2m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{x \notin \square_{m-1}\}}; \end{aligned}$$

- Excess decay estimate: if $n \in \mathbb{N}_0$ is such that $\mathcal{L}_L(\gamma, m) \leq m - n \leq (1-\gamma)^{-1}$, then for every $l \in \mathbb{N}_0$ satisfying $n \leq l \leq m - 5$ and every $x \in \square_m$,

$$(6.7) \quad \begin{aligned} & E(u, (x + \square_n) \cap \square_m) \\ & \leq C 3^{-1/4(l-n)} E(u, (x + \square_l) \cap \square_m) + C(1-\gamma)^{1/2} 3^{-l} \|u - (u)_{(x + \square_l) \cap \square_m}\|_{\underline{L}^2((x + \square_l) \cap \square_m)} \\ & \quad + C(\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{1/2l} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ & \quad + C \left((1-\gamma)(m-n) \|\nabla h\|_{L^\infty(\square_m)} + 3^{1/2l} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \right) \mathbb{1}_{\{x \notin \square_{m-1}\}}. \end{aligned}$$

At scale 3^m , Proposition 2.11 compares an $\mathbf{a}_L$ -harmonic function with a harmonic function when the coarse-graining error

$$(6.8) \quad \mathcal{E}_{1/4,\infty}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m})$$

is small.

6.1. Harmonic approximation and independence between scales. We now introduce the good event $\mathcal{G}_m(\varepsilon, s)$ at scale m , on which the coarse-graining error at that scale is deterministically bounded, and estimate the number of scales at which it fails. Proposition 6.3 bounds the coarse-graining error on a scale where the good event holds, and Proposition 6.4 shows that, among consecutive scales, the proportion of those where it fails is small. These two estimates are the input to the iteration of Lemma 6.10 in Section 6.3, and the second uses the independence of the fields across scales.

Definition 6.2 (Good event). For $s, \varepsilon \in (0, 1]$ and $m \in \mathbb{N}_0$, let the **good event** $\mathcal{G}_m(\varepsilon, s)$ be the event on which the following inequalities hold:

$$(6.9) \quad \sup_{j \geq 0} 3^{-1/8sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i \|\nabla \mathbf{g}_i\|_{L^\infty(\square_{m+1+j})} \leq \varepsilon,$$

$$(6.10) \quad \sup_{j \geq 0} 3^{-1/8sj} \left\| \prod_{i=(m-j) \vee 0}^{m+j} \exp(|\mathbf{g}_i|) + \prod_{i=m+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \right\|_{L^\infty(\square_{m+1+j})} \leq 6,$$

$$(6.11) \quad \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-1/8s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \leq \varepsilon^2.$$

For $y \in \mathbb{R}^d$, we let $\mathcal{G}_{m,y}(\varepsilon, s)$ denote the event obtained by translating the displayed quantities by y . The supremum in (6.11) is defined to be zero when there is no pair (j, n) satisfying the displayed constraints.

We call a scale m **good** if the event $\mathcal{G}_m(\varepsilon, s)$ holds: the fields at scales $m \pm j$ and the quantities J on subcubes of scale $m - j$ are small, up to weights $3^{-sj/8}$ that decay with j . A scale that is not good is called **bad**. We use the same words for the translated events $\mathcal{G}_{m,y}(\varepsilon, s)$, with respect to the center y .

Throughout this section, we use the following notation. For $m, j \in \mathbb{N}_0$ with $j \leq m$, we denote

$$(6.12) \quad \mathbf{g}_{m,j} := \sum_{i=j+1}^m \mathbf{g}_i.$$

We also note that, for $0 \leq j \leq m$,

$$(6.13) \quad \frac{\mathbf{a}_m}{\mathbf{a}_j} = \exp(\mathbf{g}_{m,j} - (m-j)\tau^2).$$

For $s, \varepsilon \in (0, 1]$, condition (6.9) implies

$$(6.14) \quad \left(\sup_{0 \leq j \leq m} 3^{-1/8s(m-j)} \|\mathbf{g}_{m,j}\|_{L^\infty(\square_m)} + \sum_{j=m}^{\infty} 3^m \|\nabla \mathbf{g}_j\|_{L^\infty(\square_m)} \right) \mathbb{1}_{\mathcal{G}_m(\varepsilon, s)} \leq C\varepsilon.$$

Proposition 6.3 (Coarse-graining on good scales). *There is a constant $C(d) < \infty$ such that, for every $s \in [64\bar{\delta}^2, 1/2]$ and $L, m \in \mathbb{N}_0$ with $L \geq m$, we have*

$$(6.15) \quad \begin{aligned} & \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}) \mathbb{1}_{\mathcal{G}_m(1,s)} \\ & \leq C \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-1/2s(m-n)} \\ & \quad \times \max_{\substack{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1}) \\ |e|=1}} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)^{1/2} \\ & \quad + Cs^{-1}\bar{\delta}^2 + C \sup_{0 \leq j \leq m} 3^{-1/8s(m-j)} \|\mathbf{g}_{m,j}\|_{L^\infty(\square_m)} \\ & \quad + C3^{-1/8sm} \left\| \sum_{i=0}^m \mathbf{g}_i \right\|_{L^\infty(\square_m)} + C \sum_{j=m}^{\infty} 3^m \|\nabla \mathbf{g}_j\|_{L^\infty(\square_m)}. \end{aligned}$$

In particular, if $\varepsilon \in [s^{-1}\bar{\delta}^2, 1]$, then (6.14) yields, on $\mathcal{G}_m(\varepsilon, s)$,

$$\mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m}) \leq C\varepsilon.$$

We will also use the following elementary inequality. If $\varepsilon \in (0, 1]$, $s > 0$, and $D \geq 0$, then every $x \in [0, \varepsilon 3^{sD/16}]$ satisfies

$$3^{-sD/2} x \leq 3^{-sD/2} \min\{x, 1\} + \varepsilon^8.$$

Indeed, the second term on the right is needed only when $x > 1$, and then $3^{-sD/16} < \varepsilon$, so that $3^{-sD/2}x \leq \varepsilon 3^{-7sD/16} \leq \varepsilon^8$.

Proposition 6.4 (Proportion of good scales). *There exists $C(d) < \infty$ such that, for every $s, \theta, \varepsilon \in (0, 1]$ satisfying*

$$(6.16) \quad \theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

and every $m_0 \in \mathbb{N}_0$ and $M \in \mathbb{N}_0$, we have the estimate

$$(6.17) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1}\{\mathcal{G}_m(\varepsilon, s)\} \leq 1 - \theta \right] \leq \exp \left(- \frac{s^6 \varepsilon^2 \theta}{C \bar{\delta}^2 |\log \bar{\delta}|} (M+1) \right).$$

Proof of Proposition 6.3. Fix $s \in [64\bar{\delta}^2, 1/2]$. Throughout the proof, we denote $\tilde{\mathbf{a}}_{L,m} := (\mathbf{b}_{L,m})_{\square_m}^{-1} \mathbf{a}_L$.

Step 1. We prove an annular decomposition, which bounds the error by the contributions of cubes lying in the annuli $\square_j \setminus \square_{j-1}$: there exists a constant $C(d) < \infty$ such that, for every $m \in \mathbb{N}_0$ and $s \in (0, 1/2]$,

$$(6.18) \quad \begin{aligned} \mathcal{E}_{s,\infty,2}(\square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{\square_m})^2 &= (1 - 3^{-2s}) \sum_{n=-\infty}^m 3^{-2s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \max_{|e|=1} J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}) \\ &\leq C \sup_{j \leq m} \sup_{n \leq j-2} 3^{-3/2s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}). \end{aligned}$$

For a domain U and $e \in \mathbb{R}^d$, write $J(U) := J(U, e, e; \tilde{\mathbf{a}}_{L,m})$. By decomposing into annular regions and interchanging the order of summation, we obtain

$$\begin{aligned} s \sum_{n=-\infty}^m 3^{-2s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} J(z + \square_n) \\ \leq s \sum_{n=-\infty}^m 3^{-2s(m-n)} \max_{n+1 \leq j \leq m} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_n) + s \sum_{n=-\infty}^m 3^{-2s(m-n)} J(\square_n) \\ \leq C \sup_{j \leq m} \sup_{n \leq j-1} 3^{-3/2s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_n) + s \sum_{n=-\infty}^m 3^{-2s(m-n)} J(\square_n). \end{aligned}$$

For the second term on the right, we use subadditivity. For every $n \leq m$, we have

$$J(\square_n) \leq \sum_{k=-\infty}^{n-1} \sum_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} \frac{|\square_k|}{|\square_n|} J(z + \square_k) \leq \sum_{k=-\infty}^{n-1} 3^{-d(n-1-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k).$$

Substituting back and interchanging the order of summation yields

$$\begin{aligned} \sum_{n=-\infty}^m 3^{-2s(m-n)} J(\square_n) &\leq \sum_{n=-\infty}^m 3^{-2s(m-n)} \sum_{k=-\infty}^{n-1} 3^{-d(n-1-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k) \\ &\leq 3^{2s} \sum_{k=-\infty}^{m-1} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k) \sum_{n=k+1}^m 3^{-(d-2s)(n-1-k)} \\ &\leq \frac{d 3^{2s}}{d-2s} \sum_{k=-\infty}^{m-1} 3^{-2s(m-k)} \max_{z \in 3^k \mathbb{Z}^d \cap (\square_{k+1} \setminus \square_k)} J(z + \square_k). \end{aligned}$$

After relabeling $k \rightarrow n$ and $k + 1 \rightarrow j$, the sum on the right becomes

$$\sum_{j=-\infty}^m 3^{-2s(m-j)} \max_{z \in 3^{j-1} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-1}) \leq \frac{C}{s} \sup_{j \leq m} 3^{-3/2s(m-j)} \max_{z \in 3^{j-1} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-1}),$$

which is of the form appearing on the right side of (6.18) with $n = j - 1$. We obtain (6.18) by combining the previous displays with subadditivity in the form

$$\max_{z \in 3^{j-1} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-1}) \leq \max_{z \in 3^{j-2} \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} J(z + \square_{j-2}).$$

Step 2. We first treat the cubes of side at most one. By (4.17), with $b = (\mathbf{b}_{L,m})_{\square_m}$, the contribution of the indices $n \leq 0$ to the left side of (6.18) is at most

$$(6.19) \quad C 3^{-2sm} \left\| \frac{\mathbf{a}_L}{(\mathbf{b}_{L,m})_{\square_m}} + \frac{(\mathbf{b}_{L,m})_{\square_m}}{\mathbf{a}_L} - 2 \right\|_{L^\infty(\square_m)}.$$

On $\mathcal{G}_m(1, s)$, the relations (6.22), (6.23), and (6.24) established below, together with (6.10), bound (6.19) by the last four terms on the right side of (6.20) below. It remains to bound the contribution of the cubes of side larger than one.

We show that there exists $C(d) < \infty$ such that

$$(6.20) \quad \begin{aligned} & \sup_{1 \leq j \leq m} \sup_{1 \leq n \leq j-1} 3^{-3/2s(m-n)} \\ & \quad \times \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}) \mathbb{1}_{\mathcal{G}_m(1,s)} \\ & \leq C \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \\ & \quad \times \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \\ & \quad + C s^{-2} \bar{\delta}^4 + C \left(\sum_{n=m}^{\infty} 3^m \|\nabla \mathbf{g}_n\|_{L^\infty(\square_m)} \right)^2 \\ & \quad + C \sup_{0 \leq n \leq m} 3^{-s(m-n)} \|\mathbf{g}_{m,n}\|_{L^\infty(\square_m)}^2 + C 3^{-sm} \left\| \sum_{i=0}^m \mathbf{g}_i \right\|_{L^\infty(\square_m)}^2. \end{aligned}$$

We apply Lemma 2.8 with $\mathbf{a} = \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n$ and $\mathbf{b} = \tilde{\mathbf{a}}_{L,m}$. For every $n \in \mathbb{N}$ with $n < m$, this gives

$$(6.21) \quad \begin{aligned} & J(z + \square_n, e, e; \tilde{\mathbf{a}}_{L,m}) \\ & \leq 2J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \\ & \quad + 3 \left(\left\| \frac{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n}{\tilde{\mathbf{a}}_{L,m}} - 1 \right\|_{L^\infty(z + \square_n)}^2 + \left\| \frac{\tilde{\mathbf{a}}_{L,m}}{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n} - 1 \right\|_{L^\infty(z + \square_n)}^2 \right) (J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) + 1). \end{aligned}$$

Put

$$\rho_{m,n} := \log \frac{\bar{\mathbf{a}}_n}{\bar{\mathbf{a}}_m} - (m - n) \tau^2.$$

By (6.13),

$$(6.22) \quad \frac{\tilde{\mathbf{a}}_{L,m}}{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n} = \frac{\mathbf{a}_L \mathbf{a}_m^{-1}}{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}} \exp(\mathbf{g}_{m,n} + \rho_{m,n}).$$

By (3.32),

$$(6.23) \quad |\rho_{m,n}| \leq \tau^2(m - n).$$

Set

$$S_m := \sum_{j=m}^{\infty} 3^m \|\nabla \mathbf{g}_j\|_{L^\infty(\square_m)}.$$

Conditions (6.9) and (6.10) in the definition of $\mathcal{G}_m(1, s)$, the mean-value theorem, and $|\exp(r) - 1| \leq |r| \exp(|r|)$ give

$$(6.24) \quad \left(\left\| \frac{\mathbf{a}_L \mathbf{a}_m^{-1}}{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}} - 1 \right\|_{L^\infty(\square_m)} + \left\| \frac{(\mathbf{a}_L \mathbf{a}_m^{-1})_{\square_m}}{\mathbf{a}_L \mathbf{a}_m^{-1}} - 1 \right\|_{L^\infty(\square_m)} \right) \mathbb{1}_{\mathcal{G}_m(1, s)} \leq C \min\{1, S_m\},$$

and also $S_m \mathbb{1}_{\mathcal{G}_m(1, s)} \leq C$. Moreover, (6.10), (6.23), $\tau^2 \leq 4\bar{\delta}^2$, and $s \geq 64\bar{\delta}^2$ imply

$$\exp(\|\mathbf{g}_{m,n}\|_{L^\infty(z+\square_n)}) \mathbb{1}_{\mathcal{G}_m(1, s)} \leq C 3^{s(m-n)/8}, \quad \exp(|\rho_{m,n}|) \leq 3^{s(m-n)/16}.$$

Applying these bounds to (6.22) and its reciprocal yields

$$\begin{aligned} & \left\| \frac{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n}{\tilde{\mathbf{a}}_{L,m}} - 1 \right\|_{L^\infty(z+\square_n)} + \left\| \frac{\tilde{\mathbf{a}}_{L,m}}{\bar{\mathbf{a}}_n^{-1} \mathbf{a}_n} - 1 \right\|_{L^\infty(z+\square_n)} \\ & \leq C 3^{3/16s(m-n)} \min\left\{1, S_m + \|\mathbf{g}_{m,n}\|_{L^\infty(z+\square_n)} + \tau^2(m-n)\right\}. \end{aligned}$$

On $\mathcal{G}_m(1, s)$, condition (6.11), combined with the subadditivity used at the end of Step 1, bounds the term $J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)$ in (6.21) by $3^{s(m-n)/8}$. Hence the square of the right side of the last display, multiplied by $J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) + 1$ and by $3^{-3s(m-n)/2}$, is bounded by

$$C 3^{-s(m-n)} \left(S_m^2 + \|\mathbf{g}_{m,n}\|_{L^\infty(z+\square_n)}^2 + (\tau^2(m-n))^2 \right).$$

Finally,

$$\sup_{r \geq 0} 3^{-sr} (\tau^2 r)^2 \leq C s^{-2} \bar{\delta}^4.$$

Combining these estimates with (6.21) proves (6.20).

Step 3. We conclude. Combining (6.18), the estimate (6.19) for cubes of side at most one, and (6.20) gives (6.15). $\square$

For each of the three conditions (6.9), (6.10), and (6.11) separately, we estimate the probability that it fails on a given proportion of the scales.

Lemma 6.5. *There exists a constant $C(d) < \infty$ such that, for every $\theta, s, \varepsilon \in (0, 1]$ satisfying $\theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2$, we have, for every $m_0 \in \mathbb{N}_0$ and $M \in \mathbb{N}_0$,*

$$(6.25) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i \|\nabla \mathbf{g}_i\|_{L^\infty(\square_{m+1+j})} > \varepsilon \right\} \geq \theta \right] \leq \exp \left(-\frac{s^6 \varepsilon^2 \theta}{C \bar{\delta}^2} (M+1) \right).$$

Lemma 6.6. *There exists a constant $C(d) < \infty$ such that, for every $m_0, M \in \mathbb{N}_0$ and $s, \theta \in (0, 1]$ satisfying $\theta \geq C s^{-2} \bar{\delta}^2$, we have the estimate*

$$(6.26) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=(m-j) \vee 0}^{m+j} \exp(|\mathbf{g}_i|) + \prod_{i=m+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \right\|_{L^\infty(\square_{m+1+j})} > 6 \right\} \geq \theta \right] \leq \exp \left(-\frac{s^2 \theta}{C \bar{\delta}^2} (M+1) \right).$$

Lemma 6.7. *There exists a constant $C(d) < \infty$ such that, if $\varepsilon, s, \theta \in (0, 1]$ satisfy the conditions*

$$\bar{\delta}^2 |\log \bar{\delta}| \leq C^{-1} s^3 \varepsilon^2 \quad \text{and} \quad \theta \geq C s^{-3} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

then we have, for every $m_0 \in \mathbb{N}_0$ and $M \in \mathbb{N}_0$, the estimate

$$(6.27) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) > \varepsilon^2 \right\} > \theta \right] \leq \exp \left(- \frac{s^3 \varepsilon^2 \theta}{C |\log \bar{\delta}| \bar{\delta}^2} (M+1) \right).$$

Proof of Proposition 6.4. Fix $s, \theta, \varepsilon \in (0, 1]$ and assume (6.16) with its constant C large enough that Lemmas 6.5, 6.6, and 6.7 are applicable with parameters $(s/8, \theta/4, \varepsilon)$ in place of (s, θ, ε) . If the good event fails on at least a proportion θ of the scales, then one of the three conditions (6.9)–(6.11) fails on at least a proportion $\theta/3$ of them. Since $\theta/3 > \theta/4$, this case is covered by the strict inequality in Lemma 6.7. A union bound over the three conditions gives (6.17). $\square$

Proof of Lemma 6.5. Interchanging the order of summation gives

$$\begin{aligned} \sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})} &\leq \sum_{j=0}^{\infty} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})} \\ &= \sum_{i \in \mathbb{N}_0} 3^{-s|i-m|} \left(\sum_{j=|i-m|}^{\infty} 3^{-s(j-|i-m|)} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})} \right). \end{aligned}$$

For a constant C_2 to be fixed and $i \in \mathbb{N}_0$, set

$$X_{m,i} := C_2 s^{-1} \varepsilon^{-1} 3^{-1/2s|m-i|} \sum_{j=|i-m|}^{\infty} 3^{-s(j-|i-m|)} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})},$$

and set $X_{m,i} = 0$ if $m < 0$ or $i < 0$. We shall apply Proposition B.2 to this extension to all indices in $\mathbb{Z}$. We have

$$\sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})} \leq C_2^{-1} s \varepsilon \sum_{i \in \mathbb{N}_0} 3^{-1/2s|m-i|} X_{m,i}.$$

By (A.4), (1.4), and (1.5),

$$\|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})} \leq \mathcal{O}_{\Gamma_2}(C \bar{\delta} (|m-i| + j + 1)^{1/2}).$$

By (A.6), there exists a constant $C_1(d)$ such that, for every $p \in [1, \infty)$,

$$\mathbb{E}[X_{m,i}^p]^{1/p} \leq C_1 C_2 s^{-5/2} \varepsilon^{-1} \bar{\delta} p^{1/2}.$$

Moreover, for $m, m', i, i' \in \mathbb{N}_0$ with $|i - i'| \geq 1$, the variables $X_{m,i}$ and $X_{m',i'}$ are independent. We fix $p := C_1^{-2} C_2^{-2} s^5 \varepsilon^2 \bar{\delta}^{-2}$. With this choice of p , if $C(d)$ is large enough in terms of C_1 and C_2 , the assumption $\theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2$, which since $\theta \leq 1$ also gives $\bar{\delta} \leq C^{-1/2} s^3 \varepsilon$, implies that $\theta \geq p^{-1}$ and $s \geq 2p^{-1}$. By the inequality displayed after the definition of $X_{m,i}$, we then have

$$\left\{ \sup_{j \geq 0} 3^{-sj} \sum_{i=(m-j) \vee 0}^{m+j} \|\mathbf{g}_i\| + 3^i |\nabla \mathbf{g}_i|_{L^\infty(\square_{m+1+j})} > \varepsilon \right\} \subseteq \left\{ \sum_{i \in \mathbb{N}_0} 3^{-1/2s|m-i|} X_{m,i} > C_2 s^{-1} \right\}.$$

For C_2 large enough, Proposition B.2 gives (6.25). $\square$

Proof of Lemma 6.6. We apply Propositions B.3 and B.4. For sufficiently large $C(d)$, set $X_n := C3^n \|\nabla \mathbf{g}_n\|_{L^\infty(\square_{n+1})}$ for $n \in \mathbb{N}_0$ and $X_n := 0$ for $n < 0$. On $\square_{k+1+j}$,

$$\prod_{i=k+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \leq \prod_{i=k+j}^{\infty} \exp\left(3^{k+j-i} (C3^i \|\nabla \mathbf{g}_i\|_{L^\infty(\square_{i+1})})\right) \leq \prod_{i=0}^{\infty} \exp(3^{-i} X_{k+i+j}).$$

The sequence (X_n) satisfies the hypotheses of Proposition B.3 with $\delta_0 := C\bar{\delta}$. If $\bar{\delta} \leq C^{-1}s\theta^{1/2}$, then that proposition gives

$$\mathbb{P}\left[\sum_{m=m_0}^{m_0+M} \mathbb{1}\left\{\sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=m+j}^{\infty} \exp(4|\mathbf{g}_i - \mathbf{g}_i(0)|) \right\|_{L^\infty(\square_{m+1+j})} > 3\right\} \geq \theta\right] \leq \exp\left(-\frac{s^2\theta}{C\bar{\delta}^2}(M+1)\right).$$

If $\bar{\delta} \leq C^{-1}s\theta^{1/2}$ for sufficiently large $C(d)$, then Proposition B.4, applied to $X_i := \mathbf{g}_i$ for $i \in \mathbb{N}_0$ and $X_i := 0$ for $i < 0$, yields

$$\mathbb{P}\left[\sum_{k=m_0}^{m_0+M} \mathbb{1}\left\{\sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=(k-j) \vee 0}^{k+j} \exp(|\mathbf{g}_i|) \right\|_{L^\infty(\square_{k+1+j})} > 3\right\} \geq \theta\right] \leq \exp\left(-\frac{s^2\theta}{C\bar{\delta}^2}(M+1)\right).$$

Apply the previous two displays with $\theta/2$ in place of θ . If the sum of the two products exceeds six, then at least one exceeds three, so a union bound gives (6.26). $\square$

Proof of Lemma 6.7. First,

$$\begin{aligned} & \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \\ & \leq \sum_{j=0}^m 3^{-s(m-j)} \sum_{n=0}^{j-2} 3^{-s(j-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n). \end{aligned}$$

We set, for $K \geq 1$ to be fixed,

$$X_{m,j} := K s^{-1} \varepsilon^{-2} \sum_{n=0}^{j-2} 3^{-s(j-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \mathbb{1}_{\{j \leq m\}}.$$

Then

$$\begin{aligned} & \left\{ \sup_{0 \leq j \leq m} \sup_{0 \leq n \leq j-2} 3^{-s(m-n)} \max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) \geq \varepsilon^2 \right\} \\ & \subseteq \left\{ \sum_{j=0}^m 3^{-s(m-j)} X_{m,j} \geq K s^{-1} \right\}. \end{aligned}$$

There is an integer $r(d) < \infty$ such that the columns $(X_{m,j})_m$, indexed by j , are $r(d)$ -dependent in the sense of (B.3). To verify this, fix one of the fields $\mathbf{g}_i$. Column j depends on $\mathbf{g}_i$ only if $i \leq j-2$, and then only through its values on the annulus $\square_j \setminus \square_{j-1}$. If $j' > j$ and $j' - j \geq r(d)$, then the distance between these annuli is at least $(3^{j'-1} - 3^j)/2 > \sqrt{d}3^{j-2} \geq \sqrt{d}3^i$, provided that $r(d)$ is large enough. Thus, by Assumption (g1), the restrictions of $\mathbf{g}_i$ to the regions used by two sets of columns at distance at least $r(d)$ are independent. Since the fields are mutually independent, so are the two sets of columns. Moreover, by the triangle inequality, stationarity, and Proposition 4.1,

there is a constant $C_1(d) < \infty$ such that, if p satisfies $2ds^{-1} \leq p \leq C_1^{-1}\bar{\delta}^{-2}|\log \bar{\delta}|^{-1}$, then

$$\begin{aligned} \mathbb{E}[X_{m,j}^p]^{1/p} &\leq Ks^{-1}\varepsilon^{-2} \sum_{n=0}^{j-2} 3^{-s(j-n)} \mathbb{E} \left[\max_{z \in 3^n \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)^p \right]^{1/p} \\ &\leq Ks^{-1}\varepsilon^{-2} \sum_{n=0}^{j-2} 3^{-(s-d/p)(j-n)} \mathbb{E} \left[\max_{|e|=1} J(\square_n, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n)^p \right]^{1/p} \leq C_1 K s^{-2} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}| p. \end{aligned}$$

Taking $p := C_1^{-1} K^{-1} s^2 \varepsilon^2 \bar{\delta}^{-2} |\log \bar{\delta}|^{-1}$ and assuming that $\bar{\delta}$ satisfies

$$\bar{\delta}^2 |\log \bar{\delta}| \leq \frac{s^3 \varepsilon^2}{2dC_1 K} \implies 2ds^{-1} \leq p = C_1^{-1} K^{-1} s^2 \varepsilon^2 \bar{\delta}^{-2} |\log \bar{\delta}|^{-1},$$

we deduce that $\mathbb{E}[X_{m,j}^p]^{1/p} \leq 1$. Assuming also $\theta \geq p^{-1}$ and applying Proposition B.2 with $r = r(d)$, we obtain (6.27) for a sufficiently large universal constant K . $\square$

6.2. Excess decay on good scales. Throughout this subsection, we denote, for $m, n \in \mathbb{Z}$ with $n \leq m$ and $x \in \square_m$,

$$(6.28) \quad U_{m,n}(x) := (x + \square_n) \cap \square_m.$$

The first lemma adapts Armstrong et al. [2026, Lemma 4.10].

Lemma 6.8 (Harmonic approximation on good scales). *There exists $C(d) < \infty$ such that, for every $s \in [512\bar{\delta}^2, 1/4]$, every $L, m, n \in \mathbb{N}_0$ with $L \geq m \geq n + 5$, every $z \in \square_m$ and $x \in U_{m,n-3}(z)$, and every $u \in H^1(\square_m)$, $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and $h \in H^{1+s}(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a}_L \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = h & \text{on } \partial \square_m, \end{cases}$$

the following holds. Let $y \in \square_m$ be such that $U_{m,n-4}(x) \subseteq y + \square_{n-2} \subseteq U_{m,n-1}(x)$. Let v be the unique solution of the Dirichlet problem

$$(6.29) \quad \begin{cases} -\Delta v = 0 & \text{in } y + \square_{n-2}, \\ v = u & \text{on } \partial(y + \square_{n-2}). \end{cases}$$

Then

$$\begin{aligned} &\|u - v\|_{L^2(U_{m,n-4}(x))} \mathbb{1}_{\mathcal{G}_{n+2,z}(1,s/8)} \\ &\leq Cs^{-3/2} \mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}) \\ &\quad \times \left(\|u - (u)_{U_{m,n}(x)}\|_{L^2(U_{m,n}(x))} + s^{-3/2} 3^n |(\nabla h)_{U_{m,n}(x)}| \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}} \right) \\ &\quad + Cs^{-15/2} (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}^{-1} 3^{(1+s)n} [\mathbf{g}]_{\underline{H}^s(U_{m,n}(x))} \\ (6.30) \quad &\quad + Cs^{-7/2} 3^{(1+s)n} [\nabla h]_{\underline{H}^s(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

Proof. Set

$$D := y + \square_{n-2}, \quad \mathbf{a}_0 := (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}.$$

The stability estimates in Lemma 2.4, (2.57), and Proposition 6.3 give, on $\mathcal{G}_{n+2,z}(1,s/8)$,

$$\begin{aligned} &\mathcal{E}_{s/3,\infty,1}(D; \mathbf{a}_L, \mathbf{a}_0) + \mathcal{E}_{s/6,\infty,2}(D; \mathbf{a}_L, \mathbf{a}_0) \leq C \mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0) \leq C, \\ (6.31) \quad &\mathbf{a}_0^{-1} \Lambda_{s/6,q}(Q; \mathbf{a}_L) + \mathbf{a}_0 \lambda_{s/6,q}^{-1}(Q; \mathbf{a}_L) \leq C, \quad q \in \{1, 2\}, \end{aligned}$$

for every cube Q of side between 3^{n-3} and 3^{n-1} which is contained in $U_{m,n}(x)$ and meets D .

Let $v_{\mathbf{g}}$ solve

$$-\mathbf{a}_0 \Delta v_{\mathbf{g}} = \nabla \cdot \mathbf{g} \quad \text{in } D, \quad v_{\mathbf{g}} = u \quad \text{on } \partial D.$$

Since constants in $\mathbf{g}$ have zero divergence, the energy estimate and the fractional Poincaré inequality yield

$$\mathbf{a}_0 \|\nabla(v_{\mathbf{g}} - v)\|_{\underline{L}^2(D)} \leq C s^{-1/2} 3^{sn} [\mathbf{g}]_{\underline{H}^s(D)}.$$

Apply Proposition 2.11 to u and $v_{\mathbf{g}}$ on the cube D , of side 3^{n-2} , with the sums in that proposition stopping at the index $n-3$, and with

$$p = 2, \quad (s_1, s, s_2) = (s/3, s/2, s).$$

The sums over scales defining the errors in that proposition, and the energy sum over k in (2.75), end at $n-3$. Extending these sums to their full ranges only increases them. Using (6.31), the preceding display, and the triangle inequality, we obtain

$$(6.32) \quad \begin{aligned} 3^{-s/2n} \|\nabla u - \nabla v\|_{\widehat{W}^{-s/2,2}(D)} \mathbb{1}_{\mathcal{G}_{n+2,z}(1,s/8)} &\leq C s^{-3/2} \mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0) \mathbf{a}_0^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(D)} \\ &\quad + C s^{-11/2} \mathbf{a}_0^{-1} 3^{sn} [\mathbf{g}]_{\underline{H}^s(D)}. \end{aligned}$$

It remains to estimate the energy on D . Partition D into its 9^d subcubes $q + \square_{n-4}$, with centers q . For each center q , let P_q be the cube obtained from $q + \square_{n-2}$ by the smallest coordinatewise translation that places it inside $\square_m$. The inclusions in the statement imply

$$P_q \subseteq U_{m,n}(x), \quad q + \square_{n-4} \subseteq P_q, \quad (q + \square_{n-3}) \cap (\partial P_q \setminus \partial \square_m) = \emptyset.$$

Indeed, the first assertion follows from

$$|\zeta - x|_{\infty} \leq 1/2 3^{n-2} + 1/2 3^{n-2} + 1/2 3^{n-1} < 1/2 3^n, \quad \zeta \in P_q,$$

and the other two follow directly from the construction of P_q .

If $q + \square_{n-3}$ does not meet $\partial \square_m$, we apply Proposition 2.6 in $q + \square_{n-3}$. Otherwise we apply Lemma 2.10 in P_q at q ; since $q + \square_{n-3}$ is separated from $\partial P_q \setminus \partial \square_m$, the only boundary condition used by the lemma is $u = h$ on $\partial \square_m$. We split h into the affine function with slope $(\nabla h)_{U_{m,n}(x)}$ and a remainder. By the Dirichlet principle for the $\mathbf{a}_L$ -harmonic extension of this affine function, the affine part contributes $C \mathbf{a}_0^{1/2} |(\nabla h)_{U_{m,n}(x)}|$, and the remainder contributes at most $C s^{-1/2} 3^{sn} [\nabla h]_{\underline{H}^s(U_{m,n}(x))}$. Using (6.31) and summing over the 9^d subcubes, we obtain

$$(6.33) \quad \begin{aligned} \mathbf{a}_0^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(D)} \mathbb{1}_{\mathcal{G}_{n+2,z}(1,s/8)} &\leq C 3^{-n} \|u - (u)_{U_{m,n}(x)}\|_{\underline{L}^2(U_{m,n}(x))} + C |(\nabla h)_{U_{m,n}(x)}| \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}} \\ &\quad + C s^{-6} \mathbf{a}_0^{-1} 3^{sn} [\mathbf{g}]_{\underline{H}^s(U_{m,n}(x))} \\ &\quad + C s^{-2} 3^{sn} [\nabla h]_{\underline{H}^s(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

Finally, since $u - v \in H_0^1(D)$ and $U_{m,n-4}(x) \subseteq D$,

$$\|u - v\|_{\underline{L}^2(U_{m,n-4}(x))} \leq C 3^{(1-s/2)n} \|\nabla u - \nabla v\|_{\widehat{W}^{-s/2,2}(D)}.$$

We insert (6.33) into (6.32) and use the previous display. The term of (6.32) involving $\mathbf{g}$ is bounded by the $s^{-15/2}$ term in (6.30), and the factors of $\mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0)$ multiplying the forcing and boundary terms coming from (6.33) are absorbed using $\mathcal{E}_{s/8,\infty,2}(z + \square_{n+2}; \mathbf{a}_L, \mathbf{a}_0) \mathbb{1}_{\mathcal{G}_{n+2,z}(1,s/8)} \leq C$, which is the first line of (6.31). This gives (6.30). $\square$

The next lemma is a version of Armstrong et al. [2026, Lemma 4.11].

Lemma 6.9 (Excess decay on good scales). *There exists a constant $C(d) < \infty$ such that, for every $s \in [512\bar{\delta}^2, 1/4]$ and $\varepsilon \in [8s^{-1}\bar{\delta}^2, 1]$, every $k \in \mathbb{N}$ and $L, m, n \in \mathbb{N}_0$ with $k \leq n$ and $L \geq m \geq n+5$, every $x, z \in \square_m$ such that $x \in U_{m,n-3}(z)$, and every $u \in H^1(\square_m)$, $\mathbf{g} \in H^s(\square_m; \mathbb{R}^d)$, and $h \in C^{1,1/2}(\square_m)$ satisfying*

$$\begin{cases} -\nabla \cdot \mathbf{a}_L \nabla u = \nabla \cdot \mathbf{g} & \text{in } \square_m, \\ u = h & \text{on } \partial \square_m, \end{cases}$$

we have the estimate

$$\begin{aligned}
& E(u, U_{m,n-k}(x)) \mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)} \\
& \leq C(3^{-1/2k} + 3^{(1+d/2)k} s^{-3/2} \varepsilon) E(u, U_{m,n}(x)) \\
& \quad + C3^{(1+d/2)k} s^{-3/2} \mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}) \\
& \quad \times \left(|\nabla \ell(u, U_{m,n}(x))| + s^{-3/2} |(\nabla h)_{U_{m,n}(x)}| \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}} \right) \\
& \quad + Cs^{-15/2} 3^{(1+d/2)k} (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}^{-1} 3^{sn} [\mathbf{g}]_{\underline{H}^s(U_{m,n}(x))} \\
(6.34) \quad & + Cs^{-3} 3^{(1+d/2)k} 3^{1/2n} [\nabla h]_{W^{1/2, \infty}(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.
\end{aligned}$$

Proof. Fix $x, z \in \square_m$ as in the statement. We may assume $k \geq 6$, since the case $k < 6$ follows by increasing $C(d)$. Choose $y \in \square_m$ so that

$$U_{m,n-4}(x) \subseteq y + \square_{n-2} \subseteq U_{m,n-1}(x).$$

Step 1. We approximate u by a harmonic function. Let v be the harmonic function in $y + \square_{n-2}$ with boundary data u :

$$\begin{cases} -\Delta v = 0 & \text{in } y + \square_{n-2}, \\ v = u & \text{on } \partial(y + \square_{n-2}). \end{cases}$$

Since $\varepsilon \leq 1$, the event $\mathcal{G}_{n+2,z}(\varepsilon, s/8)$ implies $\mathcal{G}_{n+2,z}(1, s/8)$. Lemma 6.8, applied in $y + \square_{n-2}$ on the good event $\mathcal{G}_{n+2,z}(1, s/8)$, therefore yields (6.30) with $\mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)}$ in place of $\mathbb{1}_{\mathcal{G}_{n+2,z}(1, s/8)}$ on the left side.

Step 2. We use the excess decay of harmonic functions. The function v is harmonic in $U_{m,n-4}(x)$. If $\overline{U_{m,n-4}(x)} \cap \partial \square_m \neq \emptyset$, then moreover $v = u = h$ on $\partial(y + \square_{n-2}) \cap \partial \square_m$, with $h \in C^{1,1/2}$. The Schauder estimates for harmonic functions, combined with odd reflection across the flat parts of the boundary, therefore give $v \in C^{1,1/2}(U_{m,n-5}(x))$ with the estimate, where $[\cdot]_{C^{0,1/2}(U)}$ denotes the Hölder seminorm of order $1/2$ on U ,

$$[\nabla v]_{C^{0,1/2}(U_{m,n-5}(x))} \leq C3^{-1/2n} E(v, U_{m,n-4}(x)) + C[\nabla h]_{C^{0,1/2}(U_{m,n-4}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.$$

It follows that, for every $k \geq 6$,

$$E(v, U_{m,n-k}(x)) \leq C3^{-1/2k} E(v, U_{m,n-4}(x)) + C3^{1/2(n-k)} [\nabla h]_{C^{0,1/2}(U_{m,n-2}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.$$

Step 3. We conclude. By the definition of the excess (6.1) and the triangle inequality, for $k \geq 6$,

$$\begin{aligned}
E(u, U_{m,n-k}(x)) & \leq E(v, U_{m,n-k}(x)) + 3^{-(n-k)} \|u - v\|_{\underline{L}^2(U_{m,n-k}(x))} \\
& \leq C3^{-1/2k} E(v, U_{m,n-4}(x)) + 3^{-(n-k)} \|u - v\|_{\underline{L}^2(U_{m,n-k}(x))} \\
& \quad + C3^{1/2(n-k)} [\nabla h]_{C^{0,1/2}(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.
\end{aligned}$$

For the $\|u - v\|$ term, since $U_{m,n-k}(x) \subseteq U_{m,n-4}(x)$ and $|U_{m,n}(x)|/|U_{m,n-k}(x)| \leq C3^{dk}$, we have

$$3^{-(n-k)} \|u - v\|_{\underline{L}^2(U_{m,n-k}(x))} \leq C3^{(d/2+1)k} 3^{-n} \|u - v\|_{\underline{L}^2(U_{m,n-4}(x))}.$$

The excess of v also satisfies $E(v, U_{m,n-4}(x)) \leq CE(u, U_{m,n}(x)) + C3^{-n} \|u - v\|_{\underline{L}^2(U_{m,n-4}(x))}$. Thus

$$\begin{aligned}
E(u, U_{m,n-k}(x)) & \leq C3^{-1/2k} E(u, U_{m,n}(x)) + C3^{(d/2+1)k} 3^{-n} \|u - v\|_{\underline{L}^2(U_{m,n-4}(x))} \\
& \quad + C3^{1/2(n-k)} [\nabla h]_{C^{0,1/2}(U_{m,n}(x))} \mathbb{1}_{\{\overline{U_{m,n}(x)} \cap \partial \square_m \neq \emptyset\}}.
\end{aligned}$$

To estimate the middle term, we use the bound from Step 1 together with the inequality

$$3^{-n} \|u - (u)_{U_{m,n}(x)}\|_{\underline{L}^2(U_{m,n}(x))} \leq E(u, U_{m,n}(x)) + C|\nabla \ell(u, U_{m,n}(x))|.$$

We multiply the resulting inequality by $\mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)}$. By (6.15) and the definition of $\mathcal{G}_{n+2,z}(\varepsilon, s/8)$, we have $\mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L,n+2})_{z+\square_{n+2}}) \mathbb{1}_{\mathcal{G}_{n+2,z}(\varepsilon, s/8)} \leq C\varepsilon$, so the part of the bound on $\|u - v\|$ involving $E(u, U_{m,n}(x))$ contributes $C3^{(1+d/2)k} s^{-3/2} \varepsilon E(u, U_{m,n}(x))$, the second term in the coefficient of $E(u, U_{m,n}(x))$ in (6.34). In the part involving $|\nabla \ell(u, U_{m,n}(x))|$, we keep the factor $\mathcal{E}_{s/8, \infty, 2}(z + \square_{n+2}; \mathbf{a}_L, (\mathbf{b}_{L,n+2})_{z+\square_{n+2}})$ instead of replacing it by ε ; this produces the second and third lines of (6.34). Finally, the averaged seminorm satisfies

$$3^{sn} [\nabla h]_{\underline{H}^s(U_{m,n}(x))} \leq C s^{1/2} 3^{1/2n} [\nabla h]_{W^{1/2, \infty}(U_{m,n}(x))}.$$

The forcing and boundary terms from Step 1 therefore give the last two lines of (6.34). $\square$

The following lemma iterates the excess decay over a range of scales.

Lemma 6.10. *Let $k \in \mathbb{N}$ and $\theta \in (0, 1)$ with $\theta^k \in (0, 3/5)$. Let $n, m \in \mathbb{Z}$ with $n < m$. Let $\{U_j\}_{j \in \mathbb{Z}}$ be bounded measurable sets such that, for every $j \leq m$, we have $U_{j-1} \subseteq U_j$ and U_j contains a translate of $\square_{j-2}$ and is contained in a translate of $\square_j$. Let $u \in L^2(U_m)$. Assume that $\mathcal{B} \subseteq \mathbb{Z} \cap [n, m]$ and that $\{\varepsilon_j\}_{j \in \mathbb{Z} \cap [n, m]}$ and $\{\delta_j\}_{j \in \mathbb{Z} \cap [n, m]}$ are sequences of nonnegative numbers such that*

$$(6.35) \quad E(u, U_{j-k}) \leq \theta^k E(u, U_j) + \varepsilon_j |\nabla \ell(u, U_j)| + \delta_j, \quad \forall j \in \mathbb{Z} \cap [n, m] \setminus \mathcal{B}.$$

Then there exists a constant $C(d) < \infty$ such that, with

$$A := C(k+1)(|\mathcal{B}|+1) + C \sum_{j=n}^m \varepsilon_j,$$

we have

$$(6.36) \quad 3^{-n} \|u - (u)_{U_n}\|_{\underline{L}^2(U_n)} \leq e^A \left(3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)} + \sum_{j=n}^m \delta_j \right).$$

Moreover, for every $M \geq 0$ such that $\varepsilon_j \leq M$ for every $j \in [n, m] \cap \mathbb{Z}$,

$$(6.37) \quad E(u, U_n) \leq \theta^{-C(k+1)(|\mathcal{B}|+1)} e^A \left(\theta^{m-n} E(u, U_m) + M 3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)} + (1+M) \sum_{j=n}^m \delta_j \right).$$

Proof. Write $E_j := E(u, U_j)$ and $\ell_j := \ell(u, U_j)$. The inclusions assumed for U_{j-1} and U_j , together with the definition of ℓ_j , give, for every $j \leq m$,

$$(6.38) \quad |\nabla \ell_j - \nabla \ell_{j-1}| \leq C(E_j + E_{j-1}).$$

Let $[a, b] \cap \mathbb{Z}$ be a maximal integer interval contained in $([n, m] \cap \mathbb{Z}) \setminus \mathcal{B}$, and partition it into residue classes modulo k . Summing (6.35) in each class and using $\theta^k < 3/5$ gives, for $i \in [a, b] \cap \mathbb{Z}$,

$$(6.39) \quad \sum_{j=i}^b E_j \leq C 3^{(3+3d/2)k} E_b + C \sum_{j=i}^b \varepsilon_j |\nabla \ell_j| + C \sum_{j=i}^b \delta_j,$$

$$(6.40) \quad E_i + |\nabla \ell_i| \leq \exp \left(C(k+1) + C \sum_{j=i}^b \varepsilon_j \right) \left(E_b + |\nabla \ell_b| + \sum_{j=i}^b \delta_j \right).$$

Indeed, the first inequality uses only that U_j contains a translate of $\square_{j-2}$ and is contained in a translate of $\square_j$, in order to bound the at most k terms with the largest indices by multiples of E_b . Combining the first inequality with (6.38) gives a recursive bound on $|\nabla \ell_i - \nabla \ell_b|$, from which the discrete Gronwall inequality yields the second inequality.

For every $j \leq m$, inequality (6.38) and the definition of the excess imply

$$E_{j-1} + |\nabla \ell_{j-1}| \leq C(E_j + |\nabla \ell_j|).$$

There are at most $|\mathcal{B}| + 1$ maximal integer intervals contained in $([n, m] \cap \mathbb{Z}) \setminus \mathcal{B}$, and each interval has at most k incomplete residue classes. Iterating (6.40) over these intervals and across the indices in $\mathcal{B}$ yields

$$(6.41) \quad E_n + |\nabla \ell_n| \leq e^A \left(E_m + |\nabla \ell_m| + \sum_{j=n}^m \delta_j \right).$$

The definition of the best affine approximation gives

$$E_m + |\nabla \ell_m| \leq C 3^{-m} \|u - (u)_{U_m}\|_{\underline{L}^2(U_m)} \quad \text{and} \quad 3^{-n} \|u - (u)_{U_n}\|_{\underline{L}^2(U_n)} \leq C(E_n + |\nabla \ell_n|).$$

Together with (6.41), these inequalities prove (6.36).

To prove (6.37), we again sum (6.35) separately over the residue classes modulo k , but now keep track of the powers of θ . For every maximal integer interval $[a, b] \cap \mathbb{Z}$ contained in $([n, m] \cap \mathbb{Z}) \setminus \mathcal{B}$ and every $i \in [a, b] \cap \mathbb{Z}$,

$$E_i \leq e^{C(k+1)} \theta^{-C(k+1)} \left(\theta^{b-i} E_b + M \max_{i \leq j \leq b} |\nabla \ell_j| + \sum_{j=i}^b \delta_j \right).$$

Iterating this inequality over the maximal intervals and the indices in $\mathcal{B}$, and then using (6.41) to bound the gradients, gives (6.37). $\square$

6.3. The large-scale Hölder estimate. In this subsection we prove Proposition 6.1. Lemma 6.10 iterates the excess decay estimate (6.34) of Lemma 6.9 from scale m down to scale n , at the cost of a factor controlled by the number of bad scales between n and m and by the sum of the coarse-graining errors on the good scales. The minimal scale $\mathcal{L}_L(\gamma, m)$ is built from the random variables M and N of Lemmas 6.11 and 6.12. For every $n \leq m - \mathcal{L}_L(\gamma, m)$, both the number of bad scales and the sum of the coarse-graining errors are at most C plus a small multiple of $(1 - \gamma)(m - n)$, and hence the factor is at most $C 3^{1/4(1-\gamma)(m-n)}$.

6.3.1. The minimal scale. The iteration requires bounds for

$$(6.42) \quad \sum_{k=n}^m \mathcal{E}_{1/16, \infty, 2}(\square_k; \mathbf{a}_L, (\mathbf{b}_{L,k})_{\square_k}) \mathbb{1}_{\mathcal{G}_k(\varepsilon, 1/16)}$$

and for the number of bad scales between n and m ,

$$(6.43) \quad \sum_{k=n}^m (1 - \mathbb{1}_{\mathcal{G}_k(\varepsilon, 1/16)}).$$

Propositions 6.3 and 6.4 control (6.42) and (6.43), respectively.

For $k \in \mathbb{N}_0$, $z \in \mathbb{R}^d$, and $s \in (0, 1/2]$, define

$$(6.44) \quad \begin{aligned} \mathbf{G}_{k,z}(s) := & \sup_{0 \leq j \leq k} \sup_{0 \leq l \leq j-2} 3^{-1/2s(k-l)} \max_{z' \in z + (3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1}))} \min \left\{ \max_{|e|=1} J(z' + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & + \sup_{0 \leq j \leq k} 3^{-1/8s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(z + \square_k)} + 3^{-1/8sk} \|\mathbf{g}_0\|_{L^\infty(z + \square_k)} + \sum_{j=k}^{\infty} 3^k \|\nabla \mathbf{g}_j\|_{L^\infty(z + \square_k)}. \end{aligned}$$

Here and below, a supremum over an empty set of integer indices is zero.

Lemma 6.11. *Let $s \in (0, 1/2]$. There exists $C(d) < \infty$ such that, if $s \geq C\bar{\delta}^2|\log \bar{\delta}|$, then, for every $m, n \in \mathbb{N}_0$ with $n \leq m$ and $z \in \mathbb{R}^d$,*

$$(6.45) \quad \sum_{k=n}^m \mathbf{G}_{k,z}(s) \leq Cs^{-7/2}\bar{\delta} + \mathcal{O}_{\Gamma_2}(Cs^{-7/2}\bar{\delta}|\log \bar{\delta}|^{1/2}(m-n+1)^{-1/2}).$$

Moreover, for every $\lambda > 0$ satisfying $\lambda \geq Cs^{-7/2}\bar{\delta}|\log \bar{\delta}|^{1/2}$ and every $m \in \mathbb{N}_0$, there exists a random variable $M(\lambda, s, m) \in \{-1, 0, \dots, m\}$ such that

$$(6.46) \quad 0 \leq m - M(\lambda, s, m) \leq 1 + \mathcal{O}_{\Gamma_1}(Cs^{-7}\bar{\delta}^2|\log \bar{\delta}|\lambda^{-2})$$

and, for every $n \in \mathbb{N}_0$ satisfying $n \leq M(\lambda, s, m)$, we have

$$(6.47) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s) \leq \lambda(m-n).$$

Proof. Step 1. We estimate the terms of $\mathbf{G}_{k,z}(s)$ involving the fields $\mathbf{g}_j$. By stationarity it suffices to prove (6.45) for $z = 0$. The triangle inequality gives

$$\begin{aligned} & \sum_{k=n}^m \left(\sup_{0 \leq j \leq k} 3^{-1/8s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(\square_k)} + 3^{-1/8sk} \|\mathbf{g}_0\|_{L^\infty(\square_k)} \right) \\ & \leq 2 \sum_{k=n}^m \sup_{0 \leq j \leq k} 3^{-1/8s(k-j)} \sum_{i=j}^k \|\mathbf{g}_i\|_{L^\infty(\square_k)} \\ & \leq 2 \sum_{k=n}^m \sum_{i=0}^k \sup_{0 \leq j \leq i} 3^{-1/8s(k-j)} \|\mathbf{g}_i\|_{L^\infty(\square_k)} \\ & = 2 \sum_{k=n}^m \sum_{i=0}^k 3^{-1/8s(k-i)} \|\mathbf{g}_i\|_{L^\infty(\square_k)} \\ & = 2 \left(\sum_{i=n}^m \sum_{k=i}^m + \sum_{k=n}^m \sum_{i=0}^{n-1} \right) 3^{-1/8s(k-i)} \|\mathbf{g}_i\|_{L^\infty(\square_k)}. \end{aligned}$$

By (1.4) and (1.5), together with (A.4), we obtain

$$\|\mathbf{g}_i\|_{L^\infty(\square_k)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(k-i+1)^{1/2})$$

and therefore, by independence, also

$$\begin{aligned} \sum_{k=n}^m \left(\sup_{0 \leq j \leq k} 3^{-1/8s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(\square_k)} + 3^{-1/8sk} \|\mathbf{g}_0\|_{L^\infty(\square_k)} \right) & \leq Cs^{-3/2}\bar{\delta}(m-n+1) \\ & + \mathcal{O}_{\Gamma_2}(Cs^{-3/2}\bar{\delta}(m-n+1)^{1/2}). \end{aligned}$$

For the gradient term,

$$\begin{aligned} \sum_{k=n}^m \sum_{l=k}^{\infty} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} & = \sum_{l=n}^{\infty} \sum_{k=n}^{m \wedge l} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} \\ & \leq \frac{3}{2} \sum_{l=n}^{\infty} 3^{m \wedge l} \|\nabla \mathbf{g}_l\|_{L^\infty(\square_l)} \\ & = \frac{3}{2} \sum_{l=n}^{m-1} 3^l \|\nabla \mathbf{g}_l\|_{L^\infty(\square_l)} + \frac{3}{2} \sum_{l=m}^{\infty} 3^m \|\nabla \mathbf{g}_l\|_{L^\infty(\square_l)}. \end{aligned}$$

Again by (1.4), (1.5), and independence,

$$\sum_{k=n}^m \sum_{l=k}^{\infty} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} \leq C\bar{\delta}(m-n+1) + \mathcal{O}_{\Gamma_2}(C\bar{\delta}(m-n+1)^{1/2}).$$

Thus

$$(6.48) \quad \sum_{k=n}^m \sum_{l=k}^{\infty} 3^k \|\nabla \mathbf{g}_l\|_{L^\infty(\square_k)} + \sum_{k=n}^m \left(\sup_{0 \leq j \leq k} 3^{-1/8s(k-j)} \|\mathbf{g}_{k,j}\|_{L^\infty(\square_k)} + 3^{-1/8sk} \|\mathbf{g}_0\|_{L^\infty(\square_k)} \right) \leq Cs^{-3/2}\bar{\delta}(m-n+1) + \mathcal{O}_{\Gamma_2}(Cs^{-3/2}\bar{\delta}(m-n+1)^{1/2}).$$

Step 2. We estimate the term of $\mathbf{G}_{k,z}(s)$ involving J . Bounding the supremum over j by a sum, interchanging the sums, and summing the geometric series in k , we obtain

$$(6.49) \quad \begin{aligned} & \sum_{k=n}^m \sup_{0 \leq j \leq k} \sup_{0 \leq l \leq j-2} 3^{-1/2s(k-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & \leq \sum_{j=0}^m \sum_{k=n \vee j}^m \sup_{0 \leq l \leq j-2} 3^{-1/4s(k-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & \leq 4s^{-1} \sum_{j=n}^m \sup_{0 \leq l \leq j-2} 3^{-1/4s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & \quad + 4s^{-1} \sum_{j=0}^{n-1} \sup_{0 \leq l \leq j-2} 3^{-1/4s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2}. \end{aligned}$$

By Proposition 4.1, for $\xi \in [1, C^{-1}\bar{\delta}^{-2}|\log \bar{\delta}|^{-1}]$,

$$\mathbb{E} \left[\max_{|e|=1} (J(\square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l))^{\xi/2} \right]^{1/\xi} \leq C\bar{\delta} \log(2 + \xi)^{1/2} \xi^{1/2}.$$

Hence, for every $\xi \in [1, \infty)$,

$$\mathbb{E} \left[\min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{\xi/2} \right]^{1/\xi} \leq C\bar{\delta} |\log \bar{\delta}|^{1/2} \xi^{1/2},$$

which yields

$$\min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta} |\log \bar{\delta}|^{1/2}).$$

Moreover, by Hölder's inequality, the condition $s \geq C\bar{\delta}^2 |\log \bar{\delta}|$, a union bound, and stationarity,

$$\begin{aligned} & \mathbb{E} \left[\sup_{0 \leq l \leq j-2} 3^{-1/4s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \right] \\ & \leq \mathbb{E} \left[\sup_{0 \leq l \leq j-2} 3^{-2d(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{4d/s} \right]^{s/8d} \\ & \leq \left(\sum_{l=0}^{j-2} 3^{-d(j-l)} \mathbb{E} \left[\max_{|e|=1} J(\square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l)^{4d/s} \right] \right)^{s/8d} \\ & \leq Cs^{-1/2} |\log s|^{1/2} \bar{\delta}, \end{aligned}$$

and, using (A.4),

$$\sup_{0 \leq l \leq j-2} 3^{-1/4s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min \left\{ \max_{|e|=1} J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \leq \mathcal{O}_{\Gamma_2}(Cs^{-3/2}\bar{\delta} |\log \bar{\delta}|^{1/2}).$$

Step 3. We prove (6.45). We split the indices j into the finitely many residue classes modulo the integer $r(d)$ from the proof of Lemma 6.7; as shown there, the summands with indices in the same class are independent. In each class, we apply (A.5) to the centered summands, which are $\mathcal{O}_{\Gamma_2}(Cs^{-3/2}\bar{\delta}|\log \bar{\delta}|^{1/2})$ by the last display, and combining the result with the preceding estimates we obtain

$$\begin{aligned} & \sum_{j=n}^m \sup_{0 \leq l \leq j-2} 3^{-1/4s(j-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min_{|e|=1} \left\{ \max J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & \leq Cs^{-1/2} |\log s|^{1/2} \bar{\delta} (m-n+1) + \mathcal{O}_{\Gamma_2}(Cs^{-3/2}\bar{\delta}|\log \bar{\delta}|^{1/2} (m-n+1)^{1/2}). \end{aligned}$$

Applying the last display before Step 3 to the annuli with $j < n$ and summing the resulting geometric series gives

$$\begin{aligned} & \sum_{j=0}^{n-1} \sup_{0 \leq l \leq j-2} 3^{-1/4s(n-l)} \max_{z \in 3^l \mathbb{Z}^d \cap (\square_j \setminus \square_{j-1})} \min_{|e|=1} \left\{ \max J(z + \square_l, e, e; \bar{\mathbf{a}}_l^{-1} \mathbf{a}_l), 1 \right\}^{1/2} \\ & \leq \mathcal{O}_{\Gamma_2}(Cs^{-5/2}\bar{\delta}|\log \bar{\delta}|^{1/2}). \end{aligned}$$

The last two displays, together with (6.49) and (6.48), give (6.45).

Step 4. We define $M(\lambda, s, m)$ and prove (6.46). Define

$$M(\lambda, s, m) := \min \left(\left\{ n \in \mathbb{N}_0 : n \leq m-1, \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s) > \lambda(m-n) \right\} \cup \{m\} \right) - 1.$$

The averaged estimate (6.45), stationarity, and the sub-Gaussian tail in that estimate show that, for every $0 \leq k \leq m-1$,

$$\mathbb{P} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \sum_{j=k}^m \mathbf{G}_{j,z}(s) > \lambda(m-k) \right] \leq \exp \left(C(m-k) - \frac{s^7 \lambda^2 (m-k)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right).$$

Indeed, there are at most $C3^{d(m-k)}$ centers, and we apply (6.45), translated, at each of them. The same bound is valid for $k = m$ because its right side is one. Consequently, a union bound gives, for every $\lambda \geq 2Cs^{-7/2}\bar{\delta}|\log \bar{\delta}|^{1/2}$ and $0 \leq n \leq m-1$,

$$\begin{aligned} (6.50) \quad \mathbb{P}[M(\lambda, s, m) \leq n] & \leq \sum_{k=0}^{n+1} \mathbb{P} \left[\max_{z \in 3^k \mathbb{Z}^d \cap \square_m} \sum_{j=k}^m \mathbf{G}_{j,z}(s) > \lambda(m-k) \right] \\ & \leq \sum_{k=0}^{n+1} \exp \left(C(m-k) - \frac{s^7 \lambda^2 (m-k)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right) \leq C \exp \left(- \frac{s^7 \lambda^2 (m-n-1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right). \end{aligned}$$

This implies (6.46) and completes the proof. $\square$

Lemma 6.12. *There exists $C(d) < \infty$ such that, for every $\lambda, \varepsilon, s \in (0, 1]$ satisfying*

$$(6.51) \quad \lambda \geq Cs^{-8}\varepsilon^{-2}\bar{\delta}^2|\log \bar{\delta}|$$

and for every $m \in \mathbb{N}_0$, there exists a random variable $N(\lambda, \varepsilon, s, m) \in \{-1, 0, \dots, m\}$ such that

$$(6.52) \quad 0 \leq m - N(\lambda, \varepsilon, s, m) \leq 1 + \mathcal{O}_{\Gamma_1}(Cs^{-8}\varepsilon^{-2}\lambda^{-1}\bar{\delta}^2|\log \bar{\delta}|)$$

and, for every $n \in \mathbb{N}_0$ with $n \leq N(\lambda, \varepsilon, s, m)$,

$$(6.53) \quad \min_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s)\} > (1-\lambda)(m-n).$$

Equivalently, every such n satisfies

$$(6.54) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m (1 - \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s)\}) < 1 + \lambda(m - n).$$

Proof. Define the random variable $N(\lambda, \varepsilon, s, m)$ by

$$\begin{aligned} N(\lambda, \varepsilon, s, m) &:= \min \left(\left\{ n \in \mathbb{N}_0 : n \leq m, \min_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s)\} \right. \right. \\ &\quad \left. \left. \leq (1 - \lambda)(m - n) \right\} \cup \{m + 1\} \right) - 1. \end{aligned}$$

The definition of $N(\lambda, \varepsilon, s, m)$ gives (6.53). On the event in the next display, the proportion of bad scales among the $m - n + 1$ scales from n to m is at least $\lambda/2$. Proposition 6.4, applied with $\lambda/2$ in place of θ , therefore gives

$$(6.55) \quad \mathbb{P} \left[\sum_{j=n}^m (1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)}) > \lambda(m - n) \right] \leq \exp \left(- \frac{s^8 \varepsilon^2 \lambda(m - n + 1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right).$$

By a union bound, stationarity, and the condition (6.51) with sufficiently large C , we obtain

$$\begin{aligned} (6.56) \quad \mathbb{P} \left[\max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m (1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)}) > \lambda(m - n) \right] &\leq 3^{d(m-n)} \mathbb{P} \left[\sum_{j=n}^m (1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)}) > \lambda(m - n) \right] \\ &\leq \exp \left(2d(m - n) - \frac{s^8 \varepsilon^2 \lambda(m - n + 1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right) \\ &\leq \exp \left(- \frac{s^8 \varepsilon^2 \lambda(m - n + 1)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right). \end{aligned}$$

By another union bound, for $0 \leq n \leq m - 1$,

$$(6.57) \quad \mathbb{P}[N(\lambda, \varepsilon, s, m) \leq n] \leq \sum_{k=0}^{n+1} \mathbb{P} \left[\min_{z \in 3^k \mathbb{Z}^d \cap \square_m} \sum_{j=k}^m \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)} \leq (1 - \lambda)(m - k) \right] \leq C \exp \left(- \frac{s^8 \varepsilon^2 \lambda(m - n)}{C \bar{\delta}^2 |\log \bar{\delta}|} \right).$$

This implies (6.52), completing the proof. $\square$

Proof of Proposition 6.1. Steps 1–2. We fix the parameters and the minimal scale. Fix γ in the range allowed by the statement. Choose the dimensional constants in the order indicated below and set

$$\begin{aligned} (6.58) \quad s_0 &:= 1/32, \\ \lambda &:= C_1^{-1}(1 - \gamma), \\ (6.59) \quad \varepsilon &:= C_2^{-1}(1 - \gamma)^{1/2}, \\ k &:= \left\lceil 4 \log_3 \left(4s_0^{-3/2} C_3 \right) \right\rceil, \end{aligned}$$

where $C_3(d)$ is the constant of Lemma 6.9. Choose $C_1(d) \geq 2$ sufficiently large for both (6.78) below and the absorption in the proof of (6.6) in Step 6, and then choose C_2 so that $C_1 C_2^{-8} \leq 1$ and

$$C_3 \left(3^{-k/2} + 3^{(1+d/2)k} s_0^{-3/2} C_2^{-1} \right) \leq \frac{1}{2} 3^{-k/4}.$$

Finally, choose the constant C of the statement large enough, depending on C_1 and C_2 . The lower bound on $1 - \gamma$ then ensures, for all sufficiently small $\bar{\delta}$, that the hypotheses of Lemma 6.9, Proposition 6.3, Lemma 6.11, and Lemma 6.12 hold.

For $m \in \mathbb{N}_0$, define

$$(6.60) \quad \mathcal{L}_L(\gamma, m) := \max\{m - M(\lambda, s_0, m), m - N(\lambda, \varepsilon, s_0, m)\} + k + 5,$$

where M and N are supplied by Lemmas 6.11 and 6.12. They do not involve L , so $\mathcal{L}_L(\gamma, m)$ is the same random variable for every $L \geq m$. The summand $k + 5$ ensures that every scale used in Lemma 6.9 is at most $m - 5$. Estimates (6.46) and (6.52) yield

$$(6.61) \quad \begin{aligned} \mathcal{L}_L(\gamma, m) &\leq C + \mathcal{O}_{\Gamma_1}(Cs_0^{-7}\bar{\delta}^2|\log \bar{\delta}|\lambda^{-2} + Cs_0^{-8}\varepsilon^{-2}\lambda^{-1}\bar{\delta}^2|\log \bar{\delta}|) \\ &\leq C + \mathcal{O}_{\Gamma_1}(CC_1^2(1 - \gamma)^{-2}\bar{\delta}^2|\log \bar{\delta}|). \end{aligned}$$

This is (6.3). Finally, (6.47) and (6.53) imply, whenever $n \leq m - \mathcal{L}_L(\gamma, m)$,

$$(6.62) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s_0) \leq \lambda(m - n) \quad \text{and} \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m (1 - \mathbb{1}\{\mathcal{G}_{j,z}(\varepsilon, s_0)\}) < 1 + \lambda(m - n).$$

Step 3. We prove the excess decay estimate (6.34) on good scales. Fix $m \in \mathbb{N}_0$ with $L \geq m$, and let $u \in H^1(\square_m)$ be a solution of (6.4). Fix $n \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $z \in 3^n \mathbb{Z}^d \cap \square_m$. We apply Lemma 6.9 with $s = 8s_0 = 1/4$ at the scales $j \in \mathbb{N}_0 \cap [n + k, m - 5]$, for which its hypotheses hold. For each such j , the conclusion (6.34), with j in place of n and with $x = z$, gives

$$(6.63) \quad \begin{aligned} E(u, U_{m,j-k}(z)) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} &\leq C_3(3^{-1/2k} + 3^{(1+d/2)k} s_0^{-3/2} \varepsilon) E(u, U_{m,j}(z)) \\ &\quad + C_3 3^{(1+d/2)k} s_0^{-3/2} \mathcal{E}_{s_0, 0, 2}(z + \square_{j+2}; \mathbf{a}_L, (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}) \\ &\quad \times \left(|\nabla \ell(u, U_{m,j}(z))| + s_0^{-3/2} |(\nabla h)_{U_{m,j}(z)}| \mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}} \right) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \\ &\quad + C s_0^{-15/2} 3^{(1+d/2)k} (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}^{-1} 3^{8s_0 j} [\mathbf{g}]_{H^{8s_0}(U_{m,j}(z))} \\ &\quad + C s_0^{-3} 3^{(1+d/2)k} 3^{1/2j} [\nabla h]_{W^{1/2,\infty}(U_{m,j}(z))} \mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

By our choice of k and C_2 , we have $C_3(3^{-k/2} + 3^{(1+d/2)k} s_0^{-3/2} \varepsilon) \leq 1/23^{-k/4}$. Definition 4.3 and the fact that an average lies between the infimum and the supremum give

$$\begin{aligned} &\max \left\{ \frac{(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}}{(\mathbf{b}_{L,m})_{\square_m}}, \frac{(\mathbf{b}_{L,m})_{\square_m}}{(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}} \right\} \\ &\leq \max \left\{ \frac{\bar{\mathbf{a}}_{j+2}}{\bar{\mathbf{a}}_m}, \frac{\bar{\mathbf{a}}_m}{\bar{\mathbf{a}}_{j+2}} \right\} \exp \left(\sum_{i=j+3}^m \|\mathbf{g}_i - \tau^2\|_{L^\infty(z+\square_{j+2})} + \sup_{\square_m \cup (z+\square_{j+2})}^{\text{osc}} \sum_{i=m+1}^L \mathbf{g}_i \right). \end{aligned}$$

Equations (6.12) and (6.44) give

$$\sum_{i=j+3}^m \|\mathbf{g}_i - \tau^2\|_{L^\infty(z+\square_{j+2})} \leq \tau^2(m - j - 2) + C \sum_{i=j+2}^m \mathbf{G}_{i,z}(s_0).$$

By the mean-value theorem and (6.44),

$$\sup_{\square_m \cup (z+\square_{j+2})}^{\text{osc}} \sum_{i=m+1}^L \mathbf{g}_i \leq C \sum_{i=m+1}^\infty 3^m (\|\nabla \mathbf{g}_i\|_{L^\infty(z+\square_m)} + \|\nabla \mathbf{g}_i\|_{L^\infty(\square_m)}) \leq C(\mathbf{G}_{m,z}(s_0) + \mathbf{G}_{m,0}(s_0)).$$

Applying (6.62) at the centers z and 0 , we obtain

$$\sum_{i=j+2}^m \mathbf{G}_{i,z}(s_0) + \mathbf{G}_{m,0}(s_0) \leq C\lambda(m - n).$$

The parameter choice in Steps 1–2 gives $\tau^2 \leq C\bar{\delta}^2 \leq C\lambda$. Together with (3.32), the preceding estimates show that, for every $j \in \mathbb{N}_0 \cap [n, m-5]$,

$$(6.64) \quad C^{-1} \exp(-C\lambda(m-n)) \leq (\mathbf{b}_{L,j+2})_{z+\square_{j+2}} (\mathbf{b}_{L,m})_{\square_m}^{-1} \leq C \exp(C\lambda(m-n)).$$

Moreover,

$$3^{8s_0 j} [\mathbf{g}]_{\underline{H}^{8s_0}(U_{m,j}(z))} \leq C 3^{1/2 j} [\mathbf{g}]_{W^{1/2,\infty}(U_{m,j}(z))} \leq C 3^{-1/2(m-j)} (3^{1/2 m} [\mathbf{g}]_{W^{1/2,\infty}(\square_m)}) ,$$

so, using (6.64),

$$(6.65) \quad (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}^{-1} 3^{8s_0 j} [\mathbf{g}]_{\underline{H}^{8s_0}(U_{m,j}(z))} \leq C \exp(C\lambda(m-n)) 3^{-1/2(m-j)} ((\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{1/2 m} [\mathbf{g}]_{W^{1/2,\infty}(\square_m)}).$$

The boundary datum satisfies

$$(6.66) \quad 3^{1/2 j} [\nabla h]_{W^{1/2,\infty}(U_{m,j}(z))} \leq C 3^{-1/2(m-j)} (3^{1/2 m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)}).$$

Furthermore, we use the following elementary inequality: if $D \geq 0$ and $0 \leq x \leq \varepsilon 3^{s_0 D/16}$, then

$$3^{-s_0 D/2} x \leq 3^{-s_0 D/2} \min\{x, 1\} + \varepsilon^8.$$

Indeed, the second term is needed only when $x > 1$, in which case $3^{-s_0 D/16} < \varepsilon$. By (6.11), on $\mathcal{G}_{j+2,z}(\varepsilon, s_0)$ it applies to the square root x of every quantity J on the right side of (6.15), with D the difference between the scale $j+2$ and the scale of that quantity. Proposition 6.3 and (6.44) therefore give

$$(6.67) \quad \mathcal{E}_{s_0, \infty, 2}(z + \square_{j+2}; \mathbf{a}_L, (\mathbf{b}_{L,j+2})_{z+\square_{j+2}}) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \leq C \min\{\varepsilon, s_0^{-1} \bar{\delta}^2 + \varepsilon^8 + \mathbf{G}_{j+2,z}(s_0)\}.$$

Inserting these bounds into (6.63) yields, for every $z \in 3^n \mathbb{Z}^d \cap \square_m$ and $j \in \mathbb{N}_0 \cap [n+k, m-5]$,

$$(6.68) \quad E(u, U_{m,j-k}(z)) \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)} \leq 3^{-k/4} E(u, U_{m,j}(z)) + \varepsilon_{j,z} |\nabla \ell(u, U_{m,j}(z))| + \delta_{j,z},$$

where we define

$$(6.69) \quad \varepsilon_{j,z} := C s_0^{-3/2} \min\{\varepsilon, \bar{\delta}^2 + \varepsilon^8 + \mathbf{G}_{j+2,z}(s_0)\} \mathbb{1}_{\mathcal{G}_{j+2,z}(\varepsilon, s_0)}$$

and

$$(6.70) \quad \begin{aligned} \delta_{j,z} &:= C 3^{-1/2(m-j)} \exp(C\lambda(m-n)) (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{1/2 m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ &+ C \left(\varepsilon_{j,z} \|\nabla h\|_{L^\infty(\square_m)} + 3^{-1/2(m-j)} 3^{1/2 m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \right) \mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

The geometric factor $3^{-1/2(m-j)}$ in the last term comes from (6.66). Since the sets $U_{m,j}(z)$ are nested, for $j \leq m'$,

$$\mathbb{1}_{\{\overline{U_{m,j}(z)} \cap \partial \square_m \neq \emptyset\}} \leq \mathbb{1}_{\{\overline{U_{m,m'}(z)} \cap \partial \square_m \neq \emptyset\}}.$$

By (6.62), we have

$$(6.71) \quad \sum_{j=n}^{m-2} \varepsilon_{j,z} \leq C s_0^{-3/2} \left((\bar{\delta}^2 + \varepsilon^8)(m-n) + \sum_{j=n}^{m-2} \mathbf{G}_{j+2,z}(s_0) \right) \leq C\lambda(m-n).$$

Therefore, we also obtain, for every $m', n' \in \mathbb{N}_0$ with $n \leq n' \leq m' \leq m-3$,

$$(6.72) \quad \begin{aligned} \sum_{j=n'}^{m'} \delta_{j,z} &\leq C \exp(C\lambda(m-n)) (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{1/2 m'} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ &+ C \left(\lambda(m-n) \|\nabla h\|_{L^\infty(\square_m)} + 3^{1/2 m'} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \right) \mathbb{1}_{\{\overline{U_{m,m'}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

Step 4. We set up the iteration. The inequality (6.68) is the key hypothesis (6.35) of Lemma 6.10 (with step k and $\theta = 3^{-1/4}$). For $z \in 3^n \mathbb{Z}^d \cap \square_m$, we define the set $\mathcal{B}$ of indices by

$$(6.73) \quad \mathcal{B} := \{j \in \mathbb{N}_0 \cap [n, m-2] : j < n+k \text{ or } \mathcal{G}_{j+2,z}(\varepsilon, s_0) \text{ fails}\}.$$

By (6.62) and the definition (6.73),

$$(6.74) \quad |\mathcal{B}| \leq k + \sum_{j=n+k}^{m-2} (1 - \mathbb{1}\{\mathcal{G}_{j+2,z}(\varepsilon, s_0)\}) \leq k + 1 + \lambda(m-n).$$

This and (6.71) imply, for the constant A in Lemma 6.10, that

$$e^A \leq C \exp(C\lambda(m-n)).$$

Since k is a dimensional constant, (6.74) shows that $3^{C(k+1)(|\mathcal{B}|+1)/4} e^A$ satisfies the same bound, with a larger C . We apply (6.37) with $M = C\varepsilon$, which is admissible because $\varepsilon_{j,z} \leq C\varepsilon$ by (6.69). For every $n', m' \in \mathbb{N}_0$ with $n \leq n' \leq m' \leq m-5$, Lemma 6.10 gives the following two inequalities (for $n' = m'$ they are immediate). The first inequality, from (6.36), is

$$(6.75) \quad 3^{-n'} \|u - (u)_{U_{m,n'}(z)}\|_{\underline{L}^2(U_{m,n'}(z))} \leq C \exp(C\lambda(m-n)) \left(3^{-m'} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} + \sum_{j=n'}^{m'} \delta_{j,z} \right)$$

and the second inequality, the excess decay estimate, is

$$(6.76) \quad E(u, U_{m,n'}(z)) \leq C \exp(C\lambda(m-n)) \left(3^{-m'-n'/4} E(u, U_{m,m'}(z)) + \varepsilon 3^{-m'} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} + \sum_{j=n'}^{m'} \delta_{j,z} \right).$$

Step 5. We prove the large-scale Hölder estimate (6.77). We show that, for every $n, n', m' \in \mathbb{N}_0$ with $n \leq m - \mathcal{L}_L(\gamma, m)$ and $n \leq n' \leq m' \leq m-5$ and every $z \in 3^n \mathbb{Z}^d \cap \square_m$, we have the estimate

$$(6.77) \quad \begin{aligned} & 3^{-n'} \|u - (u)_{U_{m,n'}(z)}\|_{\underline{L}^2(U_{m,n'}(z))} \\ & \leq C 3^{1/2(1-\gamma)(m-n)} \left(3^{-m'} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} + (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{1/2m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right) \\ & + C 3^{1/2(1-\gamma)(m-n)} 3^{1/2m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{\overline{U_{m,m'}(z)} \cap \partial \square_m \neq \emptyset\}}. \end{aligned}$$

By the choice of C_1 and the identity $\lambda = C_1^{-1}(1-\gamma)$,

$$(6.78) \quad (1 + \lambda(m-n)) \exp(C\lambda(m-n)) \leq C \exp(2CC_1^{-1}(1-\gamma)(m-n)) \leq C 3^{1/4(1-\gamma)(m-n)}.$$

Combining (6.75), (6.72), and (6.78) gives (6.77). For each $j \in \mathbb{N}_0$ with $j \leq n$ and each $y \in 3^j \mathbb{Z}^d \cap ((x + \square_n) \cap \square_m)$, we apply this estimate with j in place of n , with $n' = j$, $m' = m-5$, and $z = y$. If $x \in \square_{m-1}$, then $y \in x + \square_n$ and $n \leq m - \mathcal{L}_L(\gamma, m) \leq m - k - 5$, so $\overline{U_{m,m-5}(y)} \subseteq \square_m$ and the boundary indicator in (6.77) vanishes; hence this indicator is always bounded by $\mathbb{1}_{\{x \notin \square_{m-1}\}}$. Together with

$$3^{-1/2(1+\gamma)(m-j)} \leq 3^{-\gamma(m-j)},$$

this proves (6.5).

Step 6. We prove the energy estimate (6.6). We use the estimates of Lemma 2.9 and the Caccioppoli inequality of Lemma 2.10 to convert the large-scale Hölder estimate (6.77) into a gradient bound; this is possible only at good scales, where we control the coarse-grained ellipticities.

Fix $x \in \square_m$. If $m-n \leq 2k+14$, then $|(x + \square_n) \cap \square_m| \geq c3^{nd} \geq c|\square_m|$, and (6.6) follows directly from the energy in $\square_m$. We may therefore assume that $m-n > 2k+14$. Then, since $\lambda \leq 1/2$, the interval $[n+2, m-5]$ contains more than $k+1 + \lambda(m-n)$ integers, so by (6.74) it contains a scale outside the set $\mathcal{B}$ of Step 4, for the center z chosen below. Choose $y \in \square_m$ such that

$$(x + \square_{n+1}) \cap \square_m \subseteq y + \square_{n+1} \subseteq \square_m \quad \text{and} \quad (x + \square_n) \cap \partial(y + \square_{n+1}) \setminus \partial \square_m = \emptyset.$$

When $x \in \square_{m-1}$, take $y = x$. Let $z \in 3^n \mathbb{Z}^d \cap \square_m$ be a nearest neighbor of y in $3^n \mathbb{Z}^d \cap \square_m$. Then $y \in z + \square_{n+1}$ and $y + \square_{n+1} \subseteq U_{m,n+2}(z)$. We define the elements of $[n+2, m-5] \setminus \mathcal{B}$ nearest to n and to m by

$$n' := \min\{j \in \mathbb{N}_0 : j \in [n+2, m-5] \setminus \mathcal{B}\} \quad \text{and} \quad m' := \max\{j \in \mathbb{N}_0 : j \in [n', m-5] \setminus \mathcal{B}\},$$

respectively. Since $|\mathcal{B}| \leq k+1 + \lambda(m-n)$ by (6.74), we have $\max\{n'-n, m-m'\} \leq C + \lambda(m-n)$. As in Step 5, the boundary indicator in (6.77) vanishes if $x \in \square_{m-1}$, so it is bounded by $\mathbb{1}_{\{x \notin \square_{m-1}\}}$. We also fix $y' \in \square_m \cap (z + \square_{m'+1})$ such that $U_{m,m'}(z) \subseteq y' + \square_{m'} \subseteq \square_m$. By (2.52) and the last assertion of Proposition 6.3, for every $j \in [n, m-5] \setminus \mathcal{B}$ we have

$$(6.79) \quad C^{-1}(\mathbf{b}_{L,j+2})_{z+\square_{j+2}} \leq \lambda_{s_0,2}(z + \square_{j+2}; \mathbf{a}_L) \leq \Lambda_{s_0,2}(z + \square_{j+2}; \mathbf{a}_L) \leq C(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}.$$

Thus, by Lemma 2.4 and (2.59), together with (6.64), we have, for $q \in \{1, 2\}$,

$$\begin{aligned} \lambda_{1/8,q}^{-1}(y + \square_{n+1}; \mathbf{a}_L) &\leq C\lambda_{s_0,2}^{-1}(z + \square_{n+2}; \mathbf{a}_L) \leq C3^{2s_0(n'-n)}\lambda_{s_0,2}^{-1}(z + \square_{n'+2}; \mathbf{a}_L) \\ &\leq C \exp(C\lambda(m-n))(\mathbf{b}_{L,m})_{\square_m}^{-1}. \end{aligned}$$

The remaining ellipticity quantities satisfy

$$(6.80) \quad \begin{aligned} \Lambda_{1/4,q}(y + \square_{n+1}; \mathbf{a}_L)(\mathbf{b}_{L,m})_{\square_m}^{-1} + \Lambda_{1/4,q}(y' + \square_{m'}; \mathbf{a}_L)(\mathbf{b}_{L,m})_{\square_m}^{-1} + \lambda_{1/8,q}^{-1}(y' + \square_{m'}; \mathbf{a}_L)(\mathbf{b}_{L,m})_{\square_m} \\ \leq C \exp(C\lambda(m-n)). \end{aligned}$$

Moreover,

$$\left(\frac{|U_{m,m}(z)|}{|U_{m,m'}(z)|} \right)^{1/2} \leq C3^{d(m-m')/2} \leq C \exp(C\lambda(m-n)).$$

We now apply the coarse-grained Caccioppoli inequality of Proposition 2.6 with $s = 1/4$ and $t = 1/8$, or Lemma 2.10 near the boundary, together with the scaled embedding

$$[f]_{\underline{H}^{1/4}(Q)} \leq C|Q|^{1/4d}[f]_{W^{1/2,\infty}(Q)}$$

and obtain

$$\begin{aligned} \exp(-C\lambda(m-n)) \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2((x+\square_n) \cap \square_m)} \\ \leq C(\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{-n} \|u - (u)_{U_{m,n+2}(z)}\|_{\underline{L}^2(U_{m,n+2}(z))} + C(\mathbf{b}_{L,m})_{\square_m}^{-1/2} 3^{1/4n} [\mathbf{g}]_{\underline{H}^{1/4}(U_{m,n+2}(z))} \\ + C(\mathbf{b}_{L,m})_{\square_m}^{1/2} \left(|(\nabla h)_{U_{m,n+2}(z)}| + 3^{1/4n} [\nabla h]_{\underline{H}^{1/4}(U_{m,n+2}(z))} \right) \mathbb{1}_{\{x \notin \square_{m-1}\}} \\ \leq C(\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{-n} \|u - (u)_{U_{m,n+2}(z)}\|_{\underline{L}^2(U_{m,n+2}(z))} + C(\mathbf{b}_{L,m})_{\square_m}^{-1/2} 3^{1/2m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \\ + C(\mathbf{b}_{L,m})_{\square_m}^{1/2} 3^{1/2m} \|\nabla h\|_{\underline{W}^{1/2,\infty}(\square_m)} \mathbb{1}_{\{x \notin \square_{m-1}\}}. \end{aligned}$$

Furthermore, (2.63) and the fractional Poincaré inequality give

$$3^{-m} \|u - (u)_{U_{m,m'}(z)}\|_{\underline{L}^2(U_{m,m'}(z))} \leq C3^{-m} \|u - (u)_{y'+\square_{m'}}\|_{\underline{L}^2(y'+\square_{m'})} \leq C3^{-1/4m} \|\nabla u\|_{\dot{\underline{B}}_{2,2}^{-1/4}(y'+\square_{m'})}.$$

We apply Lemma 2.9 and (6.80) to obtain

$$\begin{aligned} 3^{-1/4m} \|\nabla u\|_{\dot{\underline{B}}_{2,2}^{-1/4}(y'+\square_{m'})} \\ \leq C\lambda_{1/8,2}^{-1/2}(y' + \square_{m'}; \mathbf{a}_L) \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(y'+\square_{m'})} + C\lambda_{1/8,2}^{-1}(y' + \square_{m'}; \mathbf{a}_L) 3^{1/4m} [\mathbf{g}]_{\underline{H}^{1/4}(y'+\square_{m'})} \\ \leq C \exp(C\lambda(m-n)) \left((\mathbf{b}_{L,m})_{\square_m}^{-1/2} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(\square_m)} + (\mathbf{b}_{L,m})_{\square_m}^{-1} 3^{1/2m} [\mathbf{g}]_{\underline{W}^{1/2,\infty}(\square_m)} \right). \end{aligned}$$

Combining the previous four displays with (6.77) and (6.78), and using the choice of C_1 , proves (6.6).

Step 7. We prove the excess decay estimate (6.7). Fix m, n, l satisfying $\mathcal{L}_L(\gamma, m) \leq m - n \leq (1 - \gamma)^{-1}$ and $n \leq l \leq m - 5$, and fix $x \in \square_m$. We may assume $l > n + 2$, since otherwise (6.7) follows, after enlarging C , by comparing the volumes of the sets involved. Choose $z \in 3^n \mathbb{Z}^d \cap \square_m$ such that

$$(x + \square_n) \cap \square_m \subseteq U_{m,n+1}(z) \subseteq U_{m,l-1}(z) \subseteq (x + \square_l) \cap \square_m.$$

As in Step 5, the boundary indicator in (6.72) vanishes when $x \in \square_{m-1}$, so it is bounded by $\mathbb{1}_{\{x \notin \square_{m-1}\}}$. Using

$$E(u, (x + \square_n) \cap \square_m) \leq CE(u, U_{m,n+1}(z)) \quad \text{and} \quad E(u, U_{m,l-1}(z)) \leq CE(u, (x + \square_l) \cap \square_m)$$

and

$$\|u - (u)_{U_{m,l-1}(z)}\|_{L^2(U_{m,l-1}(z))} \leq C \|u - (u)_{(x+\square_l) \cap \square_m}\|_{L^2((x+\square_l) \cap \square_m)},$$

we obtain (6.7) from (6.76), (6.72), and the assumption $\mathcal{L}_L(\gamma, m) \leq (1 - \gamma)^{-1}$, using that

$$\exp(C\lambda(m - n)) \leq C,$$

which follows from the condition $m - n \leq (1 - \gamma)^{-1} = C_1 \lambda^{-1}$. $\square$

6.4. Regularity uniformly in the cutoff. We remove the restriction $L \geq m$ from Proposition 6.1. Recall from Definition 4.3 that $\mathbf{b}_{L,n} = \bar{\mathbf{a}}_L$ for $n \geq L$. For $n \leq m$, we shall use the identity

$$(6.81) \quad \frac{\mathbf{a}_{m \wedge L}}{\mathbf{a}_{n \wedge L}} = \exp\left(\sum_{i=n+1}^m (\mathbf{g}_i - \tau^2) \mathbb{1}_{\{i \leq L\}}\right)$$

and, by (3.32),

$$(6.82) \quad 1 \leq \frac{\bar{\mathbf{a}}_{n \wedge L}}{\bar{\mathbf{a}}_{m \wedge L}} \leq \exp(2\tau^2((m \wedge L) - (n \wedge L))) \leq \exp(2\tau^2(m - n)).$$

In this subsection, for the cutoff L , we modify the good event $\mathcal{G}_m(\varepsilon, s)$ of Definition 6.2 by replacing only the quantity J in (6.11) by

$$(6.83) \quad J(z + \square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}).$$

The conditions (6.9) and (6.10) are left unchanged, and translated events are defined as before. Likewise, we modify the quantity $\mathbf{G}_{k,z}(s)$ in (6.44), for the cutoff L , by replacing its term involving J with

$$\min_{|e|=1} \left\{ \max_{|e|=1} J(z' + \square_l, e, e; \bar{\mathbf{a}}_{l \wedge L}^{-1} \mathbf{a}_{l \wedge L}), 1 \right\}^{1/2}.$$

The terms involving the fields $\mathbf{g}_j$ are unchanged. We retain the original notation. The modified event agrees with the original event when $L \geq m$, and the modified quantity agrees with the original quantity when $L \geq k$. Thus the arguments in the previous subsections still apply.

Lemma 6.13 (Moments of J uniformly in the cutoff). *There exists $C(d) < \infty$ such that, for every*

$$1 \leq p \leq C^{-1} \bar{\delta}^{-2} |\log \bar{\delta}|^{-1},$$

and every $L, n \in \mathbb{N}_0$,

$$(6.84) \quad \left\| \max_{|e|=1} J(\square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}) \right\|_{L^p(\mathbb{P})} \leq Cp \log(2 + p) \bar{\delta}^2.$$

Proof. For $n \leq L$, the estimate follows from Proposition 4.1 and the homogeneity identity

$$J(U, e, e; \bar{\mathbf{a}}_n^{-1} \mathbf{a}_n) = J(U, \bar{\mathbf{a}}_n^{-1/2} e, \bar{\mathbf{a}}_n^{1/2} e; \mathbf{a}_n).$$

If $n > L$, partition $\square_n$ into cubes P of side 3^L . By (2.22),

$$J(\square_n, e, e; \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L) \leq \sum_P J(P, e, e; \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L).$$

By Minkowski's inequality and stationarity, the L^p norm of the right side is at most that of $\max_{|e|=1} J(\square_L, e, e; \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L)$, which is covered by the case $n \leq L$. The maximum over e is handled as in Proposition 4.1. $\square$

Proposition 6.14 (Good-scale estimates uniformly in the cutoff). *There exists $C(d) < \infty$ such that, for every $L \in \mathbb{N}_0$, the following assertions hold, with $\mathcal{G}_m$ and $\mathbf{G}_{k,z}$ modified as above for the cutoff L .*

First, if $s, \varepsilon, \theta \in (0, 1]$ and

$$(6.85) \quad \theta \geq C s^{-6} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

then, for every $m_0, M \in \mathbb{N}_0$,

$$(6.86) \quad \mathbb{P} \left[\sum_{m=m_0}^{m_0+M} \mathbb{1}_{\mathcal{G}_m(\varepsilon, s)} \leq 1 - \theta \right] \leq \exp \left(- \frac{s^6 \varepsilon^2 \theta}{C \bar{\delta}^2 |\log \bar{\delta}|} (M + 1) \right).$$

Second, if $s \in (0, 1/2]$ and $s \geq C \bar{\delta}^2 |\log \bar{\delta}|$, then, for every $0 \leq n \leq m$ and $z \in \mathbb{R}^d$,

$$(6.87) \quad \sum_{k=n}^m \mathbf{G}_{k,z}(s) \leq C s^{-7/2} \bar{\delta} + \mathcal{O}_{\Gamma_2} \left(C s^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2} (m - n + 1)^{-1/2} \right).$$

Consequently, if

$$\lambda \in (0, 1] \quad \text{and} \quad \lambda \geq C s^{-7/2} \bar{\delta} |\log \bar{\delta}|^{1/2},$$

there exists $M(\lambda, s, m) \in \{-1, 0, \dots, m\}$ such that

$$(6.88) \quad 0 \leq m - M(\lambda, s, m) \leq 1 + \mathcal{O}_{\Gamma_1} (C s^{-7} \bar{\delta}^2 |\log \bar{\delta}| \lambda^{-2})$$

and, for every $n \in \mathbb{N}_0$ with $n \leq M(\lambda, s, m)$,

$$(6.89) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \mathbf{G}_{j,z}(s) \leq \lambda(m - n).$$

If, in addition,

$$\varepsilon \in (0, 1] \quad \text{and} \quad \lambda \geq C s^{-8} \varepsilon^{-2} \bar{\delta}^2 |\log \bar{\delta}|,$$

there is $N(\lambda, \varepsilon, s, m) \in \{-1, 0, \dots, m\}$ satisfying

$$(6.90) \quad 0 \leq m - N(\lambda, \varepsilon, s, m) \leq 1 + \mathcal{O}_{\Gamma_1} (C s^{-8} \varepsilon^{-2} \lambda^{-1} \bar{\delta}^2 |\log \bar{\delta}|)$$

and, for every $n \in \mathbb{N}_0$ with $n \leq N(\lambda, \varepsilon, s, m)$,

$$(6.91) \quad \max_{z \in 3^n \mathbb{Z}^d \cap \square_m} \sum_{j=n}^m \left(1 - \mathbb{1}_{\mathcal{G}_{j,z}(\varepsilon, s)} \right) < 1 + \lambda(m - n).$$

Finally, for $s \in [64 \bar{\delta}^2, 1/2]$, $\varepsilon \in [s^{-1} \bar{\delta}^2, 1]$, every $m \in \mathbb{N}_0$, and every $z \in \mathbb{R}^d$,

$$(6.92) \quad \mathcal{E}_{s,\infty,2}(z + \square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{z+\square_m}) \mathbb{1}_{\mathcal{G}_{m,z}(\varepsilon, s)} \leq C \min\{\varepsilon, s^{-1} \bar{\delta}^2 + \varepsilon^8 + \mathbf{G}_{m,z}(s)\}.$$

In particular,

$$(6.93) \quad \mathcal{E}_{s,\infty,2}(z + \square_m; \mathbf{a}_L, (\mathbf{b}_{L,m})_{z+\square_m}) \mathbb{1}_{\mathcal{G}_{m,z}(\varepsilon, s)} \leq C \varepsilon.$$

Proof. The conditions (6.9) and (6.10) in $\mathcal{G}_m$ are those of Definition 6.2. For the modified condition (6.11), the proof of Lemma 6.7 applies, with the moments of J bounded by (6.84). Indeed, the coefficient $\bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}$ has range of dependence at most $C 3^{n \wedge L} \leq C 3^n$, so $J(z + \square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L})$ depends only on the fields in the $C 3^n$ -neighborhood of $z + \square_n$. Since the quantities J enter only with $n \leq j - 2$, splitting the indices j into finitely many residue classes, as in that proof, gives the finite-range hypothesis of Proposition B.2. The moment exponent in that proof is at least Cd/s , and (6.85) ensures both the smallness condition required there and $\theta \geq p^{-1}$. This proves (6.86).

For (6.87), the terms involving the fields $\mathbf{g}_j$ are the same as those in the proof of Lemma 6.11. For the term involving J , we estimate the maximum over z as in Step 2 of that proof, using the moment of order $8d/s$ of $\min\{\max_{|e|=1} J(z + \square_n, e, e; \bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}), 1\}^{1/2}$, that is, the moment of order $4d/s$ of the same quantity without the square root. If the dimensional constant in the lower bound on s is large enough, this order lies in the range allowed by (6.84). With the same residue classes as above, the sum of these terms is then bounded exactly as in Step 3 of that proof, and the rearrangement (6.49) and the bound for the annuli below scale n are unchanged. This proves (6.87). The union bound over translated centers used to prove (6.50) gives (6.88) and (6.89). Applying the proof of Lemma 6.12 to (6.86) gives (6.90) and (6.91).

It remains to prove (6.92). By (6.11), on $\mathcal{G}_{m,z}(\varepsilon, s)$ the elementary inequality of Step 3 of the proof of Proposition 6.1, with s in place of s_0 , applies to the square root of every quantity J in the annular decomposition. For $m \leq L$, that inequality and (6.15) give (6.92). Suppose that $m > L$. We use the annular decomposition (6.18) again and, at each nonnegative scale n , compare $\bar{\mathbf{a}}_L^{-1} \mathbf{a}_L$ with $\bar{\mathbf{a}}_{n \wedge L}^{-1} \mathbf{a}_{n \wedge L}$. The two coefficients agree if $n \geq L$. If $n < L$, we apply Lemma 2.8, with the ratios of the two coefficients given by (6.81); the conditions (6.9) and (6.10) in $\mathcal{G}_m$ bound these ratios exactly as in (6.21)–(6.20), and the ratios of effective diffusivities are bounded by (6.82). The contribution of the negative scales is bounded by (6.19), with the same proof. Combining these estimates with the preceding inequality for these quantities, as in Step 3 of the proof of Proposition 6.3, gives (6.92). The definition of $\mathcal{G}_m(\varepsilon, s)$ and (6.14) give (6.93). $\square$

Lemma 6.15 (Regularity at a good scale, uniformly in the cutoff). *Let $L \in \mathbb{N}_0$. Then the following hold.*

- Harmonic approximation: in the setting of Lemma 6.8, with its hypothesis $L \geq m$ removed, its conclusion holds with the good event $\mathcal{G}_{n+2,z}$ modified as above for the cutoff L .
- Excess decay: in the setting of Lemma 6.9, with its hypothesis $L \geq m$ removed, its conclusion holds with the same modification.

Proof. In the proof of Lemma 6.8, we take for $\mathbf{a}_0$ the positive scalar $(\mathbf{b}_{L,n+2})_{z+\square_{n+2}}$ and use Proposition 6.14 in place of Proposition 6.3. The general coarse-graining estimate, its stability estimates, the interior and boundary Caccioppoli inequalities, and the estimates for the right side $\nabla \cdot \mathbf{g}$ are deterministic statements about $\mathbf{a}_L$ and $\mathbf{a}_0$. They therefore give the same estimate, with the same powers of $\mathbf{a}_0$. The bounds on $\mathbf{a}_0^{-1} \Lambda_{s,q}$ and $\mathbf{a}_0 \lambda_{s,q}^{-1}$ used there follow from (6.93) and (2.52). Finally, multiplying the Laplacian by the positive scalar $\mathbf{a}_0$ does not change its harmonic functions. The interior and reflected boundary estimates in the proof of Lemma 6.9 are consequently unchanged. $\square$

Proposition 6.16 (Cutoff-uniform large-scale regularity). *There exists $C(d) < \infty$ such that, if $\bar{\delta} \leq C^{-1}$, then for every*

$$\gamma \in \left[1/2, 1 - C\bar{\delta} |\log \bar{\delta}|^{1/2}\right]$$

and every $L, m \in \mathbb{N}_0$, there is an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ satisfying, for every $k \in \mathbb{N}$,

$$(6.94) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C\bar{\delta}^2 |\log \bar{\delta}|}\right).$$

For every $\mathbf{g} \in C^{0,1/2}(\square_m; \mathbb{R}^d)$, $h \in C^{1,1/2}(\square_m)$, and $u \in H^1(\square_m)$ satisfying (6.4), all three conclusions (6.5), (6.6), and (6.7) hold with $\mathcal{L}_L(\gamma, m)$ as the minimal scale. If $L \geq m$, one may take

$$(6.95) \quad \mathcal{L}_L(\gamma, m) = \mathcal{L}_m(\gamma, m).$$

The same statement holds on every translate $y + \square_m$, with the tail in (6.94) uniform in y .

Proof. If $L \geq m$, then this is Proposition 6.1. We may therefore suppose $L < m$. Choose $s_0, \lambda, \varepsilon$, and the step size k exactly as in Steps 1–2 of the proof of Proposition 6.1. Define

$$(6.96) \quad \mathcal{L}_L(\gamma, m) := \max\{m - M(\lambda, s_0, m), m - N(\lambda, \varepsilon, s_0, m)\} + k + 5.$$

Here M and N are the random variables of Proposition 6.14 for the cutoff L , not those of Lemmas 6.11 and 6.12. Equations (6.88) and (6.90), with the parameter choices in that proof, give (6.94). Equations (6.89) and (6.91) give the two conclusions in (6.62) for the modified $\mathbf{G}$ and $\mathcal{G}$.

For $n \leq j \leq m - 5$ and $z \in 3^n \mathbb{Z}^d \cap \square_m$, the calculation leading to (6.64), with all sums over the fields and all deterministic sums stopped at L , gives

$$(6.97) \quad C^{-1} \exp(-C\lambda(m-n)) \leq \frac{(\mathbf{b}_{L,j+2})_{z+\square_{j+2}}}{(\mathbf{b}_{L,m})_{\square_m}} \leq C \exp(C\lambda(m-n)).$$

Indeed, if $j+2 \geq L$, the numerator and the denominator both equal $\bar{\mathbf{a}}_L$. If $j+2 < L < m$, only the fields $\mathbf{g}_{j+3}, \dots, \mathbf{g}_L$ occur; they are bounded by (6.89), while (6.82) controls the ratio of effective diffusivities.

We use Lemma 6.15 in place of Lemma 6.9. With (6.97) in place of (6.64), the proofs of (6.65), (6.72), and (6.79)–(6.80), and of the coarse-grained Caccioppoli and Besov estimates in Step 6 of the proof of Proposition 6.1, apply without change. Proposition 6.14 gives the analogue of (6.67). Thus Steps 3–7 of that proof apply verbatim with $\mathcal{L}_L(\gamma, m)$ as the minimal scale; this proves the three asserted estimates. $\square$

Proposition 6.17 (Stability under multiplication by a nearly constant function). *There are $\varepsilon_*(d) > 0$, $c(d) > 0$, and $C(d) < \infty$ such that the following holds if $\bar{\delta} \leq c$. Let $L \in \mathbb{N}_0$, $m \in \mathbb{Z}$, $z \in \mathbb{R}^d$, and $j \in \mathbb{N}$. Put $B = z + \square_m$, and let B' be a cube of side 3^{m-j} contained in the concentric cube $B/2$ of half the side of B . Outside an event of probability at most*

$$(6.98) \quad C \exp\left(-\frac{c}{\bar{\delta}^2 |\log \bar{\delta}|^2}\right),$$

the following estimates hold for every positive $\vartheta \in C(B)$ satisfying

$$(6.99) \quad \inf_{b>0} \|b^{-1}\vartheta - 1\|_{L^\infty(B)} \leq \varepsilon_*$$

and every weakly $\mathbf{a}_L \vartheta$ -harmonic function $h \in H^1(B)$:

$$(6.100) \quad \operatorname{osc}_{B'} h \leq C 3^{-cj} \operatorname{osc}_B h \quad \text{and} \quad \sup_{B'} |h - (h)_B| \leq C \|h - (h)_B\|_{\underline{L}^2(B)}.$$

Proof. Multiplying ϑ by a positive constant does not change the set of $\mathbf{a}_L \vartheta$ -harmonic functions. We may therefore assume that $\|\vartheta - 1\|_\infty \leq \varepsilon_*$. Put $s_0 = 1/32$ and suppose first that $m \geq 0$. We run the proof of Proposition 6.16 for each of 4^d translates of the triadic grids $3^n \mathbb{Z}^d$, on the correspondingly translated cubes (that proposition holds on every translate of $\square_m$), with exponent $3/4$, and show that the conclusion survives with exponent $1/2$. On each cube Q on which that proof estimates the coarse-graining error, Lemma 2.8, applied with perturbation parameter ε_* and with the scalar $\mathbf{a}_0$ of Lemma 6.15, gives

$$(6.101) \quad \mathcal{E}_{s_0, \infty, 2}(Q; \mathbf{a}_L \vartheta, \mathbf{a}_0) \leq (1 + C\varepsilon_*) \mathcal{E}_{s_0, \infty, 2}(Q; \mathbf{a}_L, \mathbf{a}_0) + C\varepsilon_*^{1/2},$$

Indeed, it suffices to apply the sensitivity estimate on every subcube in the definition of $\mathcal{E}$ and to use Minkowski's inequality in the geometrically weighted sum. Consequently, on each good scale, the factor θ^k and the coefficient ε_j of $|\nabla \ell(u, U_j)|$ in (6.35) change by at most $C\varepsilon_*^{1/2}$, while the set of good scales does not depend on ϑ . Choose ε_* so that $C\varepsilon_*^{1/2} < 1/8 \log 3$. Then the iteration of Lemma 6.10, run with exponent $3/4$, still yields exponent at least $1/2$. Thus the translated large-scale Hölder estimate of Proposition 6.16 holds with exponent $1/2$, uniformly for every ϑ satisfying (6.99). This argument is deterministic whenever the good-scale conditions of Proposition 6.16 hold, so no union bound over ϑ is needed.

It remains to pass from averages to point values. We use these 4^d grids. Campanato's argument gives the first estimate down to scale one. Since B' lies in the concentric cube $B/2$, we can choose unit cubes Q with $2Q \subseteq B$ such that their concentric subcubes $Q/4$, of one quarter the side length,

cover B' . When m is bounded, we use the corresponding rescaled interior cubes instead. Moser iteration gives

$$\|h - (h)_Q\|_{L^\infty(Q/4)} \leq C \left(\frac{\sup_Q \mathbf{a}_L \vartheta}{\inf_Q \mathbf{a}_L \vartheta} \right)^C \|h - (h)_Q\|_{L^2(Q)}.$$

By Assumption (g2) and a union bound over the $O(3^{d(m-j)+})$ cubes, outside an event whose probability is bounded by (6.98), the logarithm of the largest of these coefficient ratios is at most $C\bar{\delta}\sqrt{m+1} + C\bar{\delta}^2(m+1) + C$. The resulting factor is absorbed by the Campanato decay after reducing $c(d)$.

It remains to treat cubes of side less than one. Let P be such a cube, with side r . If $r\|\nabla \log \mathbf{a}_L\|_{L^\infty(P)} \leq \varepsilon_*$, then, after dividing by the value of $\mathbf{a}_L$ at the center of P , the ellipticity ratio of $\mathbf{a}_L \vartheta$ on P is bounded by a dimensional constant. The De Giorgi oscillation estimate therefore reduces the oscillation by a fixed factor from P to the concentric triadic cube of one third its side. Iterating, we obtain decay of the oscillation like a fixed power of the side length below the scale $\varepsilon_*/(1 + \|\nabla \log \mathbf{a}_L\|_\infty)$. The estimates between this scale and scale one introduce only a fixed power of $1 + \|\nabla \log \mathbf{a}_L\|_\infty$, and the bound on $\|\nabla \log \mathbf{a}_L\|_{L^\infty(B)}$ that holds outside the exceptional event, by Assumption (g2) and a union bound, absorbs this factor into the decay from B . This proves both estimates when B' has side below one, and also proves the case $m < 0$ directly.

Fix j_0 larger than the additive constant in (6.94). If $m > j_0$, we restrict to the event that $\mathcal{L}_L(3/4, m) \leq j_0$ for each of these 4^d grids; if $m \leq j_0$, we use the preceding local estimate. For $j \leq j_0$, the oscillation bound follows from the maximum principle after increasing C , while for $j > j_0$, this event gives the Campanato estimate down to the scale of B' . By (6.94), the probability of its complement is at most $Ce^{-c/(\bar{\delta}^2 |\log \bar{\delta}|^2)}$, uniformly in L, m, z , and j . $\square$

6.5. Dirichlet homogenization. The estimates above compare a solution with an affine function. We now compare it instead with the solution of the homogenized Dirichlet problem in the unit cube, after rescaling the cutoff coefficient $\mathbf{a}_L$ from a cube at least as large as the cutoff scale.

Theorem 6.18 (Quantitative homogenization of the Dirichlet problem). *For $L, M \in \mathbb{N}_0$ with $L \leq M$, recall the rescaled coefficient of the introduction,*

$$(6.102) \quad A_{L,M}(x) := \bar{\mathbf{a}}_L^{-1} \mathbf{a}_L(3^M x), \quad x \in \square_0.$$

For every $\vartheta \in (0, 1)$ and $q \in [1, \infty)$, there are $\bar{\delta}_0(\vartheta, q, d) > 0$ and $C(\vartheta, q, d) < \infty$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there are random variables $\mathcal{Z}_{L,M} \geq 1$, $0 \leq L \leq M$, satisfying

$$(6.103) \quad \sup_{M \in \mathbb{N}_0} \sup_{0 \leq L \leq M} \|\mathcal{Z}_{L,M}\|_{L^q(\mathbb{P})} \leq C.$$

For $f \in L^2(\square_0)$ and $h \in H^2(\square_0)$, recall that $u_{L,M}$ and u_{hom} are the unique weak solutions in $H^1(\square_0)$ of

$$(6.104) \quad \begin{cases} -\nabla \cdot (A_{L,M} \nabla u_{L,M}) = f & \text{in } \square_0, \\ u_{L,M} = h & \text{on } \partial \square_0, \end{cases} \quad \begin{cases} -\Delta u_{\text{hom}} = f & \text{in } \square_0, \\ u_{\text{hom}} = h & \text{on } \partial \square_0, \end{cases}.$$

Then, almost surely, for every such f and h , the homogenization error $\mathcal{D}_{L,M}$ of the introduction satisfies

$$(6.105) \quad \mathcal{D}_{L,M} \leq \mathcal{Z}_{L,M} \bar{\delta}^\vartheta (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

If $f = 0$, then the factor $\bar{\delta}^\vartheta$ on the right side may be replaced by $C\bar{\delta}$, with the same random variable $\mathcal{Z}_{L,M}$.

Proof. Fix $s_2 \in (0, 1)$, and then choose $0 < s_1 < s < \min\{s_2, 1/2\}$ so that

$$(6.106) \quad \frac{s_2}{s_1 + s_2} > \vartheta.$$

All stochastic estimates below are in $L^r(\mathbb{P})$, for a fixed exponent r that exceeds q by a dimensional factor. Proposition 4.1 and Lemma 4.5 give, uniformly in $0 \leq L \leq M$,

$$(6.107) \quad \|\mathcal{E}_{s_1, \infty, 1}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^r(\mathbb{P})} + \|\mathcal{E}_{s_1/2, \infty, 2}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)\|_{L^r(\mathbb{P})} \leq C\bar{\delta}.$$

The admissible moment range in those estimates contains every fixed $r < \infty$ after decreasing $\bar{\delta}_0(\vartheta, q, d)$.

Write f as the divergence of a field in $H^1(\square_0; \mathbb{R}^d)$ of norm at most $C\|f\|_{L^2(\square_0)}$, for instance minus the gradient of the solution of the Dirichlet problem for the Laplacian in $\square_0$ with source f . Rescale to $\square_M$ by setting $U(y) := u_{L,M}(3^{-M}y)$ and $V(y) := u_{\text{hom}}(3^{-M}y)$, and let $G(y)$ be $\bar{\mathbf{a}}_L 3^{-M}$ times the field above evaluated at $3^{-M}y$. Then

$$-\nabla \cdot (\mathbf{a}_L \nabla U) = \nabla \cdot G, \quad -\bar{\mathbf{a}}_L \Delta V = \nabla \cdot G, \quad U - V \in H_0^1(\square_M).$$

Let $k \in \mathbb{N}$ and apply Proposition 2.11, with $p = 2$ and $n = M - k$. Definition 2.3 gives

$$\begin{aligned} \mathcal{E}_{s_1, \infty, 1}(\square_M, M - k; \mathbf{a}_L, \bar{\mathbf{a}}_L) &\leq 3^{s_1 k} \mathcal{E}_{s_1, \infty, 1}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L), \\ \mathcal{E}_{s_1/2, \infty, 2}(\square_M, M - k; \mathbf{a}_L, \bar{\mathbf{a}}_L)^2 &\leq 3^{s_1 k} \mathcal{E}_{s_1/2, \infty, 2}(\square_M; \mathbf{a}_L, \bar{\mathbf{a}}_L)^2. \end{aligned}$$

The geometric weights in the sum over k in (2.75), together with (2.7), bound that sum by $C\|\mathbf{a}_L^{1/2} \nabla U\|_{L^2(\square_M)}$. Moreover, (2.73) and (2.52) give a random variable $Y_{L,M} \geq 1$, with every fixed moment bounded uniformly in $L \leq M$, such that

$$(6.108) \quad \|A_{L,M}^{1/2} \nabla u_{L,M}\|_{L^2(\square_0)} \leq Y_{L,M} (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

Rescaling (2.75) back to $\square_0$ therefore yields

$$(6.109) \quad \begin{aligned} &\|\nabla u_{L,M} - \nabla u_{\text{hom}}\|_{H^{-s}(\square_0)} + \|A_{L,M} \nabla u_{L,M} - \nabla u_{\text{hom}}\|_{H^{-s}(\square_0)} \\ &\leq C \left(3^{s_1 k} \mathcal{E}_1 Y_{L,M} + 3^{-s_2 k} (1 + 3^{s_1 k} \mathcal{E}_2^2) \right) (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}), \end{aligned}$$

where $\mathcal{E}_1$ and $\mathcal{E}_2$ denote the two random variables in (6.107).

Take

$$k := \left\lceil \frac{\log_3(\bar{\delta}^{-1})}{s_1 + s_2} \right\rceil.$$

With $\mathcal{E}_1, \mathcal{E}_2$, and $Y_{L,M}$ as above, define

$$\begin{aligned} \mathcal{Z}_{L,M} &:= 1 + C\bar{\delta}^{-\vartheta} \left(3^{s_1 k} \mathcal{E}_1 Y_{L,M} + 3^{-s_2 k} (1 + 3^{s_1 k} \mathcal{E}_2^2) \right) \\ &\quad + C\bar{\delta}^{-1} \mathcal{E}_1 Y_{L,M}. \end{aligned}$$

Equations (6.106), (6.107), and (6.108), followed by Hölder's inequality, give (6.103). Substituting this choice of k and the definition of $\mathcal{Z}_{L,M}$ into (6.109) proves (6.105) for the two H^{-1} terms of $\mathcal{D}_{L,M}$. If $f = 0$, we apply (6.109) again, with $k = 1$. The $3^{-s_2 k}$ term is then absent, and the last term in the definition of $\mathcal{Z}_{L,M}$ gives the factor $C\bar{\delta}$.

Finally, the difference $u_{L,M} - u_{\text{hom}}$ belongs to $H_0^1(\square_0)$. For every $w \in H_0^1(\square_0)$, duality and the spectral theorem for the Dirichlet Laplacian on $\square_0$ give $\|w\|_{L^2(\square_0)} \leq C\|\nabla w\|_{H^{-s}(\square_0)}$: test with the image of w under the inverse Dirichlet Laplacian, whose gradient is bounded in $H^s(\square_0)$ by $C\|w\|_{L^2(\square_0)}$. Applying this to the difference, and using that $s < 1$ and the embedding $H^{-s}(\square_0) \hookrightarrow H^{-1}(\square_0)$, completes the proof. $\square$

Theorem 6.19 (Quantitative homogenization at a fixed cutoff). *There exists $\alpha_{\text{hom}}(d) > 0$ with the following property. Fix $L \in \mathbb{N}_0$. For $M \in \mathbb{N}_0$ with $M \geq L$, let $A_{L,M}$ be as in (6.102). For every $q \in [1, \infty)$, there is a random variable $\mathcal{Y}_{L,q} \geq 1$ satisfying*

$$(6.110) \quad \|\mathcal{Y}_{L,q}\|_{L^q(\mathbb{P})} \leq C(L, q, d, \bar{\delta})$$

such that, almost surely, for every $M \geq L$, $f \in L^2(\square_0)$ and $h \in H^2(\square_0)$, the homogenization error $\mathcal{D}_{L,M}$ of the unique weak solutions $u_{L,M}$ and u_{hom} of (6.104) satisfies

$$(6.111) \quad \mathcal{D}_{L,M} \leq \mathcal{Y}_{L,q} 3^{-\alpha_{\text{hom}}(M-L)} (\|f\|_{L^2(\square_0)} + \|h\|_{H^2(\square_0)}).$$

Proof. Consider the coefficient $A_{L,L}$. For fixed L , this is a stationary positive scalar coefficient with range of dependence at most $\sqrt{d}$, and its homogenized matrix is the identity. It satisfies the logarithmic coefficient bound

$$(6.112) \quad \|\log A_{L,L}\|_{L^\infty(\square_0)} \leq \mathcal{O}_{\Gamma_2}(C\bar{\delta}(L+1)).$$

Indeed, this follows from (3.6), Assumption (g2), and the bound $|\log \bar{\mathbf{a}}_L| \leq C\bar{\delta}^2(L+1)$ from Theorem 5.1.

We apply the results of Armstrong and Kuusi [2025, Theorem B, Theorem 4.1, and Proposition D.3] to $A_{L,L}$. Stationarity and finite range give the mixing assumption of that paper with $\beta = 0$ and $\nu = d/2$, and with Gaussian concentration. Proposition D.3 of that paper, applied to (6.112) with exponent $1/4$, gives the coarse-grained ellipticity assumption of that paper above a random scale. The probability that this scale exceeds $\sqrt{d}t$ is at most $\exp(C_L - c_L \log^2 t)$, with positive constants C_L and c_L that may depend on L ; this tail bound implies that all polynomial moments of this scale are finite.

Armstrong and Kuusi [2025, Corollary 4.3], with exponent $5/16$, gives an estimate uniform over all subcubes. Together with Definition 2.3, it yields a number $\alpha_{\text{hom}}(d) > 0$ and, for every fixed $q < \infty$, a random variable $\mathcal{W}_{L,q}$ with finite q th moment such that, for all $N \in \mathbb{N}_0$, $r \in \{1, 2\}$, and $s \in \{3/16, 3/8\}$,

$$(6.113) \quad \mathcal{E}_{s,\infty,r}(\square_N; A_{L,L}, 1) \leq \mathcal{W}_{L,q} 3^{-2\alpha_{\text{hom}}N}.$$

Indeed, by the estimate of Armstrong and Kuusi [2025, Corollary 4.3] for subcubes, the quantity $\max_{|e|=1} J(\cdot, e, e; A_{L,L})$ on a subcube at scale $k \leq N$ is bounded by a random multiple of $3^{-4\alpha_{\text{hom}}N} 3^{5/16(N-k)}$. After taking square roots, the weighted sum over scales in Definition 2.3 converges, since $s > 5/32$ for both values of s above. Below that random scale, we use (2.20) and the extrema of $A_{L,L}^{\pm 1}$ instead. By a union bound over unit cubes and (6.112), the moments of these extrema on $\square_N$ grow at most like $C_{L,q} \exp(C_{L,q}\sqrt{N+1})$. Hölder's inequality, the Gaussian-type tail bound for that scale, and the analogous tail for the scale in Armstrong and Kuusi [2025, Corollary 4.3] therefore allow us to absorb these scales into $\mathcal{W}_{L,q}$ while keeping its moments finite.

We use the estimate leading to (6.109), after rescaling $\square_M$ first by 3^L and then by 3^{M-L} . Apply Proposition 2.11 with

$$p = 2, \quad s_1 = \frac{3}{8}, \quad s = \frac{7}{16}, \quad s_2 = \frac{3}{4},$$

so that $sp' = 7/8 < 1$ and both s_1 and $s_1/2$ are covered by (6.113), and choose

$$k := \max \left\{ 1, \left\lfloor \frac{2\alpha_{\text{hom}}(M-L)}{s_1 + s_2} \right\rfloor \right\}.$$

In (6.109), we use (6.113), applied at the scale $M-L$, in place of (6.107), and we construct $Y_{L,M}$ in (6.108) from (2.73), (2.52), and now (6.113). These give $\sup_{M \geq L} \|Y_{L,M}\|_{L^r} < \infty$ for every fixed r , with constants allowed to depend on $L, r, d, \bar{\delta}$. Substituting (6.113) and the choice of k in (6.109),

and using the bound $Y_{L,M} \leq 3^{\alpha_{\text{hom}}(M-L)/25} \sup_{M' \geq L} 3^{-\alpha_{\text{hom}}(M'-L)/25} Y_{L,M'}$, we obtain the negative-Sobolev part of (6.111) with the random factor

$$\mathcal{Y}_{L,q} := C(1 + \mathcal{W}_{L,4q})^2 \left(1 + \sup_{M \geq L} 3^{-\alpha_{\text{hom}}(M-L)/25} Y_{L,M} \right).$$

Since $Y_{L,M}$ is constructed with a moment larger than $4q$, the supremum has a finite $4q$ th moment, because its $4q$ th power is at most the sum over $M \geq L$ of $3^{-4q\alpha_{\text{hom}}(M-L)/25} Y_{L,M}^{4q}$. Hölder's inequality shows that $\mathcal{Y}_{L,q}$ has a finite q th moment, which gives (6.110). The duality argument with the Dirichlet Laplacian at the end of the proof of Theorem 6.18 deduces the L^2 estimate from the estimate in a negative Sobolev norm. $\square$

6.6. Limiting coefficient and regularity. The limiting coefficient field is the almost-sure local limit

$$(6.114) \quad \tilde{\mathbf{a}}_L(x) := \frac{\mathbf{a}_L(x)}{\mathbf{a}_L(0)}, \quad \mathbf{a}(x) := \lim_{L \rightarrow \infty} \tilde{\mathbf{a}}_L(x) = \exp\left(\sum_{k=0}^{\infty} (\mathbf{g}_k(x) - \mathbf{g}_k(0))\right).$$

Dividing by $\mathbf{a}_L(0)$ is necessary, since the strong law gives $(L+1)^{-1} \log \mathbf{a}_L(0) \rightarrow -\tau^2$. This division does not affect the generator of $X^{(L)}$, and the reversible measures $(\mathbf{a}_L/\mathbf{a}_L(0)) dx$ converge locally to $\mathbf{a} dx$. By Assumption (g2) and scaling, the coefficient ratios converge locally in $C^{1,1}$. The following lemma gives bounds uniform in L .

Lemma 6.20 (The coefficients divided by their value at the origin). *There is $\bar{\delta}_0(d) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0(d)$, then there are $\kappa \in (0, 1)$ and an almost surely finite random variable C_ω such that, almost surely, for every $L \in \mathbb{N}_0$ and $x \in \mathbb{R}^d$,*

$$(6.115) \quad \sup_{\mathbf{b} \in \{\mathbf{a}\} \cup \{\tilde{\mathbf{a}}_L\}_{L \in \mathbb{N}_0}} (\mathbf{b}(x) + \mathbf{b}(x)^{-1} + |\nabla \mathbf{b}(x)|) \leq C_\omega(1 + |x|)^\kappa.$$

Moreover,

$$(6.116) \quad \tilde{\mathbf{a}}_L \longrightarrow \mathbf{a} \quad \text{locally in } C^{1,1}, \quad \nabla \log \tilde{\mathbf{a}}_L \longrightarrow \nabla \log \mathbf{a} \quad \text{locally in } C^{0,1},$$

and

$$(6.117) \quad \sup_{L \in \mathbb{N}_0} (|\nabla \log \tilde{\mathbf{a}}_L(x)| + |\nabla^2 \log \tilde{\mathbf{a}}_L(x)|) \leq C_\omega(1 + \sqrt{\log(2 + |x|)}).$$

Proof. Fix a sufficiently large dimensional constant C , and cover each ball B_{C2^n} by cubes of side 3^k . Stationarity, (1.4), (1.5), and the Borel–Cantelli lemma give a finite random variable C_ω such that, for all $n, k \in \mathbb{N}_0$ and $r \in \{1, 2\}$,

$$(6.118) \quad \|\nabla^r \mathbf{g}_k\|_{L^\infty(B_{C2^n})} \leq C_\omega 3^{-rk} \sqrt{n+k+1}.$$

Summing over k proves the local convergence (6.116) and the bound (6.117).

It remains to obtain the bound on the coefficient itself uniformly in the cutoff L . For each $z \in \mathbb{Z}^d \cap B_{C2^n}$, the variables

$$\mathbf{g}_k(z) - \mathbf{g}_k(0) \quad (0 \leq k, 3^k \leq C2^n)$$

are independent, centered, and sub-Gaussian with scale $C\bar{\delta}$. The maximal sub-Gaussian inequality, followed by a union bound over these $O(2^{dn})$ lattice points and Borel–Cantelli, yields

$$(6.119) \quad \max_{\substack{z \in \mathbb{Z}^d \cap B_{C2^n} \\ 0 \leq j, 3^j \leq C2^n}} \left| \sum_{k=0}^j (\mathbf{g}_k(z) - \mathbf{g}_k(0)) \right| \leq C\bar{\delta}n + C_\omega.$$

For $x \in B_{2^n}$, we apply (6.119) at a nearest lattice point and control the difference using (6.118); for the scales k with $3^k \leq C2^n$, this difference costs at most $C_\omega \sqrt{n+1}$. For the other scales, the mean-value theorem along the segment from 0 to x gives

$$\sum_{3^k > C2^n} |\mathbf{g}_k(x) - \mathbf{g}_k(0)| \leq C_\omega 2^n \sum_{3^k > C2^n} 3^{-k} \sqrt{n+k+1} \leq C_\omega \sqrt{n+1}.$$

Both estimates hold for every partial sum of the series. Consequently,

$$\sup_{L \in \mathbb{N}_0} \sup_{x \in B_{2^n}} |\log \tilde{\mathbf{a}}_L(x)| \leq C\bar{\delta}n + C_\omega \sqrt{n+1} + C_\omega.$$

Choosing $\bar{\delta}_0(d)$ small, bounding the square-root term by an arbitrarily small multiple of n plus a random constant, and exponentiating, we obtain (6.115) for $\tilde{\mathbf{a}}_L$ and its reciprocal. Since

$$\nabla \tilde{\mathbf{a}}_L = \tilde{\mathbf{a}}_L \nabla \log \tilde{\mathbf{a}}_L,$$

(6.117) gives the gradient bound after a final arbitrarily small increase of κ . Passing to the local limit proves the assertions for $\mathbf{a}$. $\square$

Recall that, for a large $C_0(d) < \infty$ and $\bar{\delta}$ small, the regularity exponent

$$(6.120) \quad \gamma_{\text{reg}} := 1 - C_0 \bar{\delta} |\log \bar{\delta}|^{1/2}$$

of the introduction lies in $(1/2, 1)$.

Theorem 6.21 (Large-scale Hölder regularity and a Liouville theorem). *There are $\bar{\delta}_0(d) > 0$ and $C(d) < \infty$ such that the following holds if $\bar{\delta} \leq \bar{\delta}_0(d)$. For $L = \infty$, the coefficient $\mathbf{a}_L$ stands for the limiting coefficient $\mathbf{a}$.*

- **Large-scale Hölder estimate:** *Let $\gamma \in [1/2, \gamma_{\text{reg}}]$. For every $L \in \mathbb{N}_0 \cup \{\infty\}$ and $m \in \mathbb{N}$, there is an $\mathbb{N}$ -valued random variable $\mathcal{L}_L(\gamma, m)$ satisfying*

$$(6.121) \quad \mathbb{P}[\mathcal{L}_L(\gamma, m) > k] \leq C \exp\left(-\frac{(1-\gamma)^2(k-C)_+}{C\bar{\delta}^2 |\log \bar{\delta}|}\right), \quad k \in \mathbb{N}.$$

Moreover, almost surely, for every L and m , if $u \in H^1(\square_m)$ is $\mathbf{a}_L$ -harmonic in $\square_m$, then, for every $n \in \mathbb{N}_0$ satisfying

$$n \leq m - \mathcal{L}_L(\gamma, m)$$

and every $z \in 3^n \mathbb{Z}^d$ such that $z + \square_n \subseteq \square_{m-1}$,

$$(6.122) \quad \|u - (u)_{z+\square_n}\|_{\underline{L}^2(z+\square_n)} \leq C 3^{-\gamma(m-n)} \|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)}$$

and

$$(6.123) \quad \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(z+\square_n)} \leq C 3^{(1-\gamma)(m-n)} \|\mathbf{a}_L^{1/2} \nabla u\|_{\underline{L}^2(\square_m)}.$$

- **Liouville theorem:** *Almost surely, for every $L \in \mathbb{N}_0 \cup \{\infty\}$, if $u \in H_{\text{loc}}^1(\mathbb{R}^d)$ satisfies*

$$(6.124) \quad -\nabla \cdot (\mathbf{a}_L \nabla u) = 0 \quad \text{in } \mathbb{R}^d, \quad \liminf_{R \rightarrow \infty} R^{-\gamma_{\text{reg}}} \inf_{c \in \mathbb{R}} \|u - c\|_{\underline{L}^2(B_R)} = 0,$$

then u is constant.

Proof. Fix $\gamma \in [1/2, \gamma_{\text{reg}}]$. For finite L , the two estimates are special cases of Proposition 6.16, for $\mathbf{a}_L$ -harmonic functions and interior cubes: we take $\mathbf{g} = 0$, $x = z$, and $j = n$ in (6.5), and use (6.6). Approximating the trace in $H^{1/2}(\partial \square_m)$ by smooth functions extends the conclusion from smooth to arbitrary H^1 boundary data.

For $L = \infty$, fix m and, for $N \geq m$, let $u_N \in u + H_0^1(\square_m)$ be $\tilde{\mathbf{a}}_N$ -harmonic. Proposition 6.16 supplies the same random variable $\mathcal{L}_N(\gamma, m)$ for every $N \geq m$, and multiplication of the coefficient

by $\mathbf{a}_N(0)^{-1}$ changes neither the equation nor the estimate (6.123). Lemma 6.20 and the energy identity for $u_N - u$ give

$$\|\nabla u_N - \nabla u\|_{L^2(\square_m)} \longrightarrow 0.$$

The large-scale Hölder and energy estimates therefore pass to the limit. Taking $\mathcal{L}_\infty(\gamma, m) = \mathcal{L}_N(\gamma, m)$, for any $N \geq m$, also gives (6.121).

For the Liouville assertion, fix $L \in \mathbb{N}_0 \cup \{\infty\}$ and suppose that u satisfies (6.124). By (6.121) and Borel–Cantelli, $\mathcal{L}_L(\gamma_{\text{reg}}, m) \leq m$ for all sufficiently large m . Choose a sequence of radii $R \rightarrow \infty$ and constants $c = c(R) \in \mathbb{R}$ such that

$$R^{-\gamma_{\text{reg}}} \|u - c\|_{\underline{L}^2(B_R)} \longrightarrow 0,$$

and let m be the largest integer such that $\square_m \subseteq B_R$. Since 3^m is comparable to R and $(u)_{\square_m}$ minimizes $c \mapsto \|u - c\|_{\underline{L}^2(\square_m)}$,

$$3^{-\gamma_{\text{reg}} m} \|u - (u)_{\square_m}\|_{\underline{L}^2(\square_m)} \longrightarrow 0.$$

For every fixed $z \in \mathbb{Z}^d$, we apply (6.122) with $\gamma = \gamma_{\text{reg}}$ and $n = 0$ on $z + \square_0$, which is contained in $\square_{m-1}$ once R is large enough, and let $R \rightarrow \infty$. Thus $\nabla u = 0$ almost everywhere in $\mathbb{R}^d$, so u is constant. $\square$

7. THE SCALING LIMIT

In this section we state Theorem A in the precise form in which we prove it. Sections 8 and 9 prove part (i) of Theorem 7.1 and the convergence of the measures in part (ii), and Section 10 proves the remaining assertions.

To prove almost-sure convergence, we realize all rescaled processes using one family of independent fields $(\mathbf{g}_j)_{j \in \mathbb{Z}}$, with $\mathbf{g}_j$ having the law of $\mathbf{g}_0(3^{-j} \cdot)$. Rescaling space by the factor 3^{-N} turns the field with scale 3^k into a field with scale 3^{k-N} . Accordingly, we take the rescaled potential at cutoff N to be $H + \sum_{j=0}^N \mathbf{g}_{-j}$, where

$$(7.1) \quad H(x) := \sum_{j=1}^{\infty} (\mathbf{g}_j(x) - \mathbf{g}_j(0))$$

collects the fields with scales larger than one; the series converges locally in $C^{1,1}$. Increasing N adds the field with scale 3^{-N} and leaves the others unchanged. Let $K_{N,x}^\omega$ be the law on continuous paths of the diffusion started at x and constructed from the elliptic resolvents of the generator $\bar{\mathbf{a}}_N^{-1}(\Delta + \nabla(H + \sum_{j=0}^N \mathbf{g}_{-j}) \cdot \nabla)$. Its stochastic differential description is

$$(7.2) \quad dY_t = \bar{\mathbf{a}}_N^{-1} \nabla \left(H + \sum_{j=0}^N \mathbf{g}_{-j} \right) (Y_t) dt + \sqrt{2/\bar{\mathbf{a}}_N} dW_t, \quad Y_0 = x.$$

For each N , the fields $\mathbf{g}_{k-N}$ with $k \in \mathbb{N}_0$ have the same joint law as the fields $\mathbf{g}_k(3^N \cdot)$. Thus $K_{N,x}^\omega$, viewed as a random probability measure, has the same distribution as the law of $(3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0}$ conditional on the environment, with $X_0 = 3^N x$. The diffusion (7.2) is reversible with respect to

$$\mu_N(dx) := \exp \left(H(x) + \sum_{j=0}^N (\mathbf{g}_{-j}(x) - \tau^2) \right) dx.$$

The measure $\mu_N^0 := e^{-H} \mu_N$ is stationary and has the law of $\mathbf{a}_N(3^N x) dx$. For Gaussian fields, it is the standard approximation of Gaussian multiplicative chaos: its density is obtained by exponentiating $\sum_{j=0}^N \mathbf{g}_{-j}$ and dividing by the expectation of that exponential. We write d_{weak} for a bounded complete metric on the probability measures on $C([0, \infty), \mathbb{R}^d)$ that induces weak convergence.

Theorem 7.1 (Scaling limit). *Assume (g1)–(g4). There exists $\bar{\delta}_0(d) > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there exists a continuous probability kernel $x \mapsto K_x^\omega$ such that the following statements hold.*

(i) Convergence of the path laws: *Almost surely, for every compact set $B \subseteq \mathbb{R}^d$,*

$$(7.3) \quad \lim_{N \rightarrow \infty} \sup_{x \in B} d_{\text{weak}}(K_{N,x}^\omega, K_x^\omega) = 0.$$

Almost surely, the laws $(K_x^\omega)_{x \in \mathbb{R}^d}$ define a continuous, conservative, strong Markov process Z , started at x under K_x^ω . On the same event, its semigroup maps bounded continuous functions to continuous functions, and, for every bounded open set U , the mean exit time of Z from U under K_x^ω is bounded uniformly in $x \in U$. For every $x \in \mathbb{R}^d$, the annealed law of $(3^{-N} X_{3^{2N}t/\bar{\mathbf{a}}_N})_{t \geq 0}$, with $X_0 = 3^N x$, converges to $\mathbb{E}[K_x^\omega]$.

(ii) The reversible measure: *Almost surely, the measures μ_N^0 and μ_N converge locally weakly to measures μ^0 and $\mu = e^H \mu^0$, respectively. Both limits are locally finite, nonatomic, of full support, and singular with respect to Lebesgue measure. The measure μ^0 is stationary and not deterministic, and it satisfies $\mathbb{E}[\mu^0(dx)] = dx$. Almost surely, the process Z is reversible with respect to μ , which is invariant: for every nonnegative Borel function f and every $t \geq 0$,*

$$\int \left(\int f(Z_t) dK_x^\omega \right) \mu(dx) = \int f d\mu.$$

(iii) Anomalous scaling: *There exist constants $C \geq 1$ and $\eta > 0$, depending only on d and the law of $\mathbf{g}_0$, such that, for all $\ell, m \in \mathbb{N}_0$ with $\ell \leq m$,*

$$(7.4) \quad \frac{\bar{\mathbf{a}}_m}{\bar{\mathbf{a}}_\ell} \leq C 3^{-\eta(m-\ell)}.$$

Let $\sigma_k := \inf\{t \geq 0 : Z_t - Z_0 \notin 3^{-k}\square_0\}$. For every $x \in \mathbb{R}^d$ and every integer $k \geq 0$,

$$(7.5) \quad \mathbb{E} \left[\int \sigma_k dK_x^\omega \right] \leq C 3^{-(2+\eta)k}.$$

For every $x \in \mathbb{R}^d$, almost surely in ω , the law K_x^ω is mutually singular with every mixture of Brownian laws with arbitrary covariance matrices, and K_x^ω -almost every path satisfies

$$(7.6) \quad \sup\{\gamma \geq 0 : |Z_t - Z_0| = O(t^\gamma) \text{ as } t \downarrow 0\} \leq \frac{1}{2+\eta}.$$

In dimension two, one can take $\eta = \tau^2/\log 3$.

(iv) Non-Gaussian marginals: *Almost surely, for every $x \in \mathbb{R}^d$ and every $t > 0$, the law of Z_t under K_x^ω is absolutely continuous with respect to μ , and it is not a Gaussian law.*

Theorem A follows, with $\mathbf{P}_x = K_x^\omega$: for its exit-time bound, apply (7.5) with $3^{-k-1} < r \leq 3^{-k}$.

8. UNIQUENESS OF THE LIMIT FORM

In this section we prove that the energies $\int A_N |\nabla u|^2 dx$ have a unique limit. We show that all subsequential limits of the Dirichlet forms on cubes coincide (Theorem 8.53). Section 9 pairs this limit with the limit of the reversible measures μ_N to construct the limiting diffusion. The argument has four parts.

- (1) We bound, uniformly in N , the energy that solutions carry in small balls; this includes solutions of Neumann problems whose datum is approximated by a volume source (Sections 8.1–8.2). These bounds control the change in the energy on a fixed cube when a fine-scale field is resampled. We use this control to prove compactness of the finite-volume energies (Section 8.3).

- (2) Along subsequences, the Dirichlet Green operators converge, and their limits define Dirichlet forms. These limit forms are local and consistent under restriction to subcubes, and multiplying the coefficient by a positive continuous function multiplies their energy measures by the same function (Section 8.4).
- (3) On a single event that does not depend on the solution, we bound the number of cubes on which the regularity, extension and harmonic-approximation estimates of Lemma 8.33 fail along every chain of nested cubes (Section 8.5). We integrate this bound against the energy measure of a solution. This selects cubes carrying a fixed fraction of its energy, on which these estimates hold and the solution is close to affine.
- (4) Any two subsequential limit forms are comparable: the second lies between two deterministic multiples of the first. The optimal comparison constants agree, since otherwise replacing a solution by harmonic functions on the selected cubes would improve one of them (Section 8.6). Both the energy saved by a replacement and the errors it must absorb are proportional to the gap between the optimal comparison constants. To obtain errors of this size, we apply the Efron–Stein bound of Section 8.3 to the difference between the minimal energy of the second limit form and a multiple of the minimal energy of the first, rather than to each energy separately (Proposition 8.45). Hence the two limits are proportional. The factor is one because the coarse-grained matrices of $e^{-H}A_N$ on large cubes approach the identity uniformly in N , and both limits inherit this normalization (Section 8.7).

Throughout, Assumptions (g1)–(g4) hold. Before reducing $\bar{\delta}$, we fix all exponents and tolerances and a finite list of moment orders. Bounds in $L^p(\mathbb{P})$ for the orders in this list are uniform in N .

8.1. Rescaled coefficients and energies. At fixed N , relabel the fields by replacing $\mathbf{g}_i$ with $\mathbf{g}_{i-N}(3^{-N} \cdot)$, and let $\mathbf{a}_N$ be the cutoff coefficient of (1.7) formed from the relabelled fields. The diffusion with law $K_{N,x}^\omega$ has Dirichlet form $\int A_N |\nabla u|^2 dx$ on $L^2(\mu_N)$, where $A_N := \bar{\mathbf{a}}_N^{-1} d\mu_N/dx$ (see (7.2)). In terms of the relabelled fields,

$$(8.1) \quad A_N(x) = e^{H(x)} \frac{\mathbf{a}_N(3^N x)}{\bar{\mathbf{a}}_N}, \quad A_N^0(x) = e^{-H(x)} A_N(x).$$

If H is replaced by one of its partial sums, then A_N becomes a constant multiple of a stationary cutoff coefficient, so the results of Sections 4 and 6 apply.

Lemma 8.1 (Moments of the field H). *The field H and its partial sums $\sum_{j=1}^{L'} (\mathbf{g}_j - \mathbf{g}_j(0))$ satisfy, on every compact cube Q ,*

$$\|H\|_{C^{1,1}(Q)}\|_{L^p} \leq C_Q \bar{\delta} \sqrt{p}, \quad \mathbb{E} e^{\lambda \|H\|_{C^{1,1}(Q)}} \leq 2e^{C_Q \lambda^2 \bar{\delta}^2} \quad (p \geq 2, \lambda \geq 0),$$

with the same bounds for the partial sums, uniformly in the number L' of terms. Moreover, the partial sums converge to H locally in $C^{1,1}$ almost surely.

Proof. As in the proof of Lemma 6.20, scaling, the mean value theorem and (1.5) give $\|\mathbf{g}_j - \mathbf{g}_j(0)\|_{C^{1,1}(Q)}\|_{L^p} \leq C_Q \bar{\delta} \sqrt{p} 3^{-j}$ for $j \geq 1$. Sum over j and expand the exponential. Since the sum of these norms has finite mean, the partial sums also converge in $C^{1,1}$ almost surely. $\square$

On a bounded cube Q , let

$$(8.2) \quad E_N^Q(u, v) = \int_Q A_N \nabla u \cdot \nabla v dx, \quad \Gamma_N(u) = A_N |\nabla u|^2 dx.$$

The form E_N^Q has domain $H_0^1(Q)$, and we use the same integral for functions in $H^1(Q)$ when prescribing boundary values. Let $\Lambda_{N,Q}(g)$ be the minimum of $E_N^Q(u)$ over $u \in H^1(Q)$ with $u - g \in$

$H_0^1(Q)$. Let G_N^Q be the Dirichlet Green operator, the inverse of $-\nabla \cdot (A_N \nabla)$ on $L^2(Q, dx)$ with zero Dirichlet boundary values, so that

$$(8.3) \quad \langle f, G_N^Q f \rangle = \sup_{v \in H_0^1(Q)} \left(2 \int_Q f v \, dx - E_N^Q(v) \right).$$

For the constant flux p , the quantity $|Q| p \cdot a_*^{-1}(Q; A_N) p$ is the energy $E_N^Q(v)$ of a solution v of $-\nabla \cdot (A_N \nabla v) = 0$ in Q with boundary flux $A_N \nabla v \cdot n = p \cdot n$, and it has the variational formula (2.19). Since $d\mu_N/dx = \bar{\mathbf{a}}_N A_N$, the same energy, regarded as a form on $L^2(\mu_N)$, has generator $u \mapsto \nabla \cdot (A_N \nabla u) / (\bar{\mathbf{a}}_N A_N)$, the generator of the diffusion (7.2); its resolvent is studied in Section 9.2.

8.2. Local estimates and approximation of the Neumann datum. We derive the local estimates from the coarse-grained inequalities of Section 2, the moments of Proposition 4.1, and the regularity estimates of Proposition 6.16. The constants in these estimates have the required moments, uniformly in N . These estimates also give the Morrey bound (8.10), the coercivity bound (8.5) and the bounds of Lemma 8.21 on cutoff functions, which are used in Section 9.3.

At radii above the smallest field scale 3^{-N} , we apply Proposition 6.16 after replacing H by a partial sum. At smaller radii, elliptic estimates for coefficients with small oscillation complete the bounds. At these small radii we use Lemma 8.6 for the coefficient and its logarithmic gradient. The moments of these bounds grow with N , and positive powers of 3^{-N} absorb this growth.

8.2.1. Coercivity and extension. The coarse-grained bounds of Section 4 also cover scales below the smallest field scale. We first apply them on a rescaled cube. We call a cube $z + 3^{-k}\square_0$ of a triadic mesh a **cell**.

Lemma 8.2 (Ellipticity moments on a cell). *Fix $s \in (0, 1]$, $q \in \{1, 2\}$ and a bounded cube Q . For every $p < \infty$ and all sufficiently small $\bar{\delta}$, uniformly for $0 \leq k \leq N$ and cubes $z + 3^{-k}\square_0 \subseteq Q$,*

$$(8.4) \quad \|\Lambda_{s,q}(z + 3^{-k}\square_0; A_N) + \lambda_{s,q}^{-1}(z + 3^{-k}\square_0; A_N)\|_{L^p} \leq C_{p,s,Q} \exp(C(p + p^2)\bar{\delta}^2 k).$$

In particular, on a fixed cell U these moments are bounded uniformly in N .

Proof. Choose a finite moment order $\xi \geq 4p$ large enough for Proposition 4.1 at exponent $s/2$, then decrease $\bar{\delta}$ so that its remaining hypotheses hold. The estimate (4.4), the comparison of the weighted sums at exponents $s/2$ and s , and (2.52) give

$$\sup_m \|\Lambda_{s,q}(\square_0; A_m^0) + \lambda_{s,q}^{-1}(\square_0; A_m^0)\|_{L^{2p}} < \infty.$$

By definition of A_N , the coefficient $A_N(z + 3^{-k}y)$ is the product of three factors: the scalar $\frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N} \exp(H(z) + \sum_{j < k} (\mathbf{g}_{-j}(z) - \tau^2))$, the ratio of $\exp(H + \sum_{j < k} \mathbf{g}_{-j})$ at $z + 3^{-k}y$ to its value at z , and a function of y with the law of A_{N-k}^0 . Summing the bounds on the gradients of the fields as in the proof of Lemma 8.1 bounds the oscillation of $H + \sum_{j < k} \mathbf{g}_{-j}$ over the cube $z + 3^{-k}\square_0$ by a sub-Gaussian variable of size $C_Q \bar{\delta}$. The scalar involves the values of k independent fields. By (3.33), for every fixed $v < \infty$ the sum of the scalar and its inverse, multiplied by the exponential of this oscillation, has L^v norm at most $C_{v,Q} e^{C(v+v^2)\bar{\delta}^2 k}$. Monotonicity and homogeneity of the coarse-grained ellipticity constants, followed by Hölder's inequality, prove (8.4). $\square$

Lemma 8.3 (Coercivity of the form E_N^Q). *Let Q be a fixed cube of side at most one. For every $N \geq 0$ there is a random variable K_N such that, for every $v \in H_0^1(Q)$ and every $w \in H^1(Q)$,*

$$(8.5) \quad \|v\|_{H^{3/4}(Q)}^2 \leq K_N E_N^Q(v) \quad \text{and} \quad \|w - (w)_Q\|_{H^{3/4}(Q)}^2 \leq K_N E_N^Q(w).$$

For every $p < \infty$ there is $\bar{\delta}_0(d, p) > 0$ such that $\sup_N \|K_N\|_{L^p} < \infty$ if $\bar{\delta} \leq \bar{\delta}_0$.

Proof. On $\square_0$, take $s = 1/8$. By the functional inequalities (2.61), (2.63) and (2.65), followed by $B_{2,\infty}^{7/8} \subseteq H^{3/4}$, we obtain (8.5) with $K_N = C\lambda_{1/8,1}^{-1}(\square_0; A_N)$. Lemma 8.2 bounds its L^p norms uniformly in N . For a fixed smaller cube, rescale the same inequalities and use (8.4). If its side is not triadic, then the remaining dilation lies in $[1, 3]$. Apply the same argument after replacing $\bar{\delta}$ by $C_d\bar{\delta}$, with constants depending on Q . $\square$

Lemma 8.4 (Boundary extension and bounds on grid cubes). *Let $\beta \in (1/2, 1)$ and set $\sigma = 1/2(\beta - 1/2)$. The following statements hold.*

(i) *There is $C(d, \beta) < \infty$ such that, for every cube $q = z + r\square_0$ with $r \leq 1$ and every $g \in C^\beta(\partial q)$,*

$$(8.6) \quad \Lambda_{N,q}(g) \leq C \Lambda_{\sigma/2,2}(q; A_N) r^{d-2} \|g(z + r \cdot)\|_{C^\beta(\partial \square_0)/\mathbb{R}}^2.$$

(ii) *Let Q be a bounded cube, and fix finitely many grids, each consisting of the cubes $z + r\square_0$ with $r = 3^{-k}$, $k \geq 0$, and z in $z_0 + r\mathbb{Z}^d$ for a point z_0 fixed for each grid; we call their cubes contained in Q grid cubes. For every $\eta > 0$ and $p < \infty$ there is $\bar{\delta}_0(d, \eta, p) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there are random variables K_N with $\sup_N \|K_N\|_{L^p} < \infty$ such that, for every N and every grid cube q of side $r \geq 3^{-N}$,*

$$(8.7) \quad \Lambda_{\sigma/2,2}(q; A_N) + \lambda_{\sigma/2,2}^{-1}(q; A_N) \leq K_N r^{-\eta}.$$

Proof. Step 1. We prove (8.6). The change of variables $x = z + ry$ multiplies the unnormalized energy by r^{d-2} and leaves the volume-normalized constant $\Lambda_{\sigma/2,2}$ unchanged, so we may take $q = \square_0$. The space $C^\beta(\partial \square_0)$ embeds into $H^{1/2+\sigma}(\partial \square_0)$, and the trace operator has a bounded right inverse $H^{1/2+\sigma}(\partial \square_0) \rightarrow H^{1+\sigma}(\square_0)$; let h be the resulting extension of g . Applying (2.73) with zero right side, $s = \sigma$ and boundary datum h , we find that the A_N -harmonic function v with $v = h$ on $\partial \square_0$ satisfies

$$\Lambda_{N,\square_0}(g) = \|A_N^{1/2} \nabla v\|_{L^2}^2 \leq C \sigma^{-3} \Lambda_{\sigma/2,2}(\square_0; A_N) \|\nabla h\|_{H^\sigma}^2.$$

Step 2. We prove (8.7). Since L^p norms increase with p , we may assume that $p > 1$ and $p\eta > d$. Apply Lemma 8.2 with exponents $\sigma/2$ and 2. There are at most $C3^{dk}$ grid cubes at depth k , so taking the maximum over them costs a factor $C3^{dk/p}$ in L^p . Reduce $\bar{\delta}$ so that $C(p + p^2)\bar{\delta}^2 < 1/2(\eta - d/p) \log 3$. For K_N the supremum, over $k \leq N$ and grid cubes q of depth k , of $3^{-\eta k}(\Lambda_{\sigma/2,2}(q; A_N) + \lambda_{\sigma/2,2}^{-1}(q; A_N))$, Minkowski's inequality gives

$$\|K_N\|_{L^p} \leq C_p \sum_{k \geq 0} 3^{-(\eta - d/p)k} e^{C(p+p^2)\bar{\delta}^2 k} \leq C_p.$$

$\square$

8.2.2. Morrey bounds for the Dirichlet problem. To apply the regularity estimates with a bounded volume source, we first write that source as a divergence.

Lemma 8.5 (The gradient of the Newtonian potential). *Let $f \in L^\infty$ be supported in a cube of side R , and let g be the gradient of the Newtonian potential of f , that is, the convolution of f with the gradient of the fundamental solution of the Laplacian. Then $\nabla \cdot g = f$ and, for all x and y with $|x - y| \leq R$,*

$$(8.8) \quad |g(x) - g(y)| \leq C \|f\|_\infty |x - y| (1 + \log(R/|x - y|)) \leq CR^{1/2} \|f\|_\infty |x - y|^{1/2}.$$

In particular, $[g]_{W^{1/2,\infty}} \leq CR^{1/2} \|f\|_\infty$ on that cube.

Proof. The kernel of the convolution and its gradient are bounded by $C|z|^{1-d}$ and $C|z|^{-d}$. If x is farther than $2\sqrt{d}R$ from the supporting cube, the derivative bound gives $|g(x) - g(y)| \leq C \|f\|_\infty |x - y|$ directly. Otherwise the support lies in $B_{CR}(x)$. We split the convolution according to whether the integration variable z satisfies $|z - x| \leq 2|x - y|$. The part with $|z - x| \leq 2|x - y|$ is bounded by $C \|f\|_\infty |x - y|$, both at x and at y . For $|z - x| > 2|x - y|$, the mean value theorem bounds the

difference of the kernel at $x - z$ and at $y - z$ by $C|x - y||x - z|^{-d}$. The integral over $2|x - y| < |z - x| < CR$ is at most $C|x - y|\|f\|_\infty \log(CR/|x - y|)$. The second inequality in (8.8) follows from $(|x - y|/R)^{1/2}(1 + \log(R/|x - y|)) \leq C$. $\square$

Lemma 8.6 (Pointwise extremes of the coefficient). *Let Q be a fixed cube, and set*

$$D_N = 3^{-N} \|\nabla \log A_N\|_{L^\infty(Q)}.$$

For every p there are $C_p < \infty$, $C_d < \infty$ and $c_d > 0$ such that, if $\bar{\delta} \leq c_d/p$, then

$$(8.9) \quad \|D_N\|_{L^p} \leq C_p(1 + N)^{1/2},$$

$$\|A_N\|_{L^\infty(Q)} + \|A_N^{-1}\|_{L^\infty(Q)}\|_{L^p} \leq C_p e^{(C_d \bar{\delta} + C_p \bar{\delta}^2)N}.$$

The linear term $C_d \bar{\delta} N$ in the exponent arises from taking the maximum over the cubes of side 3^{-N} in Q .

Proof. Step 1. We bound D_N . We have $\nabla \log A_N = \nabla H + \sum_{j=0}^N \nabla \mathbf{g}_{-j}$ and, in law, $\|\nabla \mathbf{g}_{-j}\|_{L^\infty(Q)}$ is at most 3^j times the maximum, over at most $C3^{dj}$ cells, of the regularity norms on unit cells from (1.5). The maximum of n sub-Gaussian variables with parameter $\bar{\delta}$ has L^p norm at most $C\bar{\delta}(\sqrt{\log n} + \sqrt{p})$. Therefore

$$\|D_N\|_{L^p} \leq 3^{-N} \|\nabla H\|_{L^\infty(Q)}\|_{L^p} + \sum_{j \leq N} 3^{j-N} C_p \bar{\delta} \sqrt{1 + j} \leq C_p \sqrt{1 + N}.$$

Step 2. We bound $\|A_N\|_{L^\infty(Q)}$ and $\|A_N^{-1}\|_{L^\infty(Q)}$. The deterministic factor $\bar{\mathbf{a}}_N^{-1} e^{-(N+1)\tau^2}$ of A_N satisfies $1 \leq e^{(N+1)\tau^2} \bar{\mathbf{a}}_N \leq e^{C\bar{\delta}^2(N+1)}$ by (2.20), Proposition 5.7 and (1.6), and $e^{\|H\|_\infty}$ has moments of every order by Lemma 8.1. By Hölder's inequality, it therefore suffices to bound the moments of $\exp(\sup_Q |\sum_{j \leq N} \mathbf{g}_{-j}|)$. Cover Q by at most $C3^{dN}$ cubes Q_i of side 3^{-N} with centers x_i . On each of them, $\sup_{Q_i} |\sum_{j \leq N} \mathbf{g}_{-j}| \leq |\sum_{j \leq N} \mathbf{g}_{-j}(x_i)| + \sum_{j \leq N} \text{osc}_{Q_i} \mathbf{g}_{-j}$, where the first term is sub-Gaussian with parameter $C\bar{\delta}\sqrt{N+1}$ and the second is sub-Gaussian with parameter $C\bar{\delta}$. Hölder's inequality gives $\mathbb{E} \exp(q \sup_{Q_i} |\sum_{j \leq N} \mathbf{g}_{-j}|) \leq C e^{Cq^2 \bar{\delta}^2(N+1)}$. For $q \geq p$, the inequality $\exp(p \sup_Q |\sum_{j \leq N} \mathbf{g}_{-j}|) \leq (\sum_i \exp(q \sup_{Q_i} |\sum_{j \leq N} \mathbf{g}_{-j}|))^{p/q}$ and Jensen's inequality bound $\mathbb{E} \exp(p \sup_Q |\sum_{j \leq N} \mathbf{g}_{-j}|)$ by $(C3^{dN} e^{Cq^2 \bar{\delta}^2(N+1)})^{p/q}$. We choose $q = \bar{\delta}^{-1} \sqrt{d \log 3/C}$, which is admissible when $\bar{\delta} \leq c_d/p$; the exponent is then at most $C_d p \bar{\delta} (N+1)$. $\square$

Proposition 8.7 (Morrey bounds at all radii). *Let $t \in (d-1, d)$ and $\alpha \in (0, 1)$, fix a finite list of moment orders, and let Q be a fixed cube of side at most one. There is $\bar{\delta}_0(d, t, \alpha, \text{list}) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there are random variables K_N with $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list, such that, almost surely, the following estimates hold for every $f \in L^\infty(Q)$ and $\varphi \in C^2(\bar{Q})$. Here $u_N \in H^1(Q)$ solves $-\nabla \cdot (A_N \nabla u_N) = f$ in Q with $u_N = \varphi$ on ∂Q .*

(i) Morrey bound: for every $x \in Q$ and $r \in (0, 1]$,

$$(8.10) \quad \Gamma_N(u_N)(B_r(x) \cap Q) \leq K_N (\|f\|_\infty + \|\varphi\|_{C^2})^2 r^t.$$

(ii) Hölder bound: $\|u_N\|_{C^\alpha(\bar{Q})} \leq K_N (\|f\|_\infty + \|\varphi\|_{C^2})$ and, if $\varphi = 0$,

$$(8.11) \quad \|G_N^Q f\|_{C^\alpha(\bar{Q})} \leq K_N \|f\|_\infty.$$

Proof. Rescale so that $Q = y_0 + \square_0$. In the coordinates $y = 3^N x$ of the relabelled fields, the domain is $3^N y_0 + \square_N$, a deterministic translate of $\square_N$. Throughout the proof, K_N denotes a random variable with $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list, which may change from line to line. Choose the exponents as follows. Take $\alpha_M \in (1/2, 1)$ with $\alpha_M > \max\{\alpha, (t+2-d)/2\}$, and set $t_0 = d-2+2\alpha_M > t$. Choose t_1 with $\max\{t, d-2+2\alpha\} < t_1 < t_0$, and require $\bar{\delta}$ to be so small that $\alpha_M \leq 1 - C\bar{\delta} |\log \bar{\delta}|^{1/2}$.

Step 1. We replace H by a partial sum $\sum_{j=1}^{L'}(\mathbf{g}_j - \mathbf{g}_j(0))$ of its series, which makes the coefficient a constant multiple of a cutoff coefficient. In Steps 1–5 we keep the notation H , A_N , u_N and Γ_N for the truncated objects; the truncation is removed in Step 6. By (8.1) and (1.7),

$$A_N(3^{-N}y) = \bar{\mathbf{a}}_N^{-1} e^{L'\tau^2} \exp\left(-\sum_{i=N+1}^{N+L'} \mathbf{g}_i(0)\right) \mathbf{a}_{N+L'}(y),$$

so $A_N(3^{-N}\cdot)$ is a random scalar multiple of the cutoff coefficient $\mathbf{a}_{N+L'}$. In the coordinates y , the rescaled function $u_N(3^{-N}\cdot)$ solves $-\nabla \cdot (\mathbf{a}_{N+L'} \nabla u_N(3^{-N}\cdot)) = \nabla \cdot g$, where g is the vector field constructed in Lemma 8.5 for the source $f(3^{-N}\cdot)$, divided by 3^{2N} and by this random scalar and extended by zero. Moreover, $u_N(3^{-N}\cdot) = \varphi(3^{-N}\cdot) \in C^{1,1/2}$ on the boundary. Since $N + L' \geq N$, the estimate (6.6) applies with $m = N$; we write $\mathcal{L} = \mathcal{L}_{N+L'}(\alpha_M, N)$ for the minimal scale in that estimate.

Step 2. We prove the Morrey bound (8.10) at the radii $3^{-N} \leq r \leq 3^{-\mathcal{L}}$. For $x \in Q$ and $0 \leq n \leq N - \mathcal{L}$, we have $B_{3^{n-N/2}}(x) \cap Q \subseteq 3^{-N}((3^N x + \square_n) \cap (3^N y_0 + \square_N))$. Substituting the definitions of the random scalar and of $\mathbf{b}_{N+L',N}$, the constants $L'\tau^2$ cancel, and the random scalar times the average $(\mathbf{b}_{N+L',N})_{3^N y_0 + \square_N}$ equals $\oint_Q e^H dx$. We square (6.6), multiply it by the volume, which is at most 3^{dn} , and return to the x -coordinates. Using Lemma 8.5 with $R = 3^N$ and bounding $\oint_Q e^H dx$ and its inverse by $e^{\|H\|_{L^\infty(Q)}}$, we obtain, for $r = 3^{n-N/2}$,

$$(8.12) \quad \Gamma_N(u_N)(B_r(x) \cap Q) \leq C r^{t_0} \left(\Gamma_N(u_N)(Q) + e^{\|H\|_{L^\infty(Q)}} (\|f\|_\infty^2 + \|\varphi\|_{C^2}^2) \right).$$

The factor $e^{\|H\|_{L^\infty(Q)}}$ has moments bounded uniformly in N and L' by Lemma 8.1. The random variable $\mathcal{L}$ does not depend on x .

Step 3. We treat the remaining large radii $3^{-\mathcal{L}} < r \leq 1$ and bound the total energy. For these radii, $\Gamma_N(u_N)(B_r \cap Q) \leq \Gamma_N(u_N)(Q) \leq 3^{t_1 \mathcal{L}} r^{t_1} \Gamma_N(u_N)(Q)$. Together with Step 2, this gives a bound $K_N r^{t_1}$ at every radius $r \geq 3^{-N}$, also when $\mathcal{L} > N$. For an order p in the list and for the larger orders used in Hölder's inequality, (6.94) gives $\mathbb{E} 3^{pt_1 \mathcal{L}} < \infty$ as soon as

$$(8.13) \quad \frac{(1 - \alpha_M)^2}{C \bar{\delta}^2 |\log \bar{\delta}|} > p t_1 \log 3,$$

which is a condition on $\bar{\delta}$ alone once α_M , p and t_1 are fixed. The global energy has moments: by (8.3), (8.5) and (8.6),

$$\Gamma_N(u_N)(Q) \leq 2\Lambda_{N,Q}(\varphi) + 2\langle f, G_N^Q f \rangle \leq 2\Lambda_{N,Q}(\varphi) + 2K_N \|f\|_{L^2}^2,$$

uniformly in the truncation, since the truncated coefficient satisfies the same estimates.

Step 4. We prove the Morrey bound for $r \leq 3^{-N}$ on balls away from ∂Q . Let $p_1 < \infty$ satisfy $d - 2d/p_1 > t_1$ and $1 - d/p_1 > \alpha$, and let ℓ_N be $3^{-N}(1 + D_N)^{-1}$ times a small constant depending on d and p_1 , chosen below. For $x_0 \in Q$ with $B_{2\ell_N}(x_0) \subseteq Q$, the oscillation of $\log A_N$ on $B_{2\ell_N}(x_0)$ is at most $2\ell_N D_N / 3^{-N}$, hence at most twice the small constant. Dividing the equation by $A_N(x_0)$ gives an equation with coefficient $A_N/A_N(x_0)$, whose logarithm is bounded by twice the small constant on the ball, and source $f/A_N(x_0)$. By Meyers [1963, Theorem 2, equation (48)], the admissible gradient exponent tends to infinity as the ellipticity ratio tends to one. Choose the small constant so that the admissible exponent exceeds p_1 , subtract the mean of u_N , and apply that estimate with initial gradient exponent 2 and a finite source exponent greater than d . After scaling to the unit ball, this gives

$$\left(\int_{B_{\ell_N}(x_0)} |\nabla u_N|^{p_1} \right)^{1/p_1} \leq C \left(\int_{B_{2\ell_N}(x_0)} |\nabla u_N|^2 \right)^{1/2} + C \ell_N A_N(x_0)^{-1} \|f\|_\infty.$$

Hölder's inequality on $B_r(x_0)$, $r \leq \ell_N$, and the fact that A_N is within a fixed factor of $A_N(x_0)$ on $B_{2\ell_N}(x_0)$ yield, for every $r \leq \ell_N$,

$$(8.14) \quad \Gamma_N(u_N)(B_r) \leq C \left(\frac{r}{\ell_N} \right)^{d-2d/p_1} \Gamma_N(u_N)(B_{2\ell_N}) + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 \ell_N^{2+2d/p_1} r^{d-2d/p_1}.$$

We divide by r^t . Since $d - 2d/p_1 > t_1 > t$, the right side is largest at $r = \ell_N$, so

$$\Gamma_N(u_N)(B_r) r^{-t} \leq C(1 + D_N)^t 3^{tN} \Gamma_N(u_N)(B_{C3^{-N}}) + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 3^{-(d+2-t)N}.$$

For $\ell_N \leq r \leq 3^{-N}$, we use $\Gamma_N(u_N)(B_r) \leq \Gamma_N(u_N)(B_{C3^{-N}}) \leq (3^{-N}/\ell_N)^t r^t 3^{tN} \Gamma_N(u_N)(B_{C3^{-N}})$. Together with Step 3 at $r = C3^{-N}$, this gives, for every $r \leq 3^{-N}$,

$$\Gamma_N(u_N)(B_r) \leq C(1 + D_N)^t 3^{-(t_1-t)N} K_N r^t + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 3^{-(d+2-t)N} r^t.$$

By (8.9), the positive powers $3^{-(t_1-t)N}$ and $3^{-(d+2-t)N}$ absorb the polynomially growing moments of D_N and the moments of $\|A_N^{-1}\|_{L^\infty(Q)}$, once $\bar{\delta}$ is small enough for the finitely many larger moment orders used in Hölder's inequality.

Step 5. We prove the Morrey bound for $r \leq 3^{-N}$ near the faces of Q , where the Dirichlet condition holds. Let x_0 be within distance $2\ell_N$ of a face, say $\{x_d = 0\}$. The function $u_N - \varphi$ vanishes on the face and solves $-\nabla \cdot (A_N \nabla(u_N - \varphi)) = f + \nabla \cdot (A_N \nabla \varphi)$, where $\|\nabla \cdot (A_N \nabla \varphi)\|_\infty \leq C A_N(x_0)(\|D^2 \varphi\|_\infty + \ell_N^{-1} \|\nabla \varphi\|_\infty)$ on the ball around x_0 on which $\log A_N$ has small oscillation, as in Step 4. We reflect $u_N - \varphi$ oddly, A_N evenly and the source oddly across the face; the reflected function solves the reflected equation, with no additional source concentrated on the face. Indeed, for a compactly supported test function on the doubled ball, the difference of the test function and its reflection, restricted to Q , lies in H^1 , has zero trace on the face, and vanishes near the other faces, so it is an admissible test function for the original equation. The map that folds the reflected domain back onto Q is 1-Lipschitz, so the oscillation of the logarithm of the reflected coefficient on a ball B centered on the face is at most that of $\log A_N$ on $B \cap \bar{Q}$. Hence (8.14) applies to the reflected function and coefficient, since the proof of (8.14) uses only the boundedness and the small oscillation of the coefficient, not its differentiability across the face. Since the energy measure of the reflected function on a symmetric set B is twice the energy measure of $u_N - \varphi$ on $B \cap Q$, and

$$\Gamma_N(u_N - \varphi)(B_{C3^{-N}} \cap Q) \leq 2\Gamma_N(u_N)(B_{C3^{-N}} \cap Q) + C \|A_N\|_{L^\infty(Q)} \|\nabla \varphi\|_\infty^2 3^{-dN},$$

we obtain

$$\begin{aligned} \Gamma_N(u_N)(B_r \cap Q) &\leq C(r/\ell_N)^{d-2d/p_1} \Gamma_N(u_N)(B_{C3^{-N}} \cap Q) + C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty^2 \ell_N^{2+2d/p_1} r^{d-2d/p_1} \\ &\quad + C \|A_N\|_{L^\infty(Q)} \|\varphi\|_{C^2}^2 (1 + D_N)^d \ell_N^{2d/p_1} r^{d-2d/p_1}. \end{aligned}$$

Dividing the last term by r^t and maximizing at $r = \ell_N$ gives $C \|A_N\|_{L^\infty(Q)} \|\varphi\|_{C^2}^2 3^{-(d-t)N} (1 + D_N)^t$. The positive power $3^{-(d-t)N}$ absorbs its moments as in Step 4. Near an edge or a corner of Q , we reflect across each face meeting the neighborhood of x_0 . On a set invariant under these reflections, the energy of the reflected solution is two to the power of the number of these faces times the energy of the solution in the part of the set inside Q . Divide by this factor to obtain the estimates on the original cube.

Step 6. We remove the truncation. By Lemma 8.1, the truncated coefficients converge uniformly on $\bar{Q}$ to the untruncated A_N almost surely. For fixed N , they are therefore bounded away from zero once the truncation is sufficiently large. Testing the difference of the two weak equations with the difference of the solutions, which lies in $H_0^1(Q)$, bounds its energy by the uniform distance between the coefficients times the product of the L^2 norms of the gradient of the untruncated solution and of the gradient of the difference. The lower bound on the coefficients then shows that the gradients converge in $L^2(Q)$. Consequently, for every x and r , the truncated energies converge to $\Gamma_N(u_N)(B_r \cap Q)$. By Steps 2–5, these energies satisfy (8.10) with random constants whose L^p norms are bounded uniformly in N and in the truncation, for every order p in the list. Their limit

inferior gives a constant for the untruncated estimate, and Fatou's lemma preserves these moment bounds.

Step 7. We prove the Hölder bounds. At radii above 3^{-N} , (6.5) is a Campanato bound at every $x \in Q$, with exponent $\alpha_M > \alpha$ and the same source norm; the large radii $3^{-L} < r \leq 1$ are handled as in Step 3, and the L^2 oscillation on Q is bounded by the energy through (8.5). At radii below 3^{-N} , the W^{1,p_1} bound of Step 4 and Morrey's inequality bound the oscillation on B_{ℓ_N} . After division by r^α , the energy on the ball of radius $C3^{-N}$, the volume source and the boundary datum contribute, respectively,

$$\begin{aligned} & CK_N^{1/2} \|A_N^{-1}\|_{L^\infty(Q)}^{1/2} 3^{-((t_1-d+2)/2-\alpha)N} (1+D_N)^C, \\ & C \|A_N^{-1}\|_{L^\infty(Q)} \|f\|_\infty 3^{-(2-\alpha)N} (1+D_N)^C, \\ & C \|\varphi\|_{C^2} 3^{-(1-\alpha)N} (1+D_N)^C. \end{aligned}$$

All three powers of 3^{-N} are positive by the choice of the parameters. Chaining balls of radius ℓ_N inside a cube of side 3^{-N} costs another fixed power of $1+D_N$. Above the smallest field scale, telescoping the cube averages costs the geometric sum $\sum_{s \leq R} s^{\alpha_M} \leq CR^{\alpha_M}$. With the same absorption of moments as in Step 4, this proves the Hölder bounds. Approximate a general bounded source by smooth sources. Remove the truncation as in Step 6: the equicontinuity just proved and the convergence in L^2 give uniform convergence of the truncated solutions. $\square$

8.2.3. Approximating the Neumann datum by a volume source. We approximate the functional $v \mapsto \int_Q p \cdot \nabla v$, which encodes the Neumann datum $p \cdot n$, by a bounded volume source near the faces.

Let $Q = (0,1)^d$, let $p \in \mathbb{R}^d$ with $|p| = 1$, and let

$$L_p(v) = \int_Q p \cdot \nabla v \, dx = \int_{\partial Q} (p \cdot n) \operatorname{Tr} v \, d\mathcal{H}^{d-1},$$

where n is the outer normal. For $0 < \epsilon < 1/8$, let f_ϵ be the sum over $i = 1, \dots, d$ of p_i times the difference of two smooth bumps of integral one, the first supported in the layer $\{\epsilon < 1 - x_i < 2\epsilon\}$ near the face $\{x_i = 1\}$ and the second in the layer $\{\epsilon < x_i < 2\epsilon\}$ near the face $\{x_i = 0\}$, both obtained from one fixed profile by dilation by ϵ , and let

$$(8.15) \quad L_{p,\epsilon}(v) = \int_Q f_\epsilon v \, dx.$$

Each summand of f_ϵ has integral zero, $f_\epsilon \in C^\infty(\overline{Q})$, $\|f_\epsilon\|_\infty \leq C\epsilon^{-1}$, and $L_{p,\epsilon}(1) = 0$.

Lemma 8.8 (Error of the volume-source approximation of the Neumann datum). *Let $v_N, v_{N,\epsilon}$ be the mean-zero Riesz representer of $L_p, L_{p,\epsilon}$ with respect to E_N^Q , so that $E_N^Q(v_N) = p \cdot a_*^{-1}(Q; A_N)p$. For every $v \in H^1(Q)$,*

$$(8.16) \quad |L_p(v) - L_{p,\epsilon}(v)| \leq C\epsilon^{1/4} \|v - (v)_Q\|_{H^{3/4}(Q)}.$$

With K_N from (8.5), this gives

$$(8.17) \quad E_N^Q(v_N - v_{N,\epsilon}) \leq CK_N \epsilon^{1/2}.$$

For every moment order q in the list, if $\bar{\delta}$ is small enough,

$$(8.18) \quad \|E_N^Q(v_N) - E_N^Q(v_{N,\epsilon})\|_{L^q} \leq C_q \epsilon^{1/4}, \quad \sup_{N,\epsilon} \|E_N^Q(v_{N,\epsilon})\|_{L^q} < \infty.$$

Proof. Subtract $(v)_Q$ and extend to $H^{3/4}(\mathbb{R}^d)$. Fourier transformation in the normal variable and Cauchy-Schwarz give, on each face,

$$\|v(\tau, \cdot) - v(0, \cdot)\|_{L^2(\mathbb{R}^{d-1})}^2 \leq \|v\|_{H^{3/4}(\mathbb{R}^d)}^2 \int_{\mathbb{R}} \frac{\min\{4, \tau^2 \xi^2\}}{(1 + \xi^2)^{3/4}} \, d\xi \leq C\tau^{1/2} \|v\|_{H^{3/4}(\mathbb{R}^d)}^2.$$

Integrate against the bump profile and sum over the faces to get (8.16). Test the difference of the two weak equations with $v_N - v_{N,\epsilon}$ and use coercivity to obtain (8.17). The reverse triangle inequality for the energy norm then gives

$$|E_N^Q(v_N) - E_N^Q(v_{N,\epsilon})| \leq CK_N^{1/2} \epsilon^{1/4} E_N^Q(v_N)^{1/2} + CK_N \epsilon^{1/2}.$$

Here $E_N^Q(v_N) = p \cdot a_*^{-1}(Q; A_N)p$ has moments of the orders in the list bounded uniformly in N by Lemma 8.2. Hölder's inequality proves (8.18). $\square$

8.2.4. Reflection across the faces. Let a be a scalar coefficient, and let u be a solution with zero conormal derivative and bounded source f . Reflect u and f evenly across a set of faces. This gives a solution $u \circ T$ of the equation with coefficient $a \circ T$ and source $f \circ T$, where T folds the reflected domain onto the original cube. The reflected equation has no additional source concentrated on the folding planes. On a set invariant under these reflections, the energy of $u \circ T$ is two to the power of the number of reflected faces times the energy of u in the part of the set inside the original cube.

Lemma 8.9 (Effect of folding on $\mathcal{E}_{s,\infty,2}$). *Let T fold the cube $z + \square_M$ across coordinate planes through z , as above. For every continuous positive coefficient a on its closure, scalar $a_0 > 0$, $s \in (0, 1/2)$ and $n \leq M$,*

$$(8.19) \quad \mathcal{E}_{s,\infty,2}(z + \square_M, n; a \circ T, a_0)^2 \leq \left(1 + \frac{3d}{3^{1-2s} - 1}\right) \mathcal{E}_{s,\infty,2}(z + \square_M, n; a, a_0)^2.$$

Proof. Write $\kappa_n(a)$ for the maximum, over grid cubes $U \subseteq z + \square_M$ of side 3^n , of $\sup_{|e|=1} J(U, e, e; a/a_0)$. On a cube meeting no folding plane, T reflects some of the coordinates, so changing variables by T shows that the energy for $a \circ T$ is at most $\kappa_n(a)$. Subdivide each cube that meets a folding plane triadically, and repeat the subdivision on the subcubes that still meet one. After j subdivisions, the subcubes that still meet a folding plane fill at most a fraction $d3^{-j}$ of the cube. By subadditivity (2.22), $\kappa_n(a \circ T) \leq \kappa_n(a) + \sum_{j \geq 1} 3d3^{-j} \kappa_{n-j}(a)$. Insert this inequality into Definition 2.3 and sum the geometric series $\sum_{j \geq 1} 3^{-(1-2s)j}$ to obtain (8.19). $\square$

Proposition 8.10 (Energy decay for the folded coefficient). *Let γ be an exponent as in Proposition 6.1, let $m, L \in \mathbb{N}_0$ with $m \leq L$, and let T fold the cube $z + \square_m$ across coordinate planes through z . Let $\mathcal{L}_L(\gamma, m)$ be the minimal scale of Section 6.3 for the cube $z + \square_m$, defined with the parameter choices (6.59) of the proof of Proposition 6.1, but with the tolerance in the good events reduced by a fixed dimensional factor. If $u \in H^1(z + \square_m)$ solves $-\nabla \cdot ((\mathbf{a}_L \circ T) \nabla u) = \nabla \cdot \tilde{g}$, with $\tilde{g} \in C^{0,1/2}(z + \square_m; \mathbb{R}^d)$, then, for $0 \leq n \leq m - \mathcal{L}_L(\gamma, m)$,*

$$(8.20) \quad \|(\mathbf{a}_L \circ T)^{1/2} \nabla u\|_{\underline{L}^2(z + \square_m)} \leq C 3^{(1-\gamma)(m-n)} \left(\|(\mathbf{a}_L \circ T)^{1/2} \nabla u\|_{\underline{L}^2(z + \square_m)} + (\mathbf{b}_{L,m})_{z + \square_m}^{-1/2} 3^{m/2} [\tilde{g}]_{W^{1/2,\infty}(z + \square_m)} \right).$$

Proof. Apply the argument proving (6.6) at z , using the averages $(\mathbf{b}_{L,j+2})_{z + \square_{j+2}}$ of the original field. The harmonic approximation and excess decay use the good event only through the quantity $\mathcal{E}_{s,\infty,2}$ of Definition 2.3, as explained in the proof of Lemma 6.15. By (8.19), folding multiplies this quantity by at most a dimensional constant. The bounds with $q = 1$ at larger exponents follow from the bound with $q = 2$ at s_0 by a weighted Cauchy–Schwarz inequality. The reduced tolerance absorbs the dimensional constant. The sum of the errors over the good scales is still bounded by (6.71). The bound (6.74) on the number of bad scales and the ratios (6.64) of these averages involve the original field and are unchanged. These bounds give the inputs to the iteration of Lemma 6.10. Finally, (2.52) supplies the coarse-grained ellipticity constants used in Step 6 of that proof. All cubes in the iteration are centered at z , hence invariant under the reflections. $\square$

Proposition 8.11 (Bounded Neumann sources). *Let $\alpha \in (0, 1)$ and $t \in (d - 1, d)$, and fix a finite list of moment orders. There is $\bar{\delta}_0(d, \alpha, t, \text{list}) > 0$ such that, if $\bar{\delta} \leq \bar{\delta}_0$, then on every fixed cube Q*

there are random variables K_N with $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list, such that the following holds for every $f \in L^\infty(Q)$ with $\int_Q f = 0$. If $u_N \in H^1(Q)$ is the mean-zero solution of

$$\int_Q A_N \nabla u_N \cdot \nabla v = \int_Q f v \quad (v \in H^1(Q)),$$

then, for every $x \in Q$ and $0 < r \leq 1$,

$$\|u_N\|_{C^\alpha(\overline{Q})} \leq K_N \|f\|_\infty, \quad \Gamma_N(u_N)(B_r(x) \cap Q) \leq K_N \|f\|_\infty^2 r^t.$$

Proof. Rescale to the unit cube. Choose $t_0 \in (\max\{t, d - 2 + 2\alpha\}, d)$ and $\eta > 0$ such that $t_1 = t_0 - \eta > \max\{t, d - 2 + 2\alpha\}$. Replace H by a partial sum as in Proposition 8.7; the identity relating the random scalar and the averages of $\mathbf{b}_{N+L'}$, over cubes to e^H gives the same averages and the same scaling of the source as in that proposition. The coercivity bound for mean-zero functions gives $E_N^Q(u_N) + \|u_N\|_2^2 \leq CK_N \|f\|_\infty^2$.

Step 1. We control the energy near the boundary by repeated reflections. At radii above 3^{-N} , we bound the energy in a target cube of side r by the energy in a cube of fixed size, through at most $d + 1$ applications of Proposition 8.10 with energy exponent t_0 . Fix a small radius of the form $3^j/2$, with j an integer. Starting at x , take ρ comparable to r , namely the radius of the smallest ball around the current center that contains the target cube. Whenever a face lies within a fixed multiple of ρ and the domain has not yet been reflected across that face, reflect across it and enlarge ρ by a dimensional factor. At most one reflection is needed in each coordinate. We stop when ρ reaches a fixed small fraction of the chosen radius. Between consecutive reflections, apply the energy decay estimate for the reflected coefficient down to the current value of ρ . The initial radius is the largest radius of the form $3^j/2$ that is at most the chosen radius and at most a fixed small fraction of the distance to the nearest face across which the domain has not been reflected. Each application contributes the t_0 th power of the ratio of its final to its initial radius. The product of these factors is at most Cr^{t_0} , where C also absorbs the bounded factors from the new reflections and the final step.

If, in the i th application of Proposition 8.10, B_i is the number of bad scales, b_i the average of $\mathbf{b}_{L,m}$ over the cube of the initial radius and R_i the initial radius, this gives

$$\Gamma_N(u_N)(B_r(x) \cap Q) \leq Ce^{C \sum_i B_i} r^{t_0} \left[E_N(u_N) + \|f\|_\infty^2 \sum_i b_i^{-1} R_i^{d+2-t_0} \right].$$

The source term follows from (8.8).

Step 2. For each N , we find one random constant that serves all radii and centers. At depth n , we run the construction of Step 1 from the points of a mesh of spacing proportional to 3^{-n} ; every ball of radius comparable to 3^{-n} lies in a target ball centered at a mesh point, once the radius of the latter is enlarged by a fixed factor. There are at most $C(n+1)^{C_d} 3^{dn}$ such constructions. The exponential tail (6.94) and Hölder's inequality over the at most $d+1$ numbers B_i of bad scales give uniformly bounded moments of $e^{C \sum_i B_i}$ of any order larger than d/η , after the moment order and then $\bar{\delta}$ are chosen. Summing over the mesh, as in Lemma 8.4(ii), bounds all these factors by $K_N r^{-\eta}$, where K_N satisfies $\sup_N \|K_N\|_{L^p} < \infty$ for every order p in the list.

The averages b_i are treated in the same way: each is taken at the initial radius of its own application, and we sum over meshes of spacing proportional to that radius. Their positive and negative moments are bounded as in the proof of Lemma 8.2. The same moment summation shows that, after the moment order and then $\bar{\delta}$ are chosen, each b_i^{-1} is at most K_N times a negative power of R_i whose exponent is smaller than $d + 2 - t_0$. Since there are at most $d + 1$ applications, Step 1 gives $\Gamma_N(u_N)(B_r \cap Q) \leq K_N \|f\|_\infty^2 r^{t_1}$ for $r \geq 3^{-N}$.

Step 3. We extend the estimates below the smallest field scale and deduce Hölder continuity. At radii below 3^{-N} , even reflection does not increase the Lipschitz constant of $\log A_N$. Apply the Meyers estimate for coefficients with small oscillation (8.14) at the radius ℓ_N of Step 4 of

Proposition 8.7, with p_1 large enough that $d - 2d/p_1 > t_1$ and $1 - d/p_1 > \alpha$. Matching this bound at radius 3^{-N} with the bound of Step 2 gives exactly the energy estimate of Step 4 of Proposition 8.7, with t_1 as chosen here. The positive powers $3^{-(t_1-t)N}$ and $3^{-(d+2-t)N}$ in that estimate absorb the moments of the coefficient bounds in (8.9). At radii above 3^{-N} the iteration of Lemma 6.10 for the reflected coefficient also bounds oscillations; at smaller radii Morrey's inequality applies. The powers of 3^{-N} in the first two terms of Step 7 of that proposition are positive because $t_1 > d - 2 + 2\alpha$. The same chaining proves the C^α bound. Finally, remove the truncation of H as in Step 6 of that proposition, by comparing the weak equations on the space of mean-zero functions and applying Fatou's lemma. $\square$

Applying Proposition 8.11 to the mean-zero source f_ϵ of (8.15), with $\|f_\epsilon\|_\infty \leq C\epsilon^{-1}$, gives, with one random constant for all ϵ ,

$$(8.21) \quad \Gamma_N(v_{N,\epsilon})(B_r(x) \cap Q) \leq K_N \epsilon^{-2} r^t.$$

8.3. Compactness of the finite-volume energies. We approximate each finite-volume energy by its conditional expectation given finitely many fields, and we bound the error by resampling one field at a time. These conditional expectations form a compact family in $L^p(\mathbb{P})$.

On a fixed cube Q , write $E_N = E_N^Q$. By a *finite-volume energy* $\mathcal{R}_N$ we mean either the minimal energy $\Lambda_{N,Q}(\varphi)$ with a nonconstant smooth boundary datum φ , or the quadratic form $\langle f, G_N^Q f \rangle$ of the Dirichlet Green operator for a nonzero smooth source f . Write $\mathcal{R}(g)$ for the same energy computed with the coefficient whose logarithm is g , so that $\mathcal{R}_N = \mathcal{R}(\log A_N)$.

8.3.1. Local perturbations of the coefficient.

Lemma 8.12 (Multiplicative perturbations). *Let A be bounded above and below by positive constants, let g be bounded and supported in $B \subseteq Q$. Write $E^g(v) = \int_Q e^g A |\nabla v|^2$ and $E = E^0$. Let u and u^g be the minimizers for the coefficients A and $e^g A$ with the same Dirichlet boundary values, or the solutions for these coefficients with the same linear right-hand side and zero Dirichlet or zero conormal boundary condition; in the conormal case, the right-hand side vanishes on constants and the solutions have mean zero. Denote their energies by $\mathcal{R}, \mathcal{R}^g$. Then*

$$(8.22) \quad \begin{aligned} E^g(u^g - u) &\leq \|g\|_\infty^2 e^{3\|g\|_\infty} \Gamma_E(u)(B), & \Gamma_{E^g}(u^g)(B) &\leq 4e^{\|g\|_\infty} \Gamma_E(u)(B), \\ |\mathcal{R}^g - \mathcal{R}| &\leq 2\|g\|_\infty e^{4\|g\|_\infty} \Gamma_E(u)(B). \end{aligned}$$

For two log coefficients g, h and any of these energies,

$$(8.23) \quad e^{-\|g-h\|_\infty} \mathcal{R}(h) \leq \mathcal{R}(g) \leq e^{\|g-h\|_\infty} \mathcal{R}(h).$$

Proof. Set $v = u^g - u$. Subtracting the Euler equations gives $E^g(v, \phi) = -\int_B (e^g - 1) d\Gamma_E(u, \phi)$. Test with v and apply Cauchy–Schwarz to obtain

$$(8.24) \quad E^g(v)^{1/2} \leq (e^{\|g\|_\infty/2} - e^{-\|g\|_\infty/2}) \Gamma_E(u)(B)^{1/2}.$$

This proves the first bound in (8.22); the triangle inequality in $L^2(B, e^g A dx)$ gives the second estimate. The Euler equations also give

$$\mathcal{R}^g - \mathcal{R} = \pm \left[\int_B (e^g - 1) d\Gamma_E(u) - E^g(v) \right],$$

with plus sign when $\mathcal{R}$ is a minimal energy with given boundary values and minus sign when it is the energy of a solution with given right-hand side. Using $|e^g - 1| \leq \|g\|_\infty e^{\|g\|_\infty}$ proves the third bound. Monotonicity in the coefficient and scaling by constants give (8.23). These arguments use only the Euler equations and the Cauchy–Schwarz inequality for energy measures, so they also apply to the regular strongly local forms constructed below. $\square$

8.3.2. Resampling one random field. We bound the change in a finite-volume energy when a field $\mathbf{g}_{-j}$ with $j \geq 0$ is replaced by an independent copy. The field has scale $r = 3^{-j}$ and range of dependence $\sqrt{d}r$, so its restrictions to boxes at mutual distance larger than $\sqrt{d}r$ are independent. We partition Q into boxes of side ℓ with $r \ll \ell \ll 1$. Delete the field on strips of width of order r around the faces of the boxes, and apply the Efron–Stein inequality to the independent restrictions of the field to the remaining interiors of the boxes. The energy in the strips controls the error caused by deleting the field there. The largest energy in one box controls the differences $X - X^{(i)}$ in the Efron–Stein inequality (8.26) below. By the Morrey bound (8.10), both are bounded by a positive power of the scale r . Exponential moments of the sup norm of the field enter through Lemma 8.12; they grow more slowly than any negative power of the scale r , so this decay absorbs them.

Lemma 8.13 (Mass near the box faces and in a single box). *Let $t > d - 1$ and $K < \infty$, and let ν be a finite positive measure on a fixed cube Q such that $\nu(B_\rho(x) \cap Q) \leq K\rho^t$ for every $x \in Q$ and $0 < \rho \leq 1$. For $4\sqrt{d}r < \ell \leq 1$, partition Q into boxes of side ℓ , and call **strips** the points within distance $4\sqrt{d}r$ of a box face. There is $C(d, t, Q) < \infty$ such that*

$$\nu(\text{strips}) \leq CK\ell^{-1}r^{t-d+1}, \quad \max_{B \text{ box}} \nu(B) \leq CK\ell^t.$$

With $\ell = r^{(t-d+1)/(t+1)}$ and sufficiently small r , both bounds are CKr^b , where $b = t(t-d+1)/(t+1) > 0$.

Proof. There are at most $C_Q\ell^{-1}$ face hyperplanes. Each strip is covered by C_Qr^{1-d} balls of radius $4\sqrt{d}r$, and each box by a dimensional number of balls of radius ℓ . Apply the bound $\nu(B_\rho(x) \cap Q) \leq K\rho^t$ and use $\ell^{-1}r^{t-d+1} = \ell^t = r^b$ for the stated choice of ℓ . $\square$

The sup norm on Q of a fine-scale field grows only logarithmically in the inverse of its scale, and its exponential moments grow more slowly than any power of the inverse scale. By (1.5), the variable $\|\mathbf{g}_0\|_{L^\infty(\square_0)}$ is sub-Gaussian with parameter $C\bar{\delta}$. Let $j \geq 0$ and $r = 3^{-j}$, and let $S = \|\mathbf{g}_{-j}\|_{L^\infty(Q)}$. The field $\mathbf{g}_{-j}$ has the law of $\mathbf{g}_0(3^j \cdot)$. Cover 3^jQ by at most Cr^{-d} unit cubes to obtain $\mathbb{P}[S > s] \leq 2Cr^{-d} \exp(-s^2/C\bar{\delta}^2)$. The union bound and $(s+a)^2 \geq s^2 + a^2$ show that the excess of S over $\bar{\delta}\sqrt{C(1+|\log r|)}$ has the tail $\exp(-s^2/C\bar{\delta}^2)$. Thus the excess is sub-Gaussian with parameter $C\bar{\delta}$, uniformly in r . Consequently, for all $k \geq 0$, $\lambda \geq 0$, $p < \infty$ and $\gamma > 0$, and every $0 < \bar{\delta} \leq 1$,

$$(8.25) \quad \|S^k e^{\lambda S}\|_{L^p} \leq C_{p,k,\lambda} \bar{\delta}^k (1 + |\log r|)^{k/2} \exp(C\lambda\bar{\delta}\sqrt{1+|\log r|} + C_p\lambda^2\bar{\delta}^2) \leq C_{p,k,\lambda,\gamma} \bar{\delta}^k r^{-\gamma}.$$

The splitting of S into a deterministic logarithmic part and a sub-Gaussian excess is needed: a union bound applied directly to the exponential moment would produce the factor Cr^{-d} .

The fluctuations are estimated with the following moment form of the Efron–Stein inequality. Let $\xi_1, \dots, \xi_m$ be independent random elements, let $p \geq 2$, and let $X = X(\xi_1, \dots, \xi_m) \in L^p$ be measurable. Write $X^{(i)}$ for the variable obtained by replacing ξ_i with an independent copy, and $\mathbb{E}_i$ for the integral over this copy alone. Then

$$(8.26) \quad \|X - \mathbb{E}X\|_{L^p} \leq C_p \left\| \left(\sum_i \mathbb{E}_i[(X - X^{(i)})^2] \right)^{1/2} \right\|_{L^p}.$$

This is a consequence of Boucheron, Bousquet, Lugosi, and Massart [2005, Theorem 2], who proved two one-sided bounds in terms of variance proxies; both proxies are dominated by the sum on the right side of (8.26). The inequality also holds conditionally on every σ -field independent of the random elements and their copies.

Lemma 8.14 (Resampling one random field). *Let $0 < \bar{\delta} \leq 1$, $t \in (d-1, d)$ and $p \geq 2$, and let $\mathcal{R}_N$ be a finite-volume energy. Assume that, for every N , the function u_N defining $\mathcal{R}_N$ (the minimizer*

with the given boundary values, or the solution with zero Dirichlet boundary values) satisfies (8.10) on Q with the constant K_N , and that

$$\sup_N \|K_N\|_{L^{3p}} + \sup_N \|\mathcal{R}_N\|_{L^{3p}} < \infty.$$

For $0 \leq j \leq N$, let $\mathcal{R}_N^{(j)}$ be the finite-volume energy obtained after replacing the field $\mathbf{g}_{-j}$ by an independent copy. Let b be the exponent of Lemma 8.13. Then there exists $C_p < \infty$, depending on p, d, t, Q , on the boundary datum or source defining $\mathcal{R}_N$ and on the two suprema in the previous display, such that, for all integers $0 \leq j \leq N$,

$$(8.27) \quad \|\mathcal{R}_N - \mathcal{R}_N^{(j)}\|_{L^p} \leq C_p \bar{\delta} 3^{-bj/8}.$$

Proof. Put $r = 3^{-j}$ and let $S = \|\mathbf{g}_{-j}\|_\infty$ on Q , as in (8.25). A prime denotes the corresponding quantity after this field is resampled. Throughout, expectations over this field are conditional on the other fields. For the finitely many scales at which the strip construction of Lemma 8.13 does not apply, the bound $|\mathcal{R}_N - \mathcal{R}'_N| \leq (S + S')e^{S+S'}\mathcal{R}_N$ and (8.25) suffice.

At the remaining scales, partition Q into boxes of side $\ell = r^{(t-d+1)/(t+1)}$. Choose smooth cutoff functions χ_i , one for each box, supported on interior subboxes that are separated by more than $\sqrt{d}r$, such that $\chi = \sum_i \chi_i$ equals one outside the strips of width $4\sqrt{d}r$. Let $\widehat{\mathcal{R}}_N$ be the energy after deleting $(1 - \chi)\mathbf{g}_{-j}$. Let $\widehat{\nu}$ be the energy measure of the corresponding solution. The restrictions of the field to these subboxes are independent. Lemmas 8.12 and 8.13 give

$$|\mathcal{R}_N - \widehat{\mathcal{R}}_N| \leq CSe^{CS}K_N r^b, \quad \max_B \widehat{\nu}(B) \leq Ce^{CS}K_N r^b, \quad \widehat{\nu}(Q) \leq e^S \mathcal{R}_N,$$

where the maximum is over the interior subboxes B . For the middle bound, apply (8.24) on the strips and the inequality that $\widehat{\nu}(B)$ is at most $2e^S \Gamma_N(u_N)(B)$ plus twice the energy, for the form with the deleted field, of the difference between the minimizers before and after the deletion. The second term bounds the change in the minimizer when replacing $\mathbf{g}_{-j}$ by $\chi\mathbf{g}_{-j}$.

Resampling the field on an interior subbox B changes $\widehat{\mathcal{R}}_N$ by at most $C(S + S')e^{C(S+S')}\widehat{\nu}(B)$. Since $\sum_B \widehat{\nu}(B)^2 \leq \max_B \widehat{\nu}(B)\widehat{\nu}(Q)$, the conditional Efron–Stein inequality (8.26) and conditional Jensen give

$$\|\widehat{\mathcal{R}}_N - \mathbb{E}_j \widehat{\mathcal{R}}_N\|_p \leq C_p r^{b/2} \|(S + S')e^{C(S+S')}K_N^{1/2}\mathcal{R}_N^{1/2}\|_p \leq C_p \bar{\delta} r^{3b/8}.$$

The last step uses Hölder’s inequality with three factors and (8.25), at a total loss of $r^{-b/8}$. The errors $\mathcal{R}_N - \widehat{\mathcal{R}}_N$ and $\widehat{\mathcal{R}}'_N - \mathcal{R}'_N$ from deleting the field on the strips are at most $C_p \bar{\delta} r^{7b/8}$ in L^p . The energy $\widehat{\mathcal{R}}'_N$, computed with the cutoff function applied to the resampled field, has the same conditional mean as $\widehat{\mathcal{R}}_N$, so

$$(8.28) \quad \mathcal{R}_N - \mathcal{R}'_N = (\mathcal{R}_N - \widehat{\mathcal{R}}_N) + (\widehat{\mathcal{R}}_N - \mathbb{E}_j \widehat{\mathcal{R}}_N) - (\widehat{\mathcal{R}}'_N - \mathbb{E}_j \widehat{\mathcal{R}}'_N) + (\widehat{\mathcal{R}}'_N - \mathcal{R}'_N).$$

Since $r^{3b/8} \leq 3^{-bj/8}$, this proves (8.27). $\square$

The following martingale bound adds up the effects of the fields on which we do not condition.

Lemma 8.15 (Summing resampling bounds). *Let $(\xi_j)_{j \in I}$ be a countable independent family and let $X \in L^p$, $1 \leq p < \infty$, be measurable with respect to this family. Write $X^{(j)}$ for the result of replacing ξ_j by an independent copy. For every $J \subseteq I$,*

$$\|X - \mathbb{E}[X \mid \sigma(\xi_j : j \in J)]\|_p \leq \sum_{j \notin J} \|X - X^{(j)}\|_p.$$

Proof. Condition first on the retained variables, and then reveal the omitted ones one at a time, in some enumeration. At the step that reveals ξ_j , the martingale difference is the conditional expectation of $X - X^{(j)}$ given the variables revealed so far. Conditional Jensen bounds its L^p norm by $\|X - X^{(j)}\|_p$. Sum and use L^p martingale convergence. $\square$

We apply this bound to approximate an energy by conditioning on finitely many fields.

Proposition 8.16 (Approximation by conditioning on finitely many fields). *Let $\ell \geq 0$ be an integer, and let b be the exponent of Lemma 8.13. Let $\mathcal{R}_N$ be a finite-volume energy satisfying the hypotheses of Lemma 8.14 for every $j \geq 0$. Then there exists $C_p < \infty$ such that, for every $N \geq \ell$,*

$$(8.29) \quad \|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_{L^p} \leq C_p \bar{\delta} 3^{-b\ell/8}.$$

If instead $\mathcal{R}_N = |Q|e \cdot a_^{-1}(Q; A_N)e$ is the Neumann energy for the constant flux e , a unit vector, assume (8.18) and (8.21) with moments of order $3p$. Then (8.29) holds with right side $C_p \bar{\delta}^{1/2} 3^{-b\ell/256}$.*

Proof. For $j \geq 0$, (8.27) bounds the resampling error by $C_p \bar{\delta} 3^{-bj/8}$, and the error is zero for $j > N$. For $j < 0$, the field H changes by the difference of the expressions $\mathbf{g}_{-j} - \mathbf{g}_{-j}(0)$ for the field and for its independent copy. This difference has norm of size $C \bar{\delta} 3^j$ and sub-Gaussian tails by Lemma 8.1. Equation (8.23) bounds the resampling error by $C_p \bar{\delta} 3^j$. Sum both geometric series using Lemma 8.15.

For the Neumann energy, first use the volume source f_ϵ . The proof of Lemma 8.14 applies because Lemma 8.12 also covers zero-conormal problems with a given right-hand side. By (8.21), the constant in the Morrey bound (8.10) for the solution with source f_ϵ is $K_N \epsilon^{-2}$; this factor enters the error from deleting the field on the strips linearly and the Efron–Stein bound through its square root. Together with (8.18), this gives

$$\|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_p \leq C_p (\epsilon^{1/4} + \bar{\delta} \epsilon^{-2} 3^{-b\ell/8} + \bar{\delta} 3^{-\ell}).$$

Choose $\epsilon = 3^{-b\ell/32}/16$ to get $C_p 3^{-b\ell/128}$. We also bound the left side by $C_p \bar{\delta}$. Remove the field H . The bound (2.41) and Proposition 4.1 put the Neumann energy for the constant flux e within $C_p \bar{\delta}$ of 1 in L^p . Restoring H costs another $C_p \bar{\delta}$ by (8.23) and Lemma 8.1. Thus $\|\mathcal{R}_N - \mathbb{E}[\mathcal{R}_N \mid \mathbf{g}_{-j}, |j| \leq \ell]\|_p \leq 2\|\mathcal{R}_N - 1\|_p \leq C_p \bar{\delta}$. The geometric mean of the two bounds proves the assertion. $\square$

8.3.3. Compactness of conditional expectations.

Proposition 8.17 (Strong compactness of the finite-volume energies). *Let Q be a cube, and let $\mathcal{R}_N$ be the minimal energy $\Lambda_{N,Q}(\varphi)$ with boundary values φ , the quadratic form $\langle f, G_N^Q f \rangle$ of the Dirichlet Green operator with source f , or the Neumann energy $|Q|e \cdot a_*^{-1}(Q; A_N)e$ for a constant flux $e \in \mathbb{R}^d$. Assume that $\mathcal{R}_N$ satisfies the conclusion of Proposition 8.16 and that $\sup_N \|\mathcal{R}_N\|_{L^q} < \infty$ for some $q > p$. Then the following statements hold.*

- (i) *The sequence $(\mathcal{R}_N)_N$ is relatively compact in $L^p(\mathbb{P})$.*
- (ii) *The same holds for $\langle f, G_N^Q g \rangle$ with two sources f and g .*
- (iii) *For any countable collection of cubes, sources, boundary data and fixed integer shifts $N \mapsto N + j$ of the cutoff level N , a common subsequence can be chosen along which all the corresponding energies converge.*

Proof. Fix ℓ and $N \geq \ell$. The log coefficient on $\bar{Q}$ is the sum of the retained potential $\sum_{-\ell \leq j < 0} (\mathbf{g}_{-j} - \mathbf{g}_{-j}(0)) + \sum_{0 \leq j \leq \ell} \mathbf{g}_{-j}$ and a remaining potential, which is independent of the retained fields. Consider the function of a potential $v \in C(\bar{Q})$ given by the expectation of the energy $\mathcal{R}$ of the log coefficient v plus the remaining potential. By independence, the conditional expectation of $\mathcal{R}_N$ given the retained fields is the value of this function at the retained potential. By (8.23), its values at two potentials v and w differ at most by the factor $e^{\|v-w\|_\infty}$. Choose a radius such that the retained potential has sup norm at most this radius with positive probability. Integrate the comparison over the event that the retained potential has sup norm at most this radius to bound the value of the function at 0, uniformly in N , by the product of $\mathbb{E}\mathcal{R}_N$ and the exponential of this radius, divided by the probability of this event. The resulting functions are finite, uniformly bounded and equicontinuous on every bounded set.

A diagonal extraction on a countable dense subset of $C(\overline{Q})$ gives pointwise convergence on the whole space. Conditional Jensen gives a uniform L^q bound on the conditional expectations, so this convergence is in $L^p(\mathbb{P})$.

For any $\varepsilon > 0$, choose ℓ so that the error in (8.29) is below $\varepsilon/2$. A finite $\varepsilon/2$ -net of the conditional expectations is then an ε -net of the energies with $N \geq \ell$. This proves (i). Polarization gives (ii); a diagonal extraction gives (iii). $\square$

8.4. Subsequential limit forms. We construct limit forms on $L^2(Q, dx)$ along a subsequence and show that they are local, compatible under restriction to subcubes, and multiplied by a continuous weight when the coefficient is.

Let $\mathcal{C}$ be the set of cubes with rational centers and triadic side lengths. It contains the cubes $\square_j$, $j \geq 0$, and all their triadic subcubes. Fix $1/2 < \beta < \alpha < 1$, $d-1 < t < d$ and $0 < \eta < 2\alpha - 1$. Choose a countable $\mathbb{Q}$ -linear family $\mathcal{D} \subseteq C_c^\infty(\mathbb{R}^d)$ such that, for each cube, the functions of $\mathcal{D}$ supported in the cube are dense in L^2 and in C_c^∞ on that cube. This also gives density in C_c^α for the C^β norm. For each finite union of closed rational cubes contained in a finite union of open rational cubes, include in $\mathcal{D}$ a smooth cutoff function, that is, a function with values in $[0, 1]$ equal to one near the first set and zero outside the second. Also include the smooth functions used to build the cutoff functions $\chi_{N,r}$ of Lemma 8.21(iii).

The **harmonic interpolant** of a function on a mesh is the function equal, on each cell, to the A_N -harmonic extension of the restriction of that function to the cell boundary.

We extract a subsequence jointly for the finite-volume energies for these cubes and data, the solutions, the environment and the random constants in (8.5)–(8.11). Include as single random variables each constant of (8.7), which is valid on all grid cubes at once, the constants of Proposition 8.7, which do not depend on the source, and $3^{-\eta N} \|A_N\|_{L^\infty(Q)}$, whose moments are bounded by Lemma 8.6. Proposition 8.17 gives L^p convergence of the energies along a subsequence. The moment and Hölder bounds give tightness of all the remaining coordinates.

Apply Prokhorov's theorem and Skorokhod's representation theorem [Kallenberg, 2002, Theorems 16.3 and 4.30] in the countable product of the Polish coordinate spaces. We obtain a further subsequence and a representation on a probability space where all coordinates converge almost surely. The energy limits retain their joint law with the environment and their measurability with respect to the random fields.

We work on the event of probability one on which all coordinates converge. The constants just included are bounded along the sequence in the Skorokhod representation. The bounds (8.7) hold for all meshes, and the bounds (8.10) hold for all sources and boundary data. Fix $Q \in \mathcal{C}$ and abbreviate $E_N = E_N^Q$, $G_N = G_N^Q$.

8.4.1. The Dirichlet Green operator and the limit form.

Proposition 8.18 (Convergence of the Dirichlet Green operators and the limit form). *Along the sequence in the Skorokhod representation, the following statements hold.*

- (i) Norm convergence: the operators G_N converge in operator norm on $L^2(Q, dx)$ to a compact, positive, self-adjoint, injective operator G .
- (ii) Limit form: the form

$$(8.30) \quad E(u) = \sup_{f \in L^2(Q)} \{2\langle f, u \rangle - \langle f, Gf \rangle\}, \quad D(E) = \{u : E(u) < \infty\} = \text{Ran } G^{1/2},$$

is a densely defined closed Dirichlet form on $L^2(Q, dx)$.

- (iii) Mosco convergence: if $u_N \rightharpoonup u$ weakly in $L^2(Q)$, then $E(u) \leq \liminf E_N(u_N)$ (the liminf inequality). Every $u \in D(E)$ has a **recovery sequence**, that is, a sequence $u_N \in H_0^1(Q)$ with $u_N \rightarrow u$ in $L^2(Q)$ and $E_N(u_N) \rightarrow E(u)$.
- (iv) Density: the functions Gf with $f \in C_c^\infty(Q)$ are dense in $D(E)$ for the form norm.

Proof. Step 1. We show that G_N converges in operator norm. By (8.5) and the energy identity,

$$\|G_N f\|_{H^{3/4}}^2 \leq K_N E_N(G_N f) = K_N \langle f, G_N f \rangle \leq K_N \|f\|_2 \|G_N f\|_2,$$

so $\|G_N\|_{L^2 \rightarrow H^{3/4}} \leq \sup_N K_N < \infty$ along the sequence in the Skorokhod representation. By Rellich's theorem the images of the unit ball under all G_N lie in a common precompact subset of L^2 . Given $\varepsilon > 0$, choose a finite ε -net for these images consisting of elements of $\mathcal{D}$, and let P be the orthogonal projection onto its span. Then $\|(I-P)G_N\| \leq \varepsilon$. By self-adjointness, $\|G_N(I-P)\| = \|(I-P)G_N\| \leq \varepsilon$. Hence $\|G_N - G_M\| \leq \|P(G_N - G_M)P\| + 4\varepsilon$. The operator $PG_N P$ is determined by the finite matrix of the bilinear form $\langle \cdot, G_N \cdot \rangle$ on the elements of the net. Polarization expresses each entry as a real linear combination of quadratic forms $\langle h, G_N h \rangle$ with $h \in \mathcal{D}$, which converge by the choice of the sequence in the Skorokhod representation. Hence (G_N) is Cauchy in operator norm.

Step 2. We show that G is injective. For $w \in H_0^1(Q)$ and $s \in \mathbb{R}$, the variational formula (8.3) gives $\langle f, G_N f \rangle \geq 2s \langle f, w \rangle - s^2 E_N(w)$, and optimizing in s gives $|\langle f, w \rangle|^2 \leq \langle f, G_N f \rangle E_N(w)$. Let $Gf = 0$ and $\phi \in \mathcal{D} \cap C_c^\infty(Q)$, and let $w_{N,r}$ be the harmonic interpolant of ϕ on the rational mesh of side $r < \text{dist}(\text{supp } \phi, \partial Q)$. By (8.6) with $\|\phi(z + r \cdot)\|_{C^\beta(\partial \square_0)/\mathbb{R}} \leq Cr \|\nabla \phi\|_\infty$,

$$E_N(w_{N,r}) = \sum_q \Lambda_{N,q}(\phi|_{\partial q}) \leq Cr^{-d} \max_q \Lambda_{\sigma/2,2}(q; A_N) r^{d-2} r^2 \|\nabla \phi\|_\infty^2,$$

which is bounded along the sequence in the Skorokhod representation at fixed r , since only finitely many cells of $\mathcal{C}$ occur. The maximum principle on each cell gives $\|w_{N,r} - \phi\|_\infty \leq Cr \|\nabla \phi\|_\infty$. Hence

$$|\langle f, \phi \rangle| \leq |\langle f, w_{N,r} \rangle| + Cr \|f\|_1 \|\nabla \phi\|_\infty,$$

and the first term tends to zero as $N \rightarrow \infty$, since $\langle f, G_N f \rangle \rightarrow \langle f, Gf \rangle = 0$. Letting $r \downarrow 0$ gives $f \perp C_c^\infty(Q)$, so $f = 0$.

Step 3. The spectral theorem gives $E(u) = \|G^{-1/2}u\|_2^2$ with domain $\text{Ran } G^{1/2}$, a dense closed form domain. For every $\lambda > 0$, Step 1 gives

$$(G_N^{-1} + \lambda I)^{-1} = G_N(I + \lambda G_N)^{-1} \longrightarrow G(I + \lambda G)^{-1} = (G^{-1} + \lambda I)^{-1}$$

in operator norm. The resolvent criterion [Kuwaie and Shioya, 2003, Theorem 2.4] therefore gives Mosco convergence on $L^2(Q, dx)$. Applying the liminf inequality to normal contractions of recovery sequences proves the Markov property of E . Finally, $E(u, Gf) = \langle u, f \rangle$; if u is orthogonal in the form norm to $G\mathcal{D}$, then $\langle (I + G)u, f \rangle = 0$ for every $f \in \mathcal{D}$, so $u = 0$. This proves (ii)–(iv). $\square$

Since $\mathcal{C}$ contains countably many cubes, the proposition gives, along the sequence in the Skorokhod representation, a limit form E^Q on every cube Q of $\mathcal{C}$; we write $E = E^Q$ when the cube is fixed.

The comparison arguments of Section 8.6 need the quadratic forms $\langle f, G_N^Q f \rangle$ to be bounded above and below with positive probability, uniformly in N and in the extracted limit. Both bounds follow from moment bounds.

Lemma 8.19 (Positive and negative moments of the quadratic forms). *Fix a cube Q of side at most one and a nonzero function $f \in C_c^\infty(Q)$. For every $p < \infty$, if $\bar{\delta}$ is sufficiently small, then*

$$\sup_N \mathbb{E}[\langle f, G_N^Q f \rangle^p + \langle f, G_N^Q f \rangle^{-p}] < \infty.$$

The same bounds hold for every extracted limit, uniformly over the subsequence and over all values of $\bar{\delta}$ in a fixed, sufficiently small interval.

Proof. The coercivity bound (8.5) gives $\langle f, G_N^Q f \rangle \leq K_N \|f\|_2^2$. For the negative moments, fix a mesh of side r and let $w_{N,r}$ be the harmonic interpolant of f on it, as in Step 2 of the proof of Proposition 8.18. The maximum principle gives $\|w_{N,r} - f\|_\infty \leq Cr \|\nabla f\|_\infty$. Fix r so small that $\langle f, w_{N,r} \rangle \geq 1/2 \|f\|_2^2$ for every N . The energy of $w_{N,r}$ has a uniformly bounded p th moment

by (8.6)–(8.7) on this fixed finite mesh. Optimize the variational formula (8.3) over the multiples of $w_{N,r}$ to obtain

$$\langle f, G_N^Q f \rangle^{-1} \leq \frac{E_N^Q(w_{N,r})}{\langle f, w_{N,r} \rangle^2} \leq 4 \|f\|_2^{-4} E_N^Q(w_{N,r}).$$

Fatou's lemma passes both moment bounds to each extracted limit. $\square$

8.4.2. Traces and localization. We will apply the following trace estimate to the energy measures of the harmonic interpolants of cutoff functions (Lemma 8.21).

Lemma 8.20 (Measure trace estimate). *Let $t > d - 1$ and $K < \infty$, and let ν be a finite positive Borel measure on $\mathbb{R}^d$ supported in $\bar{Q}$ such that $\nu(B_r(x)) \leq Kr^t$ for every $x \in \bar{Q}$ and $0 < r \leq 1$. Then the map $v \mapsto v$, from smooth functions on $\bar{Q}$ to $L^2(\nu)$, extends by completion to $H^{1/2}(Q)$, and there exists $C(d, t, Q) < \infty$ such that, for every $v \in H^{1/2}(Q)$,*

$$(8.31) \quad \int |v|^2 d\nu \leq C(d, t, Q) (K + \nu(Q)) \|v\|_{H^{1/2}(Q)}^2.$$

Moreover, if $v_N \rightarrow v$ in $L^2(Q)$ and $\sup_N \|v_N\|_{H^{3/4}(Q)} < \infty$, then $v_N \rightarrow v$ in $H^{1/2}(Q)$.

Proof. The Bessel kernel of order one is bounded by $C|z|^{1-d}$ for $|z| \leq 1$ and is bounded outside the unit ball. The assumption $\nu(B_r(x)) \leq Kr^t$ and $t > d - 1$ therefore give

$$\sup_{x \in \text{supp } \nu} \int (|x - y|^{1-d} \wedge 1) d\nu(y) \leq C \left(\nu(\bar{Q}) + K \sum_{n \geq 0} 2^{-n(t-d+1)} \right) \leq C(K + \nu(\bar{Q})).$$

Regard the Bessel potential of order $1/2$ as a map from $L^2(dx)$ to $L^2(\nu)$. The composition of this map with its adjoint has the Bessel kernel of order one as its kernel. This kernel is bounded by the integrand above. By Schur's test, the squared norm of the map is at most $C(K + \nu(\bar{Q}))$. The Bessel-potential description of $H^{1/2}(\mathbb{R}^d)$ and a bounded extension operator from the cube give (8.31) for smooth v .

The boundary of Q is ν -null: cover it with $O(r^{1-d})$ balls. Hence $\nu(\bar{Q}) = \nu(Q)$. Density of smooth functions in $H^{1/2}(Q)$ gives the asserted completion, and convergence in $L^2(dx)$ identifies it with the ordinary integral when ν has a bounded density with respect to Lebesgue measure. Finally, the same extension operator and Sobolev interpolation give

$$\|w\|_{H^{1/2}(Q)} \leq C \|w\|_{L^2(Q)}^{1/3} \|w\|_{H^{3/4}(Q)}^{2/3}.$$

Weak compactness first puts v in $H^{3/4}(Q)$; applying this inequality to $w = v_N - v$ proves the last assertion. $\square$

Along the sequence in the Skorokhod representation, recovery sequences and solutions with zero Dirichlet boundary values are bounded in $H^{3/4}$ by (8.5), and the energy measures of the cutoff functions of Lemma 8.21 below satisfy (8.10) with a common constant. Thus (8.31) controls $\int (v_N - v)^2 d\Gamma_N(\chi_N)$ uniformly in N . We use this estimate when localizing recovery sequences in Proposition 8.23.

8.4.3. Regularity, locality, and restriction to subcubes. To obtain energy measures for E , we show that E is regular and strongly local. Both properties are proved by multiplying recovery sequences by cutoff functions. Smooth cutoff functions are not suitable, since A_N has no upper bound uniform in N ; we use instead the harmonic interpolants of cutoff functions, whose energies, constants in the Morrey bounds and Hölder norms are controlled cell by cell by (8.6)–(8.11).

Lemma 8.21 (Harmonic cutoff functions). *Let $K \subseteq O$ be finite unions of closed and open rational cubes, respectively, with K compact and $\overline{O} \subseteq Q$, and let χ_N be the harmonic interpolant, on a mesh of $\mathcal{C}$, of a smooth cutoff function from $\mathcal{D}$, with values in $[0, 1]$, equal to one near K and compactly supported in O . The mesh is so fine that every cell meeting the set where this function lies strictly between 0 and 1 is contained in $O \setminus K$. Then the following statements hold.*

- (i) *The function χ_N belongs to $H_0^1(Q) \cap C(\overline{Q})$, satisfies $0 \leq \chi_N \leq 1$, equals one on a neighbourhood of K and zero outside O , and $\nabla \chi_N = 0$ on every cell on which this function is constant.*
- (ii) *Along the sequence in the Skorokhod representation, the energy $E_N(\chi_N)$, the constant in the bound (8.10) for $\Gamma_N(\chi_N)$ at all radii, and the C^α norm of χ_N are bounded. In particular, along a subsequence, χ_N converges uniformly to a continuous function χ that agrees with the original cutoff function on every cell where that function is constant.*
- (iii) *Let $r > 0$, and let $\chi_{N,r}$ be the harmonic interpolant on the mesh of side r of a smooth function $\mathbb{R}^d \rightarrow [0, 1]$ that vanishes within distance r of ∂Q , equals one at distance at least $3r$ from ∂Q and has gradient at most C/r on $\mathbb{R}^d$. Let K_N be the constant of (8.7). Then $0 \leq \chi_{N,r} \leq 1$ and*

$$E_N(\chi_{N,r}) \leq C(K_N + 3^{-\eta N} \|A_N\|_{L^\infty(Q)}) r^{-1-\eta}.$$

Proof. On a cell on which the cutoff function is constant the harmonic extension is that constant, by uniqueness, and the maximum principle gives $0 \leq \chi_N \leq 1$. Continuity across faces follows from the matching traces, and continuity up to the boundary from the Hölder bound for the cell solutions. At a fixed mesh there are finitely many cells, each with bounded $\Lambda_{\sigma/2,2}(q; A_N)$ and bounded constants in the Morrey and Hölder bounds along the sequence in the Skorokhod representation. Sum over the cells to obtain (ii). The Arzelà–Ascoli theorem gives uniform convergence.

For (iii), the maximum principle gives $0 \leq \chi_{N,r} \leq 1$ as in (i). There are $O(r^{-(d-1)})$ cells on which the function defining $\chi_{N,r}$ is not constant. On each such cell the rescaled function has C^β norm at most C . If $r \geq 3^{-N}$, then (8.6) and (8.7) bound the energy on each such cell by $C\Lambda_{\sigma/2,2}(q; A_N)r^{d-2} \leq CK_N r^{d-2-\eta}$. If $r < 3^{-N}$, the smooth function defining $\chi_{N,r}$ is itself a competitor in the minimization on each cell, which gives

$$E_N(\chi_{N,r}) \leq C\|A_N\|_{L^\infty(Q)} r^{-1} \leq C(3^{-\eta N} \|A_N\|_{L^\infty(Q)}) r^{-1-\eta}.$$

Both the factor $3^{-\eta N} \|A_N\|_{L^\infty(Q)}$ and the constant of (8.7) were included in the joint extraction, and their sum bounds the energy for all widths r . $\square$

Proposition 8.22 (Regularity). *The Dirichlet form E of Proposition 8.18 is regular.*

Proof. For $f \in \mathcal{D}$, the Hölder bound (8.11) gives $u_N = G_N f \rightarrow Gf$ uniformly on $\overline{Q}$, with zero boundary values. Let $\chi_{N,r}$ be the cutoff function of Lemma 8.21(iii). Covering the boundary layer $\{x \in Q : \text{dist}(x, \partial Q) < 3r\}$ by $O(r^{1-d})$ balls and using (8.10)–(8.11) gives

$$(8.32) \quad E_N(u_N(1-\chi_{N,r})) \leq 2\Gamma_N(u_N)(\{\text{dist}(\cdot, \partial Q) < 3r\}) + 2 \sup_{\text{dist}(\cdot, \partial Q) < 3r} |u_N|^2 E_N(\chi_{N,r}) \leq C_{\omega,f}(r^{t-d+1} + r^{2\alpha-1-\eta}).$$

At fixed r , pass to a subsequence along which $\chi_{N,r} \rightarrow \chi_r$ uniformly. The liminf inequality shows that $Gf\chi_r \in D(E) \cap C_c(Q)$ converges to Gf in the form norm as $r \downarrow 0$. Since the functions Gf are dense in $D(E)$ by Proposition 8.18(iv), $D(E) \cap C_c(Q)$ is dense in $D(E)$ for the form norm.

For density in the uniform norm, take $\phi \in \mathcal{D} \cap C_c^\infty(Q)$ and its harmonic interpolants on a mesh of side r . Their bounded energies and Hölder norms give a uniform subsequential limit $w_r \in D(E) \cap C_c(Q)$. The cell maximum principle gives $\|w_r - \phi\|_\infty \leq Cr\|\nabla \phi\|_\infty$. Since these ϕ are dense in $C_0(Q)$, the core is uniformly dense as well. $\square$

Proposition 8.23 (Strong locality on the core). *Let $u, v \in D(E) \cap C_c(Q)$, and suppose that u equals a constant c on a neighbourhood of $\text{supp } v$. Then $E(u, v) = 0$.*

Proof. Choose two cutoff functions from $\mathcal{D}$ and let ψ_N, χ_N be their harmonic interpolants from Lemma 8.21, with $\psi_N = 1$ near $\text{supp } v$, $\chi_N = 1$ near $\text{supp } \psi_N$, and $\text{supp } \chi_N \subseteq \{u = c\}^\circ$. For recovery sequences u_N, v_N of u and v , consider the sequences $c\chi_N + (1 - \chi_N)u_N$ and $\psi_N v_N$. These are still recovery sequences. Indeed, they converge in L^2 to u and v , and the only extra gradient terms are controlled by

$$\int (u_N - c)^2 d\Gamma_N(\chi_N) + \int v_N^2 d\Gamma_N(\psi_N) \longrightarrow 0.$$

On the support of $\nabla \chi_N$ we have $u = c$, and on that of $\nabla \psi_N$ we have $v = 0$. Apply the trace estimate (8.31) together with the $H^{1/2}$ convergence of the recovery sequences. Young's inequality gives $\limsup E_N(c\chi_N + (1 - \chi_N)u_N) \leq (1 + \varepsilon)E(u)$ for every $\varepsilon > 0$, and likewise for $\psi_N v_N$. The liminf inequality gives equality. Since $c\chi_N + (1 - \chi_N)u_N = c$ near $\text{supp } \psi_N$, the form E_N of the two sequences vanishes. Applying the liminf inequality to their sum and difference yields $E(u \pm v) \leq E(u) + E(v)$, hence $E(u, v) = 0$. $\square$

By regularity and Proposition 8.23, uniqueness in the Beurling–Deny representation [Fukushima et al., 2011, Section 3.2] makes the jump and killing measures vanish. Thus E is strongly local, and its energy measures $\Gamma_E(u, v)$ satisfy the chain rule, Leibniz rule, Cauchy–Schwarz inequality and locality on open sets [Fukushima et al., 2011, Theorem 3.2.2 and Lemma 3.2.5].

We normalize the energy measure by $\int \phi d\Gamma_E(u) = E(u, u\phi) - \frac{1}{2}E(u^2, \phi)$ for bounded $u \in D(E)$ and $\phi \in D(E) \cap C_c(Q)$, so that the corresponding measure at finite N is $\Gamma_N(u) = A_N |\nabla u|^2 dx$, and $\Gamma_E(u)(Q) = E(u)$ and $\Gamma_E(u, v)(Q) = E(u, v)$. The measure $\Gamma_E(u)$ is supported in $\text{supp } u$.

Proposition 8.24 (Consistency under restriction to subcubes). *Let $q \subseteq Q$ be an open cube of $\mathcal{C}$, and let E^q and E^Q be the limit forms on q and Q along the sequence in the Skorokhod representation. Then extension by zero to Q sends $D(E^q)$ onto the E^Q -closure of $\{w \in D(E^Q) \cap C_c(Q) : \text{supp } w \subseteq q\}$, and $E^q(u)$ equals the E^Q -energy of the extension of u by zero, for every $u \in D(E^q)$.*

Proof. Zero extension of a recovery sequence on q shows that the E^Q -energy of the extension of u by zero is at most $E^q(u)$. Conversely, fix $w \in D(E^Q) \cap C_c(q)$. Localize a recovery sequence on Q by a harmonic cutoff function equal to one near $\text{supp } w$ and supported in q . The trace argument in Proposition 8.23 shows that this is still a recovery sequence. The liminf inequality on q gives $E^q(w|_q) \leq E^Q(w)$. Hence equality holds on this core. Regularity of E^q and the first inequality extend equality to all of $D(E^q)$. The resulting isometry has closed image, and this image is exactly the stated closure. $\square$

8.4.4. *Continuous weights and minimal energies with given boundary values.* For $\varrho \in C(\overline{Q})$ with $\varrho > 0$, let $E_N^\varrho(u) = \int_Q \varrho d\Gamma_N(u)$ for $u \in H_0^1(Q)$, and let G_N^ϱ be its Dirichlet Green operator.

Lemma 8.25 (Form order implies measure order). *Let E, F be regular strongly local Dirichlet forms on $L^2(Q, dx)$ with $D(E) = D(F)$, let $q \subseteq Q$ be open, and let $C < \infty$ be such that $F(w) \leq CE(w)$ for every $w \in D(E) \cap C_c(Q)$ with $\text{supp } w \subseteq q$. Then $\Gamma_F(u) \leq C\Gamma_E(u)$ on q for every $u \in D(E) \cap C_c(Q)$.*

Proof. Let $v \in D(E) \cap C_c(Q)$ with $\text{supp } v \subseteq q$, and let $k \geq 1$. The functions $k^{-1}v \sin(ku)$ and $k^{-1}v \cos(ku) = k^{-1}v(\cos(ku) - 1) + k^{-1}v$ belong to $D(E) \cap C_c(Q)$ and they are supported in $\text{supp } v$. For either form, the chain and Leibniz rules show that the cross terms in the energy measures of these two functions cancel in the sum, so that

$$E(k^{-1}v \sin ku) + E(k^{-1}v \cos ku) = \int v^2 d\Gamma_E(u) + k^{-2}E(v),$$

and likewise for F . Applying $F \leq CE$ to both functions and letting $k \rightarrow \infty$ gives $\int v^2 d\Gamma_F(u) \leq C \int v^2 d\Gamma_E(u)$. For $0 \leq \phi \in C_c(q)$, choose by regularity a core function with values in $[0, 1]$ that

equals one on $\text{supp } \phi$ and is supported in q . Also choose core functions converging uniformly to $\phi^{1/2}$. Then the products of the latter with the former are admissible and their squares converge uniformly to ϕ . Hence $\int \phi d\Gamma_F(u) \leq C \int \phi d\Gamma_E(u)$. $\square$

Proposition 8.26 (Continuous weights). *Let $\varrho \in C(\overline{Q})$ be positive. Along the sequence in the Skorokhod representation, the forms E_N^ϱ Mosco-converge to a form E^ϱ such that $D(E^\varrho) = D(E)$ and*

$$(8.33) \quad E^\varrho(u, v) = \int_Q \varrho d\Gamma_E(u, v), \quad u, v \in D(E).$$

Proof. The bounds $(\inf \varrho)E_N \leq E_N^\varrho \leq (\sup \varrho)E_N$ show that the forms E_N^ϱ satisfy the $H^{3/4}$ coercivity bound (8.5) up to a constant factor, and that the order reverses for their Green operators:

$$(\sup \varrho)^{-1}G_N \leq G_N^\varrho \leq (\inf \varrho)^{-1}G_N.$$

By the finite-rank compactness argument of Proposition 8.18, every subsequence therefore has a further subsequence along which G_N^ϱ converges in norm on all cubes of $\mathcal{C}$. The reversed order makes each limit injective. The same proposition constructs their Mosco limit F , with $D(F) = D(E)$ and equivalent form norms. The two forms therefore have the same regular core. The proofs of locality and consistency under restriction to smaller cubes also apply: recovery sequences retain their $H^{3/4}$ bound, and the energy measures of the cutoff functions are at most $(\sup \varrho)\Gamma_N(\chi_N)$. Thus F is strongly local and its restrictions to smaller cubes satisfy

$$(\inf_q \varrho)E^q \leq F^q \leq (\sup_q \varrho)E^q \quad (q \in \mathcal{C}).$$

By Lemma 8.25, the same comparison holds for energy measures on q . For a core function u , take a finite partition of unity near $\text{supp } u$ subordinate to cubes on which $\text{osc } \varrho \leq \varepsilon$. Integrate against this partition of unity to obtain $|F(u) - \int \varrho d\Gamma_E(u)| \leq \varepsilon E(u)$. Let $\varepsilon \downarrow 0$, extend by density in the form norm and polarize. Every subsequential limit form is therefore the form in (8.33), which proves convergence of the whole sequence in the Skorokhod representation. $\square$

We also obtain convergence in probability of the weighted Dirichlet Green operators along the subsequence extracted above. Indeed, given a subsequence, we represent the weight, the original energy limit and the preceding coordinates jointly on one probability space. The comparison (8.23) absorbs the error in the uniform approximation of the weight. The proof of Proposition 8.26 identifies every limit with (8.33). The subsequence criterion for convergence in probability then gives the assertion, also for random positive continuous weights and for countable families of them.

Lemma 8.27 (Bounded weights and deletion of a field). *Let g be a bounded Borel function on Q . Then $E^g(u, v) = \int e^g d\Gamma_E(u, v)$ defines a regular strongly local Dirichlet form with domain $D(E)$, and $\Gamma_{E^g}(u, v) = e^g \Gamma_E(u, v)$. Moreover, for every field $\mathbf{g}_{-j}$ with $j \geq 0$, the form $e^{-\mathbf{g}_{-j}}E$, given by $(u, v) \mapsto \int e^{-\mathbf{g}_{-j}} d\Gamma_E(u, v)$, is measurable with respect to the remaining fields.*

Proof. The bounds $e^{-\|g\|_\infty}E \leq E^g \leq e^{\|g\|_\infty}E$ show that E^g is closed on $D(E)$ and has the same core as E . The chain rule for energy measures gives the Markov property, and the locality of energy measures gives strong locality. The defining identity of the energy measure and the Leibniz rule give the formula for Γ_{E^g} .

For the deletion, observe that $e^{-\mathbf{g}_{-j}}$ is continuous and bounded above and below on each fixed cube. The coefficient $e^{-\mathbf{g}_{-j}}A_N$ is independent of the deleted field for all sufficiently large N . Proposition 8.26, applied to random weights as explained after its proof, identifies the limit of the Dirichlet Green operators of $e^{-\mathbf{g}_{-j}}A_N$ with the Dirichlet Green operator of $e^{-\mathbf{g}_{-j}}E$. Hence each quadratic form $\langle f, G^Q f \rangle$ of this Dirichlet Green operator, with $f \in \mathcal{D}$ and $Q \in \mathcal{C}$, is a limit in probability of functions of the remaining fields. It is therefore measurable with respect to the remaining fields. These quadratic forms determine the form $e^{-\mathbf{g}_{-j}}E$. $\square$

Therefore, conditionally on the remaining fields, the forms corresponding to coefficients in which the deleted field is multiplied by the cutoff functions of Lemma 8.14 can be defined by weighting the fixed form $e^{-\mathbf{g}-j}E$ with the independent restrictions of the deleted field. The first part of the lemma allows these weights to be discontinuous.

Corollary 8.28 (Energy-measure convergence). *Along the sequence in the Skorokhod representation, the following statements hold, with weak convergence of finite measures on $\overline{Q}$.*

- (a) *If $u_N \in H_0^1(Q)$ is a recovery sequence for $u \in D(E)$, then $\Gamma_N(u_N) \rightarrow \Gamma_E(u)$.*
- (b) *If $u_N \rightarrow u$ in L^2 and $\sup_N E_N(u_N) < \infty$, then every weak subsequential limit of $\Gamma_N(u_N)$ is at least $\Gamma_E(u)$.*
- (c) *If (u_N) is as in (b) and (v_N) is a recovery sequence for $v \in D(E)$, then $\Gamma_N(u_N, v_N) \rightarrow \Gamma_E(u, v)$.*

Proof. Let $0 \leq \phi \in C(\overline{Q})$. For a recovery sequence, the liminf inequalities for the weighted forms with weights $1 \pm \varepsilon\phi$, together with $E_N(u_N) \rightarrow E(u)$, give both inequalities needed for (a). For (b), apply the liminf inequality for the weight $\phi + \varepsilon$, and let $\varepsilon \downarrow 0$ along a subsequence on which the measures converge weakly.

For (c), Cauchy–Schwarz bounds the total variations of the mutual energy measures. Pass to a joint subsequential limit of $\Gamma_N(u_N)$ and $\Gamma_N(u_N, v_N)$. Apply (b) to $u_N + sv_N$ and (a) to v_N . For every rational s , this shows that the measure

$$\lim_N \Gamma_N(u_N) - \Gamma_E(u) + 2s(\lim_N \Gamma_N(u_N, v_N) - \Gamma_E(u, v))$$

is nonnegative. Testing against $0 \leq \phi \in C(\overline{Q})$ and letting $s \rightarrow \pm\infty$ identifies the limit of $\Gamma_N(u_N, v_N)$ with $\Gamma_E(u, v)$. Every subsequential limit is therefore the same. $\square$

For the regular strongly local form E , the one-dimensional energy-image-density theorem and the absolutely continuous chain rule [Fitzsimmons, 2004, Proposition 1.16 and its proof] give $\Gamma_E(v)(\{v = c\}) = 0$ and $\Gamma_E(T(v)) = T'(v)^2 \Gamma_E(v)$ for Lipschitz T with $T(0) = 0$, using quasi-continuous representatives. The reference normalizes the energy measure as $2\Gamma_E$, which does not affect either assertion.

Lemma 8.29 (Zero-trace differences lie in $D(E^q)$). *Let $q \subseteq Q$ be a cube of $\mathcal{C}$, and let $v \in D(E^Q) \cap C(\overline{Q})$ vanish on ∂q . Then $v_q = v\mathbf{1}_{\overline{q}}$ belongs to the space $D(E^q) \subseteq D(E^Q)$ of functions extended by zero, and $E^Q(v_q) = \Gamma_E(v)(q)$.*

Proof. Since $v = 0$ on ∂q , for every $\epsilon > 0$ the set $\{|v| \geq \epsilon\} \cap \overline{q}$ is compact in q . Truncate v_q at height ϵ , that is, replace it by $\text{sgn}(v_q)(|v_q| - \epsilon)_+$, and multiply by a core cutoff function equal to one near this set and supported in q . The result belongs to $D(E^q)$ by consistency under restriction to smaller cubes. The chain rule and locality show that its energy measure is the restriction of $\Gamma_E(v)$ to $q \cap \{|v| > \epsilon\}$. The energy of the difference of the truncations at two heights is the mass of $\Gamma_E(v)$ on the part of q where $|v|$ lies between these heights. Since the level sets of v carry no energy, the truncations are Cauchy in the form norm and converge uniformly to v_q . Since E^q is closed, $v_q \in D(E^q)$. Letting $\epsilon \downarrow 0$ in the first identity gives $E(v_q) = \Gamma_E(v)(q)$, with $\Gamma_E(v_q)(\partial q) = 0$. $\square$

We now identify the limits of the minimal energies $\Lambda_{N,q}$ with given boundary values on cubes of $\mathcal{C}$. The candidate is the energy in q of the limit U of the harmonic interpolants of a fixed smooth datum on the 3^d cells around q .

Proposition 8.30 (Limits of the minimal energies with given boundary values). *Let $q = z + r\Box_0 \in \mathcal{C}$, let $Q = z + 3r\Box_0$ be the cube formed by the 3^d congruent cells around q , let b be a boundary datum on ∂q from $\mathcal{D}$, and fix a smooth extension of b to a function with compact support in Q . Let U_N be the function equal, on each of the 3^d cells, to the A_N -harmonic extension of the restriction of*

this extension to the cell boundary. Then, along the sequence in the Skorokhod representation, U_N converges uniformly on $\overline{Q}$ to a function U with the following properties.

- (i) Boundary values: the function U belongs to $D(E^Q) \cap C(\overline{Q})$, and $U = b$ on ∂q .
- (ii) No energy on the faces: the energy measure of U satisfies $\Gamma_E(U)(\partial q) = 0$.
- (iii) Harmonicity: for every $\phi \in D(E^q)$, extended by zero to Q , we have $E^Q(U, \phi) = 0$.
- (iv) Minimality: the minimal energies converge, and

$$\lim_N \Lambda_{N,q}(b) = \Gamma_E(U)(q) = \min\{\Gamma_E(V)(q) : V \in D(E^Q) \cap C(\overline{Q}), V = b \text{ on } \partial q\}.$$

Proof. The energy, Hölder norm and constant in the Morrey bound for U_N are bounded by summing the corresponding cell bounds. Hence every subsequence has a further subsequence with $U_N \rightarrow U$ uniformly, where $U \in D(E^Q)$ and U equals the fixed extension of b on every cell boundary. We show that (U_N) is a recovery sequence for U .

Let W_N be a recovery sequence for U , let Σ be the union of the cell boundaries, and take on each cell the cutoff function of Lemma 8.21(iii) with $r = \rho$. Their sum $\chi_{N,\rho}$ vanishes near Σ and equals one away from its 3ρ -neighborhood. Use $\chi_{N,\rho}W_N + (1 - \chi_{N,\rho})U_N$, which has the same cell-boundary values as U_N , as a competitor. Cellwise minimality and convexity give, for $\varepsilon > 0$,

$$\begin{aligned} E_N(U_N) &\leq (1 + \varepsilon)[E_N(W_N) + \Gamma_N(U_N)(\{\text{dist}(\cdot, \Sigma) < 3\rho\})] \\ &\quad + (1 + \varepsilon^{-1}) \int (W_N - U_N)^2 d\Gamma_N(\chi_{N,\rho}). \end{aligned}$$

At fixed ρ , the last term tends to zero by (8.31) and convergence in $H^{1/2}$. The term $\Gamma_N(U_N)(\{\text{dist}(\cdot, \Sigma) < 3\rho\})$ is at most $C\rho^{t-d+1}$ by the Morrey bound at every radius and a covering of the finitely many faces. Let first $N \rightarrow \infty$, then $\rho \downarrow 0$ and $\varepsilon \downarrow 0$. The liminf inequality gives $E_N(U_N) \rightarrow E(U)$.

Corollary 8.28(a) now gives $\Gamma_N(U_N) \rightarrow \Gamma_E(U)$. The common Morrey bound passes to the limit. Hence $\Gamma_E(U)(\Sigma) = 0$ and $\Lambda_{N,q}(b) = \Gamma_N(U_N)(q) \rightarrow \Gamma_E(U)(q)$. For $\phi \in D(E^q)$, a recovery sequence on q , extended by zero, is also a recovery sequence on Q by consistency under restriction to smaller cubes. The identity $E_N(U_N, \phi_N) = 0$ at finite N passes to $E(U, \phi) = 0$ by Corollary 8.28(c).

If $V = b$ on ∂q , then by Lemma 8.29 the function $w = (V - U)\mathbf{1}_{\overline{q}}$ lies in $D(E^q)$. Its energy measure does not charge ∂q . Harmonicity therefore yields $\Gamma_E(V)(q) = \Gamma_E(U)(q) + E(w) \geq \Gamma_E(U)(q)$. The same argument on each cell shows that the difference of two possible limits has zero energy on every cell. Since the energy measures do not charge the faces, coercivity forces this difference to vanish on Q . Thus the uniform convergence and all the conclusions hold along the full sequence in the Skorokhod representation. The argument applies to every fixed finite partition into cubes with a common smooth datum vanishing on the outer boundary. $\square$

Proposition 8.31 (Hölder boundary data, gluing, harmonic replacement). *Let $1/2 < \beta < \alpha < 1$ and set $\sigma = 1/2(\beta - 1/2)$. Then the following statements hold along the sequence in the Skorokhod representation.*

- (i) *Let $q = z + r\Box_0$ and $Q = z + 3r\Box_0$ be as in Proposition 8.30. For every $b \in C^\alpha(\partial q)$, the limit $\Lambda_{E,q}(b) = \lim_N \Lambda_{N,q}(b)$ exists and satisfies*

$$\Lambda_{E,q}(b) \leq C\left(\sup_N \Lambda_{\sigma/2,2}(q; A_N)\right) r^{d-2} \|b(z + r \cdot)\|_{C^\beta/\mathbb{R}}^2,$$

and all conclusions of Proposition 8.30 hold for b , with a harmonic extension $U \in D(E^Q) \cap C(\overline{Q})$.

- (ii) *Let Q be a cube of $\mathcal{C}$, and let $q_1, \dots, q_m \in \mathcal{C}$ be disjoint cells such that the closure of each q_i and the concentric cube of three times its side length lie in Q . Let $V \in D(E^Q) \cap C(\overline{Q})$ satisfy $V|_{\partial q_i} \in C^\alpha$ for every i . Let V' equal V outside $\bigcup_i q_i$ and equal, on each q_i , the*

harmonic extension U_i of $V|_{\partial q_i}$. Then $V' \in D(E^Q) \cap C(\overline{Q})$ and

$$E^Q(V') = E^Q(V) - \sum_i [\Gamma_E(V)(q_i) - \Lambda_{E,q_i}(V|_{\partial q_i})] \leq E^Q(V).$$

(iii) If moreover $\Gamma_E(V)$ charges no face ∂q_i (for instance, if V is itself a limit of functions that, at finite N , are harmonic in each cell for the cutoff coefficient and share a common smooth datum on the union of the cell boundaries), and the cells q_i partition a cube $Q' \subseteq Q$, then $\Gamma_E(V')(Q') = \sum_i \Lambda_{E,q_i}(V|_{\partial q_i})$.

Proof. (i) Extend b to a function in $C_c^\alpha(Q)$ and approximate it in C^β by functions in $\mathcal{D}$. On the fixed finite partition, apply the bound (8.6) for Hölder boundary data and the maximum principle to two approximations. The bounds show that the energy E_N of the difference of the corresponding interpolants is at most C times the squared C^β norm of the difference of the approximations. The sup norm of the difference of the interpolants is at most the sup norm of the difference of the approximations. The same bounds hold for the limits by the liminf inequality. Hence the corresponding harmonic extensions converge in energy and uniformly to a limit U with boundary values b . Harmonicity passes to the limit by energy convergence. The energy on the faces also remains zero, since energy measures depend continuously in total variation on the function in the form norm. Minimality follows again from harmonicity and Lemma 8.29. Since $\sqrt{\Lambda_{N,q}}$ is a seminorm of the datum, its reverse triangle inequality and (8.6) transfer the convergence of $\Lambda_{N,q}$ and the asserted bound from the approximations to b . The argument holds for each realization and covers random boundary data.

(ii) Consistency under restriction to subcubes extends each U_i to $D(E^Q)$, and Lemma 8.29 puts $(U_i - V)\mathbf{1}_{\overline{q_i}}$ in $D(E^{q_i})$ with zero energy on ∂q_i . Adding these differences to V shows that V' belongs to the form domain. Locality identifies its energy measure with $\Gamma_E(U_i)$ inside q_i and with $\Gamma_E(V)$ elsewhere, including the faces by Cauchy–Schwarz. Integrating gives the identity. Cell minimality gives the inequality.

(iii) The same identification and the assumption that $\Gamma_E(V)$ charges no face give the sum over the partition. For a common smooth datum, the energy measure of the limit does not charge the faces by Proposition 8.30; for a common C^α datum, this follows from the approximation in energy in (i). $\square$

Lemma 8.32 (Finite partitions with prescribed boundary values). *Let $\alpha > 1/2$, let the cubes q_i form a finite partition of a cube Q , and let $b \in C^\alpha(\overline{Q})$ vanish on ∂Q . Let v be the function equal, on each cell q_i , to the limiting harmonic extension of $b|_{\partial q_i}$. Then $v \in D(E^Q) \cap C(\overline{Q})$ and*

$$E^Q(v) \leq \sum_i \Lambda_{E,q_i}(b|_{\partial q_i}) \quad \text{and} \quad \|v - b\|_\infty \leq \sup_i \text{osc}_{q_i} b.$$

Proof. For smooth data compactly supported in Q , apply the last paragraph of the proof of Proposition 8.30 to this finite partition. The harmonic interpolants converge uniformly and in energy, and their cell energies add. For general b , choose $1/2 < \beta < \min\{\alpha, 1\}$. The zero extension of b can be approximated in C^β by $C_c^\infty(Q)$ functions. The bound (8.6) for Hölder boundary data and the maximum principle, as in Proposition 8.31(i), give convergence in energy and uniformly. This proves the energy bound, also for cells touching ∂Q . The cell maximum principle gives $\|v - b\|_\infty \leq \sup_i \text{osc}_{q_i} b$. $\square$

8.5. Estimates along chains of nested cubes. The energy measure $\Gamma_E(u)$ depends on the same fields as the regularity, extension and harmonic-approximation estimates of Lemma 8.33. To control the energy in the cells where these estimates fail, Proposition 8.38 bounds, on a single event that does not depend on u , the number of bad cells, defined below, along every chain of nested cubes. This bound can then be integrated against the energy measure.

Let E be a subsequential limit along the subsequence of cutoffs of Section 8.4, realized on the Skorokhod probability space.

8.5.1. *Continuous versions of the conditions.* Fix $s \in (0, 1]$. Let $F_{m,z}$, $P_{m,z}$ and $R_{m,z}$ be the left sides of (6.9), (6.10) and (6.11), translated to z . For $\bar{\delta} = 0$, they equal 0, 2 and 0, respectively. We use the quantity $\mathbf{G}_{m,z}(s)$ of (6.44), suppressing its fixed argument s , and the quantity $\mathcal{J}_n(P)$ of (4.1).

To measure failure of the bounds at center z and scale m , apply continuous piecewise linear functions to $F_{m,z}$, $P_{m,z}$ and $R_{m,z}$, and let $Z_{m,z}$ be their sum. Each function increases linearly from zero to one between its lower and upper thresholds, is zero below the lower threshold and one above the upper threshold, and lies between the indicators of exceeding these thresholds. The threshold pairs are $(\epsilon/2, \epsilon)$, $(6, 12)$ and $(\epsilon^2/4, \epsilon^2)$, respectively. Thus $Z_{m,z}$ takes values in $[0, 3]$; it is a continuous version of the number of the three conditions (6.9), (6.10) and (6.11) that fail at center z and scale m . The probability bounds of Lemmas 6.5, 6.6 and 6.7 are applied at the lower thresholds. Where $Z_{m,z} < 1$, each quantity lies below its upper threshold. Apply the deterministic good-scale estimates at these upper thresholds; in particular, they use the threshold 12 for $P_{m,z}$ in place of 6. Only the constants change, provided $\tau^2 \leq s \log 3/16$: the product bound 12 gives $e^{\|\mathbf{g}_{m,n}\|_\infty} \leq 12 \cdot 3^{s(m-n)/8}$, while (6.23) gives $e^{|\rho_{m,n}|} \leq 3^{s(m-n)/16}$. The squared coefficient ratio in (6.22), multiplied by the bound (6.11) and its weight, therefore still decays like $3^{-s(m-n)}$.

8.5.2. *The good event of a cell.* We now define the good event of a cell. It requires two things: the quantities $Z_{j,w}$ and $\mathbf{G}_{j,w}$ must have small averages over the scales between each of the cubes defined below and their triadic descendants; and finitely many conditions on the cell itself must hold.

Fix a rational cube Q and consider its triadic subdivision. For a cell $q = z + 3^{-k}\square_0$, evaluate the conditions below on its concentric enlargements $3^i q$, where 3^i ranges over a fixed finite list of triadic factors. Also use the cubes of the finitely many translated grids needed to cover the concentric cubes used for the harmonic approximation in (b), and all triadic descendants of these cubes. For an enlargement $U = 3^i q$ and a depth $D \geq 0$, use the centers w of its depth- D descendants and finitely many additional centers, fixed in advance, needed to compare s_E at different scales. There are at most $C_d 3^{dD}$ such centers. At each center w , consider the scales between the side of U and the side of its depth- D descendants, together with a fixed number k_1 of additional scales. Evaluate every quantity at the same cutoff level N .

Fix deterministic $\epsilon \in (0, 1)$, $\lambda \in (0, 1)$ and an initial depth k_0 at least as large as k_1 . The **good event** $G(q)$ of the cell q is the event on which, for every cube $U = 3^i q$, every $D \geq k_0$ and every center w as above at depth D , the sums of $Z_{j,w}$ and of $\mathbf{G}_{j,w}$ over the scales concerned (between U and its depth- D descendants, with the k_1 additional scales) are both smaller than λD , and on which the following three conditions hold:

- (a) the upper and lower coarse-grained ellipticity bounds (2.45) hold on q and on its finitely many concentric enlargements $3^i q$;
- (b) the error of Definition 2.3 on the cube concentric with q used for the harmonic approximation is below a prescribed tolerance ϵ_{hom} times the deterministic constant in the hypothesis of the interior harmonic approximation of Lemma 6.15;
- (c) the ratios of the scalars s_E of q and of the cubes concentric with q used for the harmonic comparison, defined in Section 8.6 and, at a finite cutoff, in Lemma 9.13, lie in $[1/2, 2]$.

Conditions (a)–(c) are finitely many conditions on scalars or matrices, and each has a continuous version given by a continuous piecewise linear function. The good event is defined with thresholds strictly smaller than those used by the deterministic estimates; the finitely many initial scales are counted as scales at which one of the conditions fails. At depth k , these conditions use the scales $N - k - D, \dots, N - k$ of the relabeled fields of (8.1), with every quantity evaluated at the same cutoff N .

Lemma 8.33 (Deterministic consequences of the good event). *Fix N and a cell q . On $G(q)$, every weak solution satisfies the following statements.*

- (i) The local Hölder estimate with source holds at exponent α , with a fixed deterministic constant. More precisely, if the iteration of Lemma 6.10 is run with a fixed exponent in $(\alpha, 1)$, then the constant in its conclusion after D scales is at most $3^{Ck_0+C(\lambda+\bar{\delta}^2+\epsilon^8)D}$, where $C(\lambda+\bar{\delta}^2+\epsilon^8)$ is smaller than the difference between this exponent and α .
- (ii) The bound (8.6) for Hölder boundary values holds on q with the factor $\Lambda_{\sigma/2,2}(q; A_N)$ of Lemma 8.4, which is comparable to the scalar $s_E(q)$ of q (see Section 8.6); this scalar is the factor by which the coefficient on q , rescaled to unit size, is a multiple of a copy of A_{N-k}^0 , up to the oscillation factor of Lemma 8.2.
- (iii) The hypothesis of the interior harmonic approximation of Lemma 6.15 holds.

Proof. On $G(q)$, we apply the iteration of Lemma 6.10 with the bad set consisting of the k_1 additional indices and the indices of the scales at which one of the conditions fails. This set has at most $2k_0+\lambda D$ elements. By (6.92) the exponent in the constant of the conclusion is at most $C(\bar{\delta}^2+\epsilon^8+\lambda)D+Ck_0$. This gives (i). Condition (a), the estimate (2.73) and a trace extension $C^\beta(\partial q) \rightarrow H^{1+\sigma}(q)$ with $0 < \sigma < \beta - 1/2$ give (ii). Condition (b) is the hypothesis in (iii). These bounds depend only on the coefficient and the deterministic geometry, so the same bounds apply to every weak solution. $\square$

Lemma 8.34 (Passage to the limit in sums over finitely many scales). *Along the subsequence that defines E , the following statements hold for each fixed finite interval of scales.*

- (i) The quantities $Z_{m,z}$ and $\mathbf{G}_{m,z}$ and each of the finitely many coarse-grained quantities in (a)–(c) converge in L^p , for each of the finitely many orders p fixed at the start of Section 8. The moment bounds are uniform in the deterministic position of the interval of scales.
- (ii) Let $A, D > 0$. If, at every cutoff, the sum of $Z_{m,z}$ or of $\mathbf{G}_{m,z}$ over the interval of scales exceeds a given threshold with probability at most e^{-AD} , then the limit of this sum exceeds any larger threshold with probability at most e^{-AD} .

Proof. We first prove (i). Split the terms that involve only the fields into finite sums at finitely many depths and geometrically weighted tails. By (1.5) the tails are bounded in L^p by their geometric weights times the moments of the fields. The other parts are geometrically weighted maxima of J . Recall that J is a fixed quadratic combination of the minimal energies $\Lambda_{N,q}$ with affine boundary values and of the coarse-grained quantities $|q|p \cdot a_*^{-1}(q; A_N)p$ of Section 8.1. These energies converge in L^p along the subsequence of Section 8.4, realized in the Skorokhod representation.

We truncate the geometrically weighted sums at a large relative depth. The finitely many retained quantities converge in L^p because their higher moments are uniformly bounded. At a relative depth j beyond the truncation there are at most $C3^{dj}$ cells. By Proposition 4.1 and the bound (3.33) on the ratios $\bar{\mathbf{a}}_m \bar{\mathbf{a}}_n^{-1}$, the quantity $\mathcal{J}$ of each cell has L^p norm at most of order $e^{C_p \bar{\delta}^2 j}$, and taking a spatial maximum costs a factor $3^{dj/p}$. We choose p with d/p below the margin of the geometric weights, and then decrease $\bar{\delta}$ so that $C_p \bar{\delta}^2$ is below the remaining margin. The tail is then exponentially small in the truncation depth, uniformly in the cutoff.

Below the smallest field scale 3^{-N} , the maximum of the coefficient on a cell whose side equals that scale controls every smaller descendant by the direct bounds (4.17), so the finer depths require no further spatial maximum. The continuous piecewise linear functions and minima are Lipschitz, and square roots satisfy $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$. Hence the quantities built from these are Cauchy in L^p .

For (ii), on the event that the limiting sum exceeds the larger threshold, a subsequence of the approximating sums that converges almost surely eventually exceeds some threshold strictly between the two thresholds, and each approximating sum exceeds that threshold with probability at most e^{-AD} . Fatou's lemma applied to the indicators gives the bound for the limit. $\square$

The **good event** $G_E(q)$ of the cell q for the limit form E is the event on which the limiting conditions (a)–(c) hold and the limiting inequalities for the sums over the scales between U and its

depth- D descendants hold for all cubes $3^i q$, depths and centers, all with strict margins relative to the conditions at finite cutoff. In particular, every ratio of the scalars s_E in (c) lies in $(1/2, 2)$. A cell q is **good** if $G_E(q)$ holds and **bad** otherwise. Because the margins are strict, each fixed finite set of these conditions holds at all sufficiently large cutoffs after passing to a further subsequence of the one defining E , in the Skorokhod representation, along which the quantities involved converge almost surely.

We apply Lemma 8.33 at each fixed depth to the already extracted solutions $u = G_E f$ with smooth f , pass the finitely many inequalities and the local energy measures to the limit along recovery sequences (8.33), and intersect over the depths. This gives, on $G_E(q)$, the local Hölder estimate with source, the bound (8.6) for Hölder boundary values, and the harmonic approximation with tolerance ϵ_{hom} . The comparison argument of Section 8.6 applies regularity only to the solutions $G_E f$ with smooth f . For the second limit form F introduced there, the argument uses only the maximum principle for harmonic extensions.

8.5.3. Events depending on finitely many fields. The conditions depend on all the fields. We approximate the quantities in the conditions by functions of the fields with indices in a finite interval around the scale of the cube, and then cover the failure of $G_E(q)$ by events that depend on such intervals and have exponentially small probabilities.

Lemma 8.35 (Approximation by functions of finitely many fields). *Let $X_{n,z}$ be one of the following: the quantity $\mathbf{G}_{n,z}$; one of the three summands of $Z_{n,z}$; or a fixed finite nonnegative combination of the quantities in (a)–(c) that vanish for $\bar{\delta} = 0$. There exist $a, c > 0$, not depending on the moment order p , such that, for every $p < \infty$, there exists $C_p < \infty$ with the following property. For every $h \geq 1$, every deterministic n and z and every cutoff, there is a random variable $\widehat{X}_{n,z}^{(h)}$, measurable with respect to the fields with indices in an interval of length at most $C(h+1)$ about n , such that*

$$(8.34) \quad \|X_{n,z} - \widehat{X}_{n,z}^{(h)}\|_{L^p} \leq C_p \bar{\delta}^c 3^{-ah}.$$

Proof. Step 1. We treat the terms that involve only the fields. Before the geometric weights are applied, the depth- j sum in $\mathbf{F}_{n,z}$ has L^p norm at most $C_p \bar{\delta} (j+1)^{3/2}$: we cover $z + \square_{n+1+j}$ by cubes of the scale of each field, use the sub-Gaussian bound for the maximum from (1.5), and sum the $2j+1$ terms. We discard the depths $j > ch$, whose total contribution is at most $C_p \bar{\delta} 3^{-sch/16}$. The tail terms of $\mathbf{G}$ (see (6.44)) carry the explicit weight 3^{k-i} .

Step 2. We treat the product term. We retain the depths $j \leq \lfloor h/2 \rfloor$ and stop the second product at $n+h$. Both $\mathbf{P}$ and its approximant are at least 2, the approximant is not larger, and the removed depths contribute only their excess over 2. Fix j . The first product factor at depth j is the exponential of $\sup_{x \in z + \square_{n+1+j}} \sum_{i=(n-j) \vee 0}^{n+j} |\mathbf{g}_i(x)|$. We cover the cube by at most $C_d 3^{d(2j+1)}$ cubes of side $3^{(n-j) \vee 0}$. On each such cube, this sum is at most the sum over i of the value of $|\mathbf{g}_i|$ at the center plus $C_d 3^{(n-j) \vee 0}$ times the gradient norm of $\mathbf{g}_i$ on the cube; these terms are independent in i and sub-Gaussian with parameter $C_d \bar{\delta}$. We take the exponential moment before the spatial maximum, which bounds the moment of order u on each cube by $\exp(C_d u \bar{\delta} (j+1) + C_d u^2 \bar{\delta}^2 (j+1))$, and a maximum over the covering cubes, taken in L^q with $q \geq p$, costs a factor equal to their number to the power $1/q$. With $e^x - 1 \leq x e^x$, choosing q large depending on d, s, p and then $\bar{\delta}$ small shows that $3^{-sj/8}$ times the L^p norm of the first product factor minus one is at most $C_{d,s,p} \bar{\delta} (j+1) 3^{-sj/16}$.

The second product factor is the exponential of $4 \sup_x \sum_{i \geq n+j} |\mathbf{g}_i(x) - \mathbf{g}_i(z)|$, which is at most a sum, over $i \geq n+j$, of sub-Gaussian variables multiplied by $C_d 3^{n+j-i}$. Hence its excess over one has L^p norm at most $C_p \bar{\delta}$, and the part with $i > n+h$ contributes $C_p \bar{\delta} 3^{j-h}$.

Summing over $j \leq h/2$ gives $C_{s,p} \bar{\delta} 3^{-h/2}$, and the discarded depths $j > h/2$ contribute $\sum_{j > h/2} 3^{-sj/8}$ times the L^p norm of the sum of the two excesses at depth j , which is at most $C_p \bar{\delta} 3^{-ah}$.

Step 3. We treat the terms built from $\mathcal{J}$. At relative depth r there are at most $C 3^{dr}$ cells. By (8.29), each cell's quantity $\mathcal{J}$ can be approximated using finitely many fields. The maximum

over the cells costs a factor $3^{dr/p}$, and the ratios $\bar{\mathbf{a}}_m \bar{\mathbf{a}}_n^{-1}$ of effective diffusivities that occur grow at most like $e^{C\bar{\delta}^2 r}$ by (3.33). Choose p with d/p below a fixed fraction of the exponent $s/8$ of the geometric weights, and decrease $\bar{\delta}$. Discard the depths $r > ch$. At each retained depth, use an interval of field indices whose radius is at least a fixed fraction of h and which is contained in the interval of length $C(h+1)$ about n . For square roots we use $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x-y|}$, which halves a and c .

Step 4. We obtain the factor $\bar{\delta}^c$ in (8.34). The deviations of the quantities defining the conditions from their values at $\bar{\delta} = 0$ satisfy

$$\|\mathbf{F}\|_p + \|\mathbf{P} - 2\|_p \leq C_p \bar{\delta}, \quad \|\mathbf{R}\|_p \leq C_p \bar{\delta}^2 \quad \text{and} \quad \|\mathbf{G}\|_p \leq C_p \bar{\delta}^{1/2}.$$

The first bound follows from Steps 1 and 2. For $\mathbf{R}$, group the cubes of (6.11) over which the supremum is taken by $r = m - n$. There are at most $C_d 3^{dr}$ such cubes, the quantity $\mathcal{J}$ of each has L^q norm $C_q \bar{\delta}^2$ by Proposition 4.1, and $\sum_{r \geq 2} 3^{-(s/8-d/q)r} < \infty$ for a fixed $q \geq p$ large enough. For $\mathbf{G}$, we take $m = n$ in Lemma 6.11 and obtain $C_p \bar{\delta}(1 + |\log \bar{\delta}|^{1/2}) \leq C_p \bar{\delta}^{1/2}$. If a truncation gives only $C_p 3^{-ah}$, without a positive power of $\bar{\delta}$, we combine it with a bound $C_p \bar{\delta}^b$, for some $b > 0$, on the distance of X from its deterministic value at $\bar{\delta} = 0$. Conditional expectation onto the fields with indices in the interval of length $C(h+1)$ about n is a contraction in L^p . Thus the distance of X from its conditional expectation is at most twice its distance from the approximation. The same bound holds with the value at $\bar{\delta} = 0$ in place of the approximation. The minimum of the two resulting bounds is at most their geometric mean $2C_p \bar{\delta}^{b/2} 3^{-ah/2}$. Since a and c are fixed by the geometric weights and by (1.5) before p is chosen, they do not depend on p . $\square$

Lemma 8.36 (Sums of Z and $\mathbf{G}$ over scales at a fixed center). *Let $\lambda > 0$ and $A < \infty$. If $\bar{\delta}$ is sufficiently small, then, for every deterministic center z , every interval $n, \dots, n+k-1$ of scales, in the coordinates of (8.1), with all quantities evaluated at a single cutoff level, and each $X \in \{Z, \mathbf{G}\}$,*

$$\mathbb{P}\left[\sum_{i=0}^{k-1} X_{n+i,z} > \frac{\lambda k}{4}\right] \leq C e^{-Ak}.$$

Proof. For $X = Z$ we apply Lemmas 6.5, 6.6 and 6.7 at the lower thresholds of the continuous piecewise linear functions. The exponential rates in these lemmas tend to infinity as $\bar{\delta} \downarrow 0$. For $X = \mathbf{G}$ we apply Lemma 6.11. Its estimate (6.45) concerns the average of the approximation errors over the interval, with an $\mathcal{O}_{\Gamma_2}$ fluctuation of size $(m-n+1)^{-1/2}$. After multiplication by the length $m-n+1$, the sum exceeds $\lambda(m-n+1)$ with probability at most $\exp(-c\lambda^2(m-n+1)\bar{\delta}^{-2}|\log \bar{\delta}|^{-1})$. The bounds are uniform in the position of the interval by the stationarity in Assumption (g1) and by Proposition 6.16. $\square$

Lemma 8.37 (Covering the failure of the good event by events depending on finitely many fields). *Let $A < \infty$. If $\bar{\delta}$ is sufficiently small, then, for every cell q at triadic depth n , there exist events $W_{q,h}$ for $h \geq 1$ such that $W_{q,h}$ is measurable with respect to the fields $\mathbf{g}_{-j}$ with $n-h \leq j \leq n+2h$, $\mathbb{P}(W_{q,h}) \leq e^{-Ah}$, and*

$$\{G_E(q) \text{ fails}\} \subseteq \bigcup_{h \geq 1} W_{q,h}.$$

The same conclusion holds for the failure of one additional scalar or matrix condition whose centered L^p norm is at most $C_p \bar{\delta}$ and whose approximation by a function of the fields with indices in an interval of length at most $C(h+1)$ about n has error at most $C_p \bar{\delta} 3^{-ah}$ in L^p , for every $h \geq 1$.

Proof. If $G_E(q)$ fails, then either one of finitely many initial conditions fails, or there are a depth $k \geq k_0$, one of at most $C e^{Ck}$ choices of a cube and a center, and a sum of Z or of $\mathbf{G}$ over an interval of k scales that exceeds λk . We approximate this sum by the sums of the approximations of Lemma 8.35 with $h = k, 2k, 4k, \dots$. By (8.34) and the triangle inequality, the approximation with

$h = 2^\ell k$ differs from the sum by at most $C_p k \bar{\delta}^c 3^{-a2^\ell k}$ in L^p . If the sum exceeds λk , then either the approximation with $h = k$ exceeds $\lambda k/2$ or, for some $\ell \geq 1$, the approximations with $h = 2^\ell k$ and $h = 2^{\ell-1} k$ differ by more than $\lambda k 2^{-\ell-2}$. The thresholds $\lambda k 2^{-\ell-2}$ for the increments sum to $\lambda k/4 < \lambda k/2$, which justifies this alternative.

The first event implies that the sum exceeds $\lambda k/4$ or differs from its approximation with $h = k$ by more than $\lambda k/4$. By Lemma 8.36, applied with a rate larger than $3A + C$, and Markov's inequality, the probability of this first event is at most $Ce^{-(3A+C)k} + (4C_p \bar{\delta}^c / \lambda)^p 3^{-apk}$. By Markov's inequality, the ℓ -th increment event has probability at most $(C_p \bar{\delta}^c)^p 2^{p(\ell+C)} 3^{-ap2^{\ell-1}k}$.

Take the union over the Ce^{Ck} spatial choices. For each h , combine into one event the events for all pairs (k, ℓ) with $2^\ell k \asymp h$. There are polynomially many such pairs. This event is measurable with respect to the fields indexed by an interval of radius at most Ch about n . After relabeling h , this interval is $[n - h, n + 2h]$. We choose p with $ap > 3A + C$, and then decrease $\bar{\delta}$ to absorb the factor $(C_p \bar{\delta}^c)^p$ and the finitely many initial conditions. For an additional centered condition, we start from its approximation by a function of the fields with indices in an interval of radius k_0 . This approximation exceeds its threshold with probability at most $(C_p \bar{\delta} / \lambda)^p$, which we make smaller than e^{-ACk_0} by decreasing $\bar{\delta}$. Telescope the increments as above. $\square$

Fix $L = 3^{H_1}$ and a bounded cube Q . A chain of length J is a sequence of J cubes of the L -adic subdivision of Q , each a child of the preceding one and the first a child of Q .

Proposition 8.38 (Bad cells along every chain). *Let $L = 3^{H_1}$, fix a bounded cube Q , and let $\theta \in (0, 1)$ and $b_* > 0$. There exists $\bar{\delta}_0 > 0$, depending on L, θ, b_* and the parameters of the good event, such that, if $\bar{\delta} \leq \bar{\delta}_0$, then there is a random variable $B_\omega < \infty$ such that, for every $t > 0$,*

$$\mathbb{P}(B_\omega > t) \leq (1 - e^{-b_*})^{-1} \exp\left(-\frac{b_* t}{1 - \theta}\right),$$

and such that, almost surely, for every $J \geq 1$ and every chain π of length J ,

$$(8.35) \quad \#\{\text{bad cells on } \pi\} \leq \theta J + B_\omega.$$

Proof. Apply Lemma 8.37 with rate A . On a fixed chain, the depths $H_1 n$, $n \in \mathbb{N}$, of the L -adic subdivision are spaced by H_1 . Enlarge the interval of field indices determining $W_{q,h}$ to the symmetric interval of radius $2h$ about its depth, set all events at other depths or odd radii to the empty event, and apply Lemma B.5. In that lemma, the rate is $A/3$ and the density is at least θ/H_1 . Thus, once $A\theta \geq CH_1$, the probability that this chain has at least θJ bad cells is at most $e^{-cA\theta J}$. There are L^{dJ} chains. Choose A with $cA\theta \geq d \log L + b_*$. The probability that some chain of length J has at least θJ bad cells is then at most $e^{-b_* J}$. These choices precede the moment order and the threshold for $\bar{\delta}$ in Lemma 8.37.

Set $B_\omega = \sup_J \max_{|\pi|=J} (\#\{\text{bad cells on } \pi\} - \theta J)_+$. If $B_\omega > t$, then for some $J > t/(1 - \theta)$ there is a chain of length J with more than θJ bad cells. Summing $e^{-b_* J}$ over these J proves the stated tail bound and (8.35). $\square$

Every point of Q lies in exactly one chain of half-open cells. Integrating (8.35) over these chains gives, on the same event, for every finite Borel measure μ on Q ,

$$(8.36) \quad \sum_{j=1}^J \mu(\text{union of the bad cells at } L\text{-adic level } j) \leq (\theta J + B_\omega) \mu(Q).$$

In particular this holds for the random energy measure $\Gamma_E(u)$.

8.6. Comparison of two subsequential limits. In this subsection we show that two subsequential limit forms differ only by a deterministic scalar factor (Theorem 8.51). Let E and F be two such forms, obtained along two subsequences of cutoffs, realized in the same Skorokhod representation

(Section 8.4), so that both are functions of the same fields. Their domains are not assumed to be equal. We compare them on the cubes of $\mathcal{C}$.

The preliminary two-sided comparison (Theorem 8.41) adapts the argument of Ding and Gwynne [2023a, Section 3.4] on the ratios of the normalizations of the two forms. In that argument, comparisons built from paths around annuli, together with upper and lower bounds on crossing distances, exclude ratios of the two normalizations that are too large or too small. Here harmonic extensions play the role of the paths. Proposition 8.40 glues harmonic extensions on cubes at the scales where the ratio r_k of the two normalizations, defined below, is small, and shows that if this happens on a positive density of scales, then F is bounded by a small multiple of E . By the variational formula (8.30), this contradicts the moment bounds of Lemma 8.19. This excludes ratios that are too small. Exchanging E and F excludes ratios that are too large. The proof of Theorem 8.41 then applies the same construction on the remaining scales, where the ratio is bounded above and below, to compare the domains and the energies in both directions.

As in the metric uniqueness proofs of Gwynne and Miller [2021, Section 6] and Ding and Gwynne [2023b, Section 4.5], we consider the optimal upper and lower comparison constants. Those proofs shorten geodesic segments to obtain probability bounds incompatible with a strict difference between the constants. They must control which annuli a field-dependent geodesic visits. The probability comparison and counting argument in Ding and Gwynne [2023b, Section 4.2] provide this control after suitable smooth changes of the Gaussian free field.

Here Proposition 8.38 bounds the number of bad cells along every chain of nested cubes, on a single event that does not depend on the solution. Its integrated form (8.36) therefore applies to the energy measure of the solution being replaced. We use harmonic extensions on disjoint cubes in place of shorter path segments. Since the solution is close to affine on good cells, the comparison of the minimal energies of E and F with affine boundary values transfers to its boundary values. Summing, over these cubes, the amount by which the minimal F -energy falls below the upper constant times the minimal E -energy improves one of the comparison constants for the forms themselves.

In this subsection a cell q is called good only when both $G_E(q)$ and $G_F(q)$ hold. Applying Proposition 8.38 to each form with half the allowed density of bad cells and adding the two counts gives (8.35) for this notion of bad cell. On a good cell we thus have the regularity and coercivity estimates for E and the bound (8.6) for Hölder boundary values for F , and the same with the roles of E and F exchanged.

The comparison is expressed through the scalars $s_E(q)$ and $s_F(q)$ of the two forms. At a cell q of side 3^{-k} and center z , the scalar $s_E(q)$ of E is

$$s_E(q) = \exp\left(H(z) + \sum_{j=0}^{k-1} (\mathbf{g}_{-j}(z) - \tau^2)\right) \lim_{N \rightarrow \infty} \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N},$$

the limit being taken along the subsequence that defines E . We define $s_F(q)$ in the same way, along the subsequence that defines F . After a further diagonal extraction, these limits exist for all k at once: since $|\bar{\mathbf{a}}_N \bar{\mathbf{a}}_{N-k}^{-1} - 1| \leq C \bar{\delta}^2 k e^{C \bar{\delta}^2 k}$ by (3.33), the ratio $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ lies in a compact subinterval of $(0, \infty)$ that depends only on k . The ratio

$$r_k = \frac{s_F(q)}{s_E(q)}$$

of these two scalars is deterministic and does not depend on z .

Choose the parameters in the following order. First fix the geometric and regularity parameters below, then L , the moment orders, and a first threshold for $\bar{\delta}$, which determines the comparison constant C_0 of Theorem 8.41. With C_0 fixed, we tighten the tolerance ϵ_{hom} of the harmonic comparison and add the comparison of the minimal energies of E and F with affine boundary values, of Corollary 8.46, whose error is proportional to the difference between the optimal comparison constants defined below. This requires only a further decrease of $\bar{\delta}$. The thresholds for sums over

intervals of scales and L stay fixed. None of these choices depends on the difference of the optimal constants.

8.6.1. Comparison from a positive density of small ratios. We first show that a positive density of scales at which $r_k \leq a$ forces $F \leq C_* a E$. Fix $\eta_0 \in (0, 1/3)$ and choose the parameters so that

$$D \geq \frac{8d}{\eta_0}, \quad \theta \leq \frac{\eta_0}{8} \quad \text{and} \quad 2(1 - \alpha) + \eta < \frac{\eta_0}{8}.$$

The last condition is equivalent to $d - 2 + 2\alpha - \eta > d - \eta_0/8$. Then choose $L = 3^{H_1}$ so large that $L^{\eta_0/3 - \eta_0/8} \geq 2C_d^{\eta_0/3}$, where C_d is the constant in the bound below on the number of children whose enlargement is not contained in the parent. We consider the L -adic levels $n \in \mathbb{N}$, which correspond to the triadic depths $k = H_1 n$; thus $r_{H_1 n}$ is the ratio r_k defined above at the depth $k = H_1 n$. For a child q of side r of a cell p of side Lr , consider its concentric enlargement of side $3r$. At most $C_d L^{d-1}$ children of each cell have an enlargement whose closure is not contained in p . A set $S \subseteq \mathbb{N}$ has **upper density** at least η if $\limsup_J J^{-1} \#(S \cap \{1, \dots, J\}) \geq \eta$.

On a good cell whose enlargement has its closure in the parent, the regularity of E and the bound (8.6) for Hölder boundary values of F combine as follows.

Lemma 8.39 (Extension on good cells). *Let Q be a cube of $\mathcal{C}$ and f a smooth function, and set $u = G_E^Q f$, where G_E^Q is the Dirichlet Green operator of E on Q .*

(i) *If $q = z + r\Box_0$ is a good child of p at depth k whose concentric enlargement of side $3r$ has its closure in p , then*

$$\inf_{c \in \mathbb{R}} \|u(z + r \cdot) - c\|_{C^\alpha(\Box_0)} \leq Cr^{2-d/2} s_E(q)^{-1/2} \sqrt{\Gamma_E(u)(p)} + Cr^2 s_E(q)^{-1} \|f\|_\infty,$$

$$\Lambda_{F,q}(u|_{\partial q}) \leq Cr_k [\Gamma_E(u)(p) + s_E(q)^{-1} r^{d+2} \|f\|_\infty^2].$$

(ii) *There is a random constant $K < \infty$ such that $\Lambda_{F,q}(u|_{\partial q}) \leq Kr^{d-2+2\alpha-\eta}$ for every cell $q \subseteq Q$ of side r .*

Proof. For (i), the local Hölder estimate with source on the good cell bounds the oscillation on q by the normalized L^2 oscillation on the concentric cube of side $3r$ and the source (Section 8.5). The coarse-grained Poincaré inequality (2.64) on that cube bounds its oscillation by its energy. Since its closure lies in p , this energy is at most $\Gamma_E(u)(p)$, giving the factor $r^{2-d/2}$ in unnormalized units. The source is controlled by (8.11): by Lemma 8.5, convolving f with the gradient of the Newtonian kernel gives a vector field with $C^{1/2}$ seminorm at most $Cr^{1/2} \|f\|_\infty$ on a cube of side r . Scaling gives the term $r^2 s_E(q)^{-1} \|f\|_\infty$. For the form F on q , the coefficient factor in (8.6) is comparable to $s_F(q) = r_k s_E(q)$. Inserting the first estimate into that bound gives the second estimate: we square the first bound and multiply by $s_F(q) r^{d-2}$.

For (ii), the global bound (8.11) for u gives $\|u(z + r \cdot) - c\|_{C^\beta(\partial\Box_0)} \leq r^\alpha [u]_\alpha$, and the bound (8.7) on grid cubes bounds the factor in (8.6) for F on q by $Kr^{-\eta}$. Then (8.6) gives the bound $Kr^{-\eta} r^{d-2} r^{2\alpha} [u]_\alpha^2$. $\square$

Proposition 8.40 (Comparison from a positive density of small ratios). *Assume that the parameters are chosen as above. There exists a deterministic $C_* < \infty$, depending on d, η_0, L and the constants of the good event, such that the following holds for every $a > 0$. Let S be a deterministic set of L -adic levels of upper density at least η_0 with $r_{H_1 n} \leq a$ for every $n \in S$. Then, on every cube of $\mathcal{C}$, $D(E) \subseteq D(F)$ and, for every $v \in D(E)$,*

$$(8.37) \quad F(v) \leq C_* a E(v).$$

Proof. Fix a cube Q of $\mathcal{C}$. Translate the L -adic grid so that its origin is a corner of Q . Since the side of Q is triadic, every sufficiently fine L -adic mesh partitions Q exactly. Fix $u = G_E f$ with smooth f , fix $c > 0$, and set $\mu = \Gamma_E(u) + c dx$. By (8.7) with $\eta < 2$, the source term in Lemma 8.39

is at most $Kr^{d+2-\eta}\|f\|_\infty^2 \leq cr^d = c|q|$ for cells of side below a number h_0 depending on u , c and the environment.

Fix an L -adic mesh of side at most h_0 that partitions Q ; all cells below are subcubes of the finitely many cubes of this mesh, and we apply Proposition 8.38 to each of them.

Step 1. We stop the subdivision at good cells that carry a fixed fraction of the mass of their parents, and bound the sum, over the stopped cells q , of the minimal F -energy $\Lambda_{F,q}(u|_{\partial q})$ with the boundary values of u . Within each cube of the mesh of side at most h_0 we follow the subdivision tree and stop at the first child q that lies at an L -adic level of S , is good, has an enlargement with closure in its parent, and satisfies

$$\mu(p) \leq L^D \mu(q).$$

By Lemma 8.39, a stopped cell satisfies $\Lambda_{F,q}(u|_{\partial q}) \leq Ca[\mu(p) + c|q|] \leq CaL^D \mu(q)$. The interiors of the stopped cells are disjoint. Hence the sum of $\Lambda_{F,q}(u|_{\partial q})$ over the stopped cells is at most $C_* a \mu(Q)$ with $C_* = CL^D$.

Step 2. We show that the sum of $\Lambda_{F,q}(u|_{\partial q})$ over the cells that are never stopped tends to zero. There are arbitrarily large depths J at which at least $3\eta_0 J/4$ of the L -adic levels below the mesh of side at most h_0 belong to S . Along a chain that is not stopped by such a depth J , every step to a good child with an enlargement in its parent, at an L -adic level of S , fails the condition $\mu(p) \leq L^D \mu(q)$. Since μ is monotone along the chain and drops by more than the factor L^D at each of these steps, while every leaf satisfies $\mu(\text{leaf}) \geq c|\text{leaf}| = cL^{-dJ}$ on the L -adic mesh of depth J , the number of these steps is at most

$$\frac{d}{D}J + \frac{\log(\mu(Q)/c)}{D \log L} \leq \frac{\eta_0}{8}J + O_{Q,c,u,\omega}(1).$$

By (8.35), at most $\theta J + B_\omega \leq \eta_0 J/8 + B_\omega$ of the steps at L -adic levels of S are bad. For all sufficiently large such J , the chain therefore has at least $3\eta_0 J/4 - \eta_0 J/4 - O(1) \geq \eta_0 J/3$ steps at L -adic levels of S to a child whose enlargement is not contained in its parent. At a step to such a child there are at most $C_d L^{d-1}$ choices of the child. At every other step there are at most L^d . Summing over the at most 2^J positions of the steps to such children, we find that the number of leaves that are never stopped is at most

$$C_Q 2^J L^{dJ} (C_d/L)^{\eta_0 J/3} \leq C_{Q,c,u,\omega} L^{(d-\eta_0/8)J},$$

where we used $L^{\eta_0/3-\eta_0/8} \geq 2C_d^{\eta_0/3}$. By Lemma 8.39(ii), the sum of $\Lambda_{F,q}(u|_{\partial q})$ over them is at most $KL^{(d-\eta_0/8)J}$ times $L^{-(d-2+2\alpha-\eta)J}$, which tends to zero because $d-2+2\alpha-\eta > d-\eta_0/8$.

Step 3. We show that $u \in D(F)$ and $F(u) \leq C_* a E(u)$. At a finite depth J , the stopped cells and the remaining leaves form a finite partition of Q into cells, on whose boundaries we take the values of u . By Lemma 8.32, the F -harmonic extensions of these boundary values glue to a function $v_J \in D(F)$ with $F(v_J) \leq C_* a \mu(Q) + o(1)$. By the maximum principle, each extension lies between the extreme boundary values of u on its cell, so $\|v_J - u\|_\infty$ is at most the modulus of continuity of u at scale h_0 . For fixed h_0 we take J large and then let $h_0 \rightarrow 0$. Then $v_J \rightarrow u$ uniformly, and the closedness of F gives $F(u) \leq C_* a (E(u) + c|Q|)$. Letting $c \downarrow 0$ gives $u \in D(F)$ and $F(u) \leq C_* a E(u)$.

Step 4. We extend the inequality to $D(E)$. Let $v \in D(E)$. By Proposition 8.18, there are smooth f_n with $G_E f_n \rightarrow v$ in the form norm of E . By Step 3, the sequence $(G_E f_n)$ is Cauchy in the form norm of F , and the closedness of F identifies its limit with v . This gives $v \in D(F)$ and (8.37). $\square$

Theorem 8.41 (Two-sided comparison). *There exists a deterministic $C_0 \geq 1$ such that, for every pair of subsequential limit forms and every $0 < \bar{\delta} \leq \bar{\delta}_*$, where $\bar{\delta}_*$ is the initial threshold for $\bar{\delta}$ chosen above, on every cube of $\mathcal{C}$,*

$$(8.38) \quad D(E) = D(F) \quad \text{and} \quad C_0^{-1}E \leq F \leq C_0 E.$$

Proof. Fix a nonzero smooth source f_0 on the unit cube. By the positive and negative moments of $\langle f_0, G_E f_0 \rangle$ and $\langle f_0, G_F f_0 \rangle$ in Lemma 8.19, choose a deterministic $A \geq 1$, uniform over the

subsequential limit forms and over $0 < \bar{\delta} \leq \bar{\delta}_*$, so that $\mathbb{P}[A^{-1} \leq \langle f_0, G_E f_0 \rangle, \langle f_0, G_F f_0 \rangle \leq A] > 0$. It suffices to choose A so that each of the four tail probabilities is less than $1/8$. Set $a_0 = (2C_* A^2)^{-1}$ and let S_- be the set of L -adic levels n with $r_{H_1 n} \leq a_0$. Since the ratios r_k are deterministic, so is S_- . If S_- has upper density at least η_0 , then Proposition 8.40 gives $D(E) \subseteq D(F)$ and $F \leq C_* a_0 E$ on $D(E)$. The variational formula (8.30) then gives, almost surely,

$$\langle f_0, G_F f_0 \rangle = \sup_{v \in D(F)} \{2\langle f_0, v \rangle - F(v)\} \geq \sup_{v \in D(E)} \{2\langle f_0, v \rangle - C_* a_0 E(v)\} = (C_* a_0)^{-1} \langle f_0, G_E f_0 \rangle = 2A^2 \langle f_0, G_E f_0 \rangle,$$

which contradicts $\mathbb{P}[A^{-1} \leq \langle f_0, G_E f_0 \rangle, \langle f_0, G_F f_0 \rangle \leq A] > 0$. Hence S_- has upper density less than η_0 . Exchanging E and F gives the same bound for the set S_+ of L -adic levels with $r_{H_1 n} \geq a_0^{-1}$. The complement of $S_- \cup S_+$ therefore has lower density greater than $1 - 2\eta_0 > \eta_0$, since $\eta_0 < 1/3$, and on this complement $a_0 < r_{H_1 n} < a_0^{-1}$. Proposition 8.40, applied in both directions with $a = a_0^{-1}$, gives $F \leq C_* a_0^{-1} E$ and $E \leq C_* a_0^{-1} F$ with equal domains. Thus (8.38) holds with $C_0 = C_*/a_0 = 2C_*^2 A^2$. $\square$

8.6.2. *The optimal constants are deterministic.* For $a > 0$ we write $aE \leq F$ if $aE^Q \leq F^Q$ on $D(E^Q) = D(F^Q)$ for every cube Q of $\mathcal{C}$, and we optimize over all these cubes:

$$m = \sup\{a : aE \leq F\}, \quad M = \inf\{a : F \leq aE\} \quad \text{and} \quad \Delta = M - m.$$

The numbers m and M are the optimal lower and upper comparison constants, and Δ is their difference. By (8.38), $C_0^{-1} \leq m \leq M \leq C_0$, and $mE \leq F \leq ME$ by monotone limits. The comparison constants are defined for each realization; the next lemma shows that they are deterministic, which Theorem 8.51 uses to conclude that the proportionality constant is deterministic.

Lemma 8.42 (The optimal comparison constants are deterministic). *The optimal comparison constants m and M are almost surely constant.*

Proof. The comparison constants are measurable: on each cube of $\mathcal{C}$, $aE \leq F$ is equivalent, by the variational formula (8.30), to $\langle f, G_F f \rangle \leq a^{-1} \langle f, G_E f \rangle$ for all sources f in the countable dense family.

Resampling one field multiplies both energy measures by the same positive continuous weight, by (8.33). By Lemma 8.25, an inequality between the two forms is equivalent to the same inequality between their energy measures. Multiplying both measures by a common weight preserves these inequalities in both directions. Thus resampling one field leaves m and M unchanged. Since these constants are bounded, Lemma 8.15, with no fields retained, gives $\|m - \mathbb{E}m\|_1 = \|M - \mathbb{E}M\|_1 = 0$. $\square$

8.6.3. *Concentration of the difference between the energies.* Assume $\Delta > 0$. We prove a concentration bound for the difference of the two energies, with an error proportional to Δ .

We first check that the minimal energies of E and F with given boundary values on a cell are taken over the same class of admissible functions. Let q be a cell and let $Q \supseteq q$ be a larger cube of $\mathcal{C}$. A boundary value ℓ on ∂q , affine or of class C^β with $\beta > 1/2$, is realized by an E -harmonic extension through a neighbourhood of ∂q , which belongs to $D(E^Q)$. By (8.38) this extension lies in $D(F^Q)$. Its restriction to q , together with the identification of the minimal energies in Propositions 8.30 and 8.31, places it in the admissible class of both $\Lambda_{E,q}(\ell)$ and $\Lambda_{F,q}(\ell)$. Thus both minimal energies below are taken over the same class of functions with the prescribed boundary values, on which $m\Lambda_{E,q} \leq \Lambda_{F,q} \leq M\Lambda_{E,q}$ as quadratic forms.

Lemma 8.43 (Difference of two minimizers). *Let u_E and u_F be the minimizers of E and F in q with the same affine boundary values, let $c \in [m, M]$, and set $D = F - cE$ and $w = u_F - u_E$. Then $F(w, v) = -D(u_E, v)$ for every $v \in D(E^q)$, and*

$$(8.39) \quad E(w) \leq \left(\frac{\Delta}{m}\right)^2 E(u_E).$$

Moreover, for all u, v in the common domain and every Borel set B ,

$$|\Gamma_D(u, v)|(B) \leq \Delta \sqrt{\Gamma_E(u)(B) \Gamma_E(v)(B)}.$$

Proof. Subtracting the Euler equations $F(u_F, v) = 0 = E(u_E, v)$ gives $F(w, v) = F(u_F, v) - F(u_E, v) = -F(u_E, v) + cE(u_E, v) = -D(u_E, v)$. Since $(m - c)E \leq D \leq (M - c)E$ and $|m - c|, |M - c| \leq \Delta$, the symmetric form D satisfies $|D(u, v)| \leq \Delta \sqrt{E(u)E(v)}$. Taking $v = w$ gives $mE(w) \leq F(w) = -D(u_E, w) \leq \Delta \sqrt{E(u_E)E(w)}$, which is (8.39).

For a Borel set A , Lemma 8.25 gives $|\Gamma_D(h)(A)| \leq \Delta \Gamma_E(h)(A)$. Polarization therefore gives, for every $t > 0$,

$$4t|\Gamma_D(u, v)(A)| \leq 2\Delta[\Gamma_E(u)(A) + t^2\Gamma_E(v)(A)].$$

Optimizing in t gives $|\Gamma_D(u, v)(A)| \leq \Delta \sqrt{\Gamma_E(u)(A) \Gamma_E(v)(A)}$. Sum this bound over a finite Borel partition of B , apply Cauchy-Schwarz, and take the supremum over the partitions to obtain the total-variation bound. $\square$

We normalize the minimal energies with affine boundary values on a cell. Fix $c \in [m, M]$. Let ℓ range over the affine boundary values in a fixed finite set that contains the coordinate functions and determines quadratic forms by polarization. Set $D_q = \sum_{i=1}^d \Lambda_{E,q}(x_i)$ and, for one of these boundary values, $\mathcal{R} = (\Lambda_{F,q}(\ell) - c\Lambda_{E,q}(\ell))/D_q$. By polarization, these quantities give the entries of the matrix B_k in (8.43). Define the normalized measures

$$\nu = \frac{1}{D_q} \sum_{\ell} [\Gamma_E(u_E^\ell) + \Gamma_E(u_F^\ell)], \quad \zeta = \frac{1}{D_q} \sum_{\ell} \Gamma_E(w^\ell),$$

where the sums run over the fixed affine boundary values and $w^\ell = u_F^\ell - u_E^\ell$. Then $\nu(q) \leq C(C_0)$ by (8.38), and $\zeta(q) \leq C(C_0)\Delta^2$ by (8.39). The measure ν inherits the Morrey bound (8.10) of the energy measures of the minimizers. For a bounded g supported in B , let $\mathcal{R}^g$ denote the quantity $\mathcal{R}$ computed for the forms E^g and F^g .

Lemma 8.44 (Perturbation of the normalized difference $\mathcal{R}$). *There exists $C(C_0) < \infty$ such that, for every bounded g supported in B ,*

$$(8.40) \quad |\mathcal{R}^g - \mathcal{R}| \leq C\|g\|_\infty e^{C\|g\|_\infty} [\Delta\nu(B) + \sqrt{\nu(B)\zeta(B)}].$$

Proof. Let u_E^g and u_F^g minimize the weighted forms E^g and F^g with the same boundary values as u_E and u_F , respectively. Put $H_q = \Lambda_{F,q}(\ell) - c\Lambda_{E,q}(\ell)$, $D = F - cE$, $D^g = F^g - cE^g$, $v_E = u_E^g - u_E$ and $v_F = u_F^g - u_F$. Subtract the weighted Euler equations to obtain, for every test function ϕ with zero boundary values on ∂q ,

$$(8.41) \quad F^g(v_F - v_E, \phi) = - \int_B (e^g - 1) [d\Gamma_F(w, \phi) + d\Gamma_D(u_E, \phi)] - D^g(v_E, \phi).$$

Test with $v_F - v_E$ and use Lemma 8.43 and the perturbation bound (8.22) for v_E . Cauchy-Schwarz gives

$$(8.42) \quad E^g(v_F - v_E) \leq C\|g\|_\infty^2 e^{C\|g\|_\infty} [\Gamma_E(w)(B) + \Delta^2 \Gamma_E(u_E)(B)].$$

By (8.22),

$$H_q^g - H_q = \int_B (e^g - 1) d[\Gamma_F(u_F) - c\Gamma_E(u_E)] - [F^g(v_F) - cE^g(v_E)].$$

Use the bilinear identities

$$\begin{aligned} \Gamma_F(u_F) - c\Gamma_E(u_E) &= \Gamma_D(u_E) + \Gamma_F(w, u_F + u_E), \\ F^g(v_F) - cE^g(v_E) &= D^g(v_E) + F^g(v_F - v_E, v_F + v_E). \end{aligned}$$

Lemma 8.43 bounds the first line, in total variation on B , by $CD_q[\Delta\nu(B) + \sqrt{\nu(B)\zeta(B)}]$. For the second, (8.22) for each form and (8.42) give

$$E^g(v_E) + E^g(v_F + v_E) \leq C\|g\|_\infty^2 e^{C\|g\|_\infty} D_q \nu(B), \quad E^g(v_F - v_E) \leq C\|g\|_\infty^2 e^{C\|g\|_\infty} D_q [\zeta(B) + \Delta^2 \nu(B)].$$

Cauchy–Schwarz and $|D^g(v_E)| \leq \Delta E^g(v_E)$ thus bound the correction by $C\|g\|_\infty^2 e^{C\|g\|_\infty} D_q [\Delta\nu(B) + \sqrt{\nu(B)\zeta(B)}]$. Absorbing $\|g\|_\infty^2$ into $\|g\|_\infty e^{C\|g\|_\infty}$ proves the required bound for $|H_q^g - H_q|/D_q$.

Finally, $|H_q| \leq C\Delta D_q$, $e^{-\|g\|_\infty} D_q \leq D_q^g \leq e^{\|g\|_\infty} D_q$, and (8.22) summed over the coordinate data gives $|D_q^g - D_q| \leq C\|g\|_\infty e^{C\|g\|_\infty} D_q \nu(B)$. Apply these bounds to $\mathcal{R}^g - \mathcal{R} = (H_q^g - H_q)/D_q^g + H_q(D_q - D_q^g)/(D_q D_q^g)$. $\square$

Proposition 8.45 (Concentration of the normalized difference B_k). *Let A_E and A_F be the coarse-grained matrices of E and F on a cell q at depth k , defined by $\Lambda_{E,q}(p \cdot x) = |q| p \cdot A_E p$, and similarly for F . Set*

$$c_k = \mathbb{E} \left[\frac{\text{tr } A_F}{\text{tr } A_E} \right], \quad B_k = \frac{A_F - c_k A_E}{\text{tr } A_E}.$$

Then $c_k \in [m, M]$ and $\mathbb{E} B_k = 0$. Moreover, there exists $a > 0$, depending only on the dimension and on the regularity exponents fixed earlier, and not on C_0 or p , such that the following holds for every $1 \leq p < \infty$. If $\bar{\delta}$ is sufficiently small, depending on p and C_0 , then, for every k and every integer $h \geq 0$,

$$(8.43) \quad \|B_k\|_{L^p} \leq C_p(C_0) \bar{\delta} \Delta \quad \text{and} \quad \|B_k - \mathbb{E}[B_k \mid \text{fields with indices within distance } h \text{ of } k]\|_{L^p} \leq C_p(C_0) \bar{\delta} \Delta 3^{-ah},$$

where $C_p(C_0)$ does not depend on k .

Proof. Step 1. The form order gives $c_k \in [m, M]$. By Assumptions (g1) and (g3), the joint law of A_E and A_F , after their common scalar factor is removed, is invariant under signed permutations of the coordinates. Thus $\mathbb{E} B_k$ is a scalar matrix. Its trace is zero by the definition of c_k , so $\mathbb{E} B_k = 0$.

Step 2. We bound the moments of the constant in the Morrey bound (8.10) for ν , uniformly in k . Rescale a cell to the unit cube and divide E by $s_E(q)$ and F by $s_F(q)$. The fields of scale smaller than the cell then have their original law, and the sum of the fields of larger scale, with its value at the cell center subtracted, has the moments of Lemma 8.1, uniformly in k . At finite cutoff, let u minimize the normalized energy with affine boundary values of slope e_1 . Test the Neumann variational formula for boundary flux $e_1 \cdot n$ with scalar multiples of u . The product of the normalized Dirichlet and Neumann energies is then at least $(\int_{\square_0} e_1 \cdot \nabla u)^2 = 1$. This inequality passes to the extracted limits. The moment bounds of Lemmas 8.2 and 8.19 therefore control each normalized minimal energy and its reciprocal. For some deterministic $A < \infty$, both normalized minimal energies with slope e_1 lie in $[A^{-1}, A]$ with positive probability. On that event, the order of the original minimal energies and their scaling give that r_k times the ratio of the two normalized energies lies in $[m, M]$, so that $m A^{-2} \leq r_k \leq M A^2$. The last bounds hold deterministically because r_k is deterministic. Also the reciprocal of the sum of the normalized minimal energies over the coordinate directions is at most the normalized energy of the Neumann problem for E with boundary flux $e_1 \cdot n$. Hölder's inequality, the Morrey bounds for the two minimizers and these reciprocal moments now give the moments, of the finitely many orders fixed at the start of Section 8, of the constant in the Morrey bound for ν , uniformly in k .

Step 3. Let $p \geq 2$, and consider one of the fields of scale smaller than the cell, of scale r . Use the cutoff functions and strips of Lemma 8.14, with boxes of side $r^{(t-d+1)/(t+1)}$ and strips of width $4\sqrt{d}r$. The restrictions of the field to the separated interior subboxes are independent by (g1), and by Lemma 8.27 the whole field can be removed from the limit forms. With b from Lemma 8.13, the original measure satisfies $\nu(\text{strips}) \leq K r^b$. By (8.40), the change of $\mathcal{R}$ caused by deleting the field on the strips is at most $C \Delta S e^{CS} (K r^b + K^{1/2} r^{b/2})$, where S is the sup norm of the

field. As in Lemma 8.14, (8.22) controls the changed minimizers, so the energy measures after the cutoff function is applied to the field satisfy

$$\max_B \hat{\nu}(B) \leq C e^{CS} K r^b, \quad \hat{\nu}(q) \leq C, \quad \hat{\zeta}(q) \leq C \Delta^2.$$

The last two bounds follow from the order between the two forms, which is preserved when the same cutoff function is applied to the field in both forms. The new ingredient in the Efron–Stein estimate is the bound on the sum of squares

$$\sum_B [\Delta \hat{\nu}(B) + \sqrt{\hat{\nu}(B) \hat{\zeta}(B)}]^2 \leq 2 \max_B \hat{\nu}(B) [\Delta^2 \hat{\nu}(q) + \hat{\zeta}(q)] \leq C \Delta^2 \max_B \hat{\nu}(B).$$

Thus the first and last terms of (8.28) in Lemma 8.14, and the conditional Efron–Stein estimate, all keep the factor Δ . By (8.25), the norms of the fields are absorbed at the cost of a fixed fraction of the exponent b . For an entry X of B_k , resampling the field at relative index j consequently gives

$$\|X - X^{(j)}\|_p \leq C_p(C_0) \bar{\delta} \Delta \begin{cases} 3^{-aj}, & j \geq 0, \\ 3^j, & j < 0. \end{cases}$$

For $j < 0$ this follows directly from (8.40) and the bound of Step 2 on the sum of the fields of larger scale, with their value at the cell center subtracted. Choose a below a fixed fraction of b . Apply Lemma 8.15 with no fields retained and then with the fields within distance h of k retained. Centering, polarization and the two geometric sums give (8.43). $\square$

By the concentration bound, we can add to the good event the requirement that the minimal energies of F and $c_k E$ with affine boundary values differ by at most a small multiple of Δ times the minimal energy of E with the same boundary values.

Corollary 8.46 (Minimal energies with affine boundary values on good cells). *Let $\epsilon > 0$. The good event of Section 8.5 can be strengthened by one additional condition, with a threshold for $\bar{\delta}$ and a bad-cell density θ that do not depend on Δ , such that, on every good cell q at depth k and for every affine boundary value ℓ ,*

$$(8.44) \quad |\Lambda_{F,q}(\ell) - c_k \Lambda_{E,q}(\ell)| \leq \epsilon \Delta \Lambda_{E,q}(\ell).$$

Proof. Apply the last assertion of Lemma 8.37 to $X = B_k/\Delta$, with the additional condition $|X| \leq \epsilon c_0$, where $c_0 > 0$ is the deterministic constant of the next paragraph. Its hypotheses are precisely (8.43). Reserve half the allowed density of bad cells for this condition. Choose the rate A of the events $W_{q,h}$ of Lemma 8.37 larger than the exponential growth rate $d \log L$ of the number of chains. Then choose the required moment order and decrease $\bar{\delta}$. Proposition 8.38 gives the strengthened good event, with choices independent of Δ .

We pass from B_k to (8.44). Condition (a) of the good event bounds the condition number of the normalized matrix A_E : the coarse-grained ellipticity constants (2.45) bound $|A_E|$ and $|A_E^{-1}|$. Hence $\lambda_{\min}(A_E) \geq c_0 \operatorname{tr} A_E$ with a deterministic $c_0 > 0$. For an affine function with slope ξ , we obtain $|\xi \cdot (A_F - c_k A_E) \xi| \leq |B_k| \operatorname{tr} A_E |\xi|^2 \leq \epsilon c_0 \Delta \operatorname{tr} A_E |\xi|^2 \leq \epsilon \Delta \xi \cdot A_E \xi$, which is (8.44). $\square$

8.6.4. Affine approximation of the boundary values.

Lemma 8.47 (Affine approximation of the boundary values). *Let $1/2 < \beta < \alpha < 1$, $0 < \gamma < 1$, $\zeta > 0$ and $\varrho > 0$, and assume that*

$$(8.45) \quad \frac{d + \zeta}{2} - \gamma \left(\frac{d - 2}{2} + \alpha + \frac{\alpha - \beta}{\alpha + d/2} (2 - \alpha) \right) < 0.$$

Let E be a subsequential limit form along the subsequence of Section 8.4, realized in the Skorokhod representation, let $f \in C_c^\infty(Q)$, $u = G_E^Q f$, $c > 0$ and $\mu = \Gamma_E(u) + c dx$. Suppose that $L = 3^{H_1}$ is chosen large, with $\lfloor \gamma H_1 \rfloor + 6 < H_1$, that ϵ_{hom} is then chosen small, and that $\bar{\delta}$ is then chosen

small, all independently of f and c . Then the following holds almost surely for cells of side below a number $h_0 > 0$, which may depend on f , c and the environment. Let q be a good child, of side r , of a cell $p \subseteq Q$ of side Lr . If the cube concentric with q of side $3^{\lfloor \gamma H_1 \rfloor + 4}r$, used in conditions (b)–(c), has its closure in p , and

$$\mu(p) \leq L^{d+\zeta} \mu(q),$$

then there is an affine function ℓ_q such that

$$(8.46) \quad \Lambda_{E,q}(u|_{\partial q} - \ell_q) \leq \varrho \mu(q).$$

Proof. Take $L = 3^{H_1}$ with $\lfloor \gamma H_1 \rfloor + 6 < H_1$ and put $S = 3^{\lfloor \gamma H_1 \rfloor + 4}r/729$. Condition (c) of the good event compares s_E at the sides $729S$ and r within a factor two. The mass bound on the parent and the estimate of the oscillation by the energy bound the oscillation of u on the cube of side S concentric with q by $CS^{2-d/2}s_E(q)^{-1/2}L^{d+\zeta/2}\sqrt{\mu(q)}$, together with the source term $CS^2s_E(q)^{-1}\|f\|_\infty$. Condition (b) and Lemma 6.15, with the harmonic comparison on the concentric cube of side $9S$ and its error measured on the concentric cube of side S , show that the $\underline{L}^2$ distance on the cube of side S between u and the Laplace-harmonic function on the cube of side $9S$ with the boundary values of u is at most

$$C\epsilon_{\text{hom}}S^{(2-d)/2}s_E(q)^{-1/2}L^{(d+\zeta)/2}\sqrt{\mu(q)} + CS^2s_E(q)^{-1}\|f\|_\infty.$$

The estimates at finite cutoff pass to the extracted solutions by uniform convergence and Corollary 8.28, since the energy measure $\Gamma_E(u)$ does not charge the faces of p . Let ℓ_q be the affine function that best approximates u in $L^2(q)$; we use it for both norms below. Approximating the Laplace-harmonic function by its first-order Taylor polynomial gains a factor $(r/S)^2$ in $L^2(q)$. Restricting the error of the harmonic approximation from the concentric cube of side S to q costs a factor $(S/r)^{d/2}$ in volume-normalized L^2 . The local Hölder bound on q gains a factor $(r/S)^\alpha$, since the affine projection is bounded on C^α . Since $L^\gamma r/27 < S \leq L^\gamma r/9$, after division by $r^{2-d/2}\sqrt{\mu(q)}/s_E(q)$ and reducing ϵ_{hom} after fixing L , the rescaled L^2 norm and C^α norm of $u - \ell_q$ on q are therefore at most

$$CL^{d+\zeta/2-\gamma d+2/2} + C\epsilon_{\text{hom}}L^{d+\zeta/2+\gamma} \quad \text{and} \quad CL^{d+\zeta/2-\gamma(d-2/2+\alpha)},$$

respectively. Interpolate between them with the exponent $(\alpha - \beta)/(\alpha + d/2)$ and apply the bound (8.6) for Hölder boundary values on the good cell, where the factor in it is comparable to $s_E(q)$. The spatial and coefficient factors cancel, giving

$$\Lambda_{E,q}(u|_{\partial q} - \ell_q) \leq C(L^{d+\zeta/2-\gamma d+2/2} + \epsilon_{\text{hom}}L^{d+\zeta/2+\gamma})^{\frac{2(\alpha-\beta)}{\alpha+d/2}} L^{2(1-\frac{\alpha-\beta}{\alpha+d/2})(d+\zeta/2-\gamma(d-2/2+\alpha))} \mu(q).$$

At $\epsilon_{\text{hom}} = 0$, half the exponent of L is the left side of (8.45), which is negative. First choose L to make this contribution smaller than $\varrho/4$. Next choose ϵ_{hom} so small that the contribution of the error of the harmonic approximation is comparably small. Then decrease $\bar{\delta}$ so that the enlarged good event has the required density of bad cells along every chain. After the same normalization, the source terms are at most $C_L c^{-1/2}s_E(q)^{-1/2}r\|f\|_{C^1} \leq K_{L,\omega}c^{-1/2}r^{1-\eta/2}\|f\|_{C^1}$, by $\mu(q) \geq cr^d$ and (8.7). They tend to zero because $\eta < 2$. A small h_0 makes them smaller than $\varrho/4$. The left side of (8.45) tends to $-(1 - \beta)/(1 + d/2) < 0$ as $\alpha, \gamma \uparrow 1$ and $\zeta \downarrow 0$, so these choices are possible in every dimension. $\square$

8.6.5. Strict improvement of a comparison constant. We now combine concentration for affine boundary values with the affine approximation of the solution's boundary values to rule out $\Delta > 0$. Let $\mathcal{U} = \{n : c_{H_1 n} \leq (m + M)/2\}$. Either this set or its complement has upper density at least $1/2$. Fix $\epsilon = 1/8$ in (8.44). On a good cell q at an L -adic level in $\mathcal{U}$, for every affine ℓ ,

$$B_q(\ell) := M\Lambda_{E,q}(\ell) - \Lambda_{F,q}(\ell) \geq (M - c_k - \epsilon\Delta)\Lambda_{E,q}(\ell) \geq \frac{3}{8}\Delta\Lambda_{E,q}(\ell).$$

Since both minimal energies are taken over the same class of admissible functions, $0 \leq B_q \leq \Delta\Lambda_{E,q}$ on all admissible boundary values. If u is replaced on q by the F -harmonic function with the same

boundary values b , then the new function has F -energy $\Lambda_{F,q}(b) = M\Lambda_{E,q}(b) - B_q(b)$ on q , which is at most $M\Gamma_E(u)(q) - B_q(b)$.

Harmonic replacements will give $F(u) \leq (M - 3\Delta/256)E(u)$ for $u = G_E f$, once the errors they introduce are controlled. Density extends this inequality to the common domain, contradicting the optimality of M . If the complementary set of L -adic levels has upper density at least $1/2$, then we interchange the forms and improve the optimal upper constant $1/m$ instead. Both the energy saved by the replacements and the approximation error are proportional to Δ , so the tolerances can be chosen independently of its size.

Fix $\beta \in (1/2, 1)$ and choose α, γ, ζ satisfying (8.45). For $L = 3^{H_1}$ as in Lemma 8.47, set $b = H_1 - \lfloor \gamma H_1 \rfloor - 4$. Then $b \geq 1$, $H_1 \geq b + 2$ and $3^{-b}Lr$ is the side of the cube concentric with q used for the harmonic comparison in Lemma 8.47. With the dimensional constant $C_d \geq 1$ of the next lemma, the closure of this cube lies in p for every child at distance at least $C_d 3^{-b}$ times the side of p from ∂p , while $C_d 3^{-b} \leq 81C_d L^{\gamma-1}$ tends to zero as L grows. Averaging over finitely many translated grids makes the mass of the children close to the boundary of their parent small, for every measure.

Lemma 8.48 (Shifted rational grids). *Let $L = 3^{H_1}$ with $H_1 \geq b + 2$, and set $M = 3^b + 1$. Translate the nested L -adic grid by the shifts $\mathcal{S} = \{0, 1/M, \dots, (M-1)/M\}^d$. Consider the children at depth n whose closure is within distance $C_d 3^{-b} L^{1-n}$ of the boundary of their parent. The dimensional constant in the counting bounds below may be larger than the constant C_d in this distance. Then, for every x and every $n \geq 1$,*

$$M^{-d} \# \{ \sigma \in \mathcal{S} : \text{the depth-}n \text{ child of } x \text{ in grid } \sigma \text{ is within distance } C_d 3^{-b} L^{1-n} \text{ of the boundary of its parent} \} \leq 2d \left(\right.$$

Consequently, for every finite Borel measure μ on Q , the average over $\sigma \in \mathcal{S}$ of the mass of the cells at depth n within distance $C_d 3^{-b} L^{1-n}$ of the boundary of their parent is at most $C_d 3^{-b} \mu(Q)$.

Proof. Fix a coordinate i . Relative to its parent at depth $n-1$, the point x has fractional coordinate $\{L^{n-1}(x_i - \sigma_i)\}$. Since $\gcd(L, M) = 1$, multiplication by L^{n-1} permutes the residues modulo M . As σ_i ranges over the M shifts, these fractional coordinates form an equally spaced set of M points with a common rotation. If the child is within this distance of the boundary of its parent in coordinate i , then x_i is within $(C_d 3^{-b} L + 1)L^{-n}$ of a face of the parent. Its fractional coordinate then lies in two intervals of total length at most $2C_d 3^{-b} + 2/L$. A rotated equally spaced set has at most $2M(C_d 3^{-b} + 1/L) + 2$ points in these two intervals. We divide by M , sum over the d coordinates, and integrate the pointwise bound against μ . $\square$

Lemma 8.49 (Good cells carry a fixed fraction of the mass). *Suppose that $\mathcal{U}$ has upper density at least $1/2$. Fix $f \in C_c^\infty(Q)$ and $c > 0$, put $u = G_E f$ and $\mu = \Gamma_E(u) + c dx$, let $\epsilon_{\text{coll}} > 0$, and assume that $\theta + L^{-\zeta} + C_d 3^{-b} + \epsilon_{\text{coll}} \leq 1/8$. Then, at L -adic levels $n \in \mathcal{U}$ of arbitrarily large depth, there are shifts $\sigma \in \mathcal{S}$ and finite families $\mathcal{Q}$ of cells at level n in the grid σ such that $\mu(\bigcup \mathcal{Q}) \geq \mu(Q)/8$. Every cell $q \in \mathcal{Q}$, of side r and with parent p , is good, satisfies $\mu(p) \leq L^{d+\zeta} \mu(q)$, and has its concentric enlargement of side $3^{\lfloor \gamma H_1 \rfloor + 4} r$ with closure in p .*

Proof. At one L -adic level, a child that violates $\mu(p) \leq L^{d+\zeta} \mu(q)$ has mass less than $L^{-(d+\zeta)} \mu(p)$, and each parent has at most L^d children. Thus the total mass of these children is at most $L^{-\zeta} \mu(Q)$. By (8.36) for each of the M^d grids, the mass of the bad cells over J L -adic levels is at most $(\theta J + B_\omega^\sigma) \mu(Q)$. The maximum over the finitely many grids of the random constants B_ω^σ is finite on the intersection of the events of Proposition 8.38 for these grids. After division by J , this maximum is absorbed in the $o(1)$ below. By Lemma 8.48, the mass of the cells within distance $C_d 3^{-b} L^{1-n}$ of the boundary of their parent, averaged over σ , is at most $C_d 3^{-b} \mu(Q)$ at every level.

We omit a neighbourhood of ∂Q of mass at most $\epsilon_{\text{coll}} \mu(Q)$. At L -adic levels of large depth, the parents of the remaining points lie inside Q . Along depths J with $\#(\mathcal{U} \cap \{1, \dots, J\}) \geq J/2 - o(J)$, we sum the mass of the good cells with enlargement in the parent that satisfy the bound on their

parents over $n \in \mathcal{U} \cap \{1, \dots, J\}$ and $\sigma \in \mathcal{S}$, and divide by $J|\mathcal{S}|$. This normalized sum is at least

$$\left[\frac{1}{2} - \epsilon_{\text{coll}} - \theta - L^{-\zeta} - C_d 3^{-b} - o(1) \right] \mu(Q) > \frac{\mu(Q)}{4}$$

for large J . A finite average that exceeds $\mu(Q)/4$ has a summand that exceeds it, and since every fixed finite set of levels contributes $o(J)$, such levels can be taken of arbitrarily large depth. $\square$

Lemma 8.50 (Harmonic replacement). *Let $u \in D(E^Q) \cap C(\overline{Q})$ and let $\mathcal{Q}$ be a finite disjoint family of cells of side at most h , whose threefold enlargements lie in Q . Assume that $u|_{\partial q} \in C^\alpha$ for every $q \in \mathcal{Q}$, with $\alpha > 1/2$, and let $u^\mathcal{Q}$ be the function equal to the E -harmonic extension of $u|_{\partial q}$ on each $q \in \mathcal{Q}$ and to u elsewhere. Then $u - u^\mathcal{Q}$ is the E -orthogonal projection of u onto $\bigoplus_q D(E^q)$, and $E(u) = E(u^\mathcal{Q}) + E(u - u^\mathcal{Q})$. If moreover $u = G_E f$, then $E(u - u^\mathcal{Q}) \leq Kh^{3/2} \|f\|_2^2$. In particular, $\sum_{q \in \mathcal{Q}} [\Gamma_E(u)(q) - \Lambda_{E,q}(u|_{\partial q})] = E(u - u^\mathcal{Q}) \rightarrow 0$ as $h \rightarrow 0$, uniformly over the choice of $\mathcal{Q}$.*

Proof. The Euler equations and strong locality identify $w = u - u^\mathcal{Q}$ as the stated orthogonal projection and give both energy identities. Recovery sequences and (8.5) give $\|v\|_{H^{3/4}(Q)}^2 \leq KE(v)$; they also show that $D(E^q) \subseteq H^{3/4}(q)$ on each cell and that its functions have zero trace on ∂q . The fractional Poincaré inequality for zero-trace functions, rescaled to q , gives

$$\|w\|_2^2 \leq Ch^{3/2} \sum_{q \in \mathcal{Q}} [w]_{H^{3/4}(q)}^2 \leq Ch^{3/2} [w]_{H^{3/4}(Q)}^2 \leq Kh^{3/2} E(w).$$

Here the seminorm is the Gagliardo double integral. Finally, for $u = G_E f$, $E(w) = E(u, w) = \langle f, w \rangle \leq \|f\|_2 \|w\|_2$, which proves the estimate. $\square$

Theorem 8.51 (Proportionality). *The optimal comparison constants coincide: $\Delta = 0$. Hence there exists a deterministic $c \in [C_0^{-1}, C_0]$ such that*

$$(8.47) \quad F = cE.$$

Proof. Suppose for a contradiction that $\Delta > 0$.

Step 1. We show that $\mathcal{U}$ having upper density at least $1/2$ contradicts the definition of M . Fix a cube Q in $\mathcal{C}$, $u = G_E f$ with smooth f , $c > 0$, $\mu = \Gamma_E(u) + c dx$ and choose $\varrho \leq 3/1024$ in Lemma 8.47. Choose the parameters in the order above, enlarging L so that $\theta + L^{-\zeta} + C_d 3^{-b} + \epsilon_{\text{coll}} \leq 1/8$ with $b = H_1 - \lceil \gamma H_1 \rceil - 4$, before fixing ϵ_{hom} and $\bar{\delta}$. Let $\mathcal{Q}$ be the family of Lemma 8.49 at an L -adic level of large depth, with maximal side h . For each cell q of this family, the triangle inequalities for $\sqrt{B_q}$ and $\sqrt{\Lambda_{E,q}}$, applied to $u|_{\partial q} = \ell_q + (u|_{\partial q} - \ell_q)$, give $B_q(u|_{\partial q}) \geq \Delta[3/16 \Lambda_{E,q}(u|_{\partial q}) - 4\varrho\mu(q)]$, by (8.46). Summing over $q \in \mathcal{Q}$ and using Lemma 8.50, we obtain

$$\begin{aligned} \sum_{q \in \mathcal{Q}} B_q(u|_{\partial q}) &\geq \Delta \left[\frac{3}{16} \sum_q \Lambda_{E,q}(u|_{\partial q}) - 4\varrho\mu(Q) \right] \\ &\geq \Delta \left[\frac{3}{16} \Gamma_E(u)(\bigcup \mathcal{Q}) - \frac{3}{16} Kh^{3/2} \|f\|_2^2 - 4\varrho(E(u) + c|Q|) \right]. \end{aligned}$$

Since $\mu(\bigcup \mathcal{Q}) \geq \mu(Q)/8$, we have $\Gamma_E(u)(\bigcup \mathcal{Q}) \geq E(u)/8 - 7c|Q|/8$, and with $\varrho \leq 3/1024$ this gives

$$\sum_{q \in \mathcal{Q}} B_q(u|_{\partial q}) \geq \Delta \left[\frac{3}{256} E(u) - Cc|Q| - o_h(1) \right].$$

Now $u \in D(F)$ by (8.38). Let v be the function equal to the F -harmonic extension of $u|_{\partial q}$ on each $q \in \mathcal{Q}$ and to u elsewhere. By strong locality and the bound $\Gamma_F \leq M\Gamma_E$ off the cells,

$$F(v) \leq M\Gamma_E(u)(Q \setminus \bigcup \mathcal{Q}) + \sum_q \Lambda_{F,q}(u|_{\partial q}) = ME(u) - \sum_q [M\Gamma_E(u)(q) - \Lambda_{F,q}(u|_{\partial q})] \leq ME(u) - \sum_q B_q(u|_{\partial q}),$$

where the last inequality uses $\Gamma_E(u)(q) \geq \Lambda_{E,q}(u|_{\partial q})$. As $h \rightarrow 0$ along the L -adic levels of Lemma 8.49, the maximum principle bounds $\|v - u\|_\infty$ by the modulus of continuity of u at

scale h , which tends to zero. Thus $v \rightarrow u$ in L^2 . Lower semicontinuity of F gives $F(u) \leq (M - 3\Delta/256)E(u) + C\Delta c|Q|$.

The tolerances and ϵ_{coll} do not depend on c . Letting $c \downarrow 0$ gives $F(u) \leq (M - 3\Delta/256)E(u)$ for every $u = G_E^Q f$ with smooth f , on every cube Q in $\mathcal{C}$. Step 4 of the proof of Proposition 8.40 extends the bound to $D(E^Q)$ for every cube Q of $\mathcal{C}$. This contradicts the definition of M as an infimum over all these cubes.

Step 2. We show that the complement of $\mathcal{U}$ having upper density at least $1/2$ contradicts the definition of m . On good cells at L -adic levels outside $\mathcal{U}$, we have $\Lambda_{F,q}(\ell) \geq (c_k - \epsilon\Delta)\Lambda_{E,q}(\ell) \geq (m + 3\Delta/8)\Lambda_{E,q}(\ell)$. We regard F as the first form and E as the second. The comparison constants of E relative to F are $1/M \leq 1/m$, with difference $1/m - 1/M$, and the gain at affine boundary values is at least $(3/8)(1/m - 1/M)$, because $m + 3\Delta/8 \leq M$. The argument of Step 1, with the solutions $G_F f$ for smooth f , improves the optimal upper comparison constant $1/m$ of this pair, which contradicts the optimality of m .

Hence $\Delta = 0$, so $F = mE$, and $c = m$ is deterministic by Lemma 8.42. $\square$

We now show that the proportionality constant c in Theorem 8.51 is one, using the fact that the homogenized matrix of the stationary coefficients A_N^0 is the identity.

8.7. Determination of the proportionality constant. The coefficients A_N^0 of (8.1) are stationary, have range of dependence at most $\sqrt{d}$, and have homogenized matrix I by (8.1).

Theorem 8.52 (Uniform qualitative homogenization). *For every unit vector e ,*

$$\lim_{k \rightarrow \infty} \sup_N \mathbb{E} J(\square_k, e, e; A_N^0) = 0.$$

Proof. Step 1. We obtain uniform moment bounds and choose cutoffs for which the expected energy changes little across a long interval of scales. Fix a sufficiently large even integer ξ . Proposition 4.1, subadditivity (2.22) and stationarity give

$$\sup_{N, k \geq 0} \left\| \sup_{|e|=1} J(\square_k, e, e; A_N^0) \right\|_{L^\xi} < \infty.$$

Moreover, the family

$$\{\Lambda_{1/4}(\square_k; A_N^0) + \lambda_{1/4}(\square_k; A_N^0)^{-1} : N, k \geq 0\}$$

is tight. This follows from Lemma 4.5, with $s = 1/8$, $r = 2$ and the initial error $C\xi \log(2 + \xi)\bar{\delta}^2$, followed by (2.52) and weighted Cauchy–Schwarz. Choose ξ large enough for that lemma and then reduce $\bar{\delta}$.

Write

$$f_N(k) = \sum_{j=1}^d \mathbb{E} J(\square_k, e_j, e_j; A_N^0) = \frac{1}{2} \mathbb{E} \text{tr} (A_N^0(\square_k) + (A_N^0)^{-1}(\square_k) - 2I).$$

By (2.21) and (3.29), $\mathbb{E} A_N^0(\square_k)$ and $\mathbb{E} (A_N^0)^{-1}(\square_k)$ decrease to I . Thus $f_N(k) \geq 0$ decreases in k , and the limit of $\sup_N f_N(k)$ as $k \rightarrow \infty$ exists. Choose N_k with $f_{N_k}(4k) \geq \sup_N f_N(4k) - k^{-1}$. Then

$$\sup_{k \leq n \leq 4k} (f_{N_k}(n) - f_{N_k}(4k)) \leq \sup_N f_N(k) - \sup_N f_N(4k) + k^{-1} \rightarrow 0.$$

Step 2. We show that the coarse-grained matrices $(A_{N_k}^0)^{-1}(\square_n)$ concentrate and that the expectations of the coarse-grained matrices $A_{N_k}^0(\square_n)$ converge along these scales. By subadditivity, the average of $(A_{N_k}^0)^{-1}(z + \square_k)$ over the scale- k subcubes of $\square_n$, minus $(A_{N_k}^0)^{-1}(\square_n)$, is positive semidefinite and has expected trace at most $2(f_{N_k}(k) - f_{N_k}(n))$. This difference tends to zero in L^1 , uniformly for $k \leq n \leq 4k$. The moment bound of Step 1 and (2.41) upgrade this to L^2 . For $n \geq 2k$, the variance of the average is at most $C3^{-d(n-k)}$, because the finite range of dependence splits the

subcubes into a dimensional number of classes of independent cubes. Consequently, after extracting a subsequence, there are deterministic matrices $P, R \geq I$ such that

$$\sup_{k \leq n \leq 4k} \|\mathbb{E}A_{N_k}^0(\square_n) - P\| \longrightarrow 0, \quad \sup_{2k \leq n \leq 4k} \|(A_{N_k}^0)_*^{-1}(\square_n) - R\|_{L^2} \longrightarrow 0.$$

For the first convergence, extract a limit of $\mathbb{E}A_{N_k}^0(\square_{4k})$ and note that $\mathbb{E}A_{N_k}^0(\square_n) - \mathbb{E}A_{N_k}^0(\square_{4k})$ is positive semidefinite with trace at most $2(f_{N_k}(n) - f_{N_k}(4k))$, which tends to zero by Step 1.

Step 3. We use the estimates of Section 4 to show that both limiting matrices equal the identity. Fix $p \in \mathbb{R}^d$ and let $v = v(\cdot, \square_{4k}, p, R^{-1}p; A_{N_k}^0)$. Apply (4.24) with $s = 1/4$ and $L_0 = 2k$. The expected gaps in the subadditivity inequality (2.21) tend to zero uniformly over $2k \leq n \leq 4k$ by Step 1. The mean-square differences of the coarse-grained matrices $(A_{N_k}^0)_*^{-1}(\square_n)$ tend to zero by Step 2 and stationarity. The weights of the two sums in (4.24) are summable, and $(A_{N_k}^0)_*^{-1}(\square_{4k})R^{-1}p - p \rightarrow 0$ in L^2 . Since the coarse-grained ellipticity constants are tight, we obtain

$$3^{-4k} \|\nabla v\|_{\hat{H}^{-1}(\square_{4k})} \longrightarrow 0 \quad \text{in probability.}$$

Now (4.23), with the same tight constants and the vanishing gaps in the subadditivity inequality, gives

$$J(\square_{4k-1}, p, R^{-1}p; A_{N_k}^0) \longrightarrow 0 \quad \text{in probability.}$$

The moment bounds make these energies uniformly integrable, so their expectations tend to zero. By (2.17), the limit of these expectations is also $1/2p \cdot (P - R^{-1})p$. Thus $P = R^{-1}$. Since $P, R \geq I$, both equal I , and $f_{N_k}(4k) \rightarrow 0$. The choice of N_k then shows that the limit of $\sup_N f_N(k)$ is zero. $\square$

Let $\Lambda_{N,Q}^0$ denote the minimal energy with given boundary values for A_N^0 . The bound (2.41) and Cauchy–Schwarz turn Theorem 8.52 into

$$(8.48) \quad \lim_{k \rightarrow \infty} \sup_N \mathbb{E} \left| \frac{\Lambda_{N,3^k \square_0}^0(p \cdot x)}{3^{dk}} - |p|^2 \right| = 0 \quad (p \in \mathbb{R}^d).$$

Theorem 8.53 (Identification of the scalar). *The scalar c in (8.47) equals one. Consequently, all subsequential limit forms coincide.*

Proof. Let $E^0 = e^{-H}E$ and $F^0 = e^{-H}F$ be the forms obtained by removing the factor e^H with the rule (8.33) for continuous weights, applied to the random continuous weight e^{-H} on a compact cube. They are subsequential limits of the same normalized sequence, whose coefficient is A_N^0 . Fix R and $p \neq 0$. By Proposition 8.30, $\Lambda_{N,R \square_0}^0(p \cdot x)$ converges along the subsequence that defines E to $\Lambda_{E^0, R \square_0}(p \cdot x)$. The uniform moments of these energies give uniform integrability. Hence

$$\mathbb{E} \left| \frac{\Lambda_{E^0, R \square_0}(p \cdot x)}{R^d} - |p|^2 \right| \leq \sup_N \mathbb{E} \left| \frac{\Lambda_{N, R \square_0}^0(p \cdot x)}{R^d} - |p|^2 \right|.$$

The same holds for F^0 along its subsequence. Letting $R \rightarrow \infty$ through triadic values, (8.48) shows that both $\mathbb{E}|\Lambda_{E^0, R \square_0}(p \cdot x)/R^d - |p|^2|$ and its analogue for F^0 tend to zero. But $F^0 = cE^0$ gives $\Lambda_{F^0, R \square_0} = c\Lambda_{E^0, R \square_0}$ on every cube, so $c|p|^2 = |p|^2$ and $c = 1$.

Since all subsequential limit forms are therefore equal, and the energies of Proposition 8.17 are strongly relatively compact, every subsequence has a further subsequence converging to the same limit. Hence these energies, for the countable family of cubes and data fixed in Section 8.4, converge along the full sequence of cutoffs. $\square$

9. CONVERGENCE OF THE PROCESS

In this section we prove Theorem 7.1(i). Combining the unique limit form of Section 8 with the limit of the reversible measures, we prove that the path laws converge in probability, and we then upgrade this convergence to almost-sure convergence.

- (1) After constructing the diffusions from every starting point (Section 9.1), we show that the measures μ_N converge and that their mass in small balls is bounded uniformly in the cutoff (Section 9.2). This bound lets us pass to the limit in the resolvent equations of the processes killed on leaving a cube, even though the sources in these equations are given by the random measures. Mollifying these sources and applying the Hölder estimates of Section 8 gives convergence in probability of the killed resolvents, uniformly in the starting point.
- (2) Together with tightness of the path laws (Section 9.3), the convergence of the killed resolvents identifies the limiting path laws and gives convergence in probability (Section 9.4).
- (3) We then compare the energies at two finite cutoffs. On a small cube, rescaling shows that the two energies differ essentially by the ratio of their deterministic normalizations. If these ratios were bounded away from one on a positive proportion of scales, the energy of a fixed affine boundary value would stay a fixed factor away from its limit, contradicting the convergence in probability. Once such scales are excluded, the comparison holds with summable error probabilities, and the Borel–Cantelli lemma gives almost-sure convergence of the finite-volume energies (Section 9.5). Bounds that hold for all cutoffs on a single event of probability one then carry this almost-sure convergence over to the Dirichlet forms, the killed resolvents and the path laws (Sections 9.6–9.7).

The conventions of Section 8 remain in force, except that Sections 9.1 and 9.3 work with the diffusions X and $X^{(L)}$ and their coefficients $\mathbf{a}$ and $\mathbf{a}_L$.

9.1. Construction of the diffusions. We construct the diffusions from their elliptic resolvents. By Lemma 6.20, the coefficient $\mathbf{a}$ is positive and locally $C^{1,1}$. The same holds for $\mathbf{a}_L$. The logarithmic derivative bound (6.117) gives at most linear growth of the drift, uniformly in L . For either coefficient, the weak solutions of

$$\lambda \mathbf{a} u - \nabla \cdot (\mathbf{a} \nabla u) = \mathbf{a} f$$

on increasing bounded cubes, with zero boundary values, converge to the whole-space resolvent. Outside large cubes, comparison with supersolutions constructed from the drift growth bound shows that the solution vanishes at infinity for compactly supported data. Contraction extends the construction to $C_0(\mathbb{R}^d)$. The maximum principle gives positivity and the bound $\|R_\lambda\| \leq \lambda^{-1}$; uniqueness gives the resolvent identity. For compactly supported smooth data, the elliptic equation gives $\lambda R_\lambda f \rightarrow f$ as $\lambda \rightarrow \infty$, and contraction extends this to $C_0(\mathbb{R}^d)$. Hille–Yosida [Kallenberg, 2002, Theorem 19.11] therefore gives a Feller semigroup. Its positive operators have transition kernels by the Riesz representation theorem.

On each bounded cube, local ellipticity and regularity give the short-time bounds needed to realize the transition kernels on continuous paths. Kolmogorov extension, followed by continuous extension from a countable dense set of times, constructs the path laws. We denote the resulting diffusions for $\mathbf{a}$ and $\mathbf{a}_L$ by X and $X^{(L)}$, respectively, initially defined up to their lifetimes. The Feller property and approximation of stopping times by dyadic times give the strong Markov property. Proposition 9.8 shows that X and every $X^{(L)}$ are conservative from every starting point.

The semigroup is symmetric on $L^2(\mathbb{R}^d, \mathbf{a} dx)$, and its resolvents agree with those of the closure of

$$(9.1) \quad f \mapsto \int_{\mathbb{R}^d} \mathbf{a} |\nabla f|^2 dx, \quad f \in C_c^\infty(\mathbb{R}^d).$$

This follows by testing the weak equation and exhausting $\mathbb{R}^d$ by cubes. Its generator is $\mathbf{a}^{-1} \nabla \cdot (\mathbf{a} \nabla)$. Restricting the form to a cube with zero boundary values identifies its resolvents with those of the

diffusion killed on leaving the cube [Fukushima et al., 2011, Theorems 4.4.2–4.4.3]. Thus the killed resolvents are the zero-boundary weak solutions.

These laws also describe the usual SDE construction. Since $\nabla \log \mathbf{a}$ is locally Lipschitz, there is a unique maximal strong solution from every $x \in \mathbb{R}^d$ of

$$(9.2) \quad dX_t = \sqrt{2} dW_t + \nabla \log \mathbf{a}(X_t) dt,$$

where W is standard Brownian motion on $\mathbb{R}^d$. Fix a bounded cube Q , $\lambda > 0$ and $f \in C_c^\infty(Q)$, and let u solve $\lambda \mathbf{a} u - \nabla \cdot (\mathbf{a} \nabla u) = \mathbf{a} f$ in Q , with zero boundary values. Interior elliptic regularity and Itô's formula, stopped first strictly inside Q , give, after exhaustion and removal of the deterministic time cutoff,

$$u(x) = \mathbf{E}_x^{\text{SDE}} \left[\int_0^{\tau_Q} e^{-\lambda t} f(X_t) dt \right], \quad x \in Q.$$

Here $\mathbf{E}_x^{\text{SDE}}$ denotes expectation for the maximal SDE solution. Thus its killed resolvents agree with those of the analytic construction. Uniqueness of Laplace transforms identifies the killed semigroups, and the Markov property identifies their path laws. Exhausting $\mathbb{R}^d$ identifies the laws up to their lifetimes, for every starting point. The same argument applies to $\mathbf{a}_L$ and to the space and time rescalings in (7.2); Proposition 9.8 makes these lifetimes infinite. For an open set U , write $\tau_U = \inf\{t \geq 0 : X_t \notin U\}$ and let $\tau_U^{(L)}$ be the exit time of $X^{(L)}$ from U . The diffusion $X^{(L)}$ has generator (1.8).

9.2. The reversible measure and resolvent convergence in probability. Write E for the unique limit form; it is measurable with respect to the random fields. Along each subsequence of the Skorokhod representation, the forms E_N^Q Mosco-converge to E^Q on $L^2(Q, dx)$, and their coercivity constants are bounded.

9.2.1. The chaos martingale and the mass of small balls. Recall the measures μ_N^0 and the reversible measures $\mu_N = e^H \mu_N^0$ from the paragraph following (7.2).

Lemma 9.1 (Positive moments of the chaos). *Let $p \geq 1$ be an integer. There exist $C_p < \infty$ and $c_p > 0$ such that, if $\bar{\delta} \leq c_p$, then, for every $N \geq 0$ and every cube q of side $r \leq 1$ contained in a fixed bounded region,*

$$(9.3) \quad \mathbb{E}[\mu_N(q)^p] \leq C_p r^{dp - C_p \bar{\delta}^2}.$$

Moreover, $N \mapsto \mu_N^0(q)$ is a nonnegative martingale, and $\mu_N(q)$ is a nonnegative martingale conditionally on H .

Proof. Since $\mathbb{E} e^{\mathbf{g}_{-j}(x)} = e^{\tau^2}$ for every x and the fields are independent, integration against fixed nonnegative test functions shows that μ_N^0 is a measure-valued martingale. The sum H of the fields with scales larger than one is independent of the fields at scales at most one, so $\mu_N = e^H \mu_N^0$ is a martingale given H . On a bounded region the factor e^{pH} is controlled by $\mathbb{E} e^{p \sup H} < \infty$, which follows from the exponential moments of the $C^{1,1}$ norm of H on compact cubes in Lemma 8.1. It therefore suffices to prove (9.3) for μ_N^0 .

For $p \geq 2$ we expand

$$\mathbb{E} \mu_N^0(q)^p = \int_{q^p} \mathbb{E} \prod_{i=1}^p \prod_{j=0}^N e^{\mathbf{g}_{-j}(x_i) - \tau^2} dx.$$

The field $\mathbf{g}_{-j}$ has range of dependence $C3^{-j}$ by Assumption (g1). If all pairwise distances $|x_i - x_k|$ exceed that range, then the p factors are independent and the normalized expectation equals one. For each of the remaining fields, Hölder's inequality gives $\mathbb{E} \prod_i e^{\mathbf{g}_{-j}(x_i) - \tau^2} \leq \mathbb{E} e^{p \mathbf{g}_0(0)} e^{-p\tau^2} \leq e^{Cp^2 \bar{\delta}^2}$ by (1.5). The number of fields for which this separation condition fails is at most $C + \sum_{i < k} \log_3^+(C/|x_i - x_k|)$, so the integrand is bounded by $C \prod_{i < k} (|x_i - x_k| \wedge 1)^{-a}$ with $a = Cp^2 \bar{\delta}^2$.

There are $p(p-1)/2$ pair factors. Hölder's inequality with exponent $p(p-1)/2$ and the bound $\int_{q \times q} |x-y|^{-ap(p-1)/2} dx dy \leq C r^{2d-ap(p-1)/2}$, valid when $ap(p-1)/2 < d$, give $\mathbb{E} \mu_N^0(q)^p \leq C_p r^{dp-ap(p-1)/2}$. Since $ap(p-1)/2 \leq C p^4 \bar{\delta}^2$, the condition $ap(p-1)/2 < d$ holds once $\bar{\delta} \leq c_p$, and (9.3) follows. $\square$

Proposition 9.2 (Mass of small balls and the limiting measure). *Let $\epsilon \in (0, 1)$, let p be an integer with $p\epsilon > d$, and assume that $\bar{\delta} \leq c_p$ and $C_p \bar{\delta}^2 < p\epsilon - d$, with c_p and C_p as in Lemma 9.1. Then, almost surely, μ_N converges locally weakly to a locally finite measure μ , and there is a random constant K_μ with $\mathbb{E} K_\mu^p < \infty$ such that, for every x in the fixed bounded region, every $0 < r \leq 1$ and every $N \geq 0$,*

$$(9.4) \quad \mu_N(B_r(x)) \leq K_\mu r^{d-\epsilon} \quad \text{and} \quad \mu(B_r(x)) \leq K_\mu r^{d-\epsilon}.$$

Almost surely, the measure μ has full support and no atoms, and $\mu(\partial Q) = 0$ for every cube Q .

Proof. For a dyadic cube q of side 2^{-k} , Doob's inequality [Kallenberg, 2002, Proposition 7.16] and (9.3) give $\mathbb{E} \sup_N \mu_N(q)^p \leq (p/(p-1))^p \sup_N \mathbb{E} \mu_N(q)^p \leq C 2^{-k(dp-C_p \bar{\delta}^2)}$. Let C be the constant in the covering below and put

$$K_\mu^p = C^p \sum_{k \geq 0} \sum_{q \text{ of side } 2^{-k}} 2^{kp(d-\epsilon)} \sup_N \mu_N(q)^p.$$

There are $C 2^{dk}$ cubes of side 2^{-k} in the region, so $\mathbb{E} K_\mu^p \leq C \sum_k 2^{k(d+C_p \bar{\delta}^2-p\epsilon)} < \infty$ by the choice of p and then of $\bar{\delta}$. A ball $B_r(x)$ with $2^{-k-1} < r \leq 2^{-k}$ is covered by a number of cubes of side 2^{-k} that depends only on d . This gives the first bound in (9.4) with one random constant for all N , x and $r \leq 1$.

Since $p > 1$, the bound (9.3) at this order gives uniform integrability of $\mu_N(C)$. The martingale convergence theorem [Kallenberg, 2002, Theorem 7.18 and Corollary 7.22] gives almost-sure and L^p convergence against every function in a countable dense subset of $C_c(\mathbb{R}^d)$. The local mass bound above gives vague relative compactness [Kallenberg, 2002, Theorem A2.3], so these limits determine a single locally finite measure μ , and $\mu_N \rightarrow \mu$ locally weakly almost surely. The bound for μ follows from the portmanteau theorem applied to open balls. The measure μ has no atoms by (9.4). To prove that $\mu(\partial Q) = 0$, cover ∂Q by $C r^{1-d}$ balls of radius r . This gives $\mu(\partial Q) \leq C K_\mu r^{1-\epsilon} \rightarrow 0$.

It remains to prove that μ has full support. Set $\mu^0 = e^{-H} \mu$. Fix an open cube Q and $k \geq 0$, and write $\mu^0 = e^{\sum_{j=0}^k \mathbf{g}_{-j} - (k+1)\tau^2} \mu^{0,(>k)}$, where $\mu^{0,(>k)}$ is the limiting measure built in the same way from the fields $\mathbf{g}_{-j}$ with $j > k$ only. Since $\mathbb{E} \mu^{0,(>k)}(\partial Q) = |\partial Q| = 0$ and the factor in front of $\mu^{0,(>k)}$ is a fixed continuous positive function, weak convergence shows that the two sides of this identity give the same mass to Q . Hence the events $\{\mu^0(Q) = 0\}$ and $\{\mu^{0,(>k)}(Q) = 0\}$ coincide up to null sets for every k , so $\{\mu^0(Q) = 0\}$ is a tail event of the independent fields. Its probability is therefore 0 or 1, and it is 0 because $\mathbb{E} \mu^0(Q) = |Q| > 0$ by uniform integrability. The positive continuous factor e^H gives the full support of μ . $\square$

9.2.2. Traces on the reversible measures. We choose $\epsilon < 1$ in (9.4), so that the trace inequality (8.31) applies with $t = d - \epsilon > d - 1$ to every μ_N and to μ , with the constant $C(K_\mu + \mu_N(Q))$. We extend functions in $H_0^1(Q)$ by zero to $\mathbb{R}^d$. For such a function w , the zero extension satisfies $\|w\|_{H^{3/4}(\mathbb{R}^d)} \leq C \|w\|_{H^{3/4}(Q)}$, so (8.5) gives

$$(9.5) \quad \|w\|_{H^{3/4}(\mathbb{R}^d)}^2 \leq K_N E_N^Q(w),$$

for every $w \in H_0^1(Q)$, with a constant that is bounded along the sequence in the Skorokhod representation. Traces on μ are defined by density: by (8.31), the map sending a smooth function on $\bar{Q}$ to its class in $L^2(\mu)$ is bounded from $H^{1/2}$ to $L^2(\mu)$, and it extends by continuity.

Lemma 9.3 (Varying-measure convergence). *Let $u_N \in H_0^1(Q)$ satisfy $\sup_N E_N^Q(u_N) < \infty$ along the sequence in the Skorokhod representation and $u_N \rightarrow u$ in $L^2(Q, dx)$. Then, for every bounded continuous function f on $\overline{Q}$,*

$$(9.6) \quad \int u_N^2 d\mu_N \longrightarrow \int u^2 d\mu \quad \text{and} \quad \int f u_N d\mu_N \longrightarrow \int f u d\mu.$$

Proof. By (9.5) the zero extensions are bounded in $H^{3/4}(\mathbb{R}^d)$, and the interpolation inequality $\|w\|_{H^{1/2}} \leq \|w\|_{L^2}^{1/3} \|w\|_{H^{3/4}}^{2/3}$ upgrades the convergence in L^2 to strong convergence in $H^{1/2}(\mathbb{R}^d)$. We write

$$\int u_N^2 d\mu_N - \int u^2 d\mu = \int (u_N - u)(u_N + u) d\mu_N + \left(\int u^2 d\mu_N - \int u^2 d\mu \right).$$

The first term tends to zero by Cauchy–Schwarz and (8.31). For the second, we approximate u in $H^{1/2}$ by smooth functions φ . By (8.31), the resulting error is small uniformly over all μ_N and μ , and $\int \varphi^2 d\mu_N \rightarrow \int \varphi^2 d\mu$ by weak convergence, since $\mu(\partial Q) = 0$. The proof of the second limit is the same. $\square$

9.2.3. The killed resolvent as a minimizer. At finite cutoff the generator is symmetric in $L^2(\mu_N)$ with form E_N^Q . Since the density of μ_N is bounded on $\overline{Q}$ at every finite N , it follows that, for $\lambda > 0$ and bounded f , the killed resolvent $u_N = R_{N,\lambda}^Q f$ is the unique minimizer over $H_0^1(Q)$ of

$$E_N^Q(u) + \lambda \int u^2 d\mu_N - 2 \int f u d\mu_N.$$

It satisfies the sub-Markov bound $\|u_N\|_\infty \leq \|f\|_\infty / \lambda$, the weak equation $E_N^Q(u_N, \varphi) = \int (f - \lambda u_N) \varphi d\mu_N$, and, by testing with u_N , the energy bound $E_N^Q(u_N) \leq \|f\|_\infty^2 \mu_N(Q) / \lambda$.

Proposition 9.4 (Passage of the minimization to the limit). *Let $f \in C(\overline{Q})$ and $\lambda > 0$, and assume that, along the sequence in the Skorokhod representation, E_N^Q Mosco-converges to E^Q and that E^Q satisfies the coercivity bound (8.5) with a finite constant. Then the functional*

$$(9.7) \quad u \longmapsto E^Q(u) + \lambda \int u^2 d\mu - 2 \int f u d\mu$$

has a unique minimizer u^ in the energy Hilbert space $(D(E^Q), E^Q)$, and every $L^2(dx)$ cluster point of the killed resolvents $u_N = R_{N,\lambda}^Q f$ equals u^* .*

Proof. The energy space is complete because $E^Q(u) \geq \|G^Q\|^{-1} \|u\|_{L^2}^2$ by the variational formula (8.30). The trace from $D(E^Q)$ to $L^2(\mu)$ is bounded, since $\int u^2 d\mu \leq C(K_\mu + \mu(Q)) \|u\|_{H^{1/2}}^2 \leq C K E^Q(u)$ by (8.31) and the coercivity of E^Q . The functional (9.7) is therefore continuous, strictly convex and coercive on the energy space, and the Riesz representation theorem gives a unique minimizer u^* , which satisfies $E^Q(u^*, v) + \lambda \int u^* v d\mu = \int f v d\mu$ for every $v \in D(E^Q)$.

To pass to the limit, note that the energies $E_N^Q(u_N)$ are bounded, so, by (8.5), a subsequence converges strongly in $L^2(dx)$. The liminf inequality of Mosco convergence and Lemma 9.3 show that the liminf of the finite-cutoff functionals at u_N is at least the value of (9.7) at the limit. Since u_N minimizes the finite-cutoff functional, recovery sequences for every $v \in D(E^Q)$, together with (9.6), show that this liminf is at most the value of (9.7) at v . Hence the limit minimizes (9.7), so it equals u^* . $\square$

Write $R_\lambda^Q f$ for the minimizer u^* .

9.2.4. *Uniform convergence of the killed resolvents.* To obtain uniform convergence we need Hölder bounds on the killed resolvents that are uniform in N . The source in the resolvent equation is the measure $(f - \lambda u_N)\mu_N$, whose density is not bounded uniformly in N , so we mollify the source before applying the Hölder estimate for bounded sources.

Proposition 9.5 (Uniform convergence of killed resolvents). *Assume the hypotheses of Proposition 9.4. Assume also that solutions with zero Dirichlet boundary values and bounded sources satisfy the Hölder estimate (8.11) with $\alpha = 1/2$ up to the boundary, with the supremum norm of the source on the right side and with a constant whose moments of the finitely many orders fixed at the start of Section 8 are finite. Let $\epsilon < 1/[8(d+2)]$ in (9.4). Then, for every bounded continuous f and every $\lambda > 0$,*

$$(9.8) \quad \sup_{\overline{Q}} |R_{N,\lambda}^Q f - R_\lambda^Q f| \longrightarrow 0 \quad \text{in probability,}$$

and the limit vanishes on ∂Q . There is a random constant K_Q with finite moments of these orders such that, for all $\lambda > 0$ and $f \in C(\overline{Q})$,

$$(9.9) \quad \|R_\lambda^Q f\|_{C^{1/4}(\overline{Q})} \leq K_Q \|f\|_\infty.$$

Proof. Let $u_N = R_{N,\lambda}^Q f$ and $\zeta_N = (f - \lambda u_N)\mu_N$, so that $|\zeta_N| \leq 2\|f\|_\infty \mu_N$ and $E_N^Q(u_N, \varphi) = \int \varphi d\zeta_N$. Fix $0 < r \leq 1$, put $R = r^{d+2} \leq 1$, and let η_R be a mollifier at scale R .

Step 1. We bound the mollified source and the error of the mollification. By (9.4),

$$|\zeta_N * \eta_R(x)| \leq CR^{-d} |\zeta_N|(B_R(x)) \leq CK_\mu R^{-\epsilon} \|f\|_\infty.$$

For the error, we first use that, by Schur's test, $\|\zeta_N\|_{H^{-1/4}}^2$ is at most $\|\zeta_N\|_{TV}$ times the supremum over x of the Bessel potential of order $1/2$ of $|\zeta_N|$ at x . Splitting the Bessel potential into dyadic annuli and using (9.4) bounds this supremum, since $\epsilon < 1/2$. Hence $\|\zeta_N\|_{H^{-1/4}} \leq K\|f\|_\infty$. Then the bound $\sup_\xi (1 + |\xi|^2)^{-1/2} \min(1, R|\xi|)^2 \leq CR$ gives

$$\|\zeta_N - \zeta_N * \eta_R\|_{H^{-3/4}} \leq CR^{1/2} \|\zeta_N\|_{H^{-1/4}} \leq KR^{1/2} \|f\|_\infty.$$

Step 2. We compare u_N with the solution for the mollified source. Let $v_{N,R} \in H_0^1(Q)$ solve $E_N^Q(v_{N,R}, \varphi) = \int \varphi (\zeta_N * \eta_R) dx$. Its source is a bounded function of size $KR^{-\epsilon} \|f\|_\infty$, so (8.11) with $\alpha = 1/2$ gives $\|v_{N,R}\|_{C^{1/2}(\overline{Q})} \leq KR^{-\epsilon} \|f\|_\infty$. The difference of the two equations and (9.5) give

$$E_N^Q(u_N - v_{N,R}) = \langle \zeta_N - \zeta_N * \eta_R, u_N - v_{N,R} \rangle \leq KR^{1/2} \|f\|_\infty \|u_N - v_{N,R}\|_{H^{3/4}} \leq KR^{1/2} \|f\|_\infty E_N^Q(u_N - v_{N,R})^{1/2},$$

hence $\|u_N - v_{N,R}\|_{L^2} \leq KR^{1/2} \|f\|_\infty$.

Step 3. We prove a Hölder bound for u_N . By Steps 1 and 2, the mean-square oscillation of u_N on $B_r(x) \cap Q$ is at most

$$K(r^{1/2}R^{-\epsilon} + r^{-d/2}R^{1/2}) \|f\|_\infty = K(r^{1/2-(d+2)\epsilon} + r) \|f\|_\infty,$$

and $1/2 - (d+2)\epsilon > 3/8 > 1/4$. Using zero extensions across ∂Q , Campanato's criterion gives $\|u_N\|_{C^{1/4}(\overline{Q})} \leq K\|f\|_\infty$ and $u_N = 0$ on ∂Q . Here K is a product of finitely many random constants with finite moments of these orders, and it is therefore tight.

Step 4. We prove convergence in probability. Fix two cutoff subsequences, and apply the Skorokhod representation of Section 8.4 to both of them jointly; the variables represented jointly include the fields $(\mathbf{g}_j)_{j \in \mathbb{Z}}$, the reversible measures, the quadratic forms $\langle f, G_N^Q f \rangle$ with $f \in \mathcal{D}$ and $Q \in \mathcal{C}$, and the constants in Steps 1–3. The constants have tight laws. The limit forms of the two sequences coincide by Theorem 8.53, and their limiting reversible measures are the same. The Arzelà–Ascoli theorem and Proposition 9.4 therefore identify both uniform limits with the unique minimizer of the same functional.

Consequently, every pair of cutoff subsequences has a further pair of subsequences along which the joint laws of the two killed resolvents converge to a law supported on the diagonal of $C(\overline{Q})^2$. Since $C(\overline{Q})$ is Polish, the Gyöngy–Krylov criterion [Gyöngy and Krylov, 1996, Lemma 1.1] gives convergence in probability on the original probability space. The joint representation with these quadratic forms and the reversible measure identifies the limit with $R_\lambda^Q f$. The constant in Step 3 is independent of λ and f , which gives (9.9) and the zero boundary values. $\square$

9.3. Tightness. In this subsection we prove Proposition 9.8 (tightness in expectation and in probability). Section 9.4 uses both forms of tightness, together with the compact containment estimate (9.23), to identify the limiting path law from its resolvents. The early-exit bound (9.21) is used again in Section 10 to bound mean exit times from below.

We first bound the probability of an early exit from a cube. After the rescaling introduced below, a cube of side 3^m becomes $\square_0$ and time is measured in units of $3^{2m}/\bar{\mathbf{a}}_{m \wedge L}$. The process started in $\square_{-2}$ leaves $\square_0$ before time t with probability at most $K\sqrt{t}$, where K has moments bounded uniformly in the cutoff, the cube size, and the position of the cube.

The proof bounds the Laplace transform of the exit time by one minus the solution of a resolvent problem with a cutoff function as source. A local maximum principle for subsolutions, with the Laplace parameter equal to $1/t$, then gives the exit-probability bound. Since pointwise ellipticity is not uniform in the cutoff and the reversible measure becomes singular in the limit, we prove this maximum principle by Moser iteration with respect to the reversible measure. The iteration uses the mass bounds, the $H^{3/4}$ coercivity bound, and cutoff functions of Lemma 9.6. The energy measures of these cutoff functions satisfy an upper bound with the same power of the ball radius as the upper mass bound for the reversible measure. The exit-probability bound shows that, with high probability, the paths stay in a large cube up to a fixed time; applied at stopping times, it also controls their modulus of continuity.

Throughout this subsection we assume (g1)–(g4) and that $\bar{\delta}$ is small enough, depending on the moment orders fixed below. We recall the diffusion $X^{(L)}$ from (1.7)–(1.8) and we set

$$Z_t^{(L)} := 3^{-L} X_{3^{2L}t/\bar{\mathbf{a}}_L}^{(L)}, \quad t \geq 0.$$

We use rescaled coordinates for $Z^{(L)}$ and physical coordinates for $X^{(L)}$. All statements of this subsection also hold for the rescaled diffusion $3^{-L} X_{3^{2L}t/\bar{\mathbf{a}}_L}$ without cutoff.

For $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, consider the cube $z + \square_m$ in physical coordinates, and set

$$j := m \wedge L, \quad c := \left(\frac{\mathbf{a}_L}{\mathbf{a}_j} \right)(z), \quad d\mu := c^{-1} \mathbf{a}_L(z + 3^m x) dx, \quad A := \bar{\mathbf{a}}_j^{-1} c^{-1} \mathbf{a}_L(z + 3^m x).$$

If $m \geq L$, then $c = 1$. In the local coordinates $x \mapsto z + 3^m x$, which rescale $z + \square_m$ to $\square_0$, and in units of time

$$(9.10) \quad \frac{3^{2m}}{\bar{\mathbf{a}}_{m \wedge L}},$$

the generator of $X^{(L)}$ becomes $\mathcal{A} = (d\mu/dx)^{-1} \nabla \cdot (A \nabla)$. It is symmetric on $L^2(\mu)$, with Dirichlet form and local energy

$$\mathcal{E}(v, w) := \int A \nabla v \cdot \nabla w \quad \text{and} \quad \mathcal{E}(v; V) := \int_V A |\nabla v|^2.$$

The factor c normalizes the product of the factors contributed by $\mathbf{g}_i$, with $j < i \leq L$, to have value one at z . By (3.4), the oscillation of the sum of these fields on the cube has uniformly bounded exponential moments, and therefore the moment bounds for the random constants below are uniform in L , m and z . For the coefficient $\mathbf{a}$ without cutoff we use instead $j = m$ and $c = \mathbf{a}(z)/\mathbf{a}_m(z)$, so that the density of μ is $c^{-1} \mathbf{a}(z + 3^m x)$, and the unit of time is $3^{2m}/\bar{\mathbf{a}}_m$. This density

is then $\mathbf{a}_m(z + 3^m x)$ multiplied by $\exp(\sum_{i>m}(\mathbf{g}_i(z + 3^m x) - \mathbf{g}_i(z)))$, and this exponential factor satisfies the same oscillation bound.

Lemma 9.6 (Local mass, coercivity and cutoff-function estimates). *Fix a bounded reference cube and a moment order $q \in [1, \infty)$. Also fix finitely many concentric pairs $V_0 \Subset V_1$ inside the reference cube. For each pair and each $s \in [0, 1]$, let V_s be the cube with the same center whose side length interpolates affinely between those of V_0 and V_1 ; the nested families below consist of the cubes $V_{k2^{-n}}$, for $n \geq 0$ and $0 \leq k \leq 2^n$. There exists a threshold $\bar{\delta}_0 > 0$, which may depend on q , such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there is a deterministic $B < \infty$, independent of q , such that, for every $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, there is a random constant $K = K_{L,m,z} \geq 1$ satisfying*

$$\sup_{L,m,z} \mathbb{E}[K_{L,m,z}^q] < \infty$$

such that, after rescaling $z + \square_m$ to $\square_0$ as in (9.10), the following statements hold.

(i) Mass bounds: for every ball $B_r(x)$ contained in the reference cube with $0 < r \leq 1$,

$$(9.11) \quad K^{-1}r^{d+1/2} \leq \mu(B_r(x)) \leq Kr^{d-1/2},$$

and the analogous bounds hold for cubes.

(ii) Coercivity in $H^{3/4}$: for each fixed interior cube V of the family with $n = 0$ and every $v \in H^1(V)$,

$$(9.12) \quad \|v\|_{H^{3/4}(V)}^2 \leq K(\mathcal{E}(v; V) + \|v\|_{L^2(V, dx)}^2),$$

and the last term can be omitted if $v \in H_0^1(V)$. For $V = V_{k2^{-n}}$, the same assertion holds with $K2^{Bn}$ in place of K .

(iii) Harmonic cutoff functions: between consecutive cubes $V_{k2^{-n}} \Subset V_{(k+1)2^{-n}}$ with gap $a \asymp 2^{-n}$ there is a cutoff function χ , equal to one on the smaller cube, vanishing near the boundary of the larger cube, and satisfying $0 \leq \chi \leq 1$, such that, for every ball $B_r(x)$ with $0 < r \leq 1$,

$$(9.13) \quad \Gamma_\chi(B_r(x)) := \int_{B_r(x)} A|\nabla \chi|^2 \leq Ka^{-B}r^{d-1/2}.$$

When needed, we replace K by a fixed power of K ; this is possible if the lemma is applied from the outset with a correspondingly larger moment order q . For a finite collection of cubes we take the maximum of their constants; the resulting moment bounds remain uniform in L, m and z .

Proof. The three estimates are local versions of estimates proved in Sections 8 and 9.2 for the rescaled coefficient A_N of (8.1): Lemma 9.1 bounds the positive moments of the mass of small cubes under the multiplicative chaos, Lemma 8.3 bounds the $H^{3/4}$ norm by the energy, Proposition 8.7 bounds the energy measure of balls at every radius for solutions of Dirichlet problems, and Lemma 8.21 constructs cutoff functions as harmonic interpolants. We give the additional arguments needed to pass to the rescaling of $z + \square_m$ to $\square_0$ and to obtain one random constant for all the nested cubes.

Step 1. We prove the mass bounds (9.11). Reindexing the fields in reverse order identifies, in law, the normalized mass of the unit cube at the scale j with $\mu_N^0(\square_0)$ from the paragraph after (7.2). Lemma 9.1, applied to the martingale $N \mapsto \mu_N^0(\square_0)$, bounds the positive moments of $\mu_N^0(\square_0)$ uniformly in N , including the field $\mathbf{g}_0$ at scale one. The bound on the second moment and $\mathbb{E}\mu_N^0(\square_0) = 1$ give $\inf_N \mathbb{P}(\mu_N^0(\square_0) \geq 1/2) > 0$, by the Cauchy–Schwarz inequality. Fix $J \in \mathbb{N}$ and choose two sufficiently separated triadic subcubes of $\square_0$ of side 3^{-J} . Since the fields are independent and have a finite range of dependence, for every $N \geq J$ we have

$$\mu_N^0(\square_0) \geq 3^{-dJ} Y(\mu_{N-J}^{0,(1)}(\square_0) + \mu_{N-J}^{0,(2)}(\square_0)),$$

where $\mu_{N-J}^{0,(1)}$ and $\mu_{N-J}^{0,(2)}$ are independent copies of μ_{N-J}^0 , independent of the random variable $Y > 0$. Here Y is the infimum over $\square_0$ of the part of the density of μ_N^0 contributed by the J coarsest

fields. By Assumption (g2), the variable $-\log Y$ has a Gaussian upper tail. Consequently, the function $G(s) := \sup_N \mathbb{P}(\mu_N^0(\square_0) < e^{-s})$ satisfies, for all sufficiently large s ,

$$G(s) \leq C e^{-cs^2} + G\left(\frac{3s}{4}\right)^2,$$

where the finitely many $N < J$ are absorbed into the first term. The lower bound on $\mathbb{P}(\mu_N^0(\square_0) \geq 1/2)$ gives $G(s) \leq \rho < 1$ for large s . Letting $s \rightarrow \infty$ in the recursion gives $G(s) \rightarrow 0$. Iterating the recursion on the intervals $[(4/3)^n s_0, (4/3)^{n+1} s_0]$ then gives $G(s) \leq C \exp(-cs^{3/2})$: since $2(3/4)^{3/2} > 1$, squaring the bound at $3s/4$ gives a stronger bound at s , and the Gaussian term in the recursion is of smaller order. Hence, for every $p < \infty$,

$$(9.14) \quad \sup_N \mathbb{E}[\mu_N^0(\square_0)^{-p}] < \infty.$$

We now pass to the triadic subcubes of the reference cube. Let P be such a subcube, of side 3^{-n} , and split the density of μ into the factors from fields of scale below and above the physical side of P . The product of the factors from fields of scale between the physical side of P and 3^j , and its reciprocal, have moments of order p bounded by $C_p 3^{C_p \bar{\delta}^2 n}$, and the oscillation of the fields above scale j has a uniformly bounded exponential moment. Combining the positive moments of $\mu_N^0(\square_0)$ with (9.14) therefore gives, for every $p < \infty$,

$$\mathbb{E} \left[\left(\frac{\mu(P)}{|P|} \right)^p + \left(\frac{\mu(P)}{|P|} \right)^{-p} \right] \leq C_p 3^{C_p \bar{\delta}^2 n}.$$

The reference cube contains $O(3^{dn})$ triadic subcubes of side 3^{-n} . Choose $p > q + 2d$, and then decrease $\bar{\delta}$ so that $p/2 > d + C_p \bar{\delta}^2$. Markov's inequality and a summation over n prove (9.11) for triadic cubes, with the asserted moments, and a finite collection of translated triadic grids gives the bounds for balls.

Step 2. We prove the $H^{3/4}$ coercivity bound (9.12). If $m \leq L$, it is the estimate of Lemma 8.3, applied with $N = m$ and with the value at z subtracted from each field of scale larger than 3^m ; the proof of that lemma is uniform in the number of these fields, by (3.4). If $m > L$, we use Lemma 4.5 to bound the coarse-grained ellipticity of the cubes of physical side at least 3^L in the fractional Poincaré inequality (2.61); the estimates for the smaller cubes are unchanged. The case $v \in H_0^1(V)$ uses (2.65).

For triadic subcubes of side 3^{-n} in the local coordinates, the fields of scale between the physical side of the subcube and 3^j contribute scalar factors. Together with (3.32), they multiply the constant by a random factor with L^p norm at most $C_p 3^{C_p \bar{\delta}^2 n}$. For subcubes of physical side less than one, the ellipticity bounds on unit cubes give a fixed power of 3^n instead, with a random factor whose moments are bounded by Lemma 8.6. Taking the maximum over the $O(3^{dn})$ triadic cubes of side 3^{-n} and summing over n with a sufficiently large geometric weight gives one constant K for all these meshes. The same summation over the $O(2^n)$ cubes $V_{k2^{-n}}$ of the nested families gives $K 2^{Bn}$ after enlarging B . The comparison (2.58) of translated cubes allows fixed enlargements and changes of grid.

Step 3. We construct the cutoff functions of (9.13). We use harmonic interpolants as in Lemma 8.21. Choose a smooth cutoff function f that equals one on a neighborhood of the smaller cube and vanishes on a neighborhood of the boundary of the larger cube, and a mesh of side $h \asymp a$ such that the cells meeting the transition region of f lie between the two prescribed cubes. Let χ be the function which, on each cell, is A -harmonic with boundary values f . The maximum principle gives $0 \leq \chi \leq 1$, and $\chi = f$ on cells where f is constant.

We apply Proposition 8.7 on each rescaled cell with zero source, smooth boundary datum f and exponent $t = d - 1/2$ in (8.10). That proposition covers radii below the physical unit scale, as well as boundary faces and corners, so its random constant already accounts for the coefficient at

scales below the unit scale. On a physical cell of side 3^k with $k \geq 0$, we apply that proposition at cutoff $k \wedge L$ and subtract from each remaining field its value at the center of the cell. If $k > L$, the same proof applies with Lemma 4.5 for the coarse-grained energy and Proposition 6.16 on cubes of side 3^k ; the contribution of scales below the smallest field scale to the moments is no larger, since $L \leq k$.

It remains to express the cell estimates after rescaling $z + \square_m$ to $\square_0$ as in (9.10). Let P be a cell of side $h = 3^{-n}$ with $n \leq m$ and physical center z_P , and put $k = m - n$ and $j_1 = k \wedge L$. In the coordinates of the cell, the coefficient A is the normalized coefficient of the cell multiplied by two factors: the exponential $\exp(\sum_{j < i \leq L} (\mathbf{g}_i(z_P) - \mathbf{g}_i(z)))$, which is controlled by the oscillation bound for these fields, and the scalar

$$\frac{\bar{\mathbf{a}}_{j_1} \mathbf{a}_j(z_P)}{\bar{\mathbf{a}}_j \mathbf{a}_{j_1}(z_P)}.$$

By (3.32) and (3.4), this scalar and its reciprocal have L^p norms at most $C_p 3^{C_p \bar{\delta}^2 n}$. After subtracting a constant from f , its rescaled C^2 norm on the cell is at most $Ch \|\nabla f\|_{C^1}$. The estimate on the cell therefore gives a random constant K_P with $\|K_P\|_{L^p} \leq C_p h^{-C_p \bar{\delta}^2}$ such that, for every $0 < r \leq h$,

$$\Gamma_\chi(B_r(x) \cap P) \leq K_P \|\nabla f\|_{C^1}^2 h^{1/2} r^{d-1/2}.$$

At each mesh size there are $O(h^{-d})$ cells. We take the maximum of the constants K_P over these cells and sum over the meshes with a sufficiently large geometric weight. This proves (9.13) when the cells have physical side at least one, or equivalently $h \geq 3^{-m}$.

If $h < 3^{-m}$, local elliptic estimates and Lemma 8.6 bound the random factor R_m that controls the coefficient on cells of side less than 3^{-m} , with $\sup_{L,m} \mathbb{E}[(3^{-B_0 m} R_m)^q] < \infty$ for a fixed B_0 . Since $h < 3^{-m}$, this factor is at most $(3^{-B_0 m} R_m) h^{-B_0}$, which is absorbed by the same summation over meshes. Finally, $\|\nabla f\|_{C^1} \leq Ca^{-2}$. The estimates for the coefficient $\mathbf{a}$ follow by locally smooth convergence of the ratios of the fields above scale j and Fatou's lemma, as in the proof of Proposition 8.7. $\square$

The regularity estimates of Section 6 concern solutions, and the coefficient A has no ellipticity bounds uniform in L , so we derive the local maximum principle for subsolutions from Lemma 9.6 alone.

Lemma 9.7 (Local bound for subsolutions). *After rescaling a cube $z + \square_m$ to $\square_0$ as in (9.10), let $V' \Subset V$ be fixed cubes. The estimates of Lemma 9.6 imply that there is a random constant K such that, for every bounded nonnegative weak subsolution w of $\nabla \cdot (A \nabla w) \geq 0$ in V ,*

$$(9.15) \quad \operatorname{ess\,sup}_{V'} w \leq K \left(\int_V w^2 d\mu \right)^{1/2}.$$

For every fixed $q < \infty$, there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then the constant K has a finite moment of order q .

Since K does not depend on w , the lemma applies with the same constant to every bounded nonnegative solution of $\nabla \cdot (A \nabla w) = \lambda w d\mu/dx$, for every $\lambda \geq 0$. We use this uniformity in λ in Step 1 of the proof of Proposition 9.8.

Proof. By a finite covering it suffices to treat a fixed concentric pair $V' \Subset V$. In both iterations of Step 3 we use the nested family $V_{k2^{-n}}$ of Lemma 9.6 associated with this pair. We apply Moser iteration with respect to μ , in which trace inequalities for measures satisfying the upper bound in (9.11) replace the Sobolev inequality.

Step 1. We prove two trace inequalities. Let ν be a measure satisfying $\nu(B_r(x)) \leq Kr^{d-1/2}$ for every ball $B_r(x)$ with $0 < r \leq 1$, and let $p_0 \geq 2$. We show that, if $s > 1/4$ and $t > d/2 - (d-1/2)/p_0$, then

$$(9.16) \quad \|v\|_{L^2(\nu)}^2 \leq CK \|v\|_{H^s}^2 \quad \text{and} \quad \|v\|_{L^{p_0}(\nu)}^2 \leq CK \|v\|_{H^t}^2.$$

For a Littlewood–Paley decomposition of v , each block v_i at frequency 2^i satisfies $\|v_i\|_{L^2(\nu)}^2 \leq CK2^{i/2}\|v_i\|_2^2$. To see this, write v_i as its convolution with a rapidly decaying reproducing kernel, apply the Cauchy–Schwarz inequality, and integrate against ν using the bound on the mass of balls. Interpolation with Bernstein’s inequality for that block gives

$$\|v_i\|_{L^{p_0}(\nu)} \leq CK^{1/p_0}2^{i(d/2-(d-1/2)/p_0)}\|v_i\|_2.$$

Summing over i with the Cauchy–Schwarz inequality, after extending v from a cube to the whole space, gives (9.16) on the fixed cubes. Below we take $s = 1/2$, $t = 3/4$ and $p_0 = 2 + 1/d$, for which $d/2 - (d-1/2)/p_0 = d/(2d+1) < 3/4$; Section 10 uses other exponents.

Step 2. We bound the $L^2(dx)$ norm of v , and the integral of v^2 against the energy measure of a cutoff function, by the energy of v and its $L^2(\mu)$ norm. Partition a cube into cubes P of side ρ , and let v_P be the average of v over P with respect to Lebesgue measure. Apply the first trace inequality, with $3/4$ in place of $1/2$, to $v - v_P$. The fractional Poincaré inequality and the lower mass bound give

$$\int_P |v - v_P|^2 d\mu \leq CK\rho[v]_{H^{3/4}(P)}^2$$

and

$$\|v\|_{L^2(dx)}^2 \leq CK\rho^{-1/2}\|v\|_{L^2(\mu)}^2 + CK^2\rho^{1/2}\|v\|_{H^{3/4}}^2.$$

We apply (9.12) to the last term and choose ρ small enough to absorb the resulting multiple of $\|v\|_{L^2(dx)}^2$ into the left side. For every $0 < \varepsilon \leq 1$ this yields

$$(9.17) \quad \|v\|_{L^2(dx)}^2 \leq \varepsilon \mathcal{E}(v) + CK^C \varepsilon^{-C} \|v\|_{L^2(\mu)}^2.$$

Next let χ be a cutoff function across a gap a . We apply the first inequality in (9.16) to $\nu = \Gamma_\chi$, which satisfies its hypothesis with Ka^{-B} in place of K by (9.13), interpolate $H^{1/2}$ between $H^{3/4}$ and $L^2(dx)$, and then use (9.12) and (9.17). For every $0 < \varepsilon \leq 1$ this gives

$$(9.18) \quad \int v^2 d\Gamma_\chi \leq \varepsilon \mathcal{E}(v; V) + CK^C \varepsilon^{-B} a^{-B} \int_V v^2 d\mu.$$

Step 3. We iterate. For $p \geq 2$, test the subsolution inequality with $\chi^2 w^{p-1}$ and put $v = w^{p/2}$. The Cauchy–Schwarz inequality and (9.18), with ε a small multiple of p^{-1} , give, for nested cubes $V_r \subseteq V_R$,

$$\mathcal{E}(v; V_r) \leq 2^{-B-2} \mathcal{E}(v; V_R) + CK^C p^B (R-r)^{-B} \int_{V_R} w^p d\mu.$$

Iterate this inequality over radii increasing from r to R with geometrically decreasing gaps. Hence

$$\mathcal{E}(w^{p/2}; V_r) \leq CK^C p^B (R-r)^{-B} \int_{V_R} w^p d\mu.$$

The second trace inequality, the $H^{3/4}$ coercivity bound (9.12) and (9.17) give, since $p_0/2 > 1$,

$$\|w\|_{L^{p_0 p/2}(V_r, \mu)} \leq (CK^C p^B (R-r)^{-B})^{1/p} \|w\|_{L^p(V_R, \mu)}.$$

We iterate with $p_i = 2(p_0/2)^i$ and geometrically decreasing gaps. The product of the resulting constants converges, which proves (9.15). For continuous functions such as $1 - u$ in the proof of Proposition 9.8, the full support of μ identifies the essential supremum with the pointwise supremum. $\square$

Proposition 9.8 (Tightness). *Let $K_{L,x}^\omega$ be the quenched law of either rescaled diffusion defined above, started at x . Equip $C([0, \infty), \mathbb{R}^d)$ with the topology of locally uniform convergence. Almost surely these processes are conservative and, for every compact set $B \subseteq \mathbb{R}^d$, the following statements hold.*

(i) Tightness in expectation: for every $\varepsilon > 0$, there is a compact set $\mathcal{K} \subseteq C([0, \infty), \mathbb{R}^d)$ such that

$$(9.19) \quad \sup_L \mathbb{E} \left[\sup_{x \in B} K_{L,x}^\omega(\mathcal{K}^c) \right] \leq \varepsilon.$$

In particular, the annealed laws $\mathbb{E}[K_{L,x}^\omega]$ with $L \in \mathbb{N}_0$ and $x \in B$ are tight.

(ii) Tightness in probability: for every $\eta > 0$, there is a compact set $\mathcal{K}_\eta$ of probability measures on $C([0, \infty), \mathbb{R}^d)$ such that

$$(9.20) \quad \sup_L \mathbb{P}[K_{L,x}^\omega \notin \mathcal{K}_\eta \text{ for some } x \in B] \leq \eta.$$

Proof. Step 1. We show that, after rescaling every cube $z + \square_m$ to $\square_0$ as in (9.10), the process started in $\square_{-2}$ leaves $\square_0$ before time t with probability at most $K\sqrt{t}$. Let χ be a cutoff function of Lemma 9.6 between $\square_{-1}$ and $\square_0$, so that χ takes values in $[0, 1]$, equals one on $\square_{-1}$, vanishes near $\partial\square_0$, and satisfies $\mathcal{E}(\chi) \leq K$. For $\lambda > 0$, let u solve

$$\lambda u - \mathcal{A}u = \lambda \chi \quad \text{in } \square_0, \quad u = 0 \quad \text{on } \partial\square_0.$$

The maximum principle gives $0 \leq u \leq 1$. Since u minimizes $\mathcal{E}(v) + \lambda \|v - \chi\|_{L^2(\mu)}^2$ over $v \in H_0^1(\square_0)$, comparison with $v = \chi$ gives

$$\int_{\square_0} |u - \chi|^2 d\mu \leq \frac{\mathcal{E}(\chi)}{\lambda} \leq \frac{K}{\lambda}.$$

On $\square_{-1}$, the nonnegative function $w = 1 - u$ satisfies $\nabla \cdot (A \nabla w) = \lambda w d\mu/dx \geq 0$, and, since $\chi = 1$ on $\square_{-1}$, the last display bounds its norm in $L^2(\square_{-1}, \mu)$. Therefore Lemma 9.7, whose constant does not depend on λ , gives

$$\sup_{\square_{-2}} (1 - u) \leq K\lambda^{-1/2}.$$

Let $\widehat{X}$ be the process obtained by rescaling $z + \square_m$ to $\square_0$, let P_x^ω and E_x^ω be its law and expectation started at x , and let $\tau_{\square_0}$ be its exit time from $\square_0$. The stopped Itô formula gives, for every $x \in \square_0$,

$$u(x) = E_x^\omega \int_0^{\tau_{\square_0}} \lambda e^{-\lambda s} \chi(\widehat{X}_s) ds \leq 1 - E_x^\omega e^{-\lambda \tau_{\square_0}}.$$

Taking $\lambda = 1/t$, we obtain

$$(9.21) \quad \sup_{x \in \square_{-2}} P_x^\omega(\tau_{\square_0} \leq t) \leq e \sup_{x \in \square_{-2}} E_x^\omega \exp\left(-\frac{\tau_{\square_0}}{t}\right) \leq K\sqrt{t}.$$

The constant K does not depend on t , and $\mathbb{E}K \leq C$. In physical coordinates, (9.21) says that the probability that $X^{(L)}$, started at any point of $z + \square_{m-2}$, leaves $z + \square_m$ before time t is at most

$$(9.22) \quad K\sqrt{t\bar{\mathbf{a}}_{m \wedge L}} 3^{-m},$$

and the same bound holds for X with m in place of $m \wedge L$. These bounds concern the processes killed on leaving a bounded cube and do not presume that the processes are conservative.

Step 2. We prove compact containment and conservativeness. For triadic $R \geq 1$, a cube of side R in rescaled coordinates has time unit $R^2 3^{2L}/\bar{\mathbf{a}}_L$ at finite cutoff, and, by (3.32), at least $R^2 3^{2L}/\bar{\mathbf{a}}_L$ for the diffusion X without cutoff. After the cube sizes are adjusted by fixed factors, (9.21) therefore gives, for every $T > 0$ and every $R \geq 1$ larger than a fixed multiple of $\max_{x \in B} |x|$,

$$(9.23) \quad \sup_L \mathbb{E} \left[\sup_{x \in B} K_{L,x}^\omega \left(\sup_{s \leq T} |Z_s| > R \right) \right] \leq C \frac{\sqrt{T}}{R}.$$

Letting $R \rightarrow \infty$ shows that, for almost every environment, the process does not explode from any starting point in B . The same estimate then gives compact containment for the laws on continuous path space.

Step 3. We bound the modulus of continuity and prove (9.19). Fix $R < \infty$ and $r > 0$, fix $k \in \mathbb{N}_0$ such that cubes of side 3^{-k} have diameter at most $r/4$, and cover the ball B_R by finitely many cubes $y + \square_{-k-2}$. For $L \geq k$, the time unit of each of the cubes $y + \square_{-k}$, measured in units of $3^{2L}/\bar{\mathbf{a}}_L$, satisfies, by (3.32),

$$3^{-2k} \frac{\bar{\mathbf{a}}_L}{\bar{\mathbf{a}}_{L-k}} \geq 3^{-(2+2\tau^2/\log 3)k} > 0.$$

We apply (9.21) to each cube of this finite cover and take the maximum of the finitely many random constants. The strong Markov property then gives, for every stopping time $S \leq T$ and every $h > 0$,

$$\sup_{L \geq k} \mathbb{P} \left(\sup_{0 \leq s \leq h} |Z_{S+s}^{(L)} - Z_S^{(L)}| > r/4, S < \tau_{B_R} \right) \leq C_{R,r} \sqrt{h}.$$

To obtain (9.19), in which the supremum over starting points lies inside the expectation, we first restrict to the event on which the maximum of the random constants of this finite cover is at most M . Its complement has probability at most $C_{R,r}/M$, uniformly in $L \geq k$. On this event, for every starting point in B_R , the probability that the path moves by more than $r/4$ within time h is at most $C_{R,r} M \sqrt{h}$, and, by the strong Markov property, the same bound holds after every stopping time before exit from B_R , conditionally on the path up to that stopping time.

Consider the successive exit times of the path from balls of radius $r/4$, the first centered at the starting point and each later one at the position of the path at the preceding exit time. Choose h_0 so small that this probability bound, with $h = h_0$, is at most $1/16$. If $n > 2T/h_0$ such exits occur before time T , then at least $n/2$ of the gaps between consecutive exit times are shorter than h_0 . The strong Markov property in the fixed environment and a union bound over the choices of these gaps bound the probability of this event by $2^n 16^{-n/2}$. Hence the number of exits before time T is tight, uniformly in the starting point.

For a fixed number of exits, the probability that some gap is shorter than h tends to zero as $h \rightarrow 0$. Since the path remains in a ball of radius $r/4$ between consecutive exit times, this gives a uniform bound on the probability that the path oscillates by more than r on an interval of length h . We choose first M , then the number of exits, then h .

For each of the finitely many $L < k$, the coefficient is smooth. Nonexplosion and continuous dependence on the starting point show that, almost surely, the laws of these diffusions started in B are uniformly tight, and dominated convergence gives the corresponding bound for the expected supremum. We apply the preceding construction along a sequence of integer time horizons and of spatial tolerances tending to zero, with summable error probabilities. Compact containment and the Arzelà–Ascoli theorem then prove (9.19).

Step 4. We prove (9.20). By (i), for every $j \geq 1$ there is a compact set $\mathcal{K}_j$ of paths for which the left side of (9.19) is at most $\eta 2^{-2j}$. Set

$$\mathcal{K}_\eta := \{P : P(\mathcal{K}_j^c) \leq 2^{-j} \text{ for every } j \geq 1\}.$$

The set $\mathcal{K}_\eta$ is closed and uniformly tight, hence compact. Markov's inequality and a union bound over j prove (9.20). $\square$

9.4. Convergence in probability of the path laws. We use the killed resolvents to identify the limiting laws on $C([0, \infty), \mathbb{R}^d)$. The ingredients are compact containment, the Markov identity at finite cutoff, and the fact that discounted integrals of finite-dimensional observations determine a continuous path law.

We record the short-time approximation for the killed resolvents. For $f \in C_0(Q)$, extend f by zero. Splitting the occupation integral at h gives

$$(9.24) \quad \sup_N \mathbb{E} \|\lambda R_{N,\lambda}^Q f - f\|_\infty \leq \sup_{|x-y| \leq r} |f(x) - f(y)| + 2\|f\|_\infty \left(e^{-\lambda h} + \sup_N \mathbb{E} \sup_{x \in \overline{Q}} K_{N,x}^\omega \left(\sup_{s \leq h} |Z_s - x| > r \right) \right).$$

The last probability tends to zero as $h \downarrow 0$ by (9.19). If a path is killed before h without moving farther than r , its starting point is within r of the boundary, where f is small. Proposition 9.5 passes this estimate to the limiting resolvents. In particular, $\mathbb{E} \|\lambda R_\lambda^Q f - f\|_\infty \rightarrow 0$ as $\lambda \rightarrow \infty$.

Proposition 9.9 (Convergence in probability). *Let $B \subseteq \mathbb{R}^d$ be compact and $\varepsilon > 0$. Then*

$$\lim_{N, N' \rightarrow \infty} \mathbb{P} \left[\sup_{x \in B} d_{\text{weak}}(K_{N,x}^\omega, K_{N',x}^\omega) > \varepsilon \right] = 0.$$

Consequently, the kernels $x \mapsto K_{N,x}^\omega$ converge in probability, locally uniformly in x , to a continuous random probability kernel $x \mapsto K_x^\omega$ that is measurable with respect to the fields.

Proof. Use the Prokhorov metric d_P associated with the complete metric on path space

$$\sum_{m \geq 1} 2^{-m} (1 \wedge \sup_{t \leq m} |z(t) - z'(t)|).$$

Write $R_{N,\lambda} f(x) = \iint_0^\infty e^{-\lambda t} f(Z_t) dt dK_{N,x}^\omega$ for the whole-space resolvent. If Q contains B , splitting at the exit time gives

$$(9.25) \quad \sup_{x \in B} |R_{N,\lambda} f(x) - R_{N,\lambda}^Q f(x)| \leq \frac{\|f\|_\infty}{\lambda} \left(e^{-\lambda T} + \sup_{x \in B} K_{N,x}^\omega(\tau_Q \leq T) \right).$$

Choose first T large and then Q large. Compact containment (9.23) makes the expectation of the right side arbitrarily small, uniformly in N . Proposition 9.5 therefore makes the whole-space resolvents Cauchy in probability, locally uniformly in x . Contraction and localization give the same conclusion for measurable random continuous sources g_N converging in probability locally uniformly to g , provided $\|g_N\|_\infty$ and $\|g\|_\infty$ are bounded by one deterministic constant. The intermediate sources in the iterated resolvents below have this bound by contraction.

For bounded continuous $f_1, \dots, f_k$ and positive integers $n_1, \dots, n_k$, consider the bounded continuous path functional

$$\int_{(0,\infty)^k} e^{-\sum_{i=1}^k n_i s_i} \prod_{i=1}^k f_i(z(s_1 + \dots + s_i)) ds.$$

The Markov property and Fubini identify its expectation with

$$R_{N,n_1}(f_1 R_{N,n_2}(f_2 \cdots R_{N,n_k} f_k))(x).$$

Iterating the preceding resolvent convergence makes these expectations Cauchy in probability, uniformly for $x \in B$. These functionals determine continuous path laws. After the substitution $u_i = e^{-s_i}$, varying the positive integers n_i gives the integrals of the finite-dimensional observations against every monomial on $[0,1]^k$. Polynomial approximation and continuity in time then determine these observations. A separating algebra of spatial test functions determines the finite-dimensional laws.

Tightness (9.19) now transfers this Cauchy property to d_P . Indeed, on a compact set of paths carrying all but a small mass, finitely many continuous tests suffice to distinguish laws at any prescribed positive d_P distance; the determining functionals approximate these tests there. Markov's inequality controls the mass outside that set, uniformly over $x \in B$. A finite union bound proves the asserted Cauchy property. The finite-cutoff kernels are continuous in x , by their Feller property

and local path tightness. Completeness of the Prokhorov metric and a subsequence with summable error probabilities on a compact exhaustion of $\mathbb{R}^d$ give a measurable continuous limiting kernel.

Finally, on the compact set of laws $\mathcal{K}_\eta$ in (9.20), every metric inducing the weak topology is uniformly equivalent to d_P . With probability at least $1 - 2\eta$, both finite-cutoff kernels take values there for every $x \in B$. Letting $\eta \downarrow 0$ transfers the assertion to d_{weak} . $\square$

9.4.1. The Markov property of the limit.

Proposition 9.10 (Properties of the limit). *Almost surely, the limiting kernel $x \mapsto K_x^\omega$ has the following properties.*

- (i) *The map $x \mapsto K_x^\omega$ is continuous in the weak topology.*
- (ii) *The laws $(K_x^\omega)_x$ are the laws of a continuous, conservative, strong Markov process Z .*
- (iii) *The process is symmetric with respect to μ : for every $t \geq 0$ and all compactly supported continuous functions f and g ,*

$$\int g(x) \int f(Z_t) dK_x^\omega \mu(dx) = \int f(x) \int g(Z_t) dK_x^\omega \mu(dx).$$

Proof. Continuity of the kernel and infinite lifetime hold by Proposition 9.9: the limiting laws are probability measures on $C([0, \infty), \mathbb{R}^d)$. The finite-cutoff Markov identity passes to the limit for bounded continuous path functionals, using locally uniform convergence of the kernels and compact containment. A monotone-class argument and approximation of stopping times from above by dyadic times then give the strong Markov property.

At finite cutoff the same identity holds for $K_{N,x}^\omega$ and μ_N , by the symmetry of the generator in $L^2(\mu_N)$. For compactly supported continuous f, g , locally uniform convergence of the kernels and local weak convergence $\mu_N \rightarrow \mu$ pass this identity to the limit. $\square$

Lemma 9.11 (Killed resolvents of the limiting process). *Almost surely, for every cube $Q \in \mathcal{C}$ and every $\lambda > 0$, the resolvent of the process of Proposition 9.10 killed on leaving Q is R_λ^Q .*

Proof. Choose a subsequence along which the whole-space kernels converge almost surely, locally uniformly in the starting point, and the killed resolvents converge uniformly on $\overline{Q}$, for a countable dense family of data and positive parameters. Fix an environment in this event. For nonnegative data, the discounted occupation functional stopped at τ_Q is lower semicontinuous on continuous path space, so the portmanteau theorem gives one inequality between its expectation under K_x^ω and $R_\lambda^Q f(x)$.

For the opposite inequality, first take $f = 1$. Fix $\varepsilon > 0$ and a smaller closed concentric cube $F \Subset Q$ so that, for all sufficiently large N , $\lambda R_{N,\lambda}^Q 1 \leq \varepsilon$ on $Q \setminus F$. Uniform convergence and the continuous zero-boundary representative of $R_\lambda^Q 1$ allow this choice. Let $\sigma_{h,F}$ be the first time on the grid $h\mathbb{N}_0$ at which the path is outside F , with $e^{-\lambda\sigma_{h,F}} = 0$. The finite-cutoff restart identity gives

$$(1 - \varepsilon) E_{N,x}^\omega e^{-\lambda\sigma_{h,F}} \leq E_{N,x}^\omega e^{-\lambda\tau_Q} = 1 - \lambda R_{N,\lambda}^Q 1(x).$$

The grid functional on the left is bounded and lower semicontinuous, since F is closed. Portmanteau passes the inequality to the limiting law. Continuity of paths gives $e^{-\lambda\tau_Q} \leq \liminf_{h \downarrow 0} e^{-\lambda\sigma_{h,F}}$; Fatou's lemma and then $\varepsilon \downarrow 0$ give $R_\lambda^Q 1(x) \leq E_x^\omega \int_0^\tau e^{-\lambda t} dt$. Together with the first inequality this proves equality for $f = 1$. Applying the first inequality to $\|f\|_\infty + f$ and $\|f\|_\infty - f$ gives equality for every bounded continuous f . Contraction, continuity in λ , and a countable intersection give the identity for all $\lambda > 0$ and all cubes on one event of probability one.

To allow $f \in C(\overline{Q})$, we multiply it by bounded compactly supported cutoff functions increasing to one in Q . The resulting data converge in $L^2(\mu)$, and the variational identity and the trace bound give the convergence of their resolvents in energy. The common $C^{1/4}$ bound then gives uniform convergence, and dominated convergence for the path integral weighted by $e^{-\lambda t}$ proves the identity for f , in particular for $f = 1$. $\square$

The identification of the killed resolvents also gives the finite mean exit times asserted in Theorem 7.1.

Corollary 9.12 (Finite mean exit times). *Almost surely, $\sup_{x \in U} \int \tau_U dK_x^\omega < \infty$ for every bounded open set U .*

Proof. Choose $Q \in \mathcal{C}$ containing $\bar{U}$. Lemma 9.11 and (9.9), with $f = 1$, give

$$\sup_{x \in Q} \int_0^{\tau_Q} e^{-\lambda t} dt dK_x^\omega = \|R_\lambda^Q 1\|_\infty \leq K_Q \quad (\lambda > 0).$$

Monotone convergence as $\lambda \downarrow 0$ and $\tau_U \leq \tau_Q$ prove the assertion. $\square$

9.5. Almost-sure convergence by comparison of two cutoffs. We compare the finite-volume energies at two cutoffs. On a small cube, rescaling to the unit cube writes the ratio of the two energies as a deterministic ratio of normalizations times a ratio of two unit-cube energies. To combine these local comparisons, we join harmonic extensions on disjoint small cubes, using the bound of Proposition 8.38 on the number of bad cells along chains of nested cubes.

Our control of the ratios of the two normalizations follows the argument of Ding and Gwynne [2023a, Section 3.4]. There, two random metrics are compared by joining paths around annuli at a positive proportion of scales into a single path. Upper and lower bounds on crossing distances then exclude ratios of the two normalizations that are too large or too small. Devlin [2024, Lemma 3.11] strengthened this argument by using convergence in probability, which rules out ratios bounded away from one on a positive proportion of scales. We apply the argument of Devlin [2024, Lemma 3.11] to the minimal energy with a fixed nonconstant affine boundary value. If the ratios were at most a fixed number less than one on a positive proportion of scales, then joining harmonic extensions on cubes of those scales would bound this energy at a finite cutoff by a fixed factor less than one times its limit, which is positive. This would contradict the convergence in probability. Interchanging the two cutoffs excludes, in the same way, ratios at least a fixed number greater than one. On the remaining scales the ratios are close to one, and Proposition 9.20 turns the comparison on these scales into bounds with summable error probabilities.

We first record how a minimal energy with given boundary values on a small cube rescales. For an integer $k \geq 0$ and $z \in \mathbb{R}^d$, let $\Theta_{k,z}$ be the map on environments that translates the fields by z , rescales them by 3^{-k} and subtracts from each field its value at z . We write $\Lambda_{m, \square_0}^{\Theta_{k,z}\omega}$ for the minimal energy on $\square_0$ with given boundary values at cutoff m in the environment $\Theta_{k,z}\omega$.

Lemma 9.13 (Rescaling of minimal energies on small cubes). *For every integer $k \geq 0$, the limit*

$$e(k) = \lim_{N \rightarrow \infty} \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N}$$

exists and is positive. Let $z \in \mathbb{R}^d$, let $q = z + 3^{-k}\square_0$, and, for $N \geq k$, let

$$s_N(q) = \frac{\bar{\mathbf{a}}_{N-k}}{\bar{\mathbf{a}}_N} \exp\left(H(z) + \sum_{j=0}^{k-1} (\mathbf{g}_{-j}(z) - \tau^2)\right)$$

be the value at the center of q of the factor in A_N coming from the normalization and the fields whose scales exceed the side length of q . Then, for every $N \geq k$ and $g \in H^1(q)$,

$$\Lambda_{N,q}(g) = 3^{-(d-2)k} s_N(q) \Lambda_{N-k, \square_0}^{\Theta_{k,z}\omega}(g(z + 3^{-k} \cdot)).$$

For each fixed z , the same identity holds almost surely for the minimal energies of the limit form, with $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ in $s_N(q)$ replaced by $e(k)$, for every boundary trace on which these are defined.

Proof. The identity at finite cutoff follows from the change of variables $x = z + 3^{-k}y$ in the minimization that defines $\Lambda_{N,q}(g)$. In the environment $\Theta_{k,z}\omega$ the fields of scale larger than 3^{-k} enter,

at scales larger than one, only through their rescaled increments about z . So all their spatial oscillations are retained, and, by stationarity and the scaling of the fields, the map $\Theta_{k,z}$ preserves their law.

To prove that the normalizations converge, fix a nonconstant affine datum. Both minimal energies in the identity converge in probability, and their limits are strictly positive. For fixed k the deterministic ratios $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ lie in a compact subset of $(0, \infty)$. Two subsequential limits of these ratios must coincide, since otherwise the same positive minimal energy of the limit form would equal one random variable multiplied by two different constants. Finally, we pass to the limit in the identity for the countable family of smooth traces of Section 8.4, and the bound (8.6) for Hölder boundary values of Lemma 8.4 gives the identity for every trace on which the minimal energy of the limit form is defined. $\square$

We compare the two cutoffs on a cell q through the trace on ∂q of a single function, which is Hölder continuous but otherwise arbitrary. The next lemma reduces this comparison to finitely many smooth traces.

Lemma 9.14 (Reduction to finitely many smooth boundary values). *Let $1/2 < \beta < \alpha < 1$, $A < \infty$ and $\eta > 0$. There exists a finite family of smooth functions on $\partial\Box_0$ with the following property. Let Λ_1 and Λ_2 be minimal energies on $\Box_0$ with given boundary values such that, for $i \in \{1, 2\}$ and every $g \in C^\beta(\partial\Box_0)$,*

$$\Lambda_i(g) \leq A\|g\|_{C^\beta/\mathbb{R}}^2,$$

and such that $|\Lambda_1(h) - \Lambda_2(h)| \leq \eta/2$ for every function h of the family. Then, for every $g \in C^\alpha(\partial\Box_0)$,

$$|\Lambda_1(g) - \Lambda_2(g)| \leq \eta\|g\|_{C^\alpha/\mathbb{R}}^2.$$

Proof. Since $\Lambda_i^{1/2}$ is a seminorm, the triangle inequality gives, for all $g, h \in C^\beta(\partial\Box_0)$,

$$|\Lambda_i(g) - \Lambda_i(h)| \leq A\|g - h\|_{C^\beta/\mathbb{R}}(\|g\|_{C^\beta/\mathbb{R}} + \|h\|_{C^\beta/\mathbb{R}}).$$

The unit ball of $C^\alpha/\mathbb{R}$ is precompact in $C^\beta/\mathbb{R}$, so for every tolerance it has a finite net consisting of smooth functions. Take the family to be a sufficiently fine such net, and apply the last display to Λ_1 and to Λ_2 . This gives the assertion for g in the unit ball, and homogeneity gives it for every g . $\square$

The number of functions in the family and their smooth norms may depend on η ; the exponents and the moment orders do not. The constant A is the one in Lemma 9.15(iii) below, and it is fixed before the family is chosen.

On a good cell we also need, at a finite cutoff, the local Hölder estimate and the bound (8.6) for Hölder boundary values that Lemma 8.39 provides for the limit form. The scalar $s_N(q)$ of Lemma 9.13 is the finite-cutoff analogue of the scalar $s_E(q)$ of Section 8.6.

Lemma 9.15 (Good cells at finite cutoff). *Fix $1/2 < \beta < \alpha < 1$, a subdivision factor L , a distance factor $b_0 \geq 3$ and a rate $B > 0$ for the exponential bounds in (i). The thresholds for the sums over intervals of scales, the finite moment orders and $\bar{\delta}_0 > 0$ can be chosen in this order so that the following holds for $\bar{\delta} \leq \bar{\delta}_0$. There are deterministic constants $C, A < \infty$, independent of the cutoff, the cell, the source and the tolerance η of Lemma 9.14, and good events $G_N(q)$ with the following properties. Here $N \geq k \geq 0$, $q = z + r\Box_0$ with $r = 3^{-k}$, and, after q is rescaled to $\Box_0$, the coefficient has cutoff $m = N - k$. The conclusions hold on the same good events both for A_N and for $A_N^0 = e^{-H}A_N$, with their respective scalars $s_N(q)$ of Lemma 9.13 and $s_N^0(q) = e^{-H(z)}s_N(q)$. Below, $s(q)$ denotes the scalar of the coefficient under consideration, and $\Gamma(u)$ denotes the energy measure of u for that coefficient.*

- (i) *The complement of $G_N(q)$ is covered by events indexed by integers $h \geq 1$, the event of index h being measurable with respect to $\sigma(\mathbf{g}_{-j} : k - h \leq j \leq k + 2h)$ and having probability at most e^{-Bh} .*

- (ii) Suppose $G_N(q)$ holds and q lies in its parent p of side Lr with $\text{dist}(q, \partial p) \geq b_0 r$. For every $f \in L^\infty(p)$ and every weak solution $u \in H^1(p)$ of $-\nabla \cdot (A \nabla u) = f$ on p , there is a constant c_q such that

$$(9.26) \quad \|u(z + r \cdot) - c_q\|_{C^\alpha(\square_0)} \leq Cr^{(2-d)/2} s(q)^{-1/2} \Gamma(u)(p)^{1/2} + Cr^2 s(q)^{-1} \|f\|_\infty.$$

- (iii) On $G_N(q)$, for every $g \in C^\beta(\partial \square_0)$, the normalized minimal energy satisfies

$$r^{2-d} s(q)^{-1} \Lambda_{N,q}(g((\cdot - z)/r)) \leq A \|g\|_{C^\beta/\mathbb{R}}^2.$$

Proof. Step 1. We define the event $G_N(q)$. It imposes the conditions of $G(q)$ on sums over intervals of scales, down to the smallest field scale, with an exponent $\alpha_1 \in (\alpha, 1)$ in place of α in the iteration of Lemma 6.10. We choose the integrability exponent used in Step 3 so that the Sobolev embedding gives a Hölder exponent larger than α_1 , then a small $\sigma > 0$, and we add the conditions $F_{0,z} \leq 1$ and $P_{0,z} \leq 12$ (notation of Section 8.5, with $s = \sigma$, so that the weights are $3^{-\sigma j/8}$) for q itself, on a fixed concentric enlargement. Their weighted tails satisfy the bounds of Lemma 8.35 and their values at $\bar{\delta} = 0$ are 0 and 2, so their failure is covered in the same way by events of the form in (i). For an interval of $D \leq m$ scales on which a bound fails, we apply, at the single fixed cutoff N , the telescoping argument from the proof of Lemma 8.37, which uses dyadically increasing intervals of field indices. Lemmas 8.36 and 8.35 are uniform at that cutoff, so the union over the at most Ce^{CD} spatial choices has the same bound. Choose the moment order used in the approximation so large that the resulting exponential rate exceeds both B and the exponential growth rate of the number of spatial choices, and then decrease $\bar{\delta}$. This gives (i) directly at finite cutoff.

Step 2. We bound the coefficient on the cell. In unit coordinates the smallest field scale is 3^{-m} . The term $j = m$ of the two conditions bounds the product of the exponentials of all fields of scale at most one by $C3^{\sigma m/8}$, and it bounds their gradients, weighted by their scales, by the same quantity. The increments $\mathbf{g}_i - \mathbf{g}_i(z)$ of the fields of scale larger than one are bounded by the condition at $j = 0$, and their gradients are summed with the geometric factors of their scales. The deterministic normalization has logarithm at most $C\tau^2(m+1)$. Denoting by $\tilde{A}$ the coefficient divided by $s(q)$, we obtain

$$\|\tilde{A}^{-1}\|_\infty + \|\tilde{A}\|_\infty + 3^{-m} \|\nabla \log \tilde{A}\|_\infty \leq C3^{C(\sigma+\tau^2)m}.$$

Step 3. We prove (ii) and (iii). Above the scale 3^{-m} , the iteration of Lemma 6.10 on the event controlling sums over intervals of scales gives (9.26) with exponent α_1 . Below 3^{-m} , we use the interior Sobolev estimate, with this exponent, of Step 4 of the proof of Proposition 8.7 on balls of radius $a/[C(1 + 3^{-m} \|\nabla \log \tilde{A}\|_\infty)]$, on which the oscillation of $\log \tilde{A}$ is small. After matching with the estimate at scale 3^{-m} , the energy term carries the factor $3^{-m(\alpha_1 - \alpha)}$ and the source term the factor $3^{-m(2-\alpha)}$, each multiplied only by fixed powers of the quantities in the last display. We choose σ , and then τ^2 , so that $C(\sigma + \tau^2) < \alpha_1 - \alpha$, where C depends only on d and the integrability exponent. The net exponents of 3^{-m} then remain positive, and (9.26) holds at all radii with a deterministic constant. Assertion (iii) is condition (a) of $G(q)$ combined with Lemma 8.4. Finally, we apply the construction separately to the coefficient with and without its factor e^H , and take the intersection of the two good events. When H is replaced by one of its partial sums, the same bounds hold uniformly in the number of terms, by the summable bounds on the increments of the fields of scale larger than one. $\square$

The comparison of Lemma 9.14 on a cell holds when the normalized minimal energies at the two cutoffs are close on the finite family of smooth boundary values of that lemma. By the convergence in probability, this comparison fails only with small probability, but the counting argument of Section 8.5 requires more: the failure must be covered by events of exponentially small probability, each measurable with respect to the fields with indices in a short interval.

Lemma 9.16 (Rare events and field indices). *Let X_m , $m \in \mathbb{N}$, be random variables with values in $[0, 1]$ that tend to zero in probability, and assume that, uniformly in m and H , they are approximated by their conditional expectations given the σ -field $\mathcal{B}_H$ generated by the fields with indices in the interval of length $C(H + 1)$ about a fixed index n , as in Lemma 8.35, in the sense that*

$$\|X_m - \mathbb{E}[X_m \mid \mathcal{B}_H]\|_{L^p} \leq C_* 3^{-aH}.$$

There is an absolute constant C such that, for every B with $a \log 3 > CB$, there exists m_0 with the following property. For every $m \geq m_0$, the event $\{X_m > 1/2\}$ is contained in a union of events indexed by integers h ; the event with index h is measurable with respect to the fields indexed by an interval of radius at most Ch about the same index n , and it has probability at most e^{-Bh} . The smallest index may depend on C_ and B .*

Proof. Let H_0 be large, to be chosen, and put $X_m^{(\ell)} = \mathbb{E}[X_m \mid \mathcal{B}_{2^\ell H_0}]$ for $\ell \geq 0$. Since the L^p errors of these approximations are summable in ℓ , we have $X_m^{(\ell)} \rightarrow X_m$ almost surely. The allowances $2^{-\ell-3}$, $\ell \geq 1$, sum to $1/8$. Hence, if $X_m > 1/2$, then either $X_m^{(0)} > 1/4$ or the increment $X_m^{(\ell)} - X_m^{(\ell-1)}$ exceeds $2^{-\ell-3}$ for some $\ell \geq 1$. By Markov's inequality, the probability of the first event is at most

$$\mathbb{P}(X_m > 1/8) + (8C_*)^p 3^{-apH_0}.$$

We first choose H_0 so large that the second term is smaller than e^{-CBH_0} with the required margin, and then use the convergence in probability to make the first term smaller than e^{-CBH_0} for all $m \geq m_0$. By Markov's inequality, the probability that the ℓ th increment exceeds its allowance is at most

$$(CC_*)^p 2^{p\ell} 3^{-ap2^{\ell-1}H_0},$$

and increasing H_0 absorbs the factor $2^{p\ell}$ uniformly in ℓ . Each of these events is measurable with respect to the fields indexed by the larger of the two intervals that determine the two conditional expectations in its definition. We index them by $h = 2^\ell H_0$ and enlarge their intervals to the form used in Lemma 8.37; the assumed margin in $a \log 3 > CB$ absorbs this fixed enlargement. $\square$

We apply Lemma 9.16 to a bounded continuous piecewise linear function, as in Section 8.5, of the largest difference between the normalized minimal energies at two cutoffs, over the family of Lemma 9.14. The moment bounds and the approximations using finitely many fields of Section 8.5 pass to the minimal energies of the limit form, so the lemma applies to this difference for every pair of sufficiently large cutoffs, with a constant C_* that may depend on the tolerance η of Lemma 9.14. The continuous piecewise linear function tends to zero in probability because the minimal energies converge in probability, and this requires no further decrease of $\bar{\delta}$. The argument is uniform over pairs of cutoffs whose minimum tends to infinity.

Lemma 9.17 (Simultaneous good L -adic depths for two cutoffs). *Fix $0 < a < b < 1$, $\theta \in (0, 1)$, $L = 3^{H_1}$ and a bounded initial cube of the subdivision. Choose the finite moment orders and the threshold $\bar{\delta}_0$ as in Lemmas 9.15 and 9.16, with the rate B of Lemma 9.15(i) large enough for the counting estimate in the proof. For every $\eta > 0$ there are $N_\eta, C_\eta < \infty$ and $c_\eta > 0$ such that, for each pair $M \geq N \geq N_\eta$, outside an event of probability at most $C_\eta 3^{-c_\eta N}$, every chain in the L -adic subdivision has at most $\theta N/H_1$ bad L -adic depths between aN and bN .*

Here an L -adic depth of a chain is good if, on the cell of the chain at that depth, Lemma 9.15 holds at both cutoffs and the two separately normalized minimal energies satisfy the conclusion of Lemma 9.14 for every C^α trace; otherwise it is bad.

Proof. The preceding application of Lemma 9.16 and the covers of Lemma 9.15(i) at the two cutoffs cover each bad depth by events of the form in Lemma 9.15(i), with a prescribed rate B , after increasing that rate by a fixed factor. As in the proof of Proposition 8.38, we apply Lemma B.5 to the consecutive L -adic depths between aN and bN . The number of these depths is $(b-a)N/H_1 + O(1)$,

while the number of chains ending at the last depth of this interval is at most $L^{d(bN/H_1+O(1))}$. Choose B so that the resulting exponential rate exceeds this growth and take a union bound. $\square$

At depth k the minimal energies at the cutoffs N and M carry the normalizations $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$ and $\bar{\mathbf{a}}_{M-k}/\bar{\mathbf{a}}_M$ of Lemma 9.13, while the random factor in the scalars $s_N(q)$ and $s_M(q)$ is common to both. The ratio of these two normalizations

$$r_{N,M}(k) = \frac{\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N}{\bar{\mathbf{a}}_{M-k}/\bar{\mathbf{a}}_M}$$

is therefore, up to the tolerance η of Lemma 9.14, the factor relating the two minimal energies on a good cell of depth k . The next lemma adapts the stopping argument of Proposition 8.40 to two finite cutoffs.

Lemma 9.18 (Comparison of two cutoffs by stopping). *Let $\vartheta > 0$. The geometry, the exponents and the finitely many moment orders can be chosen so that there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then the following holds with constants D , L and C independent of η , c , N and M . Let Q be a fixed cube of triadic side length and g a fixed smooth function on ∂Q . For every $\eta > 0$ there exist $C_\eta < \infty$ and $\gamma_\eta > 0$ with the following property. Let $N \leq M$ be cutoffs and $c \leq 2$, and assume that a deterministic set S contains at least a fraction ϑ of the L -adic depths between $N/4$ and $3N/4$, and that $r_{N,M}(k) \leq c$ for every $k \in S$. Then, outside an event of probability at most $C_\eta 3^{-\gamma_\eta N}$,*

$$\Lambda_{N,Q}(g) \leq c(1 + C_\eta L^D) \Lambda_{M,Q}(g) + C_\eta 3^{-\gamma_\eta N} \|g\|_{C^2}^2.$$

Under the same assumptions with $r_{N,M}^{-1}$ in place of $r_{N,M}$, the reverse inequality holds, with $\Lambda_{N,Q}$ and $\Lambda_{M,Q}$ interchanged.

Proof. Step 1. We define where the subdivision stops and estimate the energy on the stopped cells. Let u be the E_M -harmonic extension of g to Q , fix a small $\zeta > 0$, and let $\nu = \Gamma_M(u) + 3^{-\zeta N} dx$, so that $\nu(q) \geq 3^{-\zeta N} |q|$ for every cell q . Translate the L -adic grid so that, as in Proposition 8.40, its meshes of sufficiently small side partition Q exactly. Work on the event of Lemma 9.17 for the interval of depths $[N/4, 3N/4]$, with the fraction of bad L -adic depths chosen below. Let $J = \lfloor N/H_1 \rfloor$. We follow this subdivision from L -adic generation $\lfloor J/4 \rfloor$ (depth $H_1 \lfloor J/4 \rfloor$) to L -adic generation $\lfloor 3J/4 \rfloor$, and we stop at a child q with parent p if the depth of q belongs to S , the cell q is good and at distance at least b_0 times its side from ∂p , and $\nu(p) \leq L^D \nu(q)$. At a stopped cell q of depth k , the local Hölder estimate (9.26) and the comparison of Lemma 9.14 give

$$\Lambda_{N,q}(u|_{\partial q}) \leq r_{N,M}(k) \Lambda_{M,q}(u|_{\partial q}) + C_\eta r_{N,M}(k) \Gamma_M(u)(p).$$

Since $\Lambda_{M,q}(u|_{\partial q})$ is bounded by $\Gamma_M(u)(q)$, the stopping rule and the disjointness of the stopped cells show that the sum of $\Lambda_{N,q}(u|_{\partial q})$ over the stopped cells is at most

$$cE_M(u) + C_\eta cL^D (E_M(u) + 3^{-\zeta N} |Q|).$$

Step 2. We count the cells that are never stopped. The moment estimates of Section 8.2 are uniform in both cutoffs, so, outside an event of probability at most $C3^{-p\epsilon N}$, each of the finitely many global constants used below is at most $3^{\epsilon N}$. Consider a chain of nested cells along which no cell is stopped. At each of its steps at which the depth belongs to S and the child is good and at distance at least b_0 times its side from the boundary of its parent, ν drops by a factor larger than L^D . Since $\nu(Q)$ is controlled by the global energy and $\nu(q) \geq 3^{-\zeta N} |q|$ for every cell q of the final depth, the number of such steps is at most

$$\frac{1}{D \log L} \left(\log(3^{\zeta N} \nu(Q)) + \frac{3}{4} dN \log 3 + O(1) \right).$$

We choose D large in terms of d and ϑ , and θ , ϵ and ζ small. The chain has at least $\vartheta J/2 - O(1)$ steps at depths in S . Discarding the steps counted above and the bad steps at depths in S leaves at least $\vartheta J/3 - O(1)$ steps at depths in S at which the child is within distance b_0 times its side

of the boundary of its parent. Each cell has at most $C_d L^{d-1}$ children within that distance of its boundary, so, for L large, the number of cells at the final depth on chains where the subdivision has not stopped is at most

$$CL^{(3d/4-\vartheta/4)J}.$$

Step 3. We bound the energies on the remaining cells at the final depth. On such a cell of side r , the bound (8.6) for Hölder boundary values, together with the bounds (8.7) on grid cubes, bounds the energy by $K_N[u]_\alpha^2 r^{d-2+2\alpha-\eta_0}$. We choose the fixed exponents so that $2 - 2\alpha + \eta_0 < \vartheta/8$. The sum of the energies on these cells is then at most $C3^{3\epsilon N} L^{-\vartheta J/8}$, which decays exponentially in N if ϵ is small.

Step 4. We compare the minimal energies with boundary values g . The stopped cells and remaining cells at the final depth form a finite partition of Q . Joining the E_N -harmonic extensions of the traces of u on the cells of this partition gives a competitor with boundary values g . The Dirichlet principle then gives the asserted upper bound for $\Lambda_{N,Q}(g)$, and all excluded events have exponentially small probability. The reverse inequality follows from the same proof with the cutoffs interchanged. $\square$

The same construction compares the energies of solutions with a volume source.

Lemma 9.19 (Comparison for volume sources). *Assume the hypotheses of Lemma 9.18. There is an event of probability at least $1 - C_\eta 3^{-\gamma_\eta N}$ on which the following holds. Let u_M solve $-\nabla \cdot A_M \nabla u_M = f$ on Q , either with a smooth Dirichlet datum g , in which case we put $B = \|f\|_\infty + \|g\|_{C^2}$, or with zero conormal derivative and mean zero, in which case we assume that $\int_Q f = 0$ and put $B = \|f\|_\infty$. Then there is a trial function v_N , with the same Dirichlet datum in the first case and with mean zero in the second, such that*

$$\|v_N - u_M\|_\infty \leq C_\eta 3^{-\gamma_\eta N} B \quad \text{and} \quad E_N^Q(v_N) \leq c(1 + C_\eta L^D) E_M^Q(u_M) + C_\eta 3^{-\gamma_\eta N} B^2.$$

The same holds with A_N and A_M replaced by $A_N^0 = e^{-H} A_N$ and $A_M^0 = e^{-H} A_M$.

Proof. We use the partition of the proof of Lemma 9.18, now with $\nu = \Gamma_M(u_M) + 3^{-\zeta N} B^2 dx$ in place of the measure of that proof. On a good cube q of side r whose parent p satisfies $\text{dist}(q, \partial p) \geq b_0 r$, the deterministic estimate with source adds to the local comparison the term

$$C_\eta r_{N,M}(k) s_M(q)^{-1} r^{d+2} \|f\|_\infty^2.$$

On the event of the preceding proof, the bounds (8.7) on grid cubes give $s_M(q)^{-1} \leq 3^{\epsilon N} r^{-\eta_0}$. Since the side of every cube used in the comparison is at most $C3^{-N/4}$, this source term is bounded by $C_\eta c 3^{-\zeta N} B^2 |q|$ if $\epsilon + \zeta < (2 - \eta_0)/4$, so it is absorbed in the lower bound $3^{-\zeta N} B^2 |q|$ for $\nu(q)$. The global estimates of Proposition 8.7 and Proposition 8.11 bound the energy by $3^{\epsilon N} B^2$ and the C^α norm by $3^{\epsilon N} B$, after these constants are added to the finite list of constants required to be at most $3^{\epsilon N}$.

On each element of the partition we take the A_N -harmonic extension of the trace of u_M . By the maximum principle, its distance from u_M is at most twice the oscillation of u_M on that element. All elements have side at most $C3^{-N/4}$, so this distance is at most $C3^{-(\alpha/4-\epsilon)N} B$. In the Neumann case we first subtract the mean. This proves both inequalities.

For A_N^0 and A_M^0 , the estimates with source follow from the same regularity argument. On each cube used in the comparison the common weight e^{-H} has ratio of supremum to infimum at most $\exp(C3^{-N/4} \|H\|_{C^1(Q)})$. We add $e^{\|H\|_{C^1(Q)}}$, which has every finite moment, to the constants required to be at most $3^{\epsilon N}$; this ratio is then $1 + O(3^{-N/5})$ for large N .

By the variational definition of the minimal energy with given boundary values, the local comparison transfers to the coefficients $A_N^0 = e^{-H} A_N$ and A_M^0 with an exponentially small additional error. The ratio of their scalars $s_N^0(q)$ and $s_M^0(q)$ is still $r_{N,M}(k)$, because the value of the weight

at the center of the cell cancels. After summation over the stopped cells, the additional error is at most $C3^{-N/5}E_M^0(u_M)$, which is absorbed in the exponential remainder on this event. The rest of the construction is unchanged. $\square$

Proposition 9.20 (Almost-sure convergence of minimal energies). *Let Q be a cube of triadic side length and g a smooth function on ∂Q . Then $\Lambda_{N,Q}(g) \rightarrow \Lambda_{E,Q}(g)$ almost surely as $N \rightarrow \infty$ along the full sequence of integers.*

Proof. Step 1. We show that the normalizations are close to their limits on most depths k with $N/4 \leq k \leq 3N/4$. Put $r_N(k) = (\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N)/e(k)$, and fix $\varepsilon > 0$ and a density $\vartheta < 1/3$. Suppose that, for infinitely many N , at least a fraction ϑ of the depths in this interval satisfy $r_N(k) \leq 1 - \varepsilon$. For each such N , since $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_M \rightarrow e(k)$, the same depths satisfy $r_{N,M}(k) \leq 1 - \varepsilon/2$ for all sufficiently large M . We apply Lemma 9.18 to a fixed nonconstant affine datum, with η so small that the multiplicative error $C\eta L^D$ is less than $\varepsilon/4$, and we let $M \rightarrow \infty$ first. Along these N , the lemma bounds the minimal energy with this boundary value at cutoff N by a fixed factor strictly smaller than one times its limit, up to an error that tends to zero in probability. This contradicts the convergence in probability of this minimal energy to its limit, which is strictly positive. The reverse comparison excludes in the same way a fraction ϑ of depths with $r_N(k) \geq 1 + \varepsilon$. Hence, for all sufficiently large N , more than a fraction $1 - 2\vartheta$ of the depths in this interval satisfy $1 - \varepsilon \leq r_N(k) \leq 1 + \varepsilon$.

Step 2. We prove an exponential bound and conclude. For each large N , we apply the two comparisons of Lemma 9.18, with S the set of depths found in Step 1, and let $M \rightarrow \infty$. Taking η small relative to ε gives, with constants depending on ε and on the datum,

$$\mathbb{P}[|\Lambda_{N,Q}(g) - \Lambda_{E,Q}(g)| > C\varepsilon\Lambda_{E,Q}(g) + C_\varepsilon 3^{-\gamma_\varepsilon N}] \leq C_\varepsilon 3^{-\gamma_\varepsilon N}.$$

The right side is summable in N , so the Borel–Cantelli lemma for $\varepsilon = 1/j$, $j \in \mathbb{N}$, gives the almost-sure convergence. $\square$

The same argument applies to the quantities (ii)–(iv) of the next proposition.

Proposition 9.21 (Summable error probabilities). *Let Q be a cube of triadic side length, and let Z_N be one of the following quantities on Q , computed for the coefficient A_N or for A_N^0 :*

- (i) *the minimal energy $\Lambda_{N,Q}(g)$ with a fixed smooth boundary value g ;*
- (ii) *the quadratic form $\langle f, G_N^Q f \rangle$ of (8.3) for a fixed bounded source f ;*
- (iii) *the quadratic form $\sup_{v \in H^1(Q)} \{2 \int_Q f v - E_N^Q(v)\}$ for a fixed bounded source f with $\int_Q f = 0$;*
- (iv) *the Neumann energy $|Q|p \cdot a_*^{-1}(Q; A_N)p$ of Section 8.1, for the constant flux given by a fixed vector p .*

There exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then Z_N converges in probability to a limit Z , and, for every $\varepsilon > 0$, there exist $C_\varepsilon < \infty$, $c_\varepsilon > 0$ and a deterministic N_ε such that, for every $N \geq N_\varepsilon$,

$$(9.27) \quad \mathbb{P}[|Z_N - Z| > C_\varepsilon Z + C_\varepsilon 3^{-c_\varepsilon N}] \leq C_\varepsilon 3^{-c_\varepsilon N}.$$

The finitely many moment orders and the threshold $\bar{\delta}_0$ do not depend on ε . For the coefficients A_N^0 , the constants are uniform over deterministic translations of Q . In particular, for every countable family of such quantities, all of them converge almost surely along the full sequence of cutoffs.

Proof. For the minimal energies of A_N , the bound (9.27) was proved in the proof of Proposition 9.20. For A_N^0 , the convergence in probability also follows from Proposition 8.26 on continuous weights, applied with the weight e^{-H} , and the last paragraph of the proof of Lemma 9.19 gives the same comparison between two cutoffs.

Consider next a volume source f , let u_M be the solution at cutoff M , and write $Z_M = E_M(u_M) = \int_Q f u_M$. Let a be the factor multiplying $E_M^Q(u_M)$ in Lemma 9.19. Inserting $a^{-1}v_N$ in the variational

formula for Z_N , we obtain from the two bounds of that lemma, when $1/2 \leq a \leq 3$,

$$Z_N \geq a^{-1} Z_M - C_\varepsilon 3^{-c_\varepsilon N} \|f\|_\infty^2.$$

The construction with the cutoffs interchanged gives the opposite inequality. Step 1 of the proof of Proposition 9.20 has shown that, for every prescribed tolerance and all large N , a positive proportion of the depths k with $N/4 \leq k \leq 3N/4$ have $r_N(k)$ within that tolerance of one. For fixed large N the two comparisons therefore hold with $a = 1 + C\varepsilon$, uniformly in all sufficiently large M .

In case (iii), this argument also shows that a limit in probability exists, without using a Neumann limit form. Indeed, we compare two sufficiently large cutoffs M and M' with the same fixed N . The two exceptional probabilities tend to zero as $N \rightarrow \infty$, and, since the quantities have uniformly bounded moments, the multiplicative error can then be made arbitrarily small by reducing ε . Hence (Z_M) is Cauchy in probability. In case (ii) the limit is the quadratic form of the Dirichlet Green operator of the unique limit form. Letting $M \rightarrow \infty$ with N fixed gives (9.27) in cases (ii) and (iii).

For (iv), (8.18) and case (iii) first give the Cauchy property in probability. To obtain the rate, take $h = 3^{-vN}$ with $0 < v < c_\varepsilon/4$ in the two-cutoff comparison. The exceptional events of that comparison do not depend on the source, and the factor $\|f_h\|_\infty^2 \leq C3^{2vN}$ is absorbed by the remainder. Fix a moment order $q \geq 1$ before decreasing $\bar{\delta}$. Markov's inequality and (8.18) show that, uniformly for $M \geq N$, the energy at cutoff M and the corresponding energy for the smoothed source f_h of Lemma 8.8 differ by more than $3^{-vN/s}$ with probability at most $C_q 3^{-qvN/s}$. The same bound holds at cutoff N . Let $M \rightarrow \infty$ with N fixed in the two comparisons, decreasing c_ε if necessary. This proves (9.27).

All the choices that depend on ε are spatial cutoffs, finite nets, lengths of the initial intervals of field indices and the exponent v of the smoothing width; none of them requires a further moment order or a further decrease of $\bar{\delta}$. The last assertion follows from the Borel–Cantelli lemma. $\square$

9.6. Almost-sure bounds uniform in the cutoff. We now make the local estimates uniform over all cutoffs on one event of probability one. On cells of depth at most θN , Proposition 9.21 compares the finite-cutoff energies with their limits, with error probabilities summable over the cells and over N . For the deeper cells, the moment bounds imply that, almost surely, the random constants in the estimates are smaller than any fixed exponential in N once N is large. The margins in the spatial exponents absorb this growth.

Lemma 9.22 (Coarse-grained ellipticity bounds uniform in the cutoff). *Let Q be a cube of triadic side length and let s be one of the positive fractional orders used above. Then, almost surely, for $q \in \{1, 2\}$,*

$$\sup_N (\Lambda_{s,q}(Q; A_N) + \lambda_{s,q}(Q; A_N)^{-1}) < \infty.$$

In particular, the coercivity bound (8.5) and the bound (8.6) for Hölder boundary values on Q hold with one finite random constant for all N . The same conclusions hold with A_N^0 in place of A_N .

Proof. It suffices to consider a unit cube.

Step 1. We bound the finite-cutoff energies of the cells of depth at most θN by their limits. Denote by $Z_m(z)$ the finite family consisting of the minimal energies with the affine boundary values $x \mapsto e \cdot x$ and the Neumann energies for the fluxes e , for $e \in \{e_i, e_i + e_j\}$, computed for A_m^0 in the environment shifted to z , at cutoff m . We apply (9.27) with one fixed small tolerance to this family on every grid cube of depth $k \leq \theta N$, whose cutoff after rescaling to the unit cube is $m = N - k$. There are at most $C3^{dk}$ such cubes. We choose $\theta > 0$ so small that $d\theta < c(1 - \theta)/2$, where c is the exponent in (9.27) at this tolerance; then the sum over these cubes of the exceptional probabilities is summable in N . Hence, almost surely, for all sufficiently large N , each of these finite-cutoff energies is bounded by $C(1 + Z(\Theta_{k,z}\omega))$, where Z is its limit. All the entries are nonnegative, and polarization recovers the coarse-grained matrices from this finite family.

Step 2. We bound the coarse-grained matrices of the cells of depth at most θN . For a cube $q = z + 3^{-k}\square_0$, separating the scalar $s_N(q)$ of Lemma 9.13 gives

$$A_N(z + 3^{-k}y) = s_N(q) \exp\left(\left[H + \sum_{j=0}^{k-1} \mathbf{g}_{-j}\right](z + 3^{-k}y) - \left[H + \sum_{j=0}^{k-1} \mathbf{g}_{-j}\right](z)\right) A_{N-k}^{0, \Theta_{k,z}\omega}(y).$$

The deterministic ratio in $s_N(q)$ and its inverse are at most $e^{C\tau^2 k}$. Hence the factors in front of $A_{N-k}^{0, \Theta_{k,z}\omega}$ and their inverses are bounded by

$$e^{C\tau^2 k} \exp\left(\left|H(z) + \sum_{j=0}^{k-1} \mathbf{g}_{-j}(z)\right| + \text{osc}_q\left[H + \sum_{j=0}^{k-1} \mathbf{g}_{-j}\right]\right),$$

whose L^p norm is at most $C_p e^{C_p \bar{\delta}^2 k}$, by the independence of the fields and the bound on their gradients weighted by their scales. The limits Z have the same moment bounds as their approximants. We choose $\rho < s/4$, a moment order p with $p\rho > 2d$, and then $\bar{\delta}$ small enough. A union bound over all k and z gives a random $K_\omega < \infty$ such that, for every grid cell of depth k , the product of the expression in the last display and $1 + Z(\Theta_{k,z}\omega)$ is at most $K_\omega 3^{\rho k}$. Since both coarse-grained matrices are monotone in the coefficient, $a(q; A_N) + a_*^{-1}(q; A_N) \leq CK_\omega 3^{\rho k}$ on the cells of depth at most θN , with the same K_ω for all sufficiently large N .

Step 3. We bound the matrices of the cells of depth greater than θN and sum over all depths. For $\theta N < k \leq N$, the proof of Lemma 8.4, applied only to the matrices of the cells, gives the bound $K_N 3^{\rho k}$, where the moments of K_N of the finitely many orders fixed at the start of Section 8 are bounded uniformly in N . By the Borel–Cantelli lemma, $K_N \leq 3^\xi N$ for all large N , for every fixed $\xi > 0$. The contribution of these cells to either sum defining $\Lambda_{s,q}$ or $\lambda_{s,q}$ is at most $CK_N 3^{-(s-\rho)\theta N}$, which is bounded if $\xi < (s-\rho)\theta$. The cells below the smallest field scale contribute at most $C3^{-sN}(\sup_Q A_N + \sup_Q A_N^{-1})$, which is bounded almost surely after the fixed decrease of $\bar{\delta}$, by Lemma 8.6 and Markov’s inequality. The cells of depth at most θN contribute at most $CK_\omega \sum_{k \geq 0} 3^{-(s-\rho)k}$. For $q = 1$, weighted Cauchy–Schwarz bounds the square of the sum of the square roots of the matrices by this sum of matrices; for $q = 2$, the definition is already a sum of matrices, with the smaller weight 3^{-2sk} . Finally, the coercivity bound and the bound (8.6) for Hölder boundary values follow from these bounds as in the proofs of Lemmas 8.3 and 8.4. $\square$

Lemma 9.23 (Regularity uniform in the cutoff). *Let $0 < \alpha < 1$ and $d-1 < t < d$ be strictly smaller than the exponents fixed in the moment estimates of Proposition 8.7 and Proposition 8.11. Then, for every fixed cube Q , almost surely the estimates of Proposition 8.7 and Proposition 8.11 with the exponents α and t hold on Q with one finite random constant for all N .*

Proof. The regularity estimates follow from the iteration of Lemma 6.10 on the good events $G(q)$ of the cells, defined in Section 8.5. We show that, above the scale $3^{-\theta N}$ with $\theta > 0$ small, these events can be controlled by limiting quantities that do not depend on N (Steps 1–5), and we use the moment bounds below $3^{-\theta N}$ (Step 6).

Step 1. We pass the conditions defining $G(q)$ to the limit. Each coarse-grained matrix at scale $n = N - k$, in the coordinates of (8.1), that enters the conditions defining $G(q)$ in Section 8.5 is a unit-cube energy of $A_{N-k}^{0, \Theta_{k,z}\omega}$, in which the fields of scale larger than the cell side are absent. Hence the variational energy $\mathcal{J}_n$ contains no ratio $r_N(k)$. Using the limits of these unit-cube energies given by Proposition 9.21, we define the limiting versions of the quantities $\mathbf{F}$, $\mathbf{P}$, $\mathbf{R}$ and $\mathbf{G}$ in these conditions. By Fatou’s lemma their weighted tails satisfy the bounds of Lemma 8.35, and sums over finite intervals of scales satisfy the bounds of Lemma 8.34.

Step 2. We show that, almost surely for all large N , the retained finite-cutoff energies are within a small relative error of their limits and the discarded tails are small. Fix $\eta > 0$ below the margins

between the lower and upper thresholds of the continuous piecewise linear functions. We truncate each quantity in the conditions defining $G(q)$ at relative depth $[\theta'N]$, and we consider intervals of scales of length at most $[\theta N]$ at all the centers needed in the regularity proof, including the nearest points on the faces and the centers of the initial balls of the iterations. There are at most $CN^C 3^{C_d(\theta+\theta')N}$ retained unit-cube energies, all at cutoff at least $(1-\theta-\theta')N-C$ after rescaling. By (9.27), the probability that one of them differs from its limit by more than the relative error η is at most

$$C_\eta N^C 3^{C_d(\theta+\theta')N - c_\eta(1-\theta-\theta')N}.$$

The discarded tails, at finite cutoff or in the limit, have L^p norms at most $C_p 3^{-a[\theta'N]}$, so a union bound over the centers used in the conditions defining $G(q)$ bounds the probability that one of them exceeds η by

$$C_\eta N^C 3^{C_d\theta N - ap\theta'N}.$$

We choose θ' so small that the first exponent is negative, and then θ so small that both are. Both bounds are summable in N .

Step 3. We compare the finite and the limiting sums over intervals of scales. The relative error of the unit-cube energies controls the continuous piecewise linear functions: polarization gives $|\mathcal{J}_N - \mathcal{J}_\infty| \leq C\eta(1 + \mathcal{J}_\infty)$ on each retained normalized cell, and weighting gives the same inequality for the condition on $\mathbf{R}$. A bounded continuous piecewise linear function of this quantity changes by at most $C\eta$, and the square-root terms of the sums over intervals of scales change by at most $C\sqrt{\eta}$. The terms of $\mathbf{G}$ involving the fields $\mathbf{g}_j$ at a retained depth coincide once their finite interval of field indices lies below N , and their omitted tails satisfy the geometric bound of Step 2. Consequently, almost surely for all large N , every retained sum over an interval of scales at finite cutoff is bounded by the corresponding limiting sum plus $C\sqrt{\eta}$ per scale.

Step 4. We bound, uniformly over the choices of center and initial radius, the number of scales after which the limiting sums over intervals of scales satisfy the thresholds. For a fixed choice π of center and initial radius, let B_π be the least integer after which all limiting sums over intervals of scales satisfy the stricter thresholds of the iteration. The bounds of Lemma 8.36 on the sums of the quantities $\mathbf{F}$, $\mathbf{P}$, $\mathbf{R}$ and $\mathbf{G}$ at a fixed center, summed over the lengths of the intervals of scales, give $\mathbb{P}(B_\pi > b) \leq Ce^{-Ab}$, uniformly in π . At target depth n , the meshes of centers and initial radii used in the proof of Proposition 8.11 involve at most $C(n+1)^C 3^{C_d n}$ choices of centers, initial radii and faces, edges and corners. We fix a small $\xi_0 > 0$ before decreasing $\bar{\delta}$, and choose the rate A so that $A\xi_0 > C_d \log 3 + 2$. A union bound over the choices π at depth n gives

$$\mathbb{P}\left[\max_\pi B_\pi > \xi_0 n + b\right] \leq C(n+1)^C 3^{C_d n} e^{-A(\xi_0 n + b)}.$$

Summing over n shows that, almost surely, there is one finite B_ω such that $B_\pi \leq \xi_0 n + B_\omega$ for every choice π at every depth n .

Step 5. We transfer these bounds to finite cutoffs and prove the estimates at radii $r \geq 3^{-\theta N}$. We take η so small that the loss per scale of Step 3 is below the strict margin of the iteration. Then, for finite cutoffs and depths $n \leq \theta N$, the same numbers B_π apply with slightly weaker thresholds. The estimate of Lemma 8.9 transfers the original field bounds to the reflected coefficients near faces, edges and corners. The reflection across faces is applied at most $d+1$ times, and the numbers B_π of these applications add up, so their total contribution is bounded by a common random constant times $3^{C\xi_0 n}$.

We next show that the averages b_i at the initial radius of each application are bounded by a random constant times a small negative power of the initial radius. Let θ_0 be the value of θ in the proof of Lemma 9.22, so that the cells of depth at most $\theta_0 N$ were compared with their limits there, and decrease θ to at most $\theta_0/2$. For an initial radius at depth $k \leq \theta N$, we split its coarse-grained

sum at descendant depth $\theta_0 N$. For $0 < \rho < s$, the part with descendant depth at most $\theta_0 N$ is at most $CK_\omega 3^{\rho k}$, and the part with descendant depth greater than $\theta_0 N$ is at most

$$CK_N 3^{\rho k} 3^{-(s-\rho)(\theta_0 N - k)} \leq C 3^{\rho k} 3^{(\xi - (s-\rho)\theta_0/2)N},$$

where $K_N \leq 3^{\xi N}$ for large N , as in the proof of Lemma 9.22. We choose $\xi < (s - \rho)\theta_0/2$; the contribution below the smallest field scale satisfies the same bound by Lemma 8.6. The average at its own scale involves only the value of $H + \sum_{j < k} \mathbf{g}_{-j}$ at the center and the deterministic ratio $\bar{\mathbf{a}}_{N-k}/\bar{\mathbf{a}}_N$. Hence the energy in the initial ball, the coercivity constant and the constant in (8.6) are bounded by a random constant times a small negative power of the initial radius.

We apply the iteration of Lemma 6.10 with exponents strictly larger than α and t , and choose ξ_0 and ρ so small that the margins in the exponents absorb both losses. The iteration gives one random constant for the oscillation and energy bounds at all radii $r \geq 3^{-\theta N}$.

Step 6. We treat the radii $r \leq 3^{-\theta N}$. We choose exponents $\alpha_0 > \alpha$ and $t_0 > t$ for which the Morrey bounds at all radii of Proposition 8.7 and Proposition 8.11 hold. Their constants satisfy $K_N \leq 3^{\xi N}$ almost surely for all large N , for every fixed $\xi > 0$. For $r \leq 3^{-\theta N}$,

$$K_N r^{\alpha_0 - \alpha} \leq 3^{(\xi - \theta(\alpha_0 - \alpha))N} \quad \text{and} \quad K_N r^{t_0 - t} \leq 3^{(\xi - \theta(t_0 - t))N}.$$

We choose ξ smaller than $\theta(\alpha_0 - \alpha)$ and $\theta(t_0 - t)$; then these quantities are bounded. In the oscillation estimates at radii larger than $3^{-\theta N}$, point values are compared with averages over balls of radius $3^{-\theta N}$, using the estimate at small radii just proved. $\square$

Proposition 9.24 (Almost-sure convergence of energies and resolvents). *Almost surely, for every cube $Q \in \mathcal{C}$, $G_N^Q \rightarrow G^Q$ in operator norm on $L^2(Q, dx)$, E_N^Q Mosco-converges to E^Q , and, for every $f \in C(\bar{Q})$ and $\lambda > 0$,*

$$\|R_{N,\lambda}^Q f - R_\lambda^Q f\|_{C(\bar{Q})} \rightarrow 0.$$

On each fixed cube, the $C^{1/4}(\bar{Q})$ norms of $R_{N,\lambda}^Q f$ are bounded by one finite random constant, uniformly in N , in λ in a compact subset of $(0, \infty)$, and in f with $\|f\|_\infty \leq 1$.

Proof. Intersect the events of Lemmas 9.22 and 9.23, Proposition 9.2, and Proposition 9.21 for a countable dense family of bounded sources and their pairwise sums. Coercivity makes the images of the L^2 unit ball under G_N^Q bounded in $H^{3/4}$ uniformly in N , and polarization gives convergence of the matrix elements of G_N^Q on the dense family. The finite-rank argument and the variational formula (8.3) of Proposition 8.18 give norm and Mosco convergence to the limits already identified in probability.

All the constants in Proposition 9.5 are now bounded along the full sequence. Its variational and Arzelà–Ascoli argument therefore proves uniform convergence and the Hölder bound. $\square$

9.7. Almost-sure convergence of the path laws.

Proposition 9.25 (Almost-sure quenched convergence). *Almost surely, for every compact $B \subseteq \mathbb{R}^d$,*

$$\lim_{N \rightarrow \infty} \sup_{x \in B} d_{\text{weak}}(K_{N,x}^\omega, K_x^\omega) = 0.$$

Proof. Intersect the full-probability events of Proposition 9.24, Proposition 9.10 and Lemma 9.11 over the countable family of cubes, and fix an environment in this event. The killed occupation resolvents of $K_{N,x}^\omega$ converge to those of K_x^ω , uniformly on each closed cube.

We first obtain path tightness in this fixed environment. The image of a compact starting set under the continuous kernel $x \mapsto K_x^\omega$ is a compact family of laws on continuous path space. Its paths have infinite lifetime, so their probability of leaving a large cube before a fixed time, or of moving a fixed distance in a sufficiently short time, tends to zero uniformly in the starting point. Uniform convergence of the killed resolvents transfers these bounds to all sufficiently large cutoffs.

For compact containment, compare $R_{N,1}^Q 1$ with its limiting counterpart on a sufficiently large cube: the identity $1 - R_{N,1}^Q(x) = E_{N,x}^\omega e^{-\tau_Q}$ bounds exits before a fixed time. For short exits, choose a continuous function equal to zero near the starting point and one near the boundary of a surrounding smaller cube. The limiting occupation resolvent is small near the starting point and bounded below on that boundary. The strong Markov identity at exit, together with convergence of these resolvents, bounds the short-exit probability of the cutoff process. A finite cover gives the bound uniformly over a compact starting set.

Apply this short-exit bound at successive exits from small balls. The strong Markov property bounds the number of exits on a finite time interval; it also makes the probability of a very short gap tend to zero. Thus the moduli of continuity are tight. Together with compact containment, Arzelà–Ascoli gives tightness on continuous path space.

Now let $N_j \rightarrow \infty$ and $x_j \rightarrow x$. Localization as in (9.25) and killed-resolvent convergence give convergence of the whole-space discounted occupation integrals. Tightness and uniqueness of the Laplace transform give convergence of the one-time laws at every time. The Markov identity then gives convergence of the finite-dimensional laws, and approximation by cylinder functions on compact sets of paths gives $K_{N_j, x_j}^\omega \rightarrow K_x^\omega$. If convergence were not uniform on a compact B , a sequence of offending starting points would have a convergent subsequence, contradicting this conclusion and continuity of $x \mapsto K_x^\omega$. This proves the result in d_P ; tightness transfers it to every metric inducing the weak topology. $\square$

Proof of Theorem 7.1(i). The relabeling (8.1) and the changes of space and time transform the weak elliptic resolvent equations of the physical process into those defining $K_{N,x}^\omega$. Uniqueness of the weak solutions identifies the resolvents and hence the semigroups; the resulting finite-dimensional distributions identify the continuous path laws.

Proposition 9.25 gives the almost-sure convergence (7.3) of the quenched laws, and Proposition 9.9 shows that the limiting kernel is measurable with respect to the fields. Proposition 9.10 gives a continuous kernel, which is the family of laws of a continuous, conservative, strong Markov process with a μ -symmetric semigroup. Since the kernel is continuous, the semigroup maps bounded continuous functions to continuous functions. Proposition 9.2 shows that $\mu = \lim_N \mu_N$ is locally finite, has no atoms and has full support, and Corollary 9.12 gives the finite mean exit times. Finally, integrating the quenched convergence against $\mathbb{P}$ and using that the path functionals are bounded gives the convergence of the annealed laws. $\square$

10. THE LIMITING PROCESS AND MEASURE

In this section we prove the remaining assertions of Theorem 7.1. The mean-exit-time bound of Lemma 10.2 and power-law decay of the effective diffusivities between arbitrary scales bound the mean time to leave a cube of side 3^{-k} by $C3^{-(2+\eta)k}$, uniformly in the cutoff level L . These bounds pass to the limit and show that the limiting paths leave small cubes faster than Brownian paths. Consequently, their pointwise Hölder exponent is at most $(2+\eta)^{-1}$, and the annealed path laws are mutually singular with every Brownian law; for each deterministic starting point, so is the quenched path law in almost every environment.

We then prove that the reversible measure is invariant and singular with respect to Lebesgue measure, and that the transition laws are absolutely continuous with respect to it. Together with the bounds on its mass in small balls, this shows that no transition law is Gaussian.

Throughout, we assume (g1)–(g4) at weak disorder and rescale cubes to $\square_0$ as in Section 9.3, with unit of time (9.10). We write P_x^ω and E_x^ω for the law and expectation, in the environment ω , of the diffusion under consideration ($X^{(L)}$, X , or their rescalings to $\square_0$) started at x .

10.1. Power-law decay and mean exit times. We first extend the power-law decay of the effective diffusivities in Proposition 5.6 to the ratio between two arbitrary scales.

Lemma 10.1 (Power-law decay between arbitrary scales). *There exist constants $C < \infty$ and $\eta > 0$, depending only on d and the law of $\mathbf{g}_0$, such that, for every $\ell, m \in \mathbb{N}_0$ with $\ell \leq m$,*

$$(10.1) \quad \frac{\bar{\mathbf{a}}_m}{\bar{\mathbf{a}}_\ell} \leq C 3^{-\eta(m-\ell)},$$

where in dimension two one can take $\eta = \tau^2/\log 3$.

Proof. In dimension two, use (5.2). For $d \geq 3$, we start the gluing argument of Proposition 5.6 at scale ℓ .

We first prove for Δ_ℓ^π the energy bound that Proposition 4.1 gives for cubes. The Whitney cubes of Δ_ℓ^π of side $3^{\ell-i}$ occupy at most a proportion $C 3^{-i}$ of its volume. For $i \leq \ell$, use on each cube the minimizer of the energy with coefficient $\mathbf{a}_{\ell-i}$ among functions equal to $p \cdot x$ on the boundary of the cube. The factors contributed by the fields in $\mathbf{a}_\ell$ that are absent from $\mathbf{a}_{\ell-i}$ are independent of this minimizer and have mean one. By (3.32), the expected energy density is bounded by $C \bar{\mathbf{a}}_\ell e^{2\tau^2 i} |p|^2$. For $i > \ell$, the affine competitor gives the same bound since $\bar{\mathbf{a}}_\ell \geq e^{-2\tau^2(\ell+1)}$. Join these competitors, which all equal $p \cdot x$ on the boundaries of the Whitney cubes. Uniformly in ℓ , π and p , we obtain

$$\mathbb{E}[p \cdot \mathbf{a}_\ell(\Delta_\ell^\pi)p] \leq C \bar{\mathbf{a}}_\ell |p|^2 \sum_{i \geq 0} 3^{-i} e^{2\tau^2 i} \leq C \bar{\mathbf{a}}_\ell |p|^2.$$

Take R and $q_R < 1$ from (5.21). Keep the fields with indices at most ℓ and with indices $\ell + R, \ell + 2R, \dots$, and omit the others. The induction (5.22), started at scale ℓ from the bound for Δ_ℓ^π above, and reinsertion of the omitted fields by (5.23) give

$$\bar{\mathbf{a}}_m \leq C \bar{\mathbf{a}}_\ell q_R^{\lfloor (m-\ell)/R \rfloor}.$$

This proves (10.1) with $\eta = -\log q_R/(R \log 3)$. $\square$

Lemma 10.2 (Mean exit times). *There exists $C < \infty$ such that, for every $L, m \in \mathbb{N}_0$ and every $z \in \mathbb{R}^d$, the cube $U = z + \square_m$ satisfies*

$$(10.2) \quad \mathbb{E} \left[\sup_{x \in U} E_x^\omega \tau_U^{(L)} \right] \leq C 3^{2m} / \bar{\mathbf{a}}_{m \wedge L}.$$

More generally, there is $c > 0$ such that the following holds. For every $q < \infty$, there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there are random constants $K \geq 1$, one for each $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, with $\mathbb{E}[K^q]$ bounded uniformly in L, m and z , such that the process obtained by rescaling $z + \square_m$ to $\square_0$ satisfies

$$\sup_{x \in \square_0} E_x^\omega \tau_{\square_0} \leq K \quad \text{and} \quad \inf_{x \in \square_{-2}} E_x^\omega \tau_{\square_0} \geq c K^{-2}.$$

The same assertions hold for the diffusion X without cutoff, with $\bar{\mathbf{a}}_m$ in (10.2).

Proof. We rescale $z + \square_m$ to $\square_0$ as in Section 9.3, for either the diffusion $X^{(L)}$ or the diffusion X ; here μ has a positive, locally $C^{1,1}$ density. The mass bound (9.11), the trace inequality (9.16) and the coercivity bound (9.12) for functions with zero boundary values give, on every cube U that is fixed in the rescaled coordinates,

$$(10.3) \quad \|v\|_{L^{2d/(d-1)}(U, \mu)}^2 \leq K \mathcal{E}(v; U), \quad v \in H_0^1(U).$$

Here K may depend on U and has the asserted uniform moment bounds.

Let u solve $-\nabla \cdot (A \nabla u) = d\mu/dx$ in $\square_0$ with zero boundary values. Testing with $(u - h)_+$ gives, for $k > h \geq 0$,

$$\mu(\{u > k\}) \leq \left(\frac{K}{k - h} \right)^{2d/(d-1)} \mu(\{u > h\})^{(d+1)/(d-1)}.$$

Iterate this inequality along the thresholds equal to $1 - 2^{-i}$ times a suitable multiple of $K\mu(\square_0)^{1/d}$ to obtain $\|u\|_\infty \leq CK\mu(\square_0)^{1/d}$, which by (9.11) is bounded by another random constant K with the asserted moments.

We identify u with the mean exit time. The semigroup of the diffusion killed on leaving $\square_0$ is associated with the part of the Dirichlet form on $\square_0$ [Fukushima et al., 2011, Theorems 4.4.2–4.4.3]. Interpolate (10.3) with $L^1(\mu)$ to obtain a Nash inequality with respect to μ . On each fixed cube, it bounds the killed semigroup from $L^1(\mu)$ to $L^\infty(\mu)$ by a constant times t^{-d} . We adapt the eigenfunction construction for symmetric operators in Bass [1998, Chapter VII, Section 6, Theorem 6.3 and Proposition 6.4] to $L^2(\square_0, \mu)$. Interior elliptic regularity of the eigenfunctions, and uniform convergence of the eigenfunction expansion on compact sets of positive times and interior points, give a kernel $p_{\square_0}(t, x, y)$ that is jointly continuous at positive times and interior points.

Since the drift of the rescaled stochastic differential equation is locally bounded, Itô's formula applied to the squared distance from the starting point, stopped on exit from a small ball, bounds the probability of leaving that ball before time t by a quantity tending to zero as $t \downarrow 0$, uniformly over starting points in a compact subset of a fixed cube.

The local Nash bounds, this exit estimate and continuity of the semigroup on compactly supported continuous functions show, by approximation in $L^1(\mu)$ on an enclosing cube, that the semigroup of the diffusion without killing maps bounded Borel functions to continuous functions at each positive time. The exit estimate also gives continuity in the starting point of the transition probabilities of the diffusion killed on leaving $\square_0$. For each fixed positive time and Borel subset of $\square_0$, the killed transition probability and the integral of $p_{\square_0}$ over that subset are continuous and agree μ -almost everywhere. They therefore agree at every interior starting point, since μ has full support.

Coercivity gives a positive spectral gap. Together with the bound on the killed semigroup, this gives $P_x^\omega(\tau_{\square_0} > t) \leq Ce^{-ct}$ for $t \geq 1$, uniformly in $x \in \square_0$. Integrate the survival probability in time to obtain a function of the starting point that is continuous at interior points; as an element of $L^2(\mu)$, it equals the inverse of the positive killed generator applied to 1. This function is the continuous representative of u at interior points. With this representative,

$$u(x) = \int_0^\infty \int_{\square_0} p_{\square_0}(t, x, y) \mu(dy) dt = \int_0^\infty P_x^\omega(\tau_{\square_0} > t) dt = E_x^\omega \tau_{\square_0}, \quad x \in \square_0.$$

The bound on $\|u\|_\infty$ obtained above therefore holds at every interior point for this representative. This proves the upper bound after rescaling, and restoring the time unit (9.10) gives (10.2). The lower bound follows from (9.21): taking $t = (2K)^{-2}$ gives $P_x^\omega(\tau_{\square_0} > t) \geq 1/2$ for every $x \in \square_{-2}$. $\square$

10.2. Annealed and quenched path laws. For a deterministic starting point x , the annealed law $\mathbb{E}[K_x^\omega]$ of the limiting process is the average over ω of its quenched law K_x^ω .

Let $\mathbf{P}$ be a subsequential limit of the annealed laws of the rescaled diffusions $X^{(L)}$ or X in Proposition 9.8, started at deterministic points that converge in rescaled coordinates; in particular, $\mathbf{P}$ may be $\mathbb{E}[K_x^\omega]$. We use the exit times σ_k of Theorem 7.1(iii) and write $Z_t^* = \sup_{0 \leq s \leq t} |Z_s - Z_0|$.

Theorem 10.3 (Annealed and quenched singularity). *There exists $C < \infty$ such that every law $\mathbf{P}$ as above satisfies, for every $k \in \mathbb{N}_0$,*

$$(10.4) \quad E_{\mathbf{P}} \sigma_k \leq C 3^{-(2+\eta)k},$$

where η is the exponent of Lemma 10.1. Every such annealed law $\mathbf{P}$ is mutually singular with every mixture of Brownian laws with arbitrary covariance matrices. For each deterministic x , the quenched law K_x^ω has the same singularity for $\mathbb{P}$ -almost every ω . More generally, every subsequential limit in distribution of the random quenched laws of either family in Proposition 9.8, started at deterministic points that converge in rescaled coordinates, is almost surely mutually singular with every such mixture of Brownian laws.

This proves (7.5) and the mutual singularity in Theorem 7.1(iii).

Proof. Step 1. We prove (10.4). For $L \geq k$, the mean exit-time bound (10.2), applied at scale $L - k$ with time divided by $3^{2L}/\bar{\mathbf{a}}_L$, and Lemma 10.1 give, for either family,

$$\mathbb{E}\sigma_k(Z^{(L)}) \leq C3^{-2k} \frac{\bar{\mathbf{a}}_L}{\bar{\mathbf{a}}_{L-k}} \leq C3^{-(2+\eta)k}.$$

The exit time σ_k is a lower semicontinuous function of the path for the topology of locally uniform convergence. The portmanteau theorem, applied to this nonnegative lower semicontinuous functional, gives (10.4).

Step 2. We prove annealed singularity. Consider the event that $3^{2k}\sigma_k \rightarrow 0$ as $k \rightarrow \infty$. By (10.4),

$$E_{\mathbf{P}} \sum_{k \geq 0} 3^{2k}\sigma_k \leq C \sum_{k \geq 0} 3^{-\eta k} < \infty,$$

so this event has $\mathbf{P}$ -probability one. For a Brownian law with a nonzero finite covariance matrix, Brownian scaling gives $3^{2k}\sigma_k \stackrel{\text{law}}{=} \sigma_0 > 0$. Apply Fatou's lemma to the events that $3^{2k}\sigma_k \leq \varepsilon$. For every $\varepsilon > 0$, the probability of the event under this law is at most the probability that $\sigma_0 \leq \varepsilon$. Letting $\varepsilon \downarrow 0$ shows that the event has probability zero. When the covariance matrix is zero, the law of the constant path also gives the event probability zero.

Step 3. We prove the quenched assertions. For each deterministic x , Step 2 with $\mathbf{P} = \mathbb{E}[K_x^\omega]$ shows that the average over ω of the K_x^ω -probability of the event is one. Since this probability lies in $[0, 1]$, it equals one for $\mathbb{P}$ -almost every ω .

For the general assertion, convergence in distribution of the random quenched laws implies weak convergence of their averages. The average of the random limiting law is therefore an annealed subsequential limit and gives the event mass one by Step 2. The argument of the preceding paragraph then shows that the random limiting law gives the event mass one, almost surely. $\square$

For every law $\mathbf{P}$ as above, the exit-time bound (10.4) gives lower bounds on displacement at small times and an upper bound on the pointwise Hölder exponent defined in (7.6).

Corollary 10.4 (Small-time behavior of the limiting paths). *The following statements hold.*

(i) Oscillation: for every $\varepsilon > 0$, there is $c_\varepsilon > 0$ such that, $\mathbf{P}$ -almost surely, for all sufficiently small $t > 0$,

$$(10.5) \quad Z_t^* \geq c_\varepsilon \left(\frac{t}{(\log(1/t))^{1+\varepsilon}} \right)^{1/(2+\eta)}.$$

In particular, $\mathbf{P}$ -almost surely, $Z_t^*/\sqrt{t} \rightarrow \infty$ as $t \downarrow 0$.

(ii) Hölder exponent: $\mathbf{P}$ -almost surely, the pointwise Hölder exponent at zero is at most $1/(2+\eta)$, and at most $(2 + \tau^2/\log 3)^{-1}$ in dimension two.

(iii) Exit times: for every $\varepsilon > 0$, $\mathbf{P}$ -almost surely, for all sufficiently large k ,

$$(10.6) \quad \sigma_k \leq k^{1+\varepsilon} 3^{-(2+\eta)k}.$$

(iv) Moments of the running maximum: for every $p > 0$, there is $c_p > 0$ such that, for every $0 < t \leq 1$,

$$(10.7) \quad E_{\mathbf{P}}(Z_t^*)^p \geq c_p t^{p/(2+\eta)}.$$

For each deterministic x , parts (i)–(iii) also hold K_x^ω -almost surely for $\mathbb{P}$ -almost every ω , by Step 3 of the proof of Theorem 10.3. Parts (i) and (ii) give (7.6) in Theorem 7.1.

Proof. Markov's inequality gives $\mathbf{P}(\sigma_k > k^{1+\varepsilon} 3^{-(2+\eta)k}) \leq Ck^{-1-\varepsilon}$, so the Borel–Cantelli lemma proves (iii). For t sufficiently small, choose k with

$$k^{1+\varepsilon} 3^{-(2+\eta)k} \leq t < (k-1)^{1+\varepsilon} 3^{-(2+\eta)(k-1)}.$$

Then $Z_t^* \geq 3^{-k}/2$ and k is comparable to $\log(1/t)$, proving (i). Part (i) implies (ii), since $|Z_t - Z_0| = O(t^\gamma)$ would give $Z_t^* = O(t^\gamma)$ and contradict (i) for $\gamma > 1/(2+\eta)$.

For (iv), enlarge C in (10.4) to at least one and take the smallest $k \geq 0$ with $C3^{-(2+\eta)k} \leq t/2$. Then $\mathbf{P}(\sigma_k \leq t) \geq 1/2$ and 3^{-k} is comparable to $t^{1/(2+\eta)}$, giving the claimed moment bound. $\square$

Corollary 10.5 (Exit moments and small displacements). *For every $p > 0$, there exists $\bar{\delta}_0 > 0$ such that, if $0 < \bar{\delta} \leq \bar{\delta}_0$, then there are $0 < c_p \leq C_p < \infty$ such that, for every $L, m \in \mathbb{N}_0$ and $z \in \mathbb{R}^d$, the cube $U = z + \square_m$ satisfies*

$$(10.8) \quad \mathbb{E} \left[\sup_{x \in U} E_x^\omega (\tau_U^{(L)})^p \right] \leq C_p 3^{2mp} \bar{\mathbf{a}}_{m \wedge L}^{-p}.$$

For starting points in the inner cube, the matching lower bound is

$$\mathbb{E} \left[\inf_{x \in z + \square_{m-2}} E_x^\omega (\tau_U^{(L)})^p \right] \geq c_p 3^{2mp} \bar{\mathbf{a}}_{m \wedge L}^{-p}.$$

The same bounds hold for X with m in place of $m \wedge L$. Moreover, every $\mathbf{P}$ as above satisfies $E_{\mathbf{P}} \sigma_k^p \leq C_p 3^{-p(2+\eta)k}$ for every $k \in \mathbb{N}_0$, and

$$\mathbf{P}(Z_t^* \leq \varepsilon t^{1/(2+\eta)}) \leq C_p \varepsilon^{p(2+\eta)}, \quad 0 < t, \varepsilon \leq 1.$$

Consequently, for every $0 < q < p(2+\eta)$,

$$E_{\mathbf{P}}(Z_t^*)^{-q} \leq C_{p,q} t^{-q/(2+\eta)}, \quad 0 < t \leq 1.$$

Proof. After rescaling $z + \square_m$ to $\square_0$ as in the proof of Lemma 10.2, Markov's inequality and the strong Markov property give, for each environment,

$$\sup_{x \in \square_0} P_x^\omega (\tau_{\square_0} > 2nK) \leq 2^{-n}, \quad n \in \mathbb{N}_0.$$

Integrating yields $\sup_x E_x^\omega \tau_{\square_0}^p \leq C_p K^p$. For the lower bound, (9.21) and $\mathbb{E}K \leq C$ give, for a sufficiently small deterministic $c > 0$,

$$\mathbb{E} \left[\inf_{x \in \square_{-2}} E_x^\omega \tau_{\square_0}^p \right] \geq c^p \mathbb{E} \left[\inf_{x \in \square_{-2}} P_x^\omega (\tau_{\square_0} > c) \right] \geq \frac{c^p}{2}.$$

Taking expectations in the upper bound and restoring the time unit (9.10) proves both bounds. Rescaling as in Step 1 of the proof of Theorem 10.3, Lemma 10.1 and the portmanteau theorem for σ_k^p give the exit-moment bound for $\mathbf{P}$.

For $0 < r \leq 1/12$, choose $k \geq 0$ with $4r \leq 3^{-k} < 12r$. If $Z_t^* \leq r$, then $\sigma_k > t$, so Markov's inequality gives

$$\mathbf{P}(Z_t^* \leq r) \leq t^{-p} E_{\mathbf{P}} \sigma_k^p \leq C_p t^{-p} r^{p(2+\eta)}.$$

Set $r = \varepsilon t^{1/(2+\eta)}$ to obtain the stated probability bound. Integrating it against $q\varepsilon^{-q-1} d\varepsilon$ on $(0, 1)$ proves the negative-moment bound. $\square$

10.3. Singularity of the measure and of the transition laws. Recall the limiting measures μ^0 and $\mu = e^H \mu^0$ of Theorem 7.1(ii), constructed in Proposition 9.2 and its proof. For Gaussian fields, $\tau^2 = 1/2 \mathbb{E}[\mathbf{g}_0(0)^2]$, so μ^0 is Gaussian multiplicative chaos, and μ is its product with the positive, locally $C^{1,1}$ function e^H . Existence of the limit, moment bounds, absence of atoms, full support, and singularity with respect to Lebesgue measure are standard properties of Gaussian multiplicative chaos in the subcritical regime, in which the variance of the field is small enough for the limiting measure to be nonzero. Moreover, the densities of the approximating measures diverge at almost every point with respect to the limiting measure; see Kahane [1985] and Rhodes and Vargas [2014, Sections 2 and 4.1]. The proofs in Section 9.2 and below also apply to the non-Gaussian fields allowed by (g1)–(g4).

The measures $\mathbf{a}_N(3^N x) dx$ have the laws of μ_N^0 and therefore converge in distribution to μ^0 . The measures μ^0 and μ are nonatomic and have full support, and Proposition 9.2 with $\epsilon = 1/2$ gives, on each bounded set of centers, a random constant K such that, for every x in that set,

$$(10.9) \quad \mu^0(B_r(x)) + \mu(B_r(x)) \leq Kr^{d-1/2}, \quad 0 < r \leq 1.$$

For every finite moment order, K has a finite moment of that order at weak disorder, with a threshold depending on the order, by the same proposition and by Lemma 8.1 for $e^{\pm H}$.

Theorem 10.6 (The limiting measures). *The following statements hold.*

- (i) Translation invariance: the law of μ^0 is invariant under translations, μ^0 is not deterministic, and it satisfies $\mathbb{E}[\mu^0(dx)] = dx$. Its restrictions to two sets at distance greater than $\sqrt{d}$ are independent.
- (ii) Singularity: almost surely, the density of μ_N^0 at x tends to zero as $N \rightarrow \infty$ for Lebesgue-almost every x , whereas it tends to infinity for μ^0 -almost every and μ -almost every x . In particular, both limiting measures are singular with respect to Lebesgue measure.
- (iii) Invariance: almost surely, the reversible measure μ is invariant for the limiting process: for every nonnegative Borel function f and every $t \geq 0$,

$$(10.10) \quad \iint f(Z_t) dK_x^\omega \mu(dx) = \int f d\mu.$$

Proof. Write ρ_N for the density of μ_N^0 . Lemma 9.1 gives uniform integrability of the masses $\mu_N^0(U)$ on bounded cubes U , and hence

$$\mathbb{E}[\mu^0(dx) \mid \mathbf{g}_0, \dots, \mathbf{g}_{-N}] = \rho_N(x) dx, \quad \mathbb{E}\mu^0(dx) = dx.$$

The strong law and Fubini's theorem give, for $\mathbb{P} \otimes dx$ -almost every (ω, x) ,

$$(10.11) \quad \frac{\log \rho_N(x)}{N+1} = \frac{1}{N+1} \sum_{i=0}^N \mathbf{g}_{-i}(x) - \tau^2 \longrightarrow -\tau^2 < 0.$$

For the behavior at a point sampled from μ^0 , fix a bounded cube U and consider the probability measure $|U|^{-1} \mathbb{P}(d\omega) \mu^{0,\omega}(dx)$ on $\Omega \times U$. This is the change of measure used for Gaussian chaos in Rhodes and Vargas [2014, Section 4.1]; here independence of the fields suffices. The conditional-expectation identity above and stationarity show that, under this measure, the variables $\mathbf{g}_{-j}(x)$ for $j \geq 0$ are independent, and each has the law of $\mathbf{g}_0(0)$ tilted by $e^{\mathbf{g}_0(0) - \tau^2}$.

Their common mean is $\mathbb{E}[\mathbf{g}_0(0)e^{\mathbf{g}_0(0) - \tau^2}] > \tau^2$: under the tilted law, $e^{-\mathbf{g}_0(0)}$ has expectation $e^{-\tau^2}$, and Jensen's inequality for the exponential is strict because $\mathbf{g}_0(0)$ is not almost surely constant. The strong law therefore gives

$$\frac{\log \rho_N(x)}{N+1} \longrightarrow \mathbb{E}[\mathbf{g}_0(0)e^{\mathbf{g}_0(0) - \tau^2}] - \tau^2 > 0$$

almost surely under this probability measure. Thus $\rho_N \rightarrow \infty$ at μ^0 -almost every point, almost surely. The positive factor e^H gives the same conclusion for μ . Together with (10.11), this proves (ii). Since $\mathbb{E}\mu^0(dx) = dx$, singularity also shows that μ^0 is not deterministic. Translation invariance of the law and independence of restrictions to sets at distance greater than $\sqrt{d}$ pass from the fields to μ^0 , which is a measurable function of them.

Finally, symmetry and conservativeness from Proposition 9.10 give (10.10). $\square$

The Sobolev estimate (10.3) gives absolute continuity of the transition laws at positive times with respect to μ . The measure μ is carried by a Lebesgue-null set and gives no mass to affine hyperplanes, so these transition laws are singular with respect to every Gaussian law.

Theorem 10.7 (Quenched transition laws). *Almost surely, for every $x \in \mathbb{R}^d$ and every $t > 0$,*

$$(10.12) \quad K_x^\omega(Z_t \in \cdot) \ll \mu$$

and the law of Z_t under K_x^ω is singular with respect to Lebesgue measure and every Gaussian law. For every bounded open set U , there is $C_U < \infty$ such that, for every $x \in U$ and $t > 0$,

$$K_x^\omega(Z_t \in \cdot, t < \tau_U) \leq C_U t^{-d} \mu|_U.$$

Moreover, almost surely, for every starting point, the quenched joint law of the positions at any finite set of positive times is mutually singular with every Gaussian law, and the quenched path law on any interval $[s, t]$, $0 \leq s < t$, is mutually singular with the law of every continuous Gaussian process on that interval.

Proof. Fix a cube U of a deterministic countable exhaustion of $\mathbb{R}^d$ and let $S_{N,t}$ be the symmetric killed semigroup of E_N^U on $L^2(U, \mu_N)$. By the relabeling (8.1), the coefficient of E_N^U and the measure μ_N have the joint law of the coefficient and the reversible measure after rescaling of X at scale N . Hence (10.3) applies on U with constants K_N satisfying $\sup_N \mathbb{E} K_N < \infty$.

We use Nash's argument. For nonzero $f \in L^1(U, \mu_N) \cap L^2(U, \mu_N)$, interpolation of (10.3) with L^1 , the energy identity and L^1 contraction give

$$\frac{d}{dt} \|S_{N,t} f\|_{L^2(U, \mu_N)}^2 \leq -2K_N^{-1} \|S_{N,t} f\|_{L^2(U, \mu_N)}^{2+2/d} \|f\|_{L^1(U, \mu_N)}^{-2/d}.$$

Integration, symmetry and the semigroup property yield

$$(10.13) \quad \|S_{N,t} f\|_\infty \leq C(K_N/t)^d \|f\|_{L^1(U, \mu_N)}, \quad t > 0.$$

For $f \in C_c(U)$, the function $S_{N,t} f$ is continuous, so this bound holds at every point of U .

By Proposition 9.25, the whole-space path laws converge almost surely, locally uniformly in the starting point. Fatou's lemma gives $\liminf_N K_N < \infty$ almost surely. For each such environment, choose a subsequence on which K_N is bounded. If $f \in C_c(U)$ is nonnegative, the path functional $z \mapsto f(z(t)) \mathbf{1}_{\{t < \tau_U\}}$ is lower semicontinuous: a path that stays in U up to time t has a positive distance from its complement. The portmanteau theorem, (10.13), and local weak convergence $\mu_N \rightarrow \mu$ therefore give the same bound for the killed transition law of K_x^ω . Positive compactly supported tests give the asserted measure domination. The subsequence depends only on the environment and U , so the bound holds for every $x \in U$ and $t > 0$.

Enclosing an arbitrary bounded open set in one of these cubes and comparing exit times gives the bound on that set. The limiting process is conservative and has continuous paths, so each path remains in some cube of the exhaustion up to time t . Monotone convergence therefore proves (10.12) for all x and $t > 0$ on a single event of probability one.

By Theorem 10.6, a Lebesgue-null set carries μ and hence every transition law, proving singularity with respect to Lebesgue measure and every nondegenerate Gaussian. A bounded subset of an affine hyperplane is covered by $O(r^{1-d})$ balls of radius r , so (10.9) bounds its μ -mass by a constant times $r^{1/2}$, which tends to zero as $r \downarrow 0$. Thus every affine hyperplane is μ -null. Every degenerate Gaussian, including a point mass, is supported on such a hyperplane, which has zero probability under every transition law by (10.12). This proves mutual singularity with every Gaussian law.

Since the marginals of a Gaussian law are Gaussian, and two laws are mutually singular as soon as their marginals at one time are, projecting onto a single positive time gives the assertions for joint laws and path laws. $\square$

APPENDIX A. MOMENTS OF RANDOM VARIABLES WITH LOG-NORMAL TAILS

Stochastic bounds throughout the paper are stated in the Orlicz notation of [Armstrong et al. \[2019, Appendix A\]](#). We use exactly the convention of [Armstrong et al. \[2019, Appendix A, p. 416, \(A.1\)–\(A.2\)\]](#): the subscript refers to the Orlicz function $\exp(t^\sigma)$, and for $\sigma > 0$ and $A > 0$ we write

$$X \leq \mathcal{O}_{\Gamma_\sigma}(A) \quad \text{if} \quad \mathbb{E}[\exp((A^{-1} \max\{X, 0\})^\sigma)] \leq 2,$$

and

$$(A.1) \quad X = \mathcal{O}_{\Gamma_\sigma}(A) \quad \text{if} \quad X \leq \mathcal{O}_{\Gamma_\sigma}(A) \quad \text{and} \quad -X \leq \mathcal{O}_{\Gamma_\sigma}(A).$$

If $X \leq \mathcal{O}_{\Gamma_\sigma}(A)$, then, by [Armstrong et al. \[2019, Lemma A.1\]](#),

$$(A.2) \quad \mathbb{P}[X \geq tA] \leq 2 \exp(-t^\sigma) \quad (t \geq 0).$$

Conversely, the bound $\mathbb{P}[X \geq tA] \leq \exp(-t^\sigma)$ for every $t \geq 0$ implies $X \leq \mathcal{O}_{\Gamma_\sigma}(2^{1/\sigma}A)$.

By [Armstrong et al. \[2019, Lemma A.4\]](#), for every $\sigma \geq 1$, every finite index set I , and random variables X_k satisfying $X_k \leq \mathcal{O}_{\Gamma_\sigma}(a_k)$ for every $k \in I$,

$$(A.3) \quad \sum_{k \in I} X_k \leq \mathcal{O}_{\Gamma_\sigma} \left(\sum_{k \in I} a_k \right).$$

The same conclusion holds for a countable family of nonnegative X_k with $\sum_k a_k < \infty$, by monotone convergence. The tail bound (A.2), a union bound, and the converse part of [Armstrong et al. \[2019, Lemma A.1\]](#) give, for every $s > 0$, $\sigma > 0$ and $k \in \mathbb{N}$, and all random variables $X_1, \dots, X_k$ satisfying $X_i \leq \mathcal{O}_{\Gamma_s}(\sigma)$ for every i ,

$$(A.4) \quad \max_{i \in \{1, \dots, k\}} X_i \leq \mathcal{O}_{\Gamma_s}(C_s(\log(2k))^{1/s} \sigma).$$

If, in addition, $s \in (1, 2]$, the variables are independent and mean zero, and $X_i = \mathcal{O}_{\Gamma_s}(\sigma)$, then, by [Armstrong et al. \[2019, Lemmas A.7 and A.9\]](#),

$$(A.5) \quad \sum_{i=1}^k X_i = \mathcal{O}_{\Gamma_s}(C_s k^{1/2} \sigma).$$

Lemma A.1. *There exists a universal constant $C < \infty$ such that, for every $\sigma > 0$ and every random variable X satisfying $X = \mathcal{O}_{\Gamma_2}(\sigma)$, the following estimates hold. For every $k > 0$,*

$$(A.6) \quad \mathbb{E}[|X|^k] \leq C \sigma^k \Gamma(k/2 + 1),$$

for every $t > 0$,

$$(A.7) \quad \mathbb{E}[\exp(tX) - (1 + tX)] \leq C \sigma^2 t^2 \exp(C \sigma^2 t^2),$$

and, for every $m \geq 1$ and $t > 0$,

$$(A.8) \quad \mathbb{E}[|\exp(tX) - 1|^m] \leq (C m^{1/2} \sigma t)^m \exp(C(m \sigma t)^2).$$

Proof. By (A.2), applied to X and $-X$,

$$\mathbb{P}[|X| \geq r\sigma] \leq 4e^{-r^2} \quad (r \geq 0).$$

The layer-cake formula gives

$$\mathbb{E}[|X|^k] \leq 4k\sigma^k \int_0^\infty r^{k-1} e^{-r^2} dr = 4\sigma^k \Gamma\left(\frac{k}{2} + 1\right),$$

which is (A.6). Moreover, Young's inequality and (A.1) give

$$\mathbb{E}[e^{\lambda|X|}] \leq 4e^{\lambda^2 \sigma^2 / 2} \quad (\lambda \geq 0).$$

The elementary inequalities

$$|e^u - 1 - u| \leq \frac{1}{2}u^2 e^{|u|} \quad \text{and} \quad |e^u - 1|^m \leq |u|^m e^{m|u|},$$

together with the Cauchy–Schwarz inequality, (A.6) and the bound on $\mathbb{E}[e^{\lambda|X|}]$, yield

$$\mathbb{E}[|e^{tX} - 1 - tX|] \leq C\sigma^2 t^2 e^{C\sigma^2 t^2}$$

and

$$\mathbb{E}[|e^{tX} - 1|^m] \leq t^m \mathbb{E}[|X|^{2m}]^{1/2} \mathbb{E}[e^{2mt|X|}]^{1/2} \leq (Cm^{1/2}\sigma t)^m e^{C(m\sigma t)^2}.$$

These are (A.7) and (A.8). □

APPENDIX B. CONCENTRATION INEQUALITIES

In this appendix we collect concentration inequalities used in the body of the paper: Rosenthal's inequality for sums of independent random variables, used in Sections 3 and 4, and bounds on the proportion of scales at which independent or finitely dependent random variables are large, used in Sections 6 and 8.5.

We use Rosenthal's inequality in the form proved by [Boucheron et al. \[2013, Theorem 15.11\]](#).

Lemma B.1 (Rosenthal's inequality). *For every $p \in [2, \infty)$, $N \in \mathbb{N}$ and every sequence $X_1, \dots, X_N$ of independent random variables satisfying $\mathbb{E}[X_i] = 0$, we have the estimate*

$$(B.1) \quad \mathbb{E} \left[\left| \sum_{i=1}^N X_i \right|^p \right]^{1/p} \leq 6p^{1/2} \left(\sum_{i=1}^N \mathbb{E}[X_i^2] \right)^{1/2} + 4p \mathbb{E} \left[\max_{i \in \{1, \dots, N\}} |X_i|^p \right]^{1/p}.$$

Proposition B.2 (Concentration across finitely dependent columns). *Let $p \in [1, \infty)$, $r \in \mathbb{N}$ and $s \in [1/p, 1]$. Let $\{X_{k,j} : k \in \mathbb{Z}, j \in \mathbb{Z}\}$ be a collection of nonnegative random variables satisfying, for every $k, j \in \mathbb{Z}$, the moment bound*

$$(B.2) \quad \mathbb{E}[X_{k,j}^p] \leq 1,$$

and r -dependence across columns: for every two subsets $I, J \subseteq \mathbb{Z}$ with $\text{dist}(I, J) \geq r$,

$$(B.3) \quad \{X_{k,j} : k \in \mathbb{Z}, j \in I\} \quad \text{and} \quad \{X_{k,j} : k \in \mathbb{Z}, j \in J\} \quad \text{are independent}.$$

There exists a universal constant $C > 0$ such that, for every $m_0 \in \mathbb{Z}$, $m \in \mathbb{N}_0$ and $\theta \in (0, 1]$,

$$(B.4) \quad \mathbb{P} \left[\frac{1}{m+1} \sum_{k=m_0}^{m_0+m} \mathbb{1} \left\{ \sum_{j \in \mathbb{Z}} 3^{-s|k-j|} X_{k,j} > 6s^{-1} C^{1/p} \theta^{-1/p} \right\} > \theta \right] \leq \exp \left(-\frac{sp\theta}{16r} (m+1) \right).$$

Proof. This is [Armstrong et al. \[2026, Proposition D.1, version 2\]](#). $\square$

Proposition B.3 (Concentration across scales). *There is a universal constant $C < \infty$ with the following property. Let $\delta_0 \in (0, 1]$, and let $(X_i)_{i \in \mathbb{Z}}$ be independent nonnegative random variables satisfying $X_i = \mathcal{O}_{\Gamma_2}(\delta_0)$ for every $i \in \mathbb{Z}$. Then, for every $m_0 \in \mathbb{Z}$, $M \in \mathbb{N}_0$, $\theta \in (0, 1]$, and $s \in (0, 1]$ such that $\delta_0 \leq C^{-1}(s\theta)^{1/2}$,*

$$(B.5) \quad \mathbb{P} \left[\frac{1}{M+1} \sum_{k=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \exp \left(\sum_{i=0}^{\infty} 3^{-i} X_{k+i+j} \right) > 3 \right\} \geq \theta \right] \leq \exp \left(-\frac{s\theta}{C\delta_0^2} (M+1) \right).$$

Proposition B.4 (Concentration across scales for random fields). *Let $\delta_0 \in (0, 1]$, and let $(X_i)_{i \in \mathbb{Z}}$ be independent real-valued random fields. Assume that X_i is $3^i \mathbb{Z}^d$ -stationary and $\|X_i\|_{L^\infty(\square_i)} \leq \mathcal{O}_{\Gamma_2}(\delta_0)$ for every $i \in \mathbb{Z}$. There exists $C(d) < \infty$ such that, for every $m_0 \in \mathbb{Z}$, $M \in \mathbb{N}_0$, $h \in \mathbb{N}$, and $s, \theta \in (0, 1]$ satisfying $\delta_0 \leq C^{-1}s(\theta \wedge h^{-1})^{1/2}$,*

$$(B.6) \quad \mathbb{P} \left[\frac{1}{M+1} \sum_{k=m_0}^{m_0+M} \mathbb{1} \left\{ \sup_{j \geq 0} 3^{-sj} \left\| \prod_{i=k-j}^{k+j} \exp(|X_i|) \right\|_{L^\infty(\square_{k+h+j})} > 3 \right\} \geq \theta \right] \leq \exp \left(-\frac{s^2\theta}{C\delta_0^2} (M+1) \right).$$

Propositions B.3 and B.4 are deduced from the next lemma.

Lemma B.5 (Selection of rare intervals). *There are universal constants $c > 0$ and $C < \infty$ such that the following holds. Let $(\xi_i)_{i \in \mathbb{Z}}$ be independent random elements and let $\lambda > 0$. For $k \in \mathbb{Z}$ and $j \in \mathbb{N}_0$, let $E_{k,j}$ be an event that depends only on the variables ξ_i with indices $i \in [k-j, k+j] \cap \mathbb{Z}$,*

and assume that $\mathbb{P}[E_{k,j}] \leq e^{-\lambda(j+1)}$. If $0 < \theta \leq 1$ and $\lambda\theta \geq C$, then every integer interval I of cardinality $N \geq 1$ satisfies

$$(B.7) \quad \mathbb{P}[\#\{k \in I : E_{k,j} \text{ occurs for some } j\} \geq \theta N] \leq e^{-c\lambda\theta N}.$$

Proof. A union bound over $k \in I$ and $j \geq N$ shows that the probability that $E_{k,j}$ occurs for some $k \in I$ and $j \geq N$ is at most $e^{-c\lambda N}$. On the complementary event, for each $k \in I$ such that $E_{k,j}$ occurs for some j , choose one such j , necessarily $j < N$, and take the corresponding interval of indices, $[k, k+j]$ or $[k-j, k+j]$. Retain these intervals greedily in decreasing order of length, discarding those at distance less than two from a retained interval. Every discarded interval has length at most that of the interval which discards it, and lies in its enlargement by that length on each side. Hence the retained intervals have total cardinality at least $\theta N/3$.

The retained intervals are separated by at least one missing integer and lie in an interval of cardinality at most $3N$. Their union determines the intervals and their starting indices, so there are at most 2^{3N} configurations. For each configuration, independence and the fact that $j+1$ is at least half the cardinality of the corresponding interval give probability at most $e^{-\lambda\theta N/6}$. Thus the required probability is at most $e^{-c\lambda N} + e^{CN-\lambda\theta N/6}$. Taking the constant in the assumption $\lambda\theta \geq C$ large enough absorbs the counting factor e^{CN} and the sum of the two terms. $\square$

Proof of Proposition B.3. Call an index k bad if the event in the indicator in (B.5) occurs. Every bad index k admits some $j \geq 0$ such that

$$(B.8) \quad X_{k+j} > \frac{\log 3}{3}(1 + sj).$$

Indeed, if $X_{k+j} \leq \frac{\log 3}{3}(1 + sj)$ for every $j \geq 0$, then, for every $j \geq 0$,

$$\sum_{i \geq 0} 3^{-i} X_{k+j+i} \leq \frac{\log 3}{3} \left(\frac{3}{2}(1 + sj) + \frac{3}{4}s \right) \leq \frac{3}{4}(1 + sj) \log 3,$$

which excludes the event in (B.5). The event (B.8) depends only on the index $k+j$ and has probability at most

$$2e^{-c(1+sj)^2/\delta_0^2} \leq e^{-c_1 s(j+1)/\delta_0^2},$$

since $(1+sj)^2 \geq s(j+1)$. We apply Lemma B.5 to these events on the $M+1$ indices $m_0, \dots, m_0+M$, with $\lambda = c_1 s/\delta_0^2$, which gives

$$\mathbb{P}[\#\{\text{bad indices}\} \geq \theta(M+1)] \leq \exp\left(-\frac{s\theta}{C\delta_0^2}(M+1)\right).$$

This is (B.5). $\square$

Proof of Proposition B.4. Let $E_{k,j}$ be the event that the expression at index j inside the supremum in (B.6) exceeds three. It depends only on $(X_i)_{k-j \leq i \leq k+j}$. Cover $\square_{k+h+j}$ by its $3^{d(h+2j)}$ triadic cubes of scale $k-j$. On each of these cubes, the logarithm of the product is bounded above by the sum of the suprema of $|X_i|$ on its scale- i parent cubes. These $2j+1$ random variables are independent, have means at most $C\delta_0$, and, after centering, have sub-Gaussian parameters at most $C\delta_0$. By (A.5), a union bound, and $1+sj \geq s(j+1)$,

$$(B.9) \quad \mathbb{P}[E_{k,j}] \leq \exp\left(C(h+j+1) - \frac{c(1+sj)^2}{\delta_0^2(j+1)}\right) \leq \exp\left(-\frac{c_1 s^2}{\delta_0^2}(j+1)\right).$$

Here $\delta_0 \leq C^{-1}s$ absorbs the means and the term proportional to $j+1$, while $\delta_0^2 h \leq C^{-2}s^2$ absorbs the cost of the union bound over the covering cubes at $j=0$, and hence at every j . Lemma B.5, with $\lambda = c_1 s^2/\delta_0^2$ and $I = \{m_0, \dots, m_0+M\}$, now gives (B.6). $\square$

APPENDIX C. BESOV NORM COMPARISONS

In this appendix we prove two statements about Besov norms recorded in Section 2.1: the duality estimate (2.13), and the comparison (2.10) of Besov and fractional Sobolev seminorms.

The fractional Sobolev–Poincaré inequality states that, for each $s \in (0, 1]$ and $p \in [1, d/s]$, there exists $C(s, p, d) < \infty$ such that, for every $m \in \mathbb{Z}$ and $f \in W^{s,p}(\square_m)$,

$$(C.1) \quad \|f - (f)_{\square_m}\|_{\underline{L}^{pd/(d-sp)}(\square_m)} \leq C 3^{sm} [f]_{\underline{W}^{s,p}(\square_m)}.$$

For $s \in (0, 1)$, this follows from the Sobolev embedding of Di Nezza et al. [2012, Theorem 6.7], applied to $f - (f)_{\square_m}$, the fractional Poincaré inequality, and rescaling. For $s = 1$, it is the classical Sobolev–Poincaré inequality in a cube.

Lemma C.1 (Besov duality). *Let $m \in \mathbb{Z}$, $s \in (0, 1]$, and $q, r \in [1, \infty]$, with $q^{-1} + (q')^{-1} = r^{-1} + (r')^{-1} = 1$, and assume that either $s < 1$ or $(s, q) = (1, 1)$. Then (2.13) holds for all functions f, g on $\square_m$ such that $fg \in L^1(\square_m)$ and the right side of (2.13) is finite.*

Proof. Let E_n denote averaging on the partition into scale- n triadic cubes. At finite depth $n < m$,

$$(C.2) \quad E_n g = (g)_{\square_m} + \sum_{k=n+1}^m (E_{k-1} g - E_k g).$$

On a scale- k parent Q and its child Q' , one has $|(g)_{Q'} - (g)_Q| \leq 3^d \|g - (g)_Q\|_{L^1(Q)}$. Pairing (C.2) with f and applying Hölder's inequality first to the cube averages and then to the sum over scales gives

$$\left| \int_{\square_m} f E_n g - (f)_{\square_m} (g)_{\square_m} \right| \leq C s^{-1} \|f\|_{\dot{B}_{r,q}^{-s}(\square_m)} [g]_{B_{1,q',r'}^s(\square_m)}.$$

To see this, insert the factor $3^{s(k-1)} 3^{-s(k-1)}$ before applying Hölder's inequality to the sum over scales. Replacing the partition averages by the overlapping averages in (2.8) costs a factor depending only on d , and the normalizations of the two sums over scales give the factor s^{-1} , also at the endpoint exponents 1 and ∞ . The term of the negative norm at the top scale gives

$$|(f)_{\square_m} (g)_{\square_m}| \leq s^{-1} \|f\|_{\dot{B}_{r,q}^{-s}(\square_m)} 3^{-sm} |(g)_{\square_m}|.$$

If g is bounded, then $E_n g \rightarrow g$ almost everywhere and $|f E_n g| \leq \|g\|_{\infty} |f|$; dominated convergence proves (2.13) in this case.

For general g , put $a = (g)_{\square_m}$ and, for $M > 0$, define

$$T_M(t) = (-M) \vee (t \wedge M), \quad g_M = a + T_M(g - a) - (T_M(g - a))_{\square_m}.$$

The mean of g_M is a . Since T_M is one-Lipschitz, every cube Q appearing in (2.8) satisfies

$$\|g_M - (g_M)_Q\|_{L^1(Q)} \leq 2 \|T_M(g - a) - T_M((g)_Q - a)\|_{L^1(Q)} \leq 2 \|g - (g)_Q\|_{L^1(Q)}.$$

Thus $\|g_M\|_{B_{1,q',r'}^s} \leq 2 \|g\|_{B_{1,q',r'}^s}$. Moreover $g_M \rightarrow g$ almost everywhere and

$$|f g_M| \leq |f g| + \left(2|a| + \int_{\square_m} |g - a| \right) |f|.$$

Applying the bounded case to g_M and then dominated convergence proves (2.13). $\square$

Lemma C.2. *There exists a constant $C(d) < \infty$ such that, for every $s \in (0, 1)$, $p \in [1, \infty)$, $m \in \mathbb{Z}$ and $u \in L^p(\square_m)$,*

$$(C.3) \quad C^{-1} [u]_{\underline{W}^{s,p}(\square_m)} \leq [u]_{\underline{B}_{p,p}^s(\square_m)} \leq C [u]_{\underline{W}^{s,p}(\square_m)}.$$

Proof. Let $\mathcal{Q}_n$ be the set of cubes of side 3^n with centers in $3^{n-1}\mathbb{Z}^d$ which are contained in $\square_m$. Their number is between $|\square_m|3^{-nd}$ and $3^d|\square_m|3^{-nd}$, and their overlap multiplicity is bounded by 4^d . For every such cube, Jensen's inequality and the triangle inequality give

$$\int_Q |u - (u)_Q|^p \leq \int_Q \int_Q |u(x) - u(y)|^p dx dy \leq 2^p \int_Q |u - (u)_Q|^p.$$

Average over $\mathcal{Q}_n$, multiply by $s3^{-ns}$, and sum over $n \leq m$. Tonelli's theorem shows that $[u]_{B_{p,p}^s(\square_m)}^p$ is comparable, within factors $C(d)^p$, to

$$\frac{s}{|\square_m|} \int_{\square_m} \int_{\square_m} |u(x) - u(y)|^p \sum_{n \leq m} 3^{-n(d+sp)} \sum_{Q \in \mathcal{Q}_n} \mathbb{1}_Q(x) \mathbb{1}_Q(y) dx dy.$$

For distinct $x, y \in \square_m$, write $r = |x - y|_\infty$. The points x and y lie in a common cube of scale n only if $3^n \geq r$, so bounded overlap and a geometric sum bound the displayed kernel above by $C(d)r^{-d-sp}$. Conversely, if $r \leq 3^{m-1}$, choose $n \leq m$ with $3r \leq 3^n < 9r$. The mesh spacing 3^{n-1} ensures that x, y lie in a common $Q \in \mathcal{Q}_n$. For $r > 3^{m-1}$, use the top cube $\square_m$ instead. The kernel is therefore bounded below by $9^{-d-sp}r^{-d-sp}$. Taking the p th root proves (C.3), with a constant independent of s and p . $\square$

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